



(12) **United States Patent**
Garner et al.

(10) **Patent No.:** **US 12,402,619 B2**
(45) **Date of Patent:** ***Sep. 2, 2025**

- (54) **PERFORMING MULTIPLE ACTIONS ON PLANT OBJECTS ON A MOVING PLATFORM**
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- (72) Inventors: **Curtis Dale Garner**, Modesto, CA (US); **Gabriel Thurston Sibley**, Menlo Park, CA (US); **Patrick Christopher Leger**, Belmont, CA (US); **Wisit Jirattigalachote**, Palo Alto, CA (US)
- (73) Assignee: **Verdant Robotics, Inc.**, Hayward, CA (US)
- (*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 53 days.
- This patent is subject to a terminal disclaimer.

(21) Appl. No.: **18/497,889**

(22) Filed: **Oct. 30, 2023**

(65) **Prior Publication Data**
US 2024/0057580 A1 Feb. 22, 2024

Related U.S. Application Data

- (63) Continuation of application No. 17/722,285, filed on Apr. 15, 2022, now Pat. No. 11,800,862, which is a continuation of application No. 17/544,152, filed on Dec. 7, 2021, now Pat. No. 11,844,343, which is a continuation of application No. 17/073,244, filed on Oct. 16, 2020, now Pat. No. 11,694,434.
- (51) **Int. Cl.**
A01M 7/00 (2006.01)
A01C 23/00 (2006.01)

- A01C 23/04** (2006.01)
A01M 9/00 (2006.01)
- (52) **U.S. Cl.**
CPC **A01M 7/0089** (2013.01); **A01C 23/007** (2013.01); **A01C 23/04** (2013.01); **A01M 9/0092** (2013.01); **A01M 7/0032** (2013.01)
- (58) **Field of Classification Search**
CPC A01M 7/0089; A01M 9/0092; A01M 7/0032; A01C 23/007; A01C 23/04
See application file for complete search history.

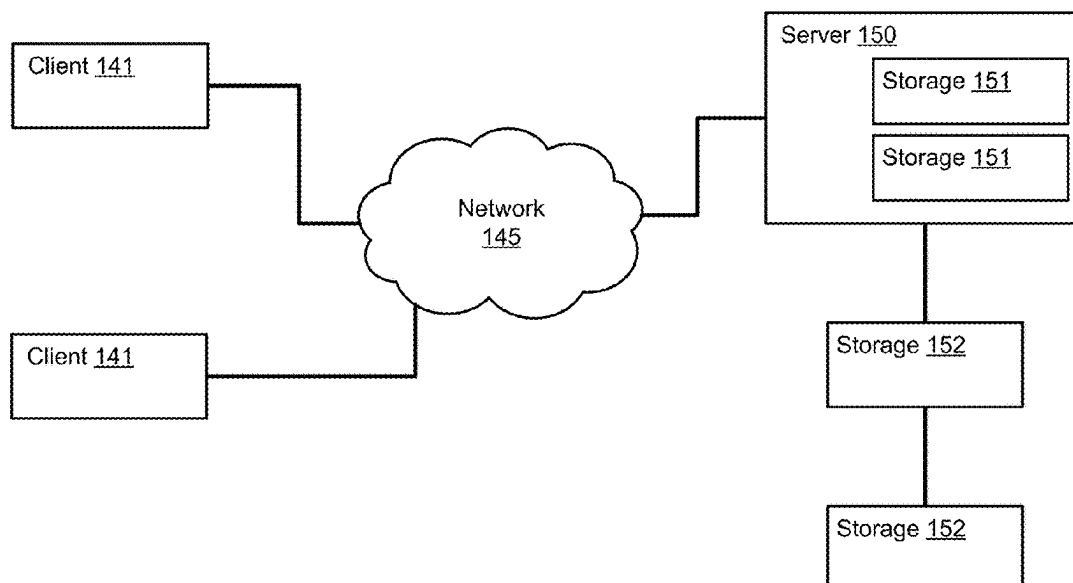
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Primary Examiner — Qingzhang Zhou
(74) *Attorney, Agent, or Firm* — Cognition IP, P.C.;
Edward Steakley

- (57) **ABSTRACT**
Various embodiments of methods, systems, and computer program products described herein are directed to a treatment system. An example method includes identifying a type of a first target agricultural object, based on the identified type of the first target agricultural object, determining a first spray profile for treating the first target agricultural object, sending instructions of a first treatment parameter to a treatment unit, and based on the first spray profile, activating the treatment unit to emit a first fluid at the first target agricultural object, wherein the first fluid regulator is moved to an opened position for an amount of time such that the first fluid is emitted, via the first spraying tip, as a first fluid projectile at the first target agricultural object for a first spray duration.

20 Claims, 81 Drawing Sheets



(56)

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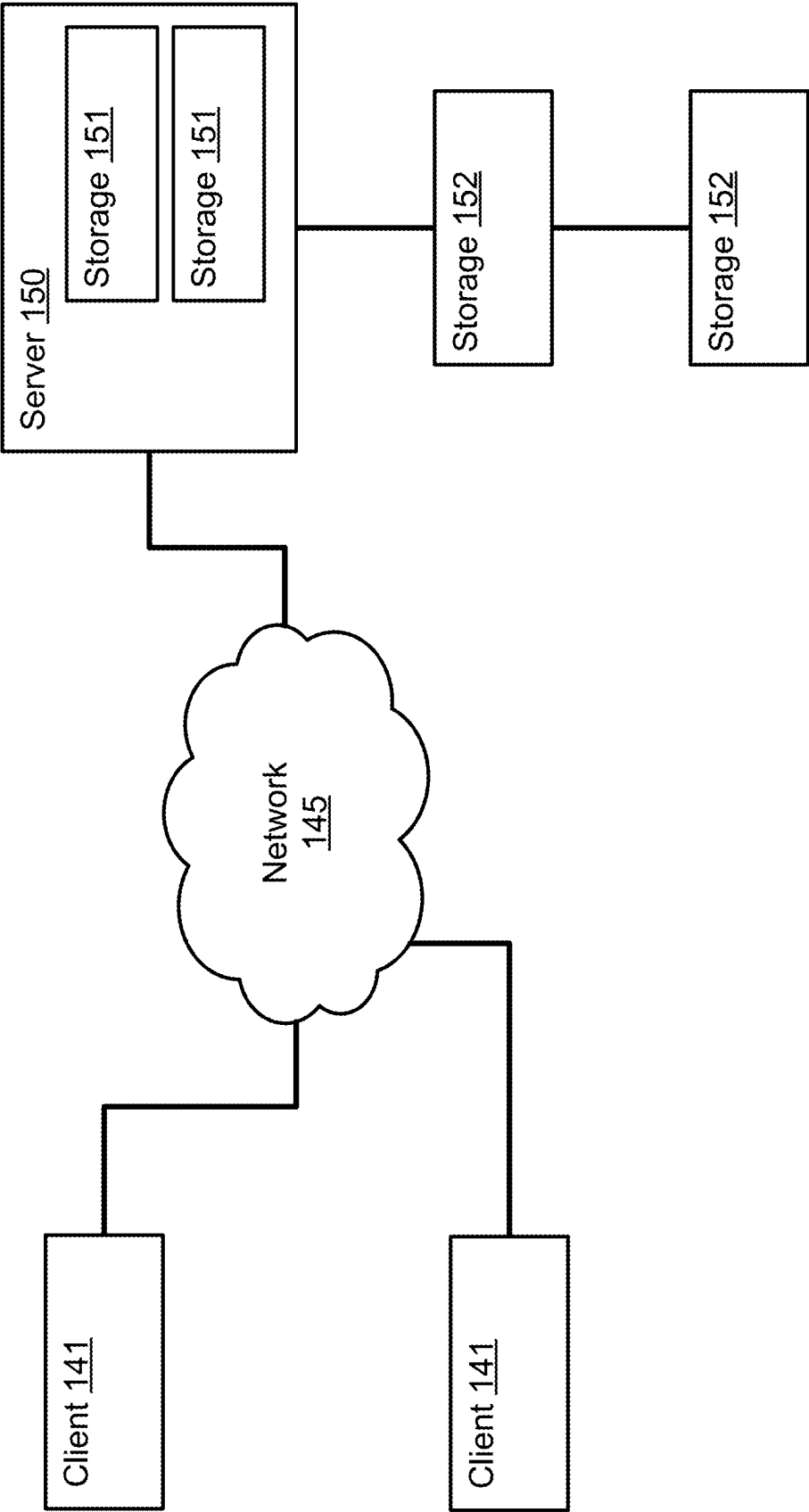


FIG. 1

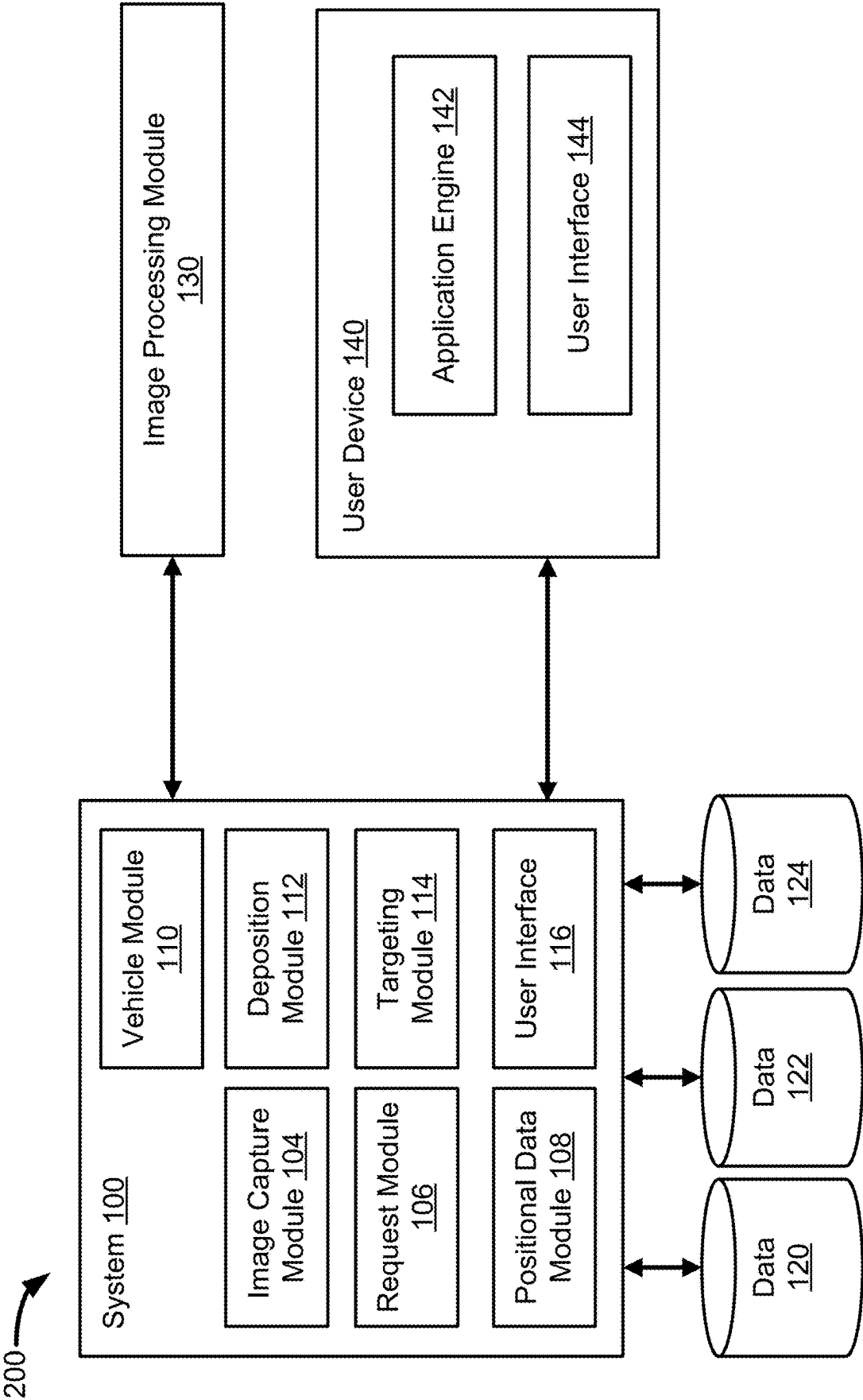


FIG. 2

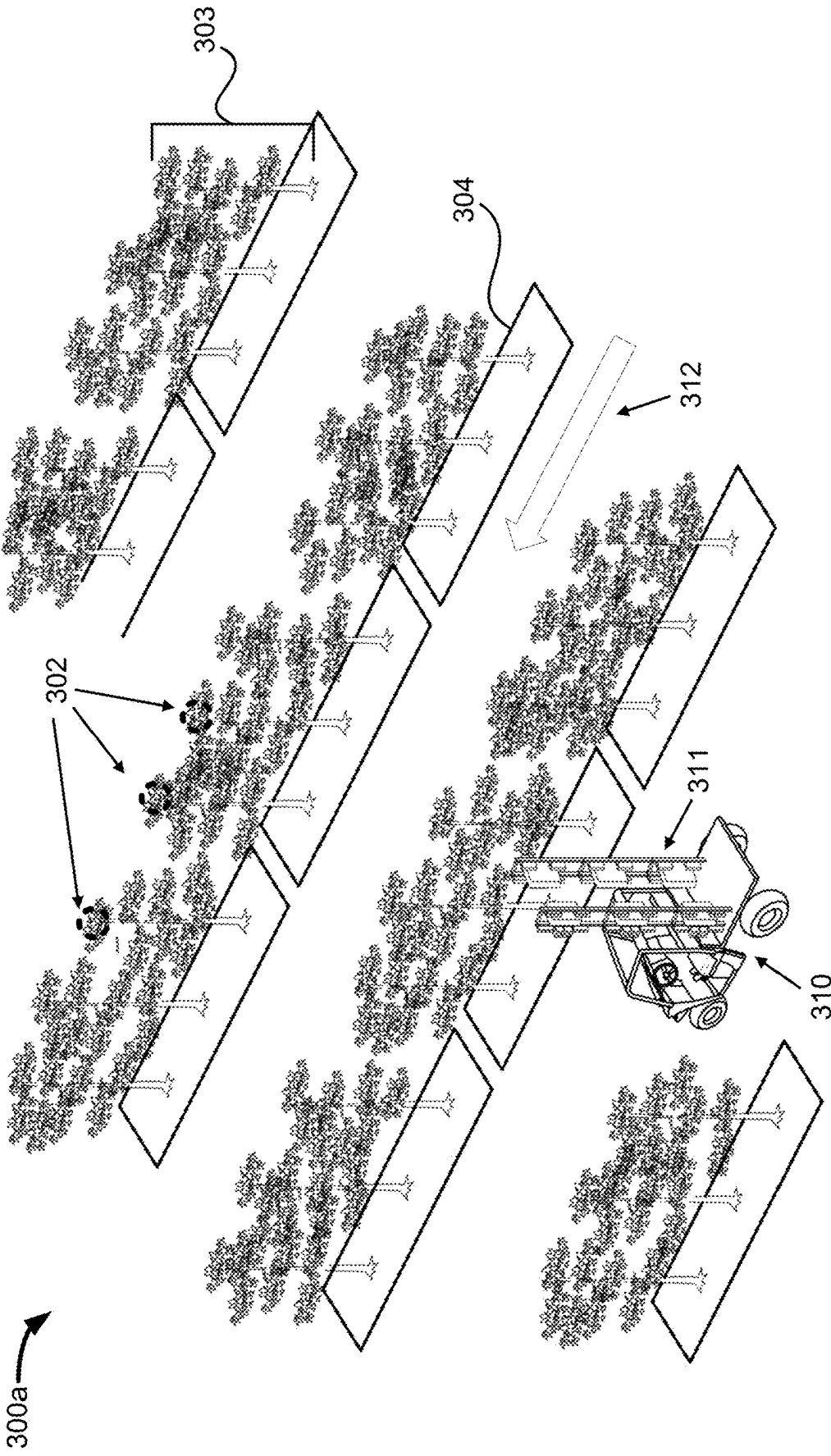


FIG. 3A

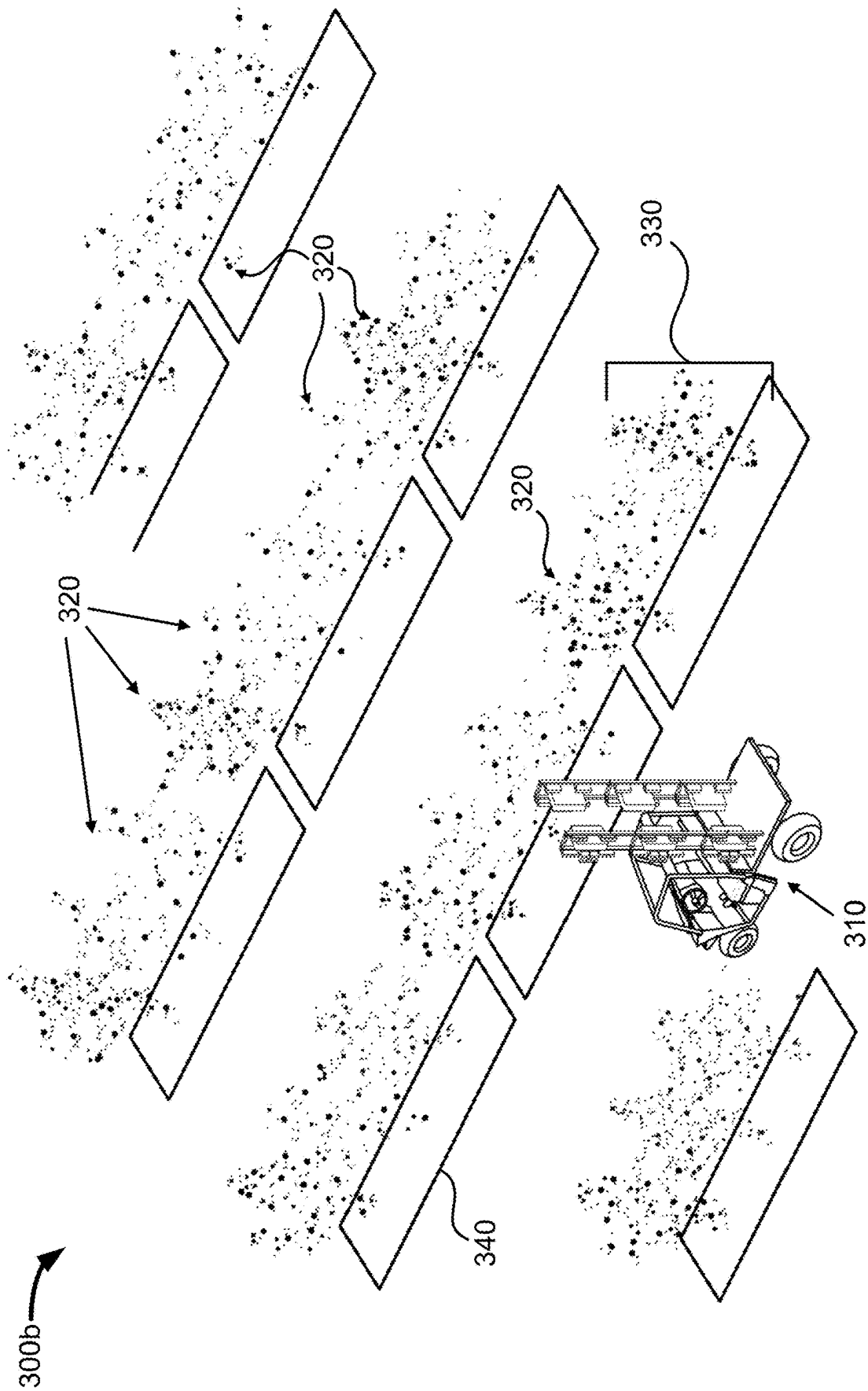


FIG. 3B

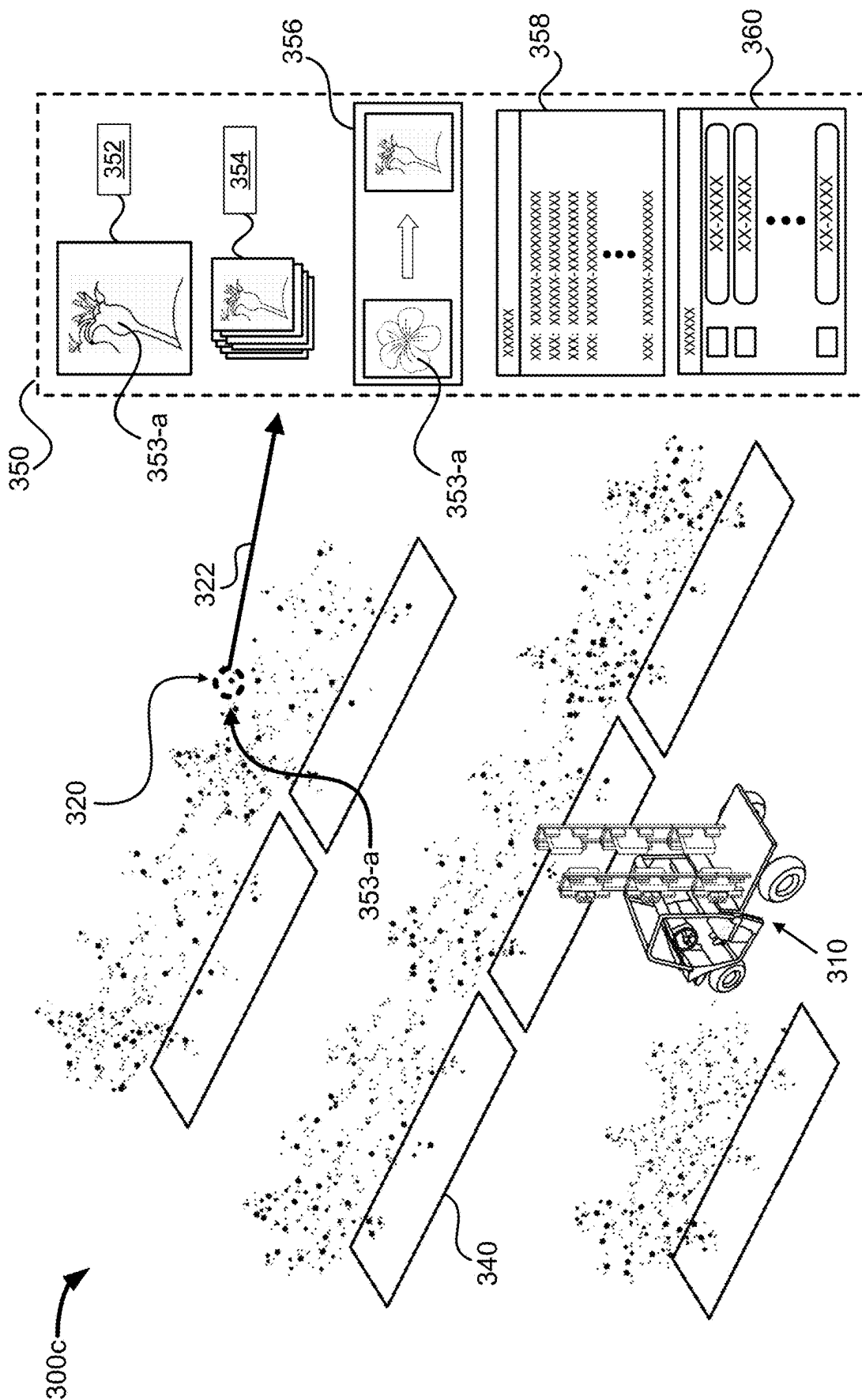
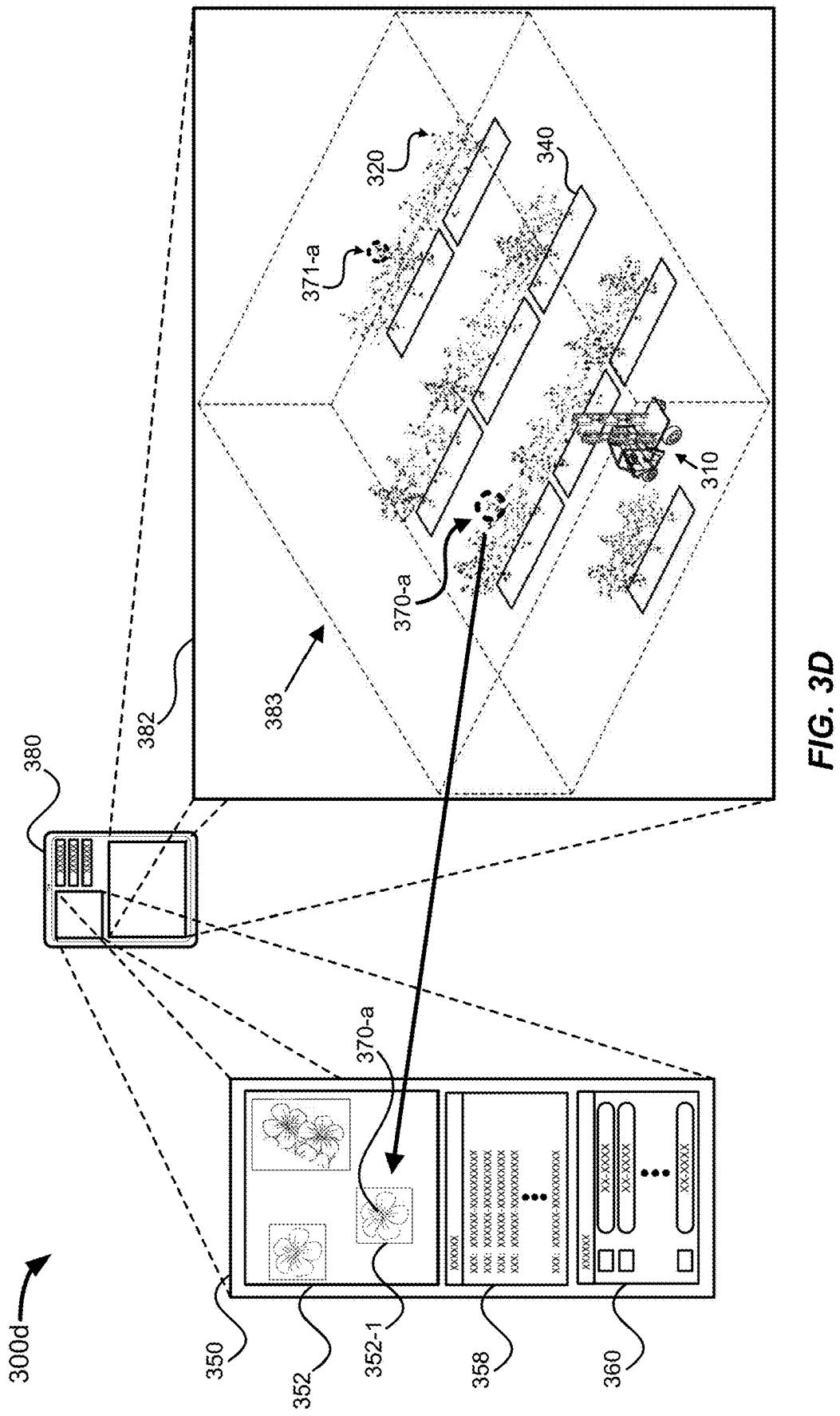
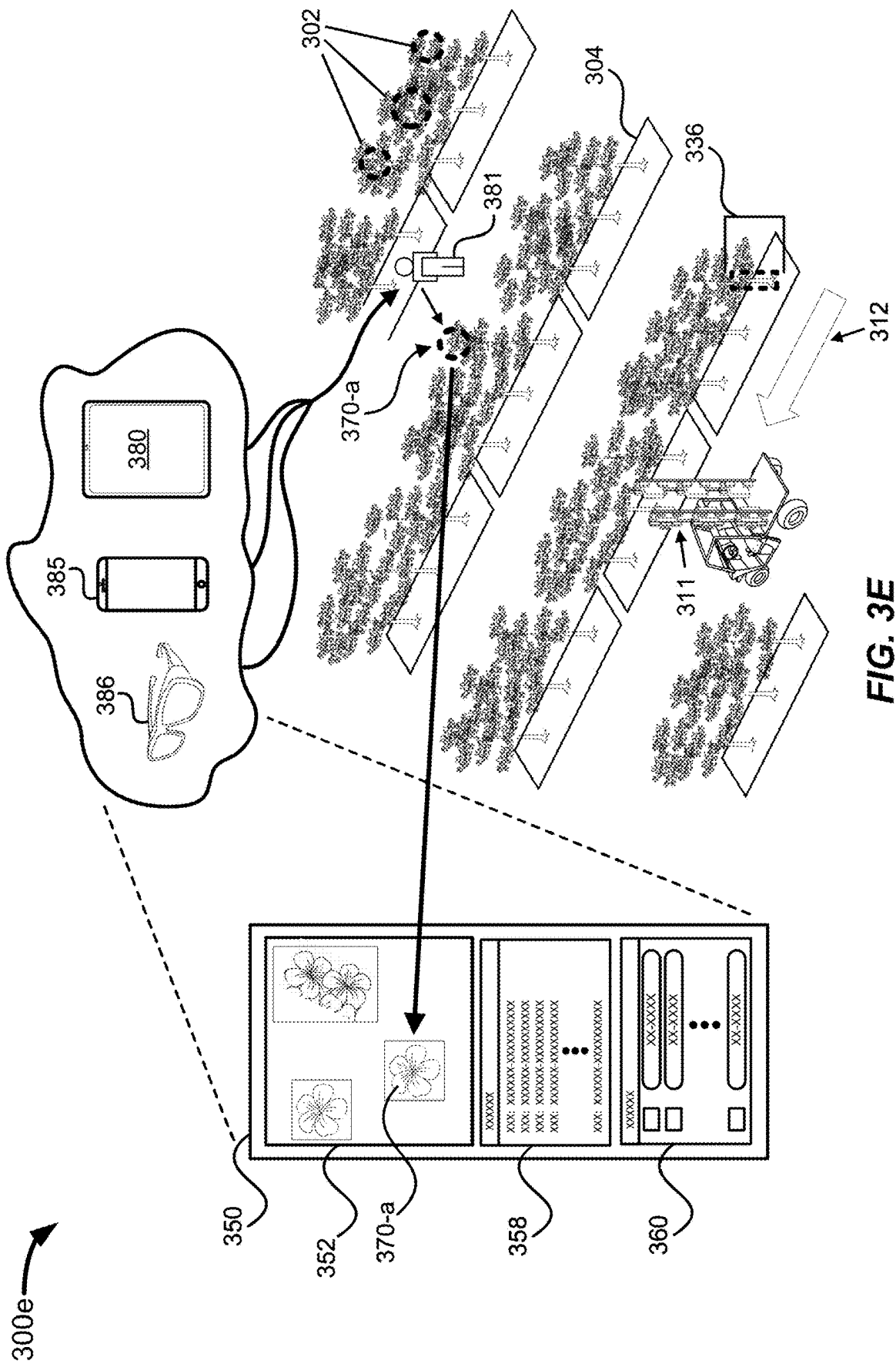


FIG. 3C





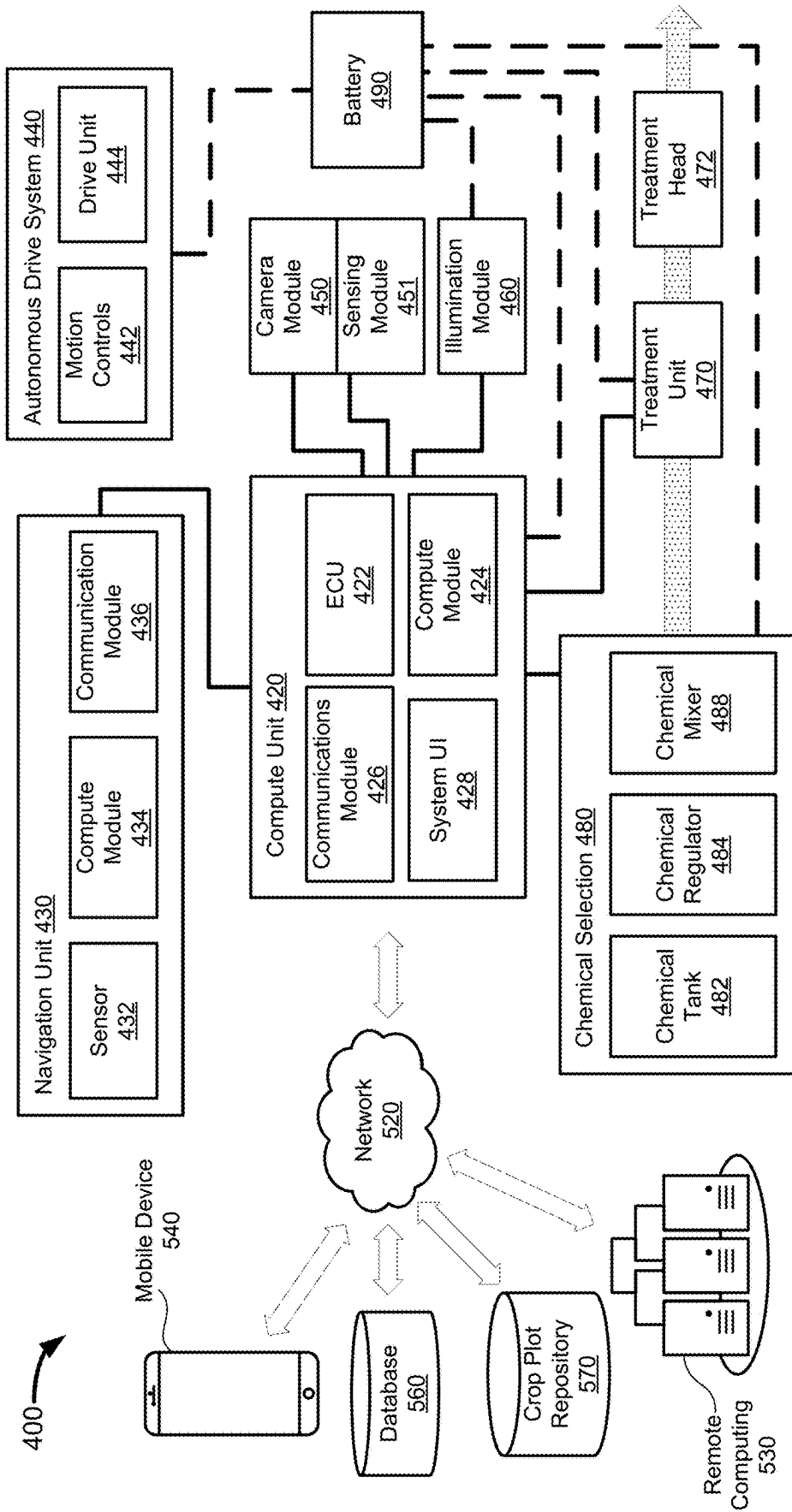


FIG. 4

402

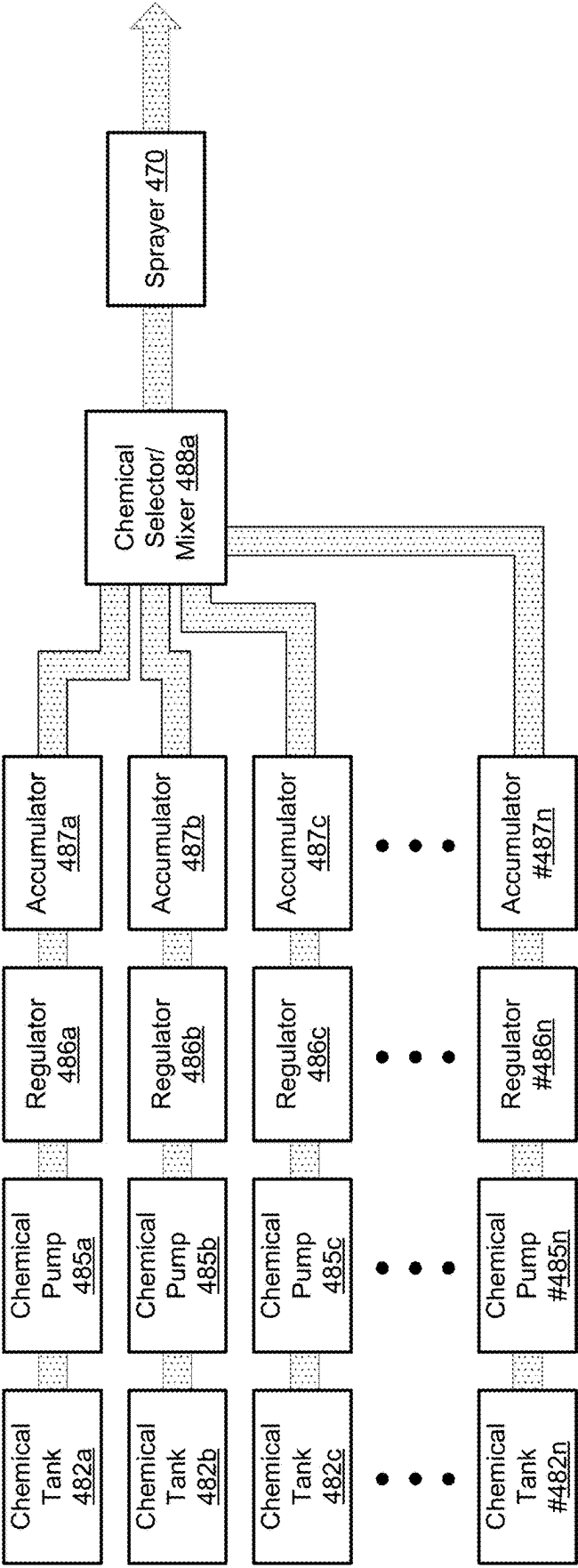
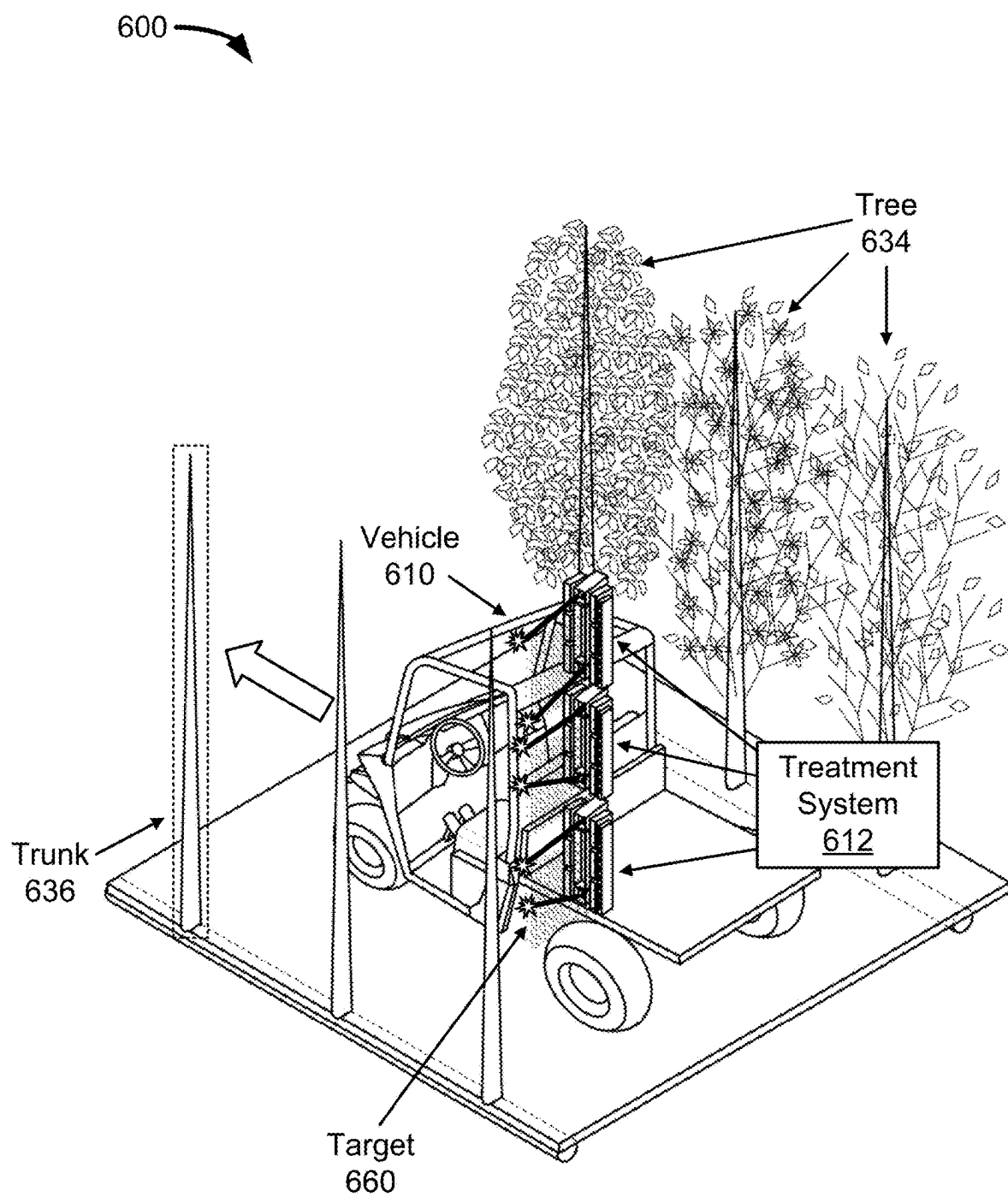


FIG. 5

**FIG. 6**

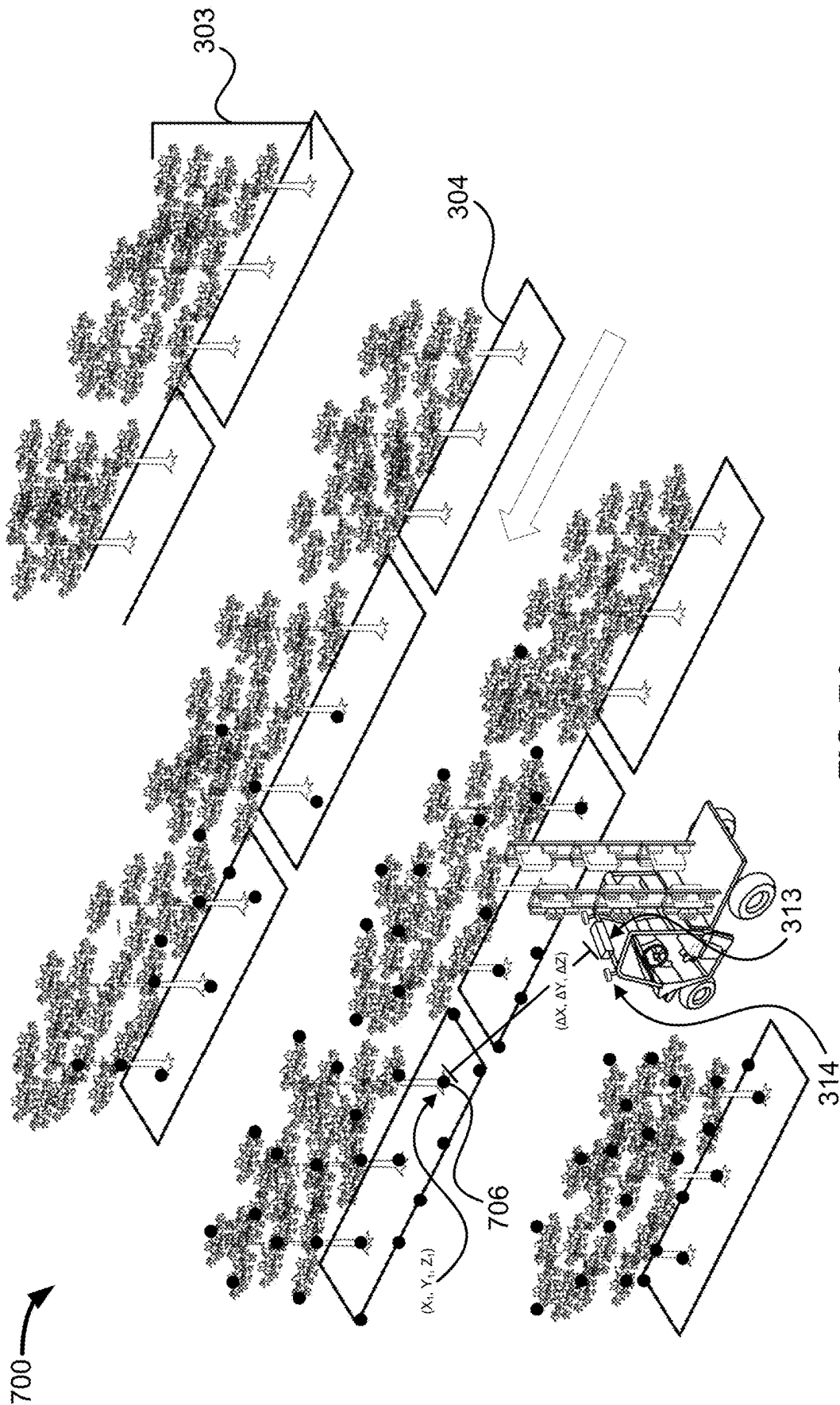
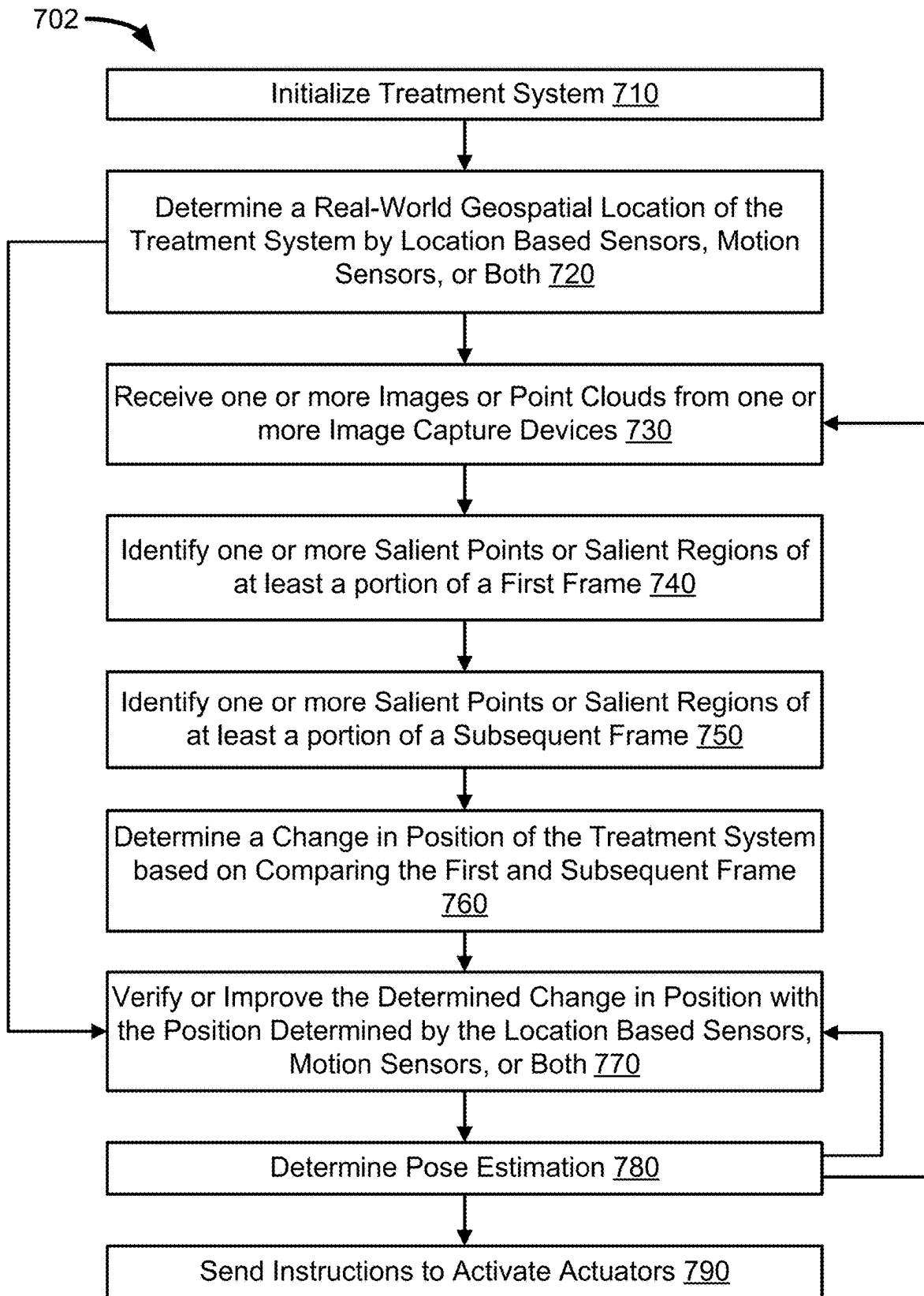
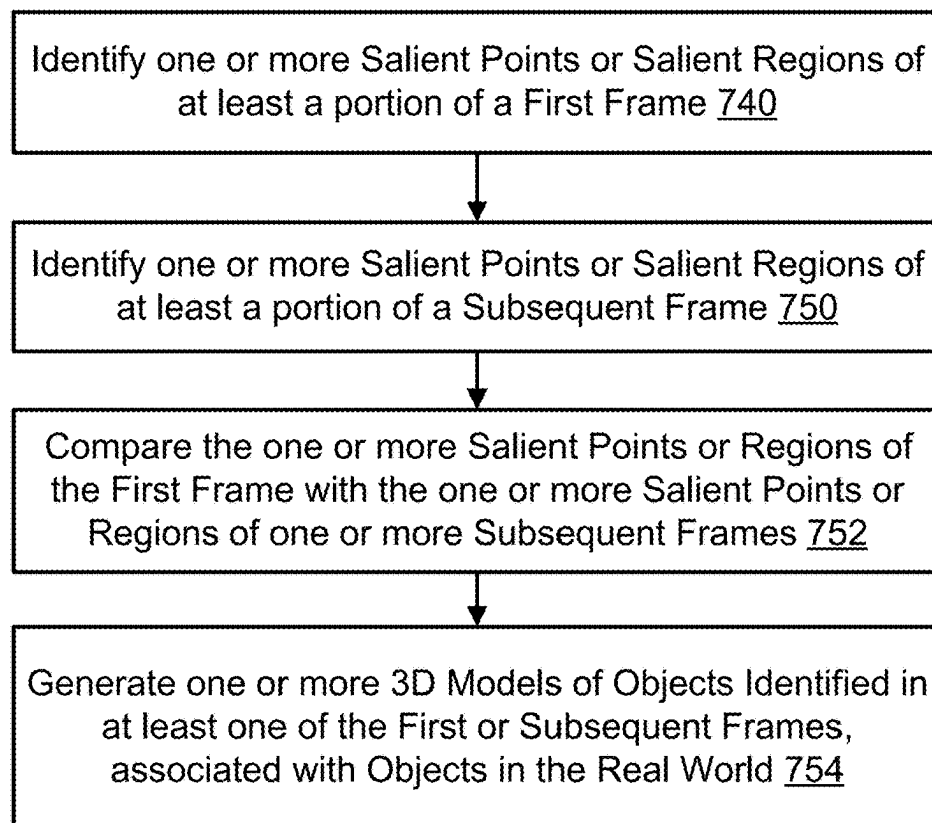
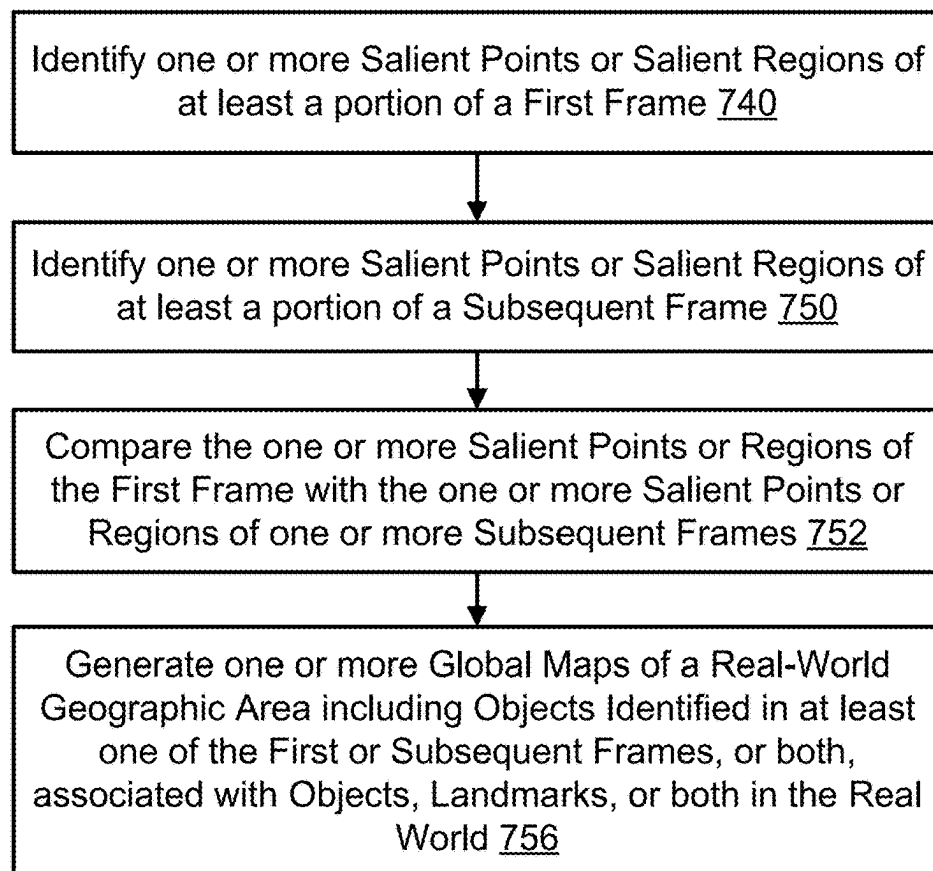


FIG. 7A

**FIG. 7B**

702 **FIG. 7C**

702 **FIG. 7D**

800

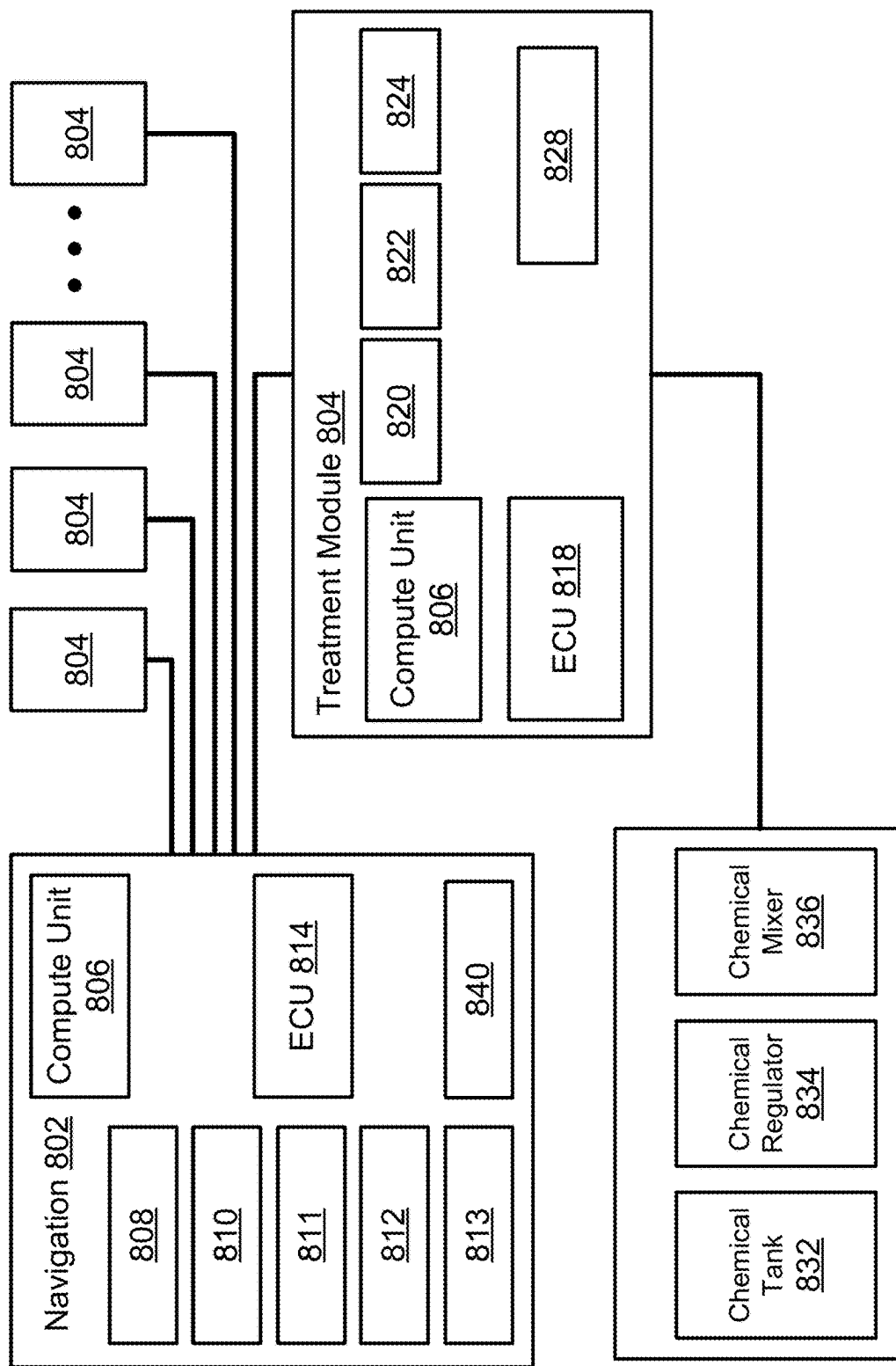


FIG. 8

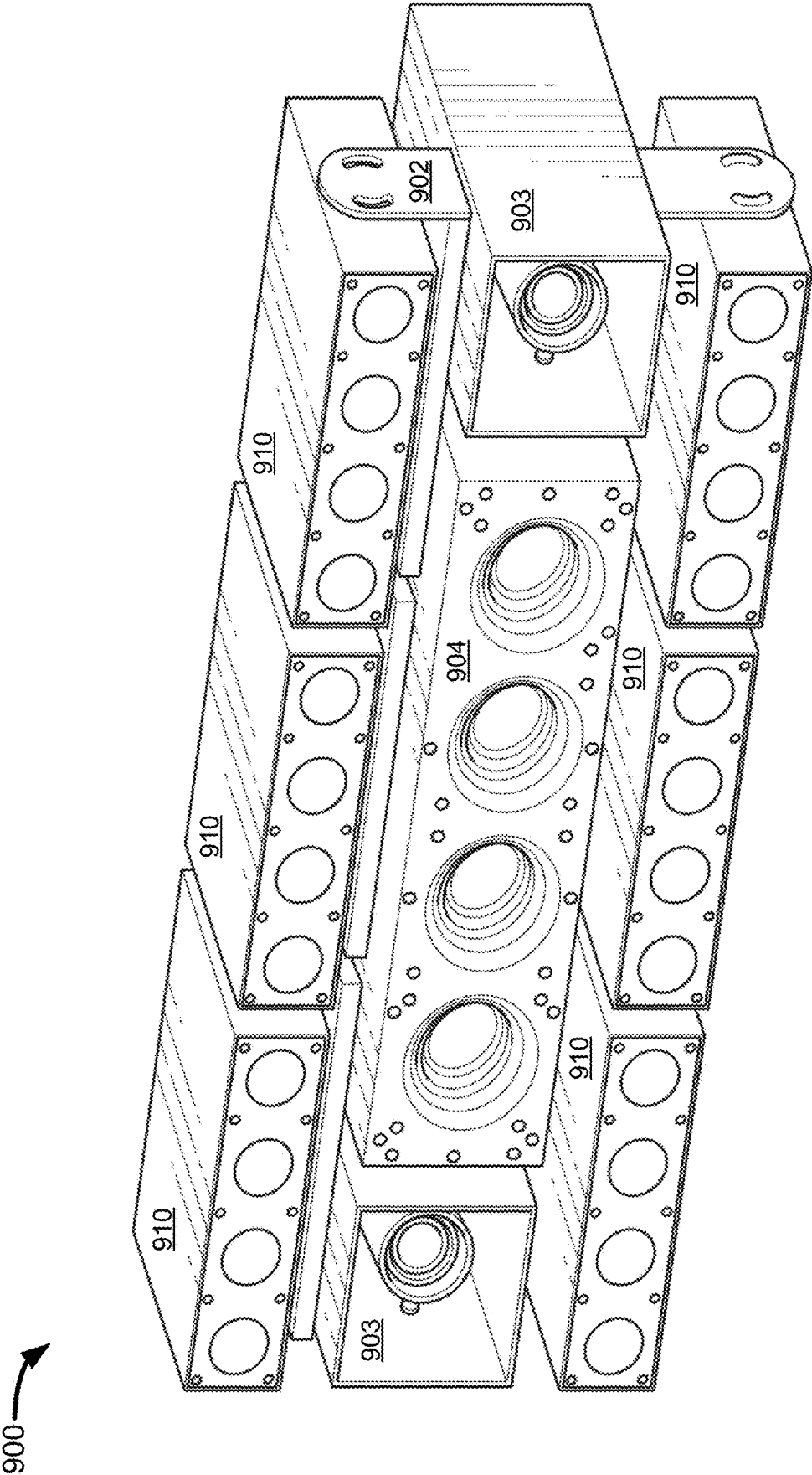


FIG. 9A

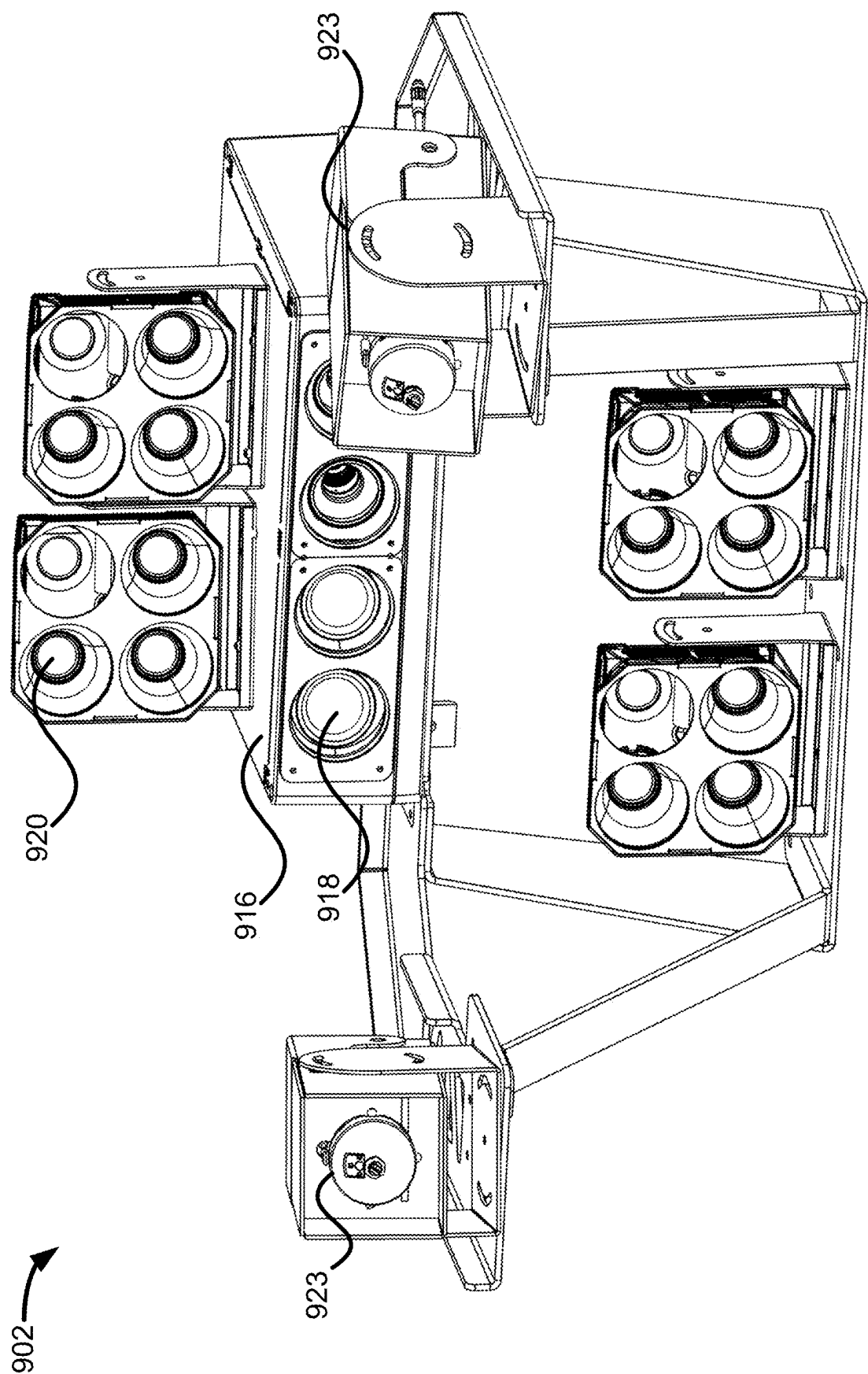
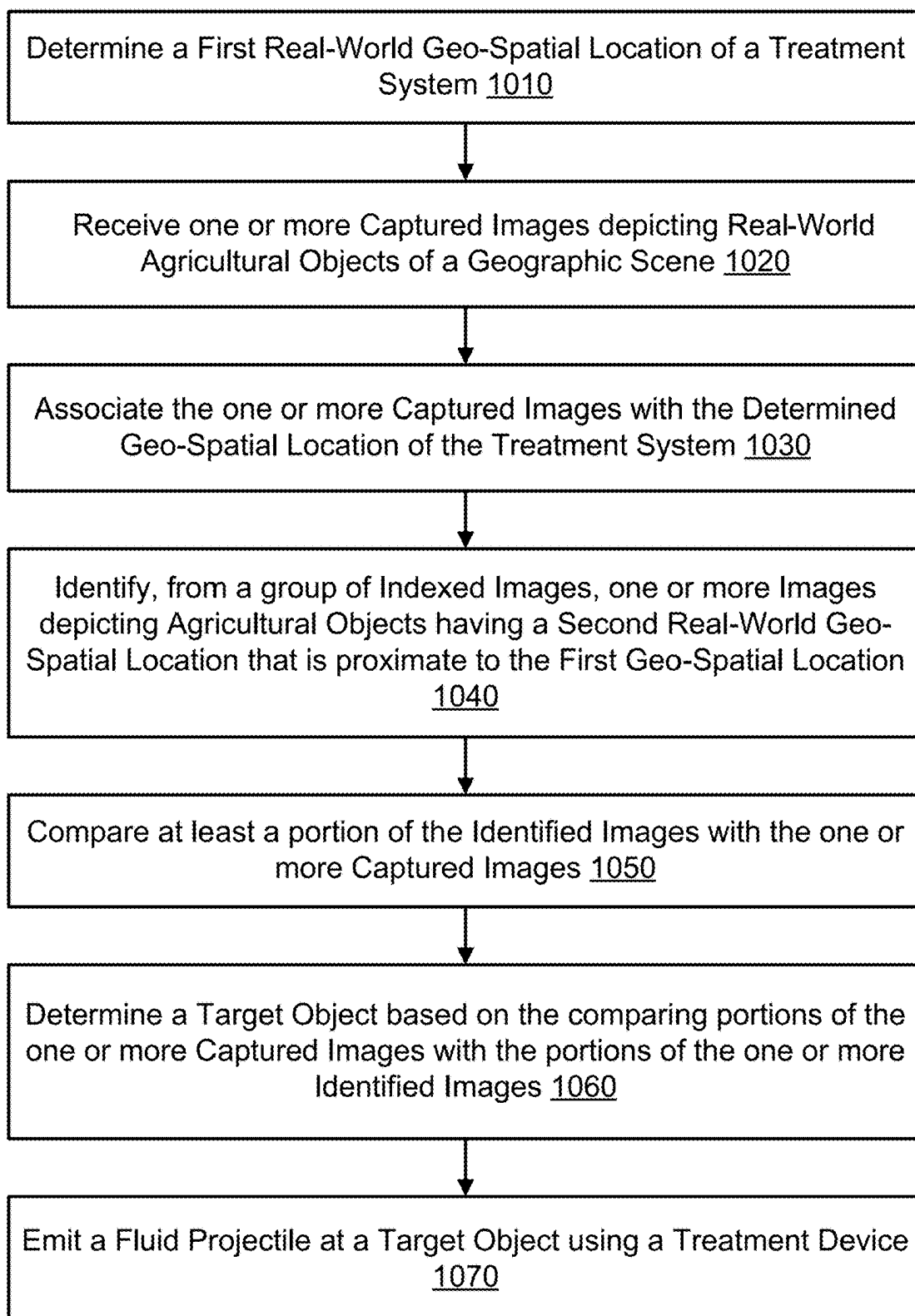
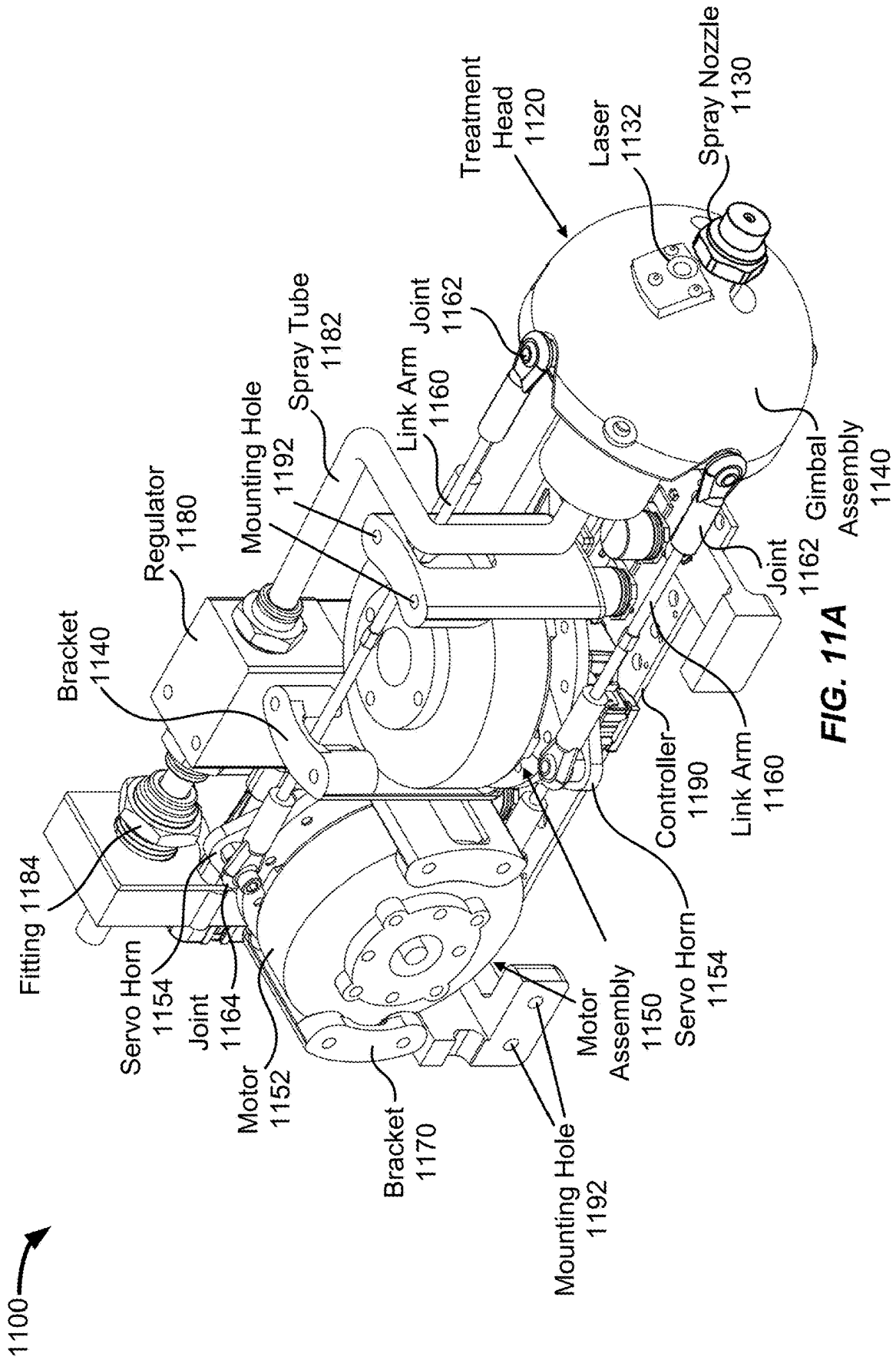


FIG. 9B

1000 **FIG. 10**



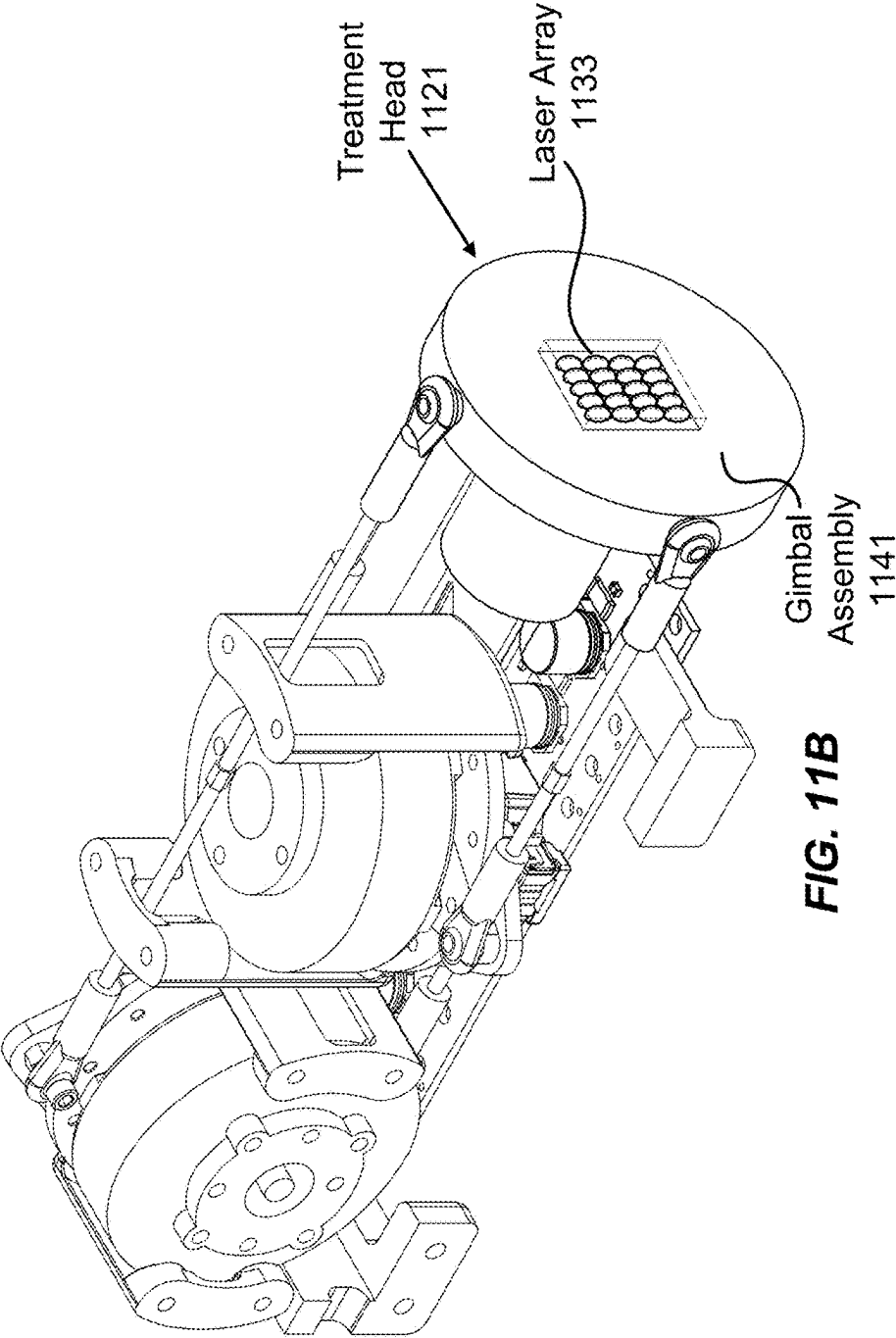


FIG. 11B

1100

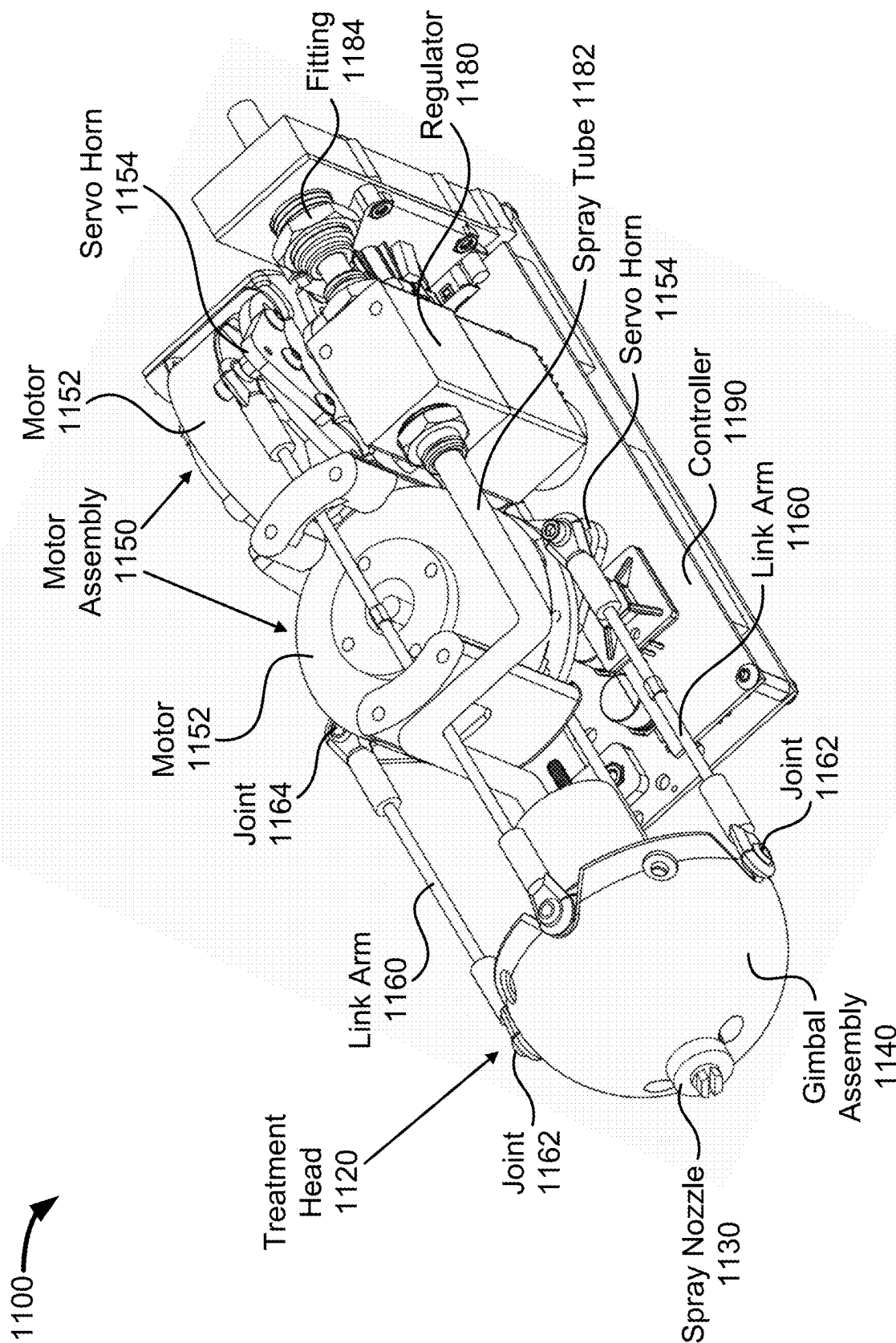
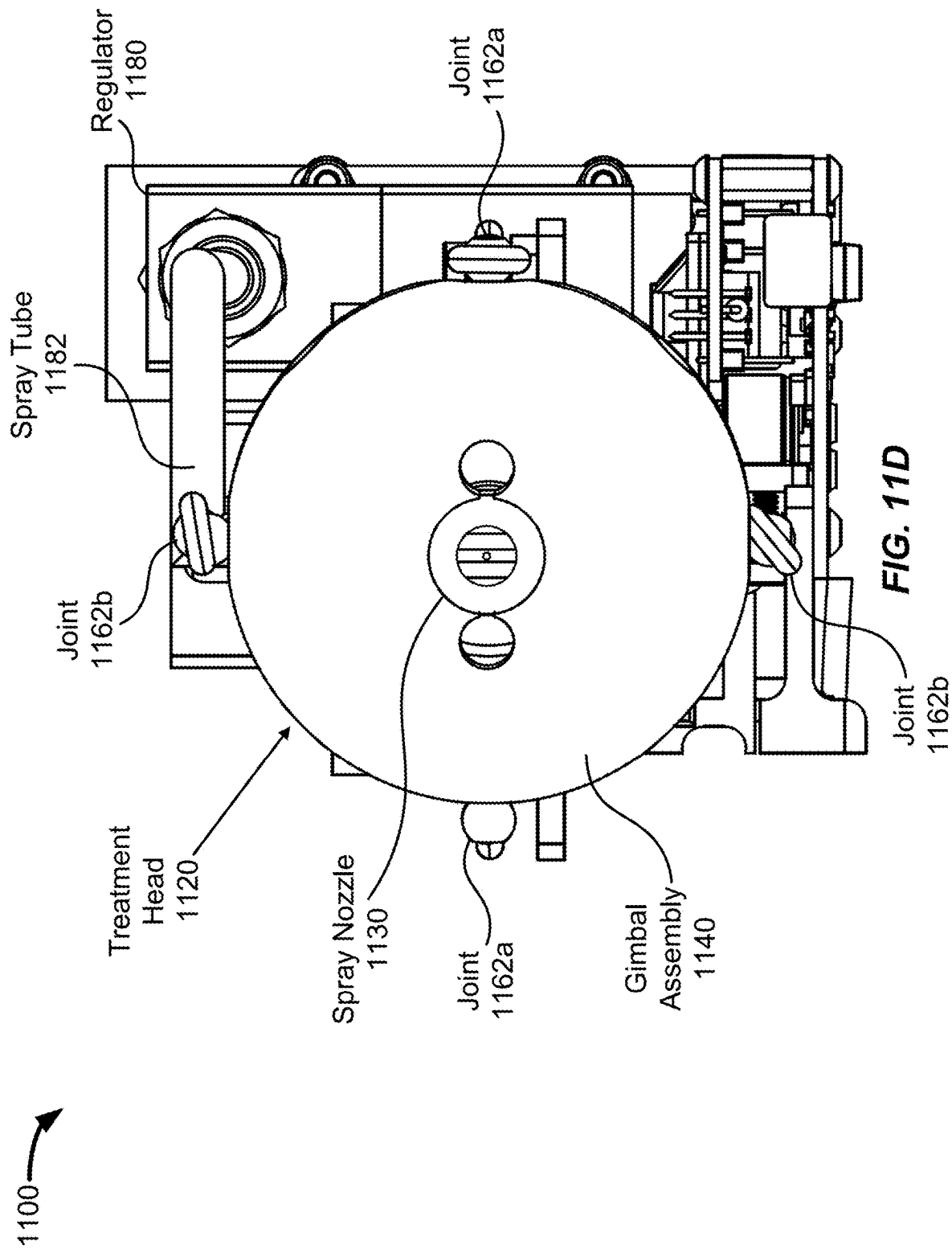


FIG. 11C



1100

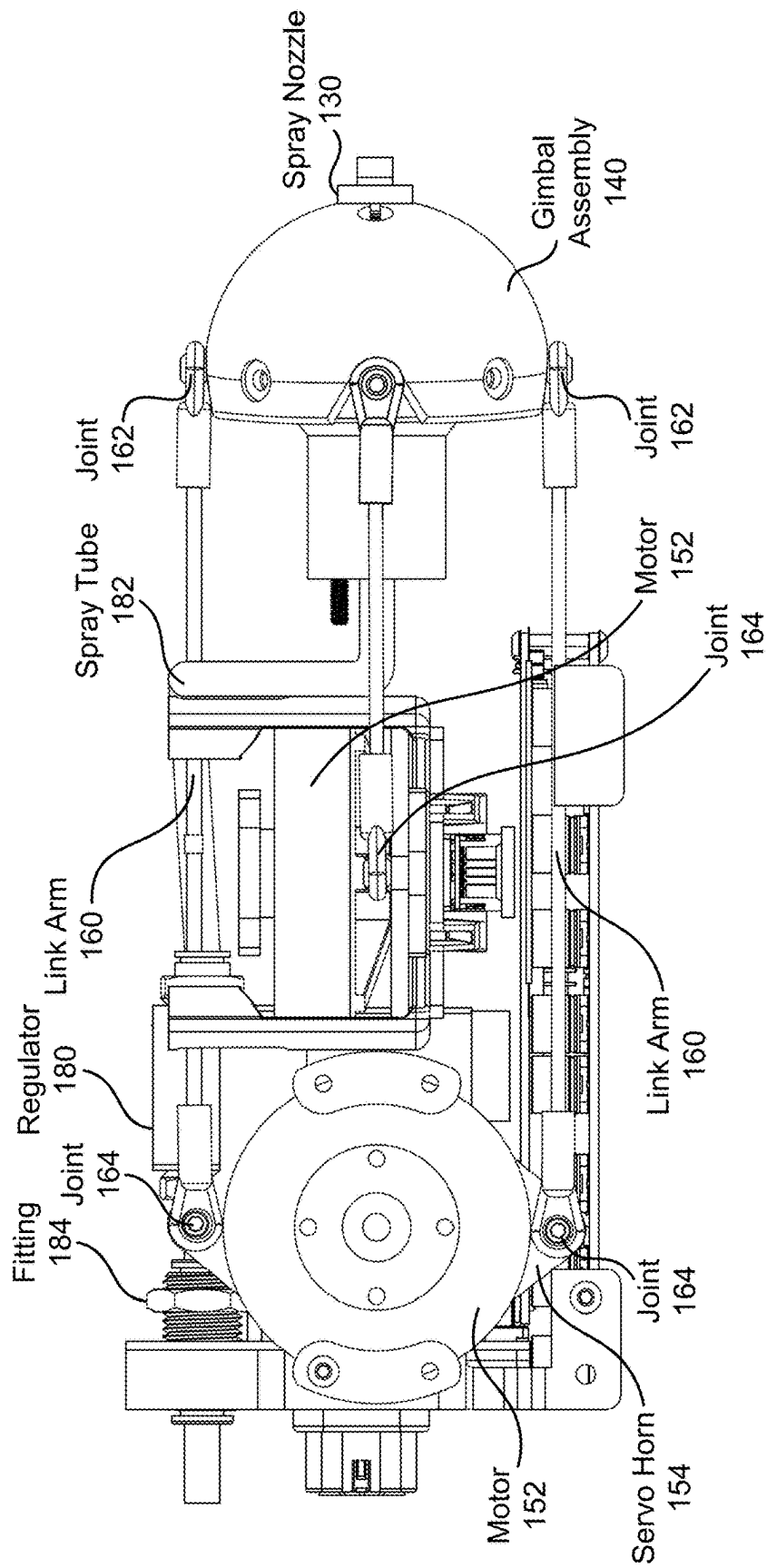


FIG. 11E

1100

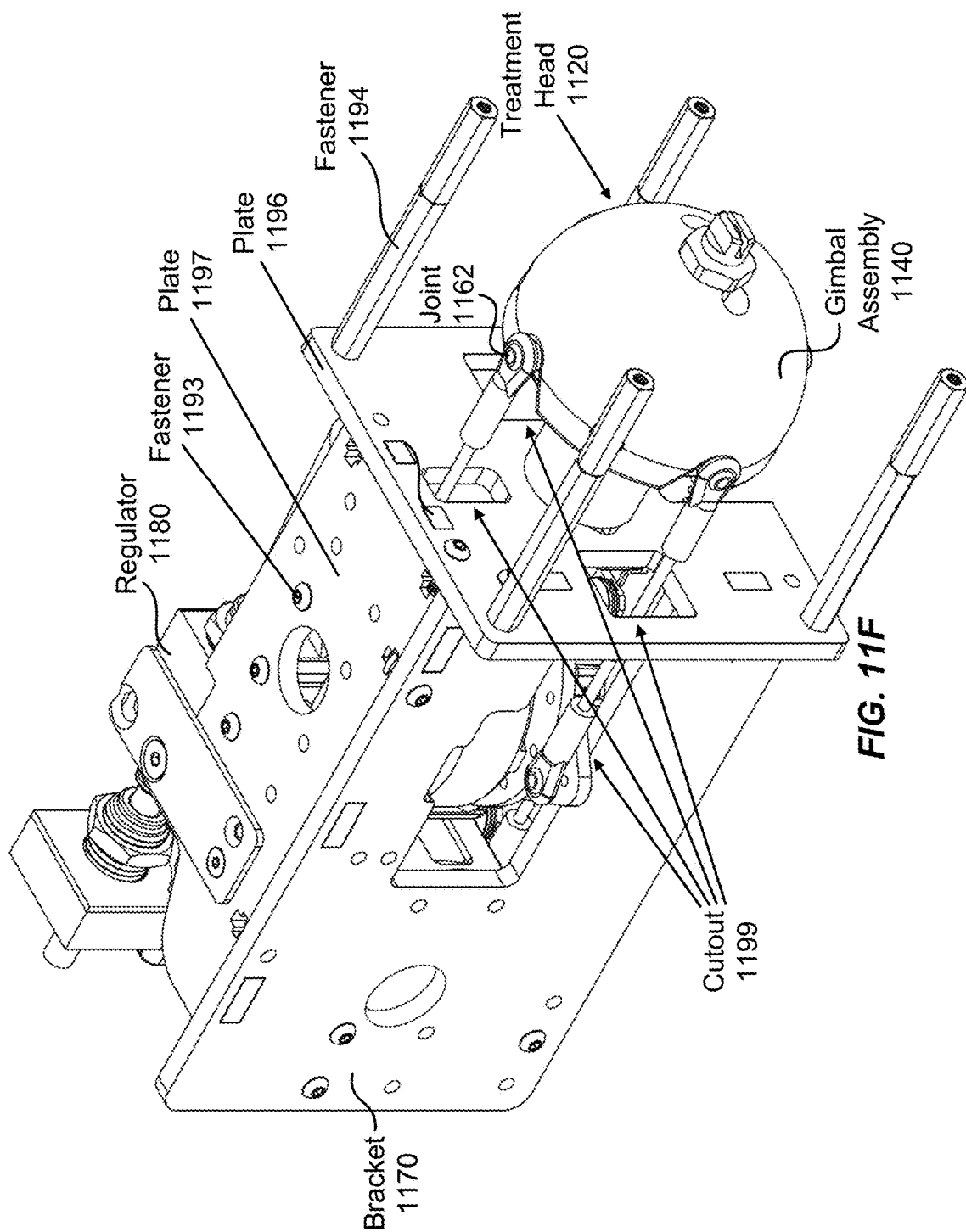
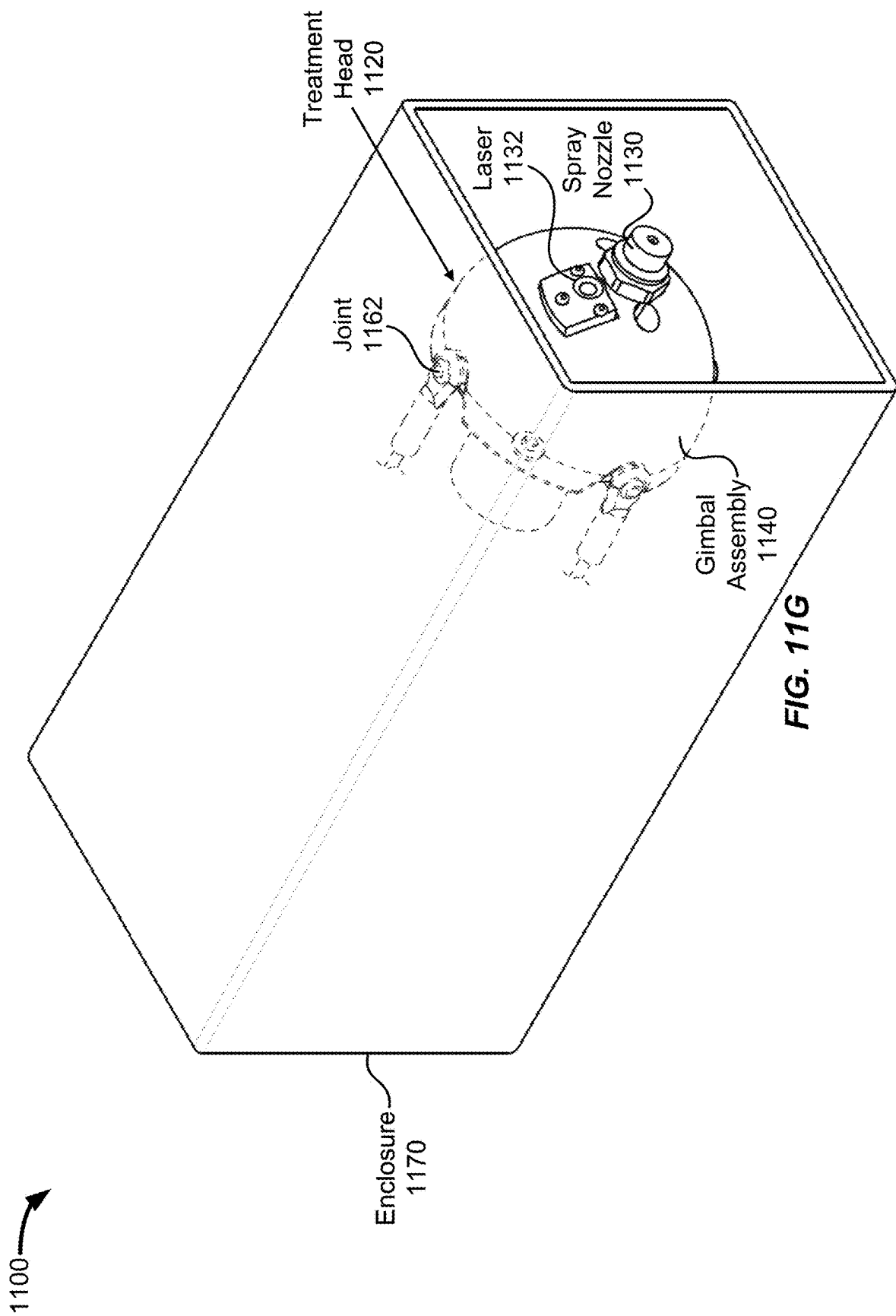
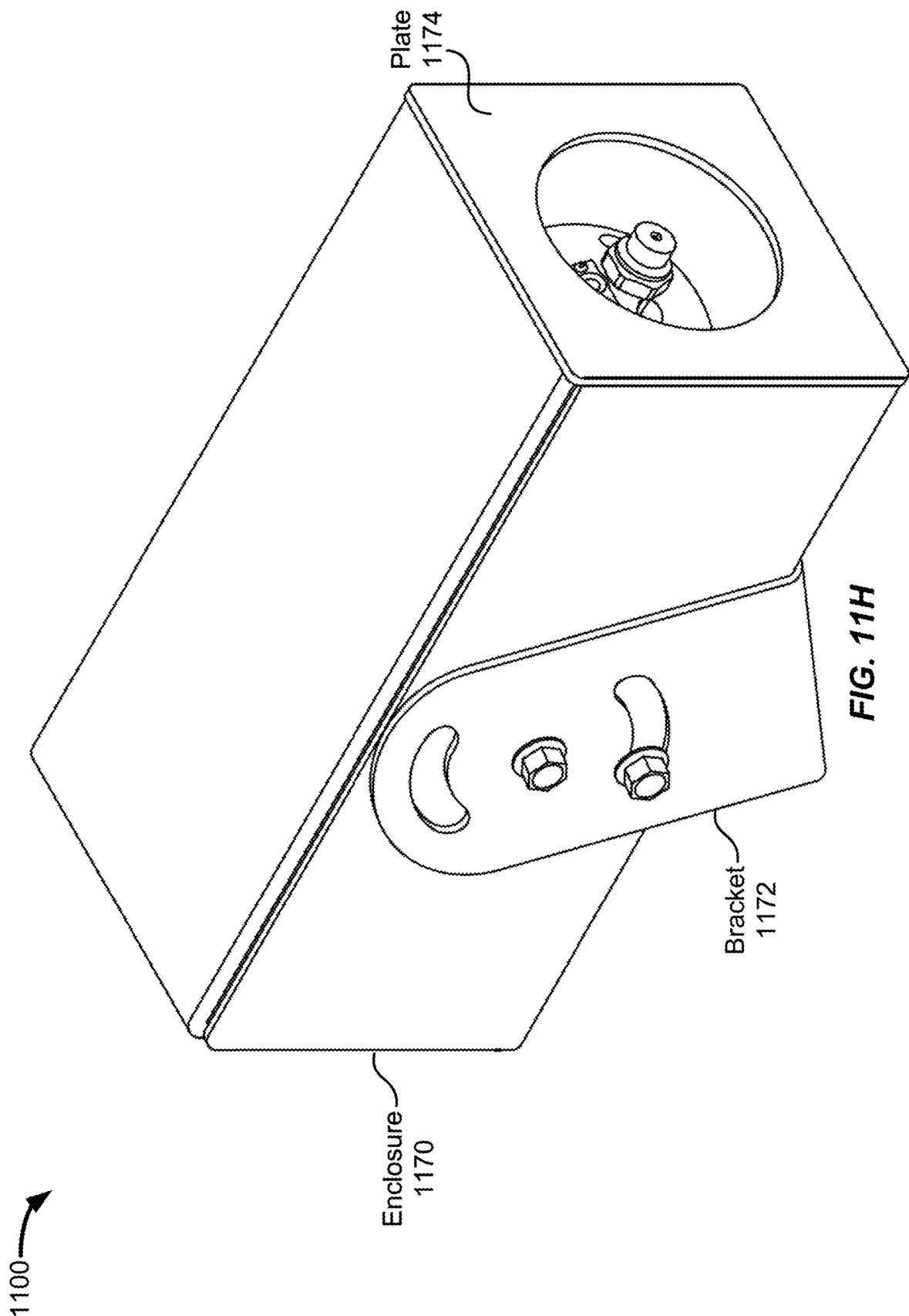
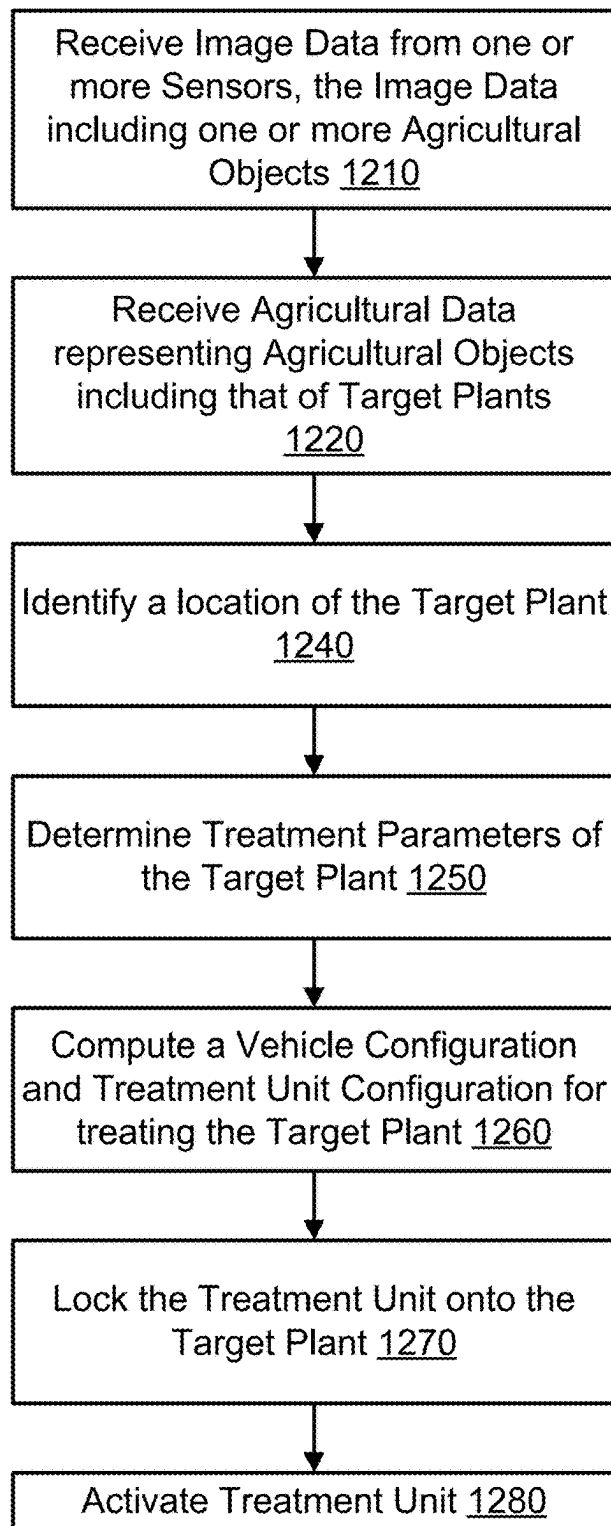
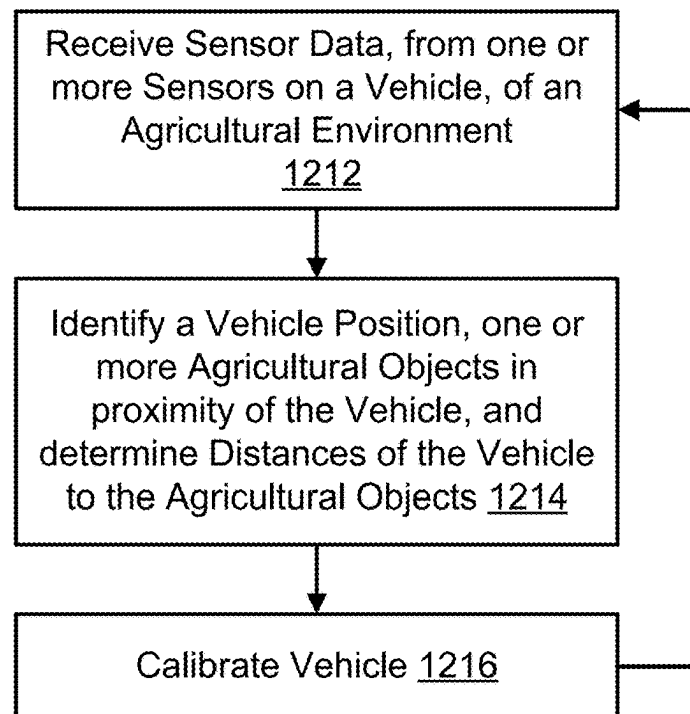


FIG. 11F





1200 **FIG. 12A**

1200 **FIG. 12B**

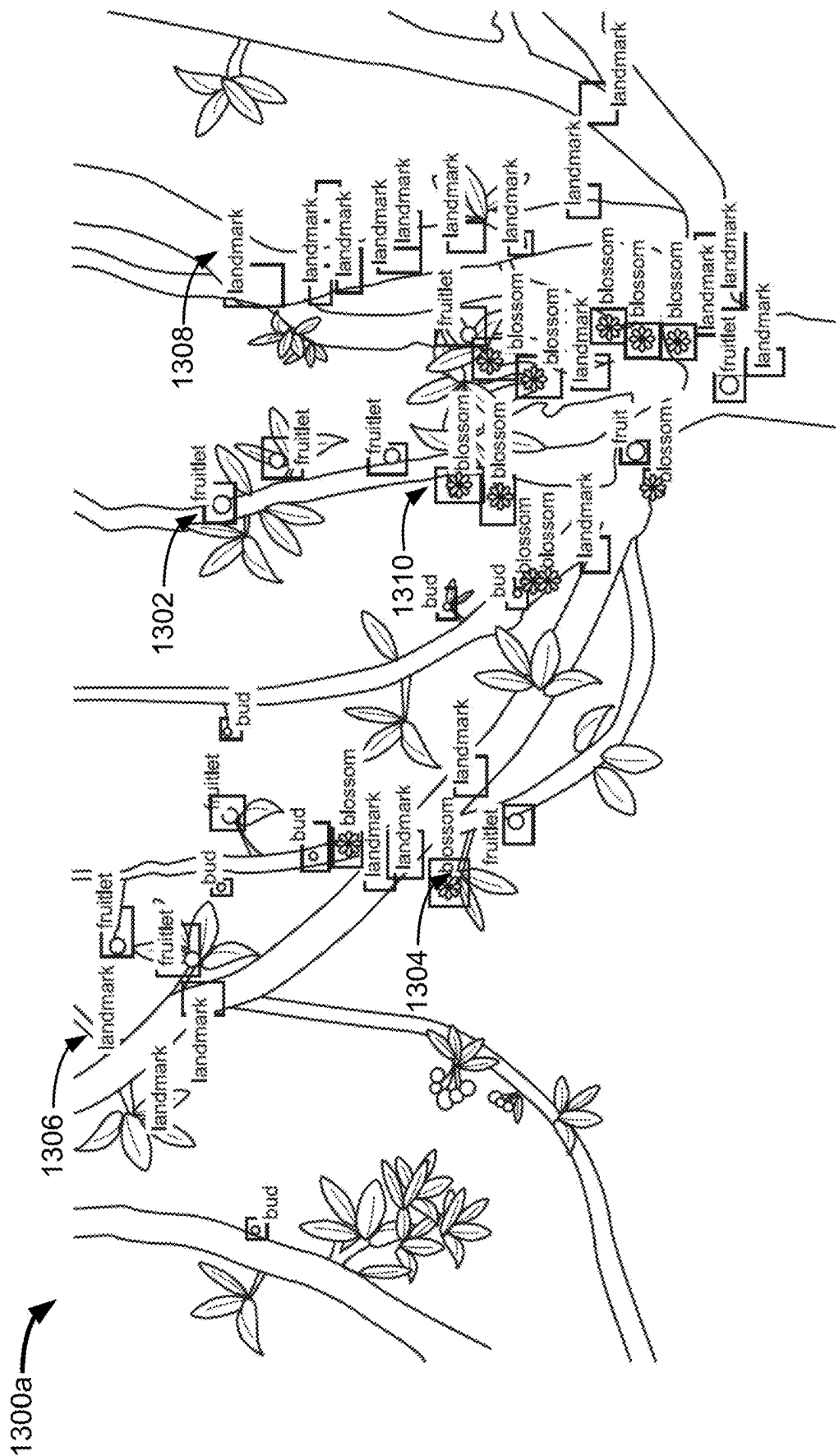


FIG. 13A

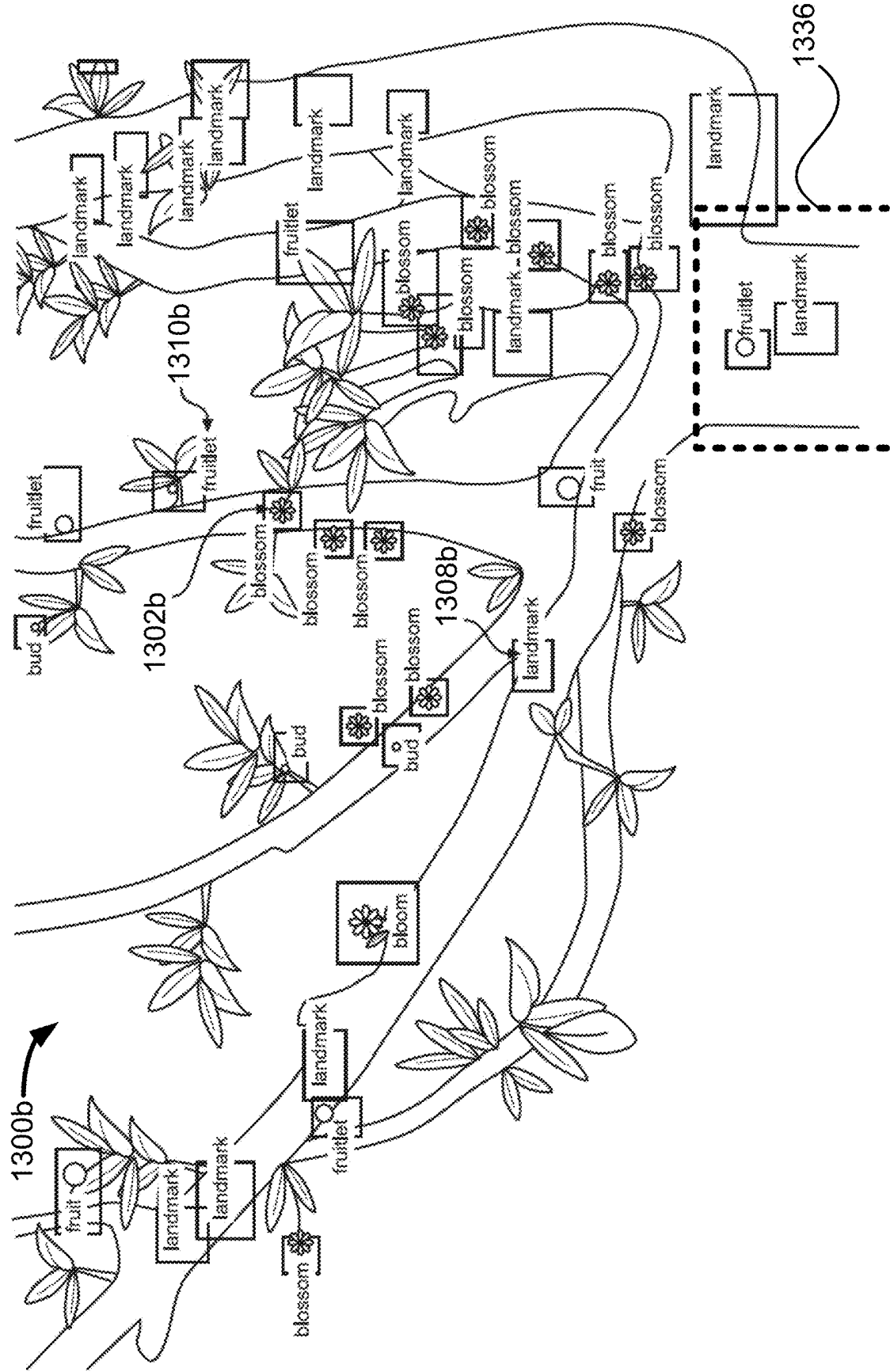
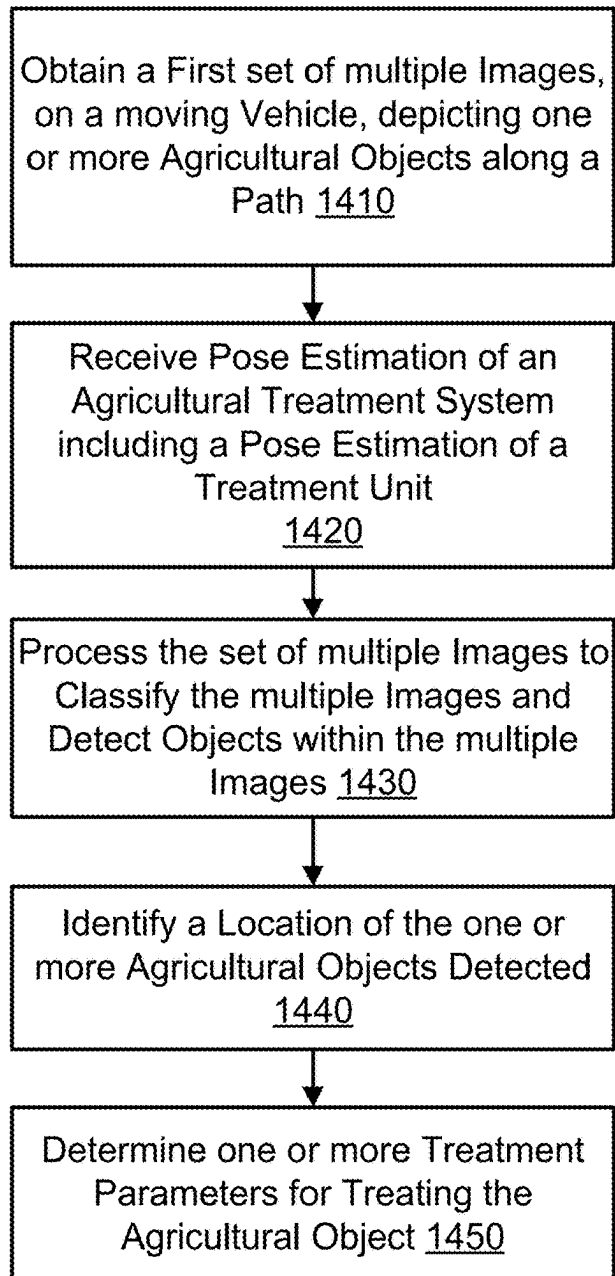


FIG. 13B

1400 **FIG. 14A**

1400 

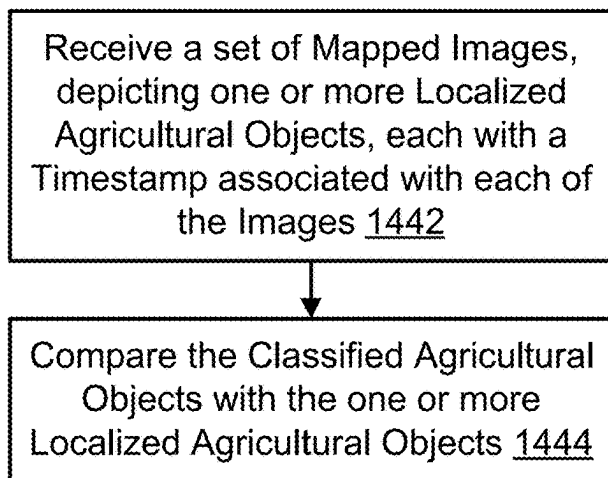
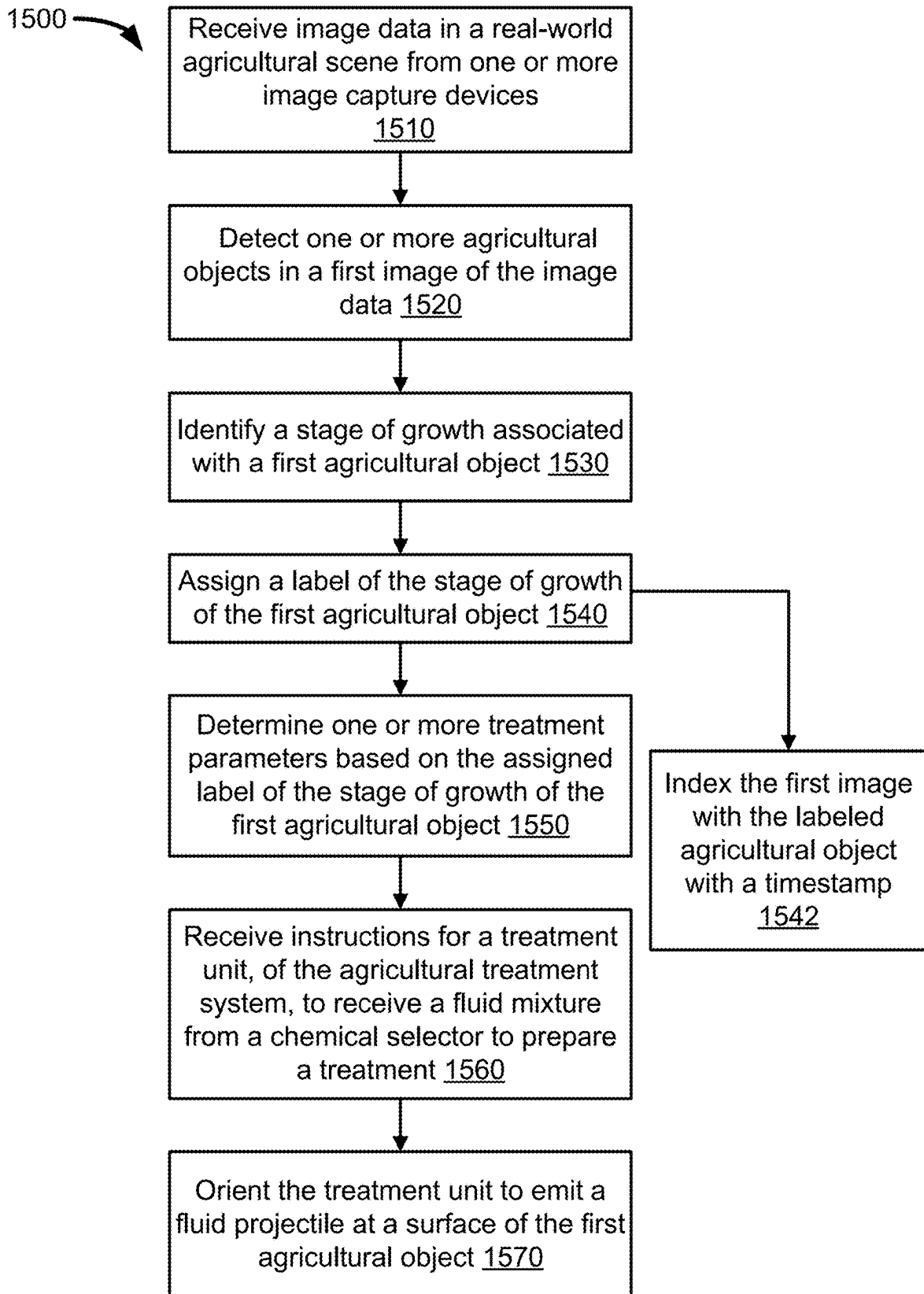


FIG. 14B

**FIG. 15**

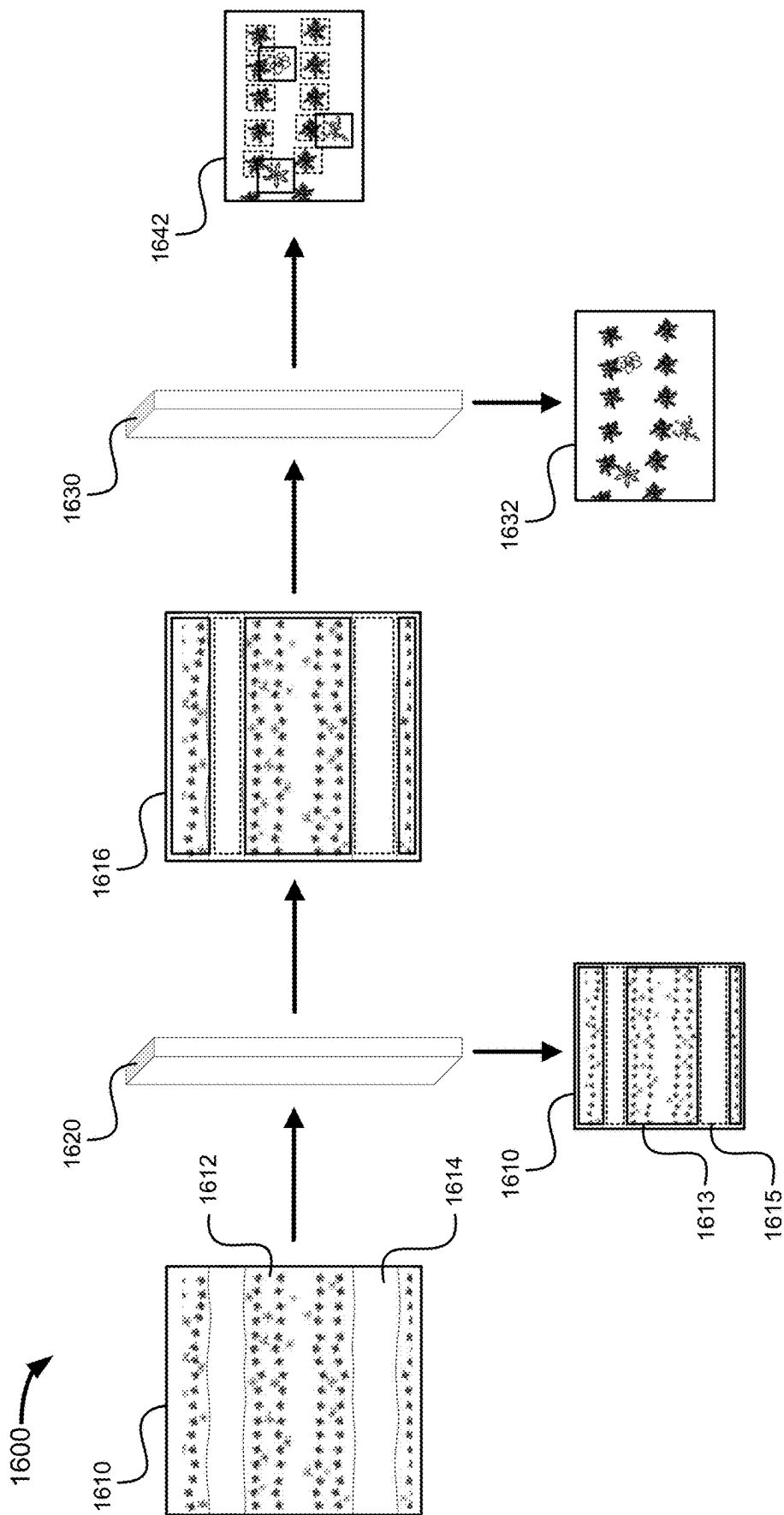
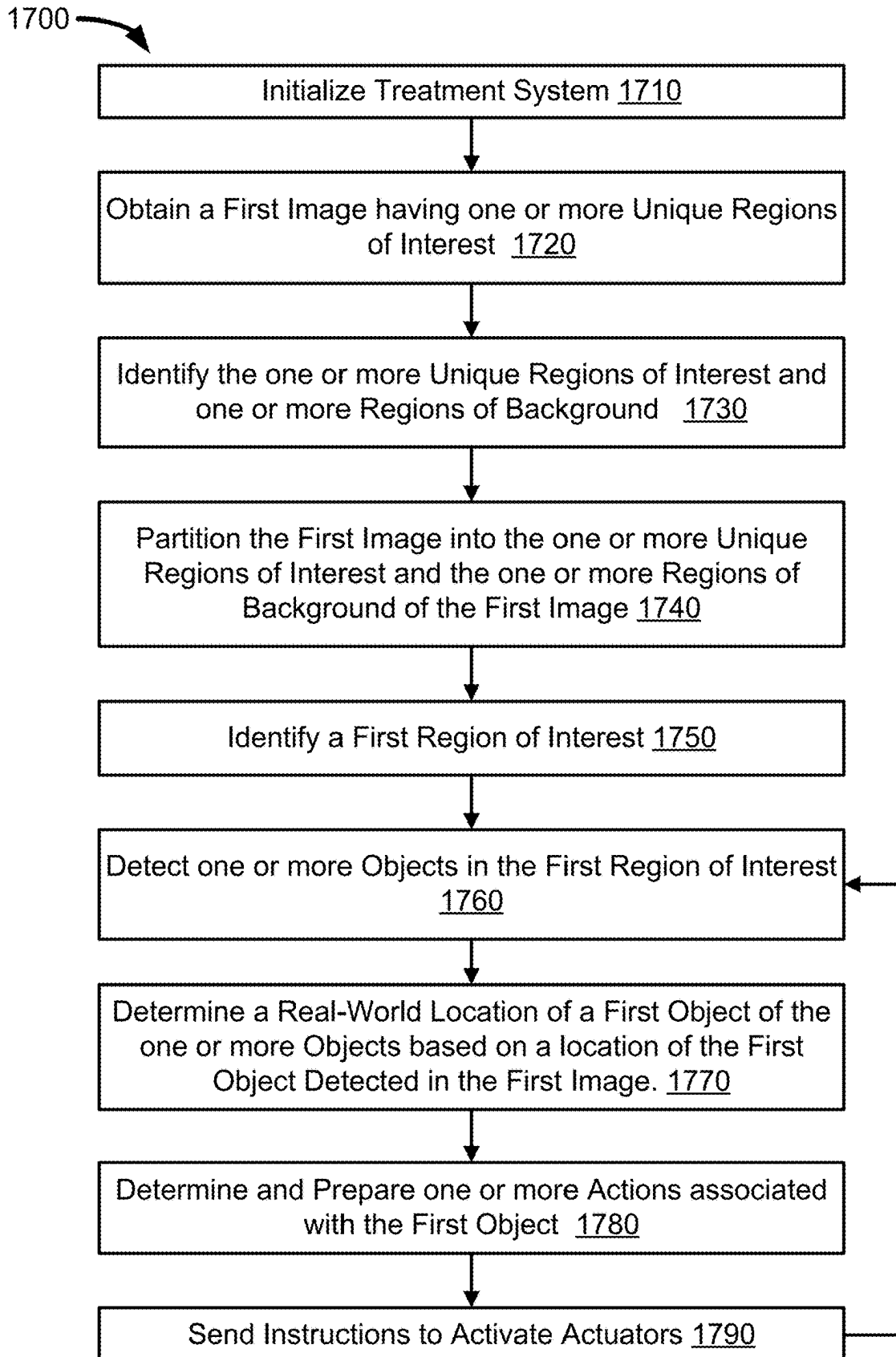
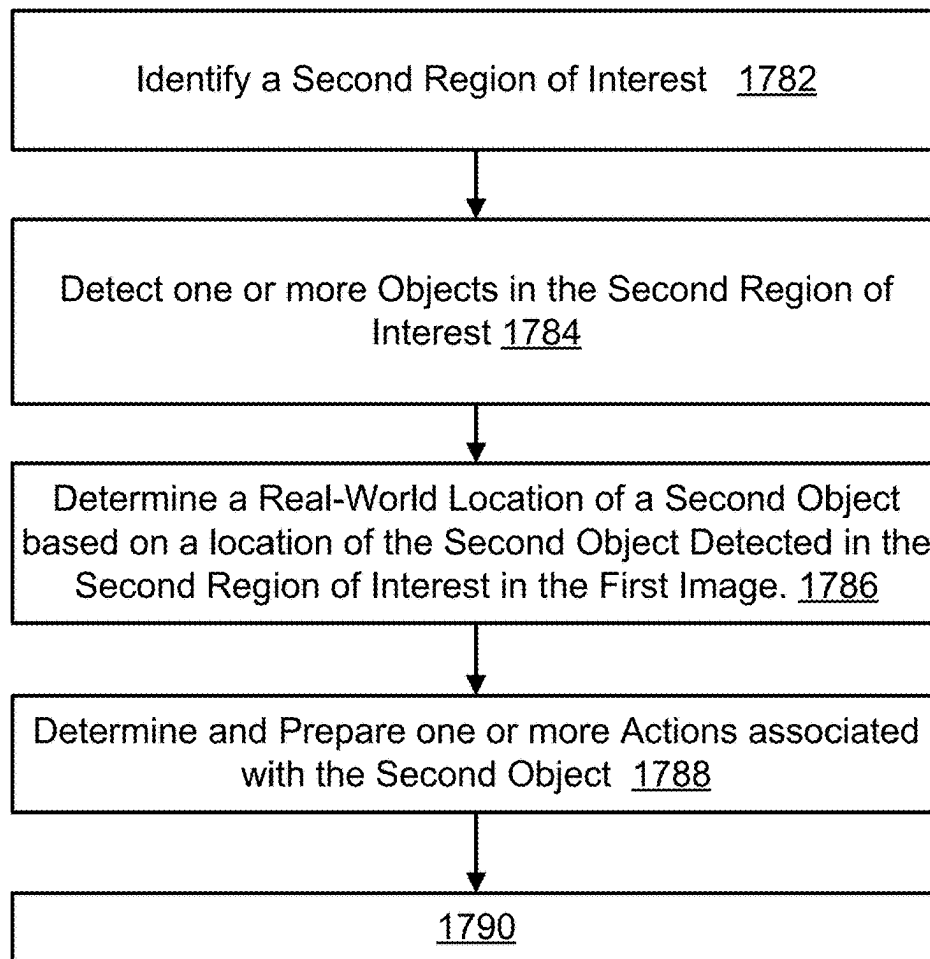


FIG. 16

**FIG. 17A**

1700 **FIG. 17B**

1800

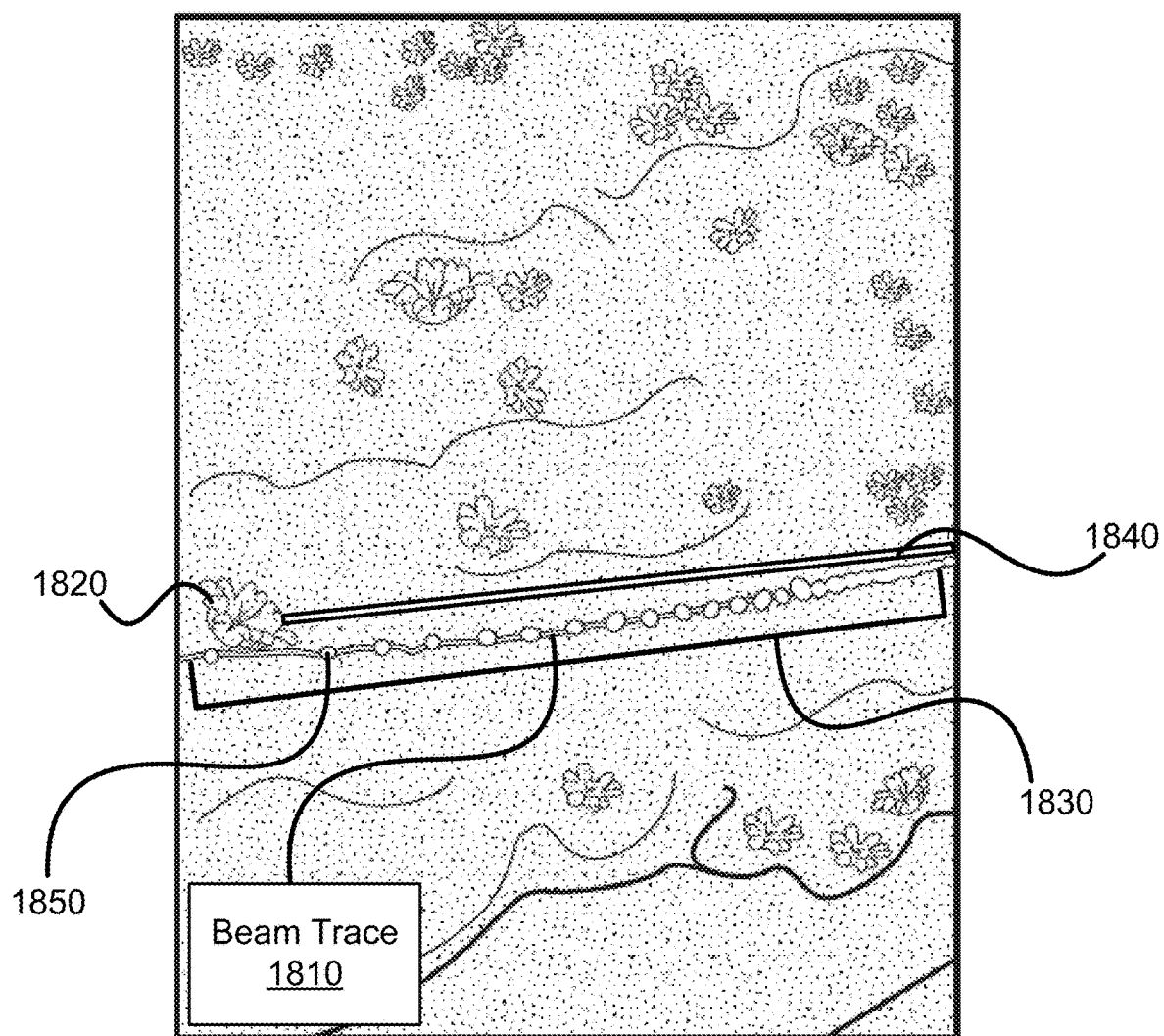


FIG. 18A

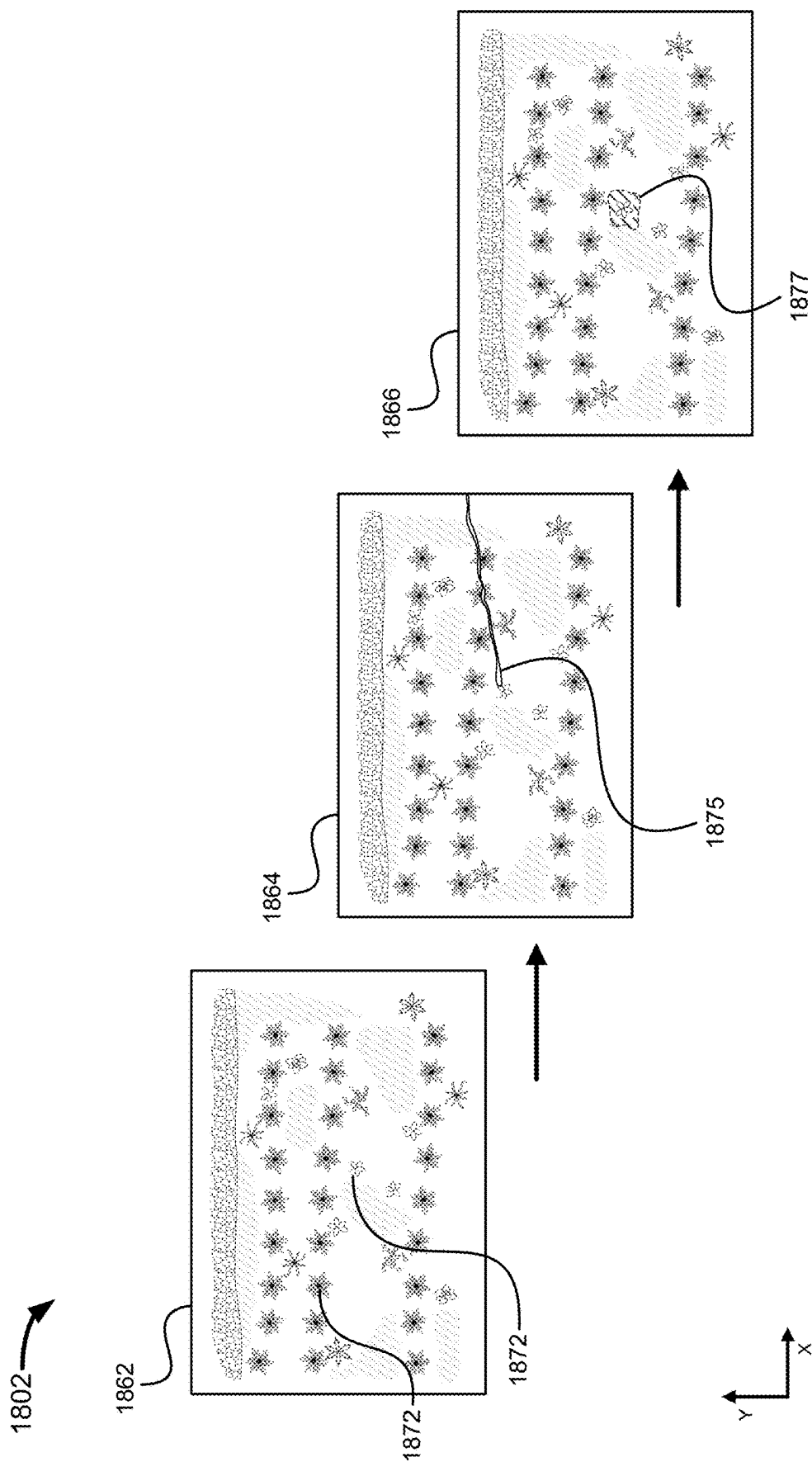


FIG. 18B

1803

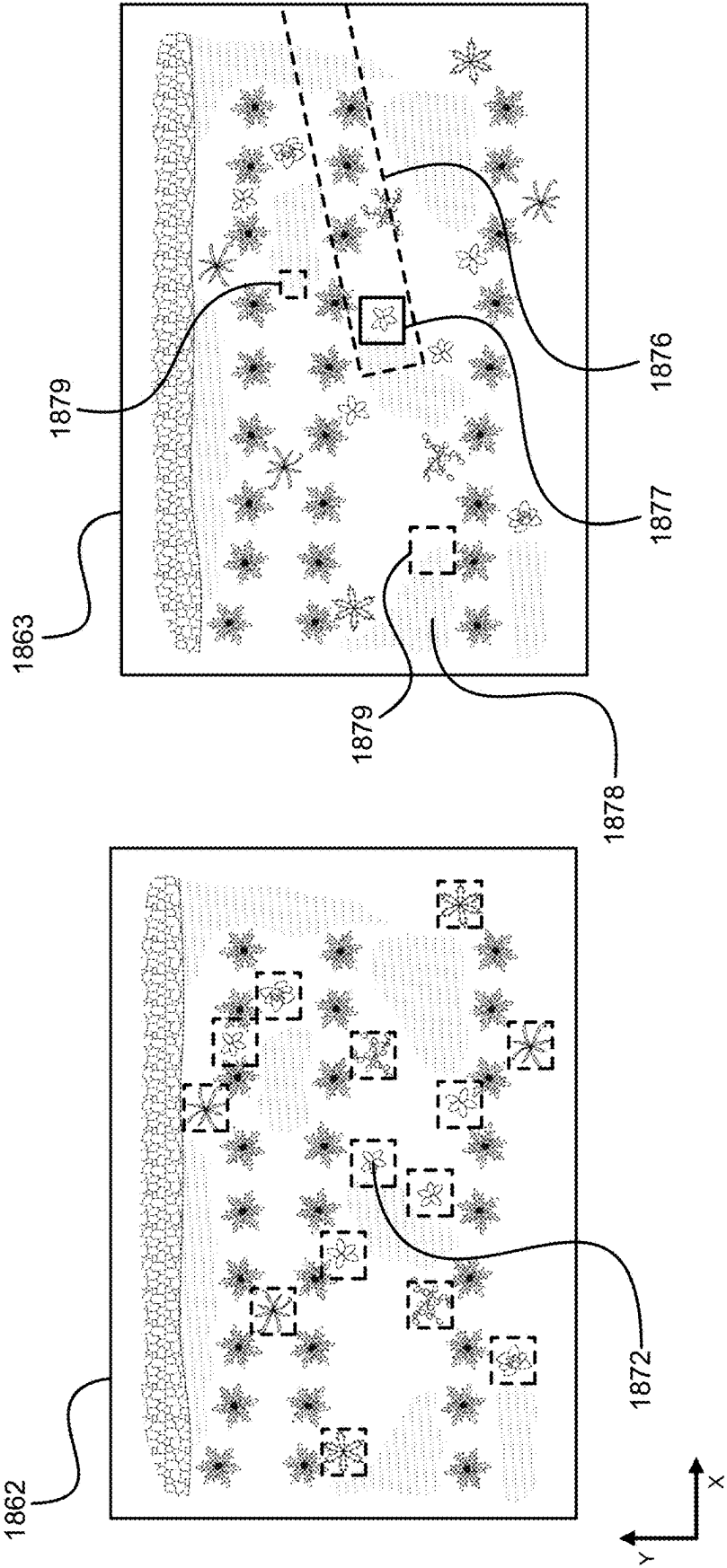
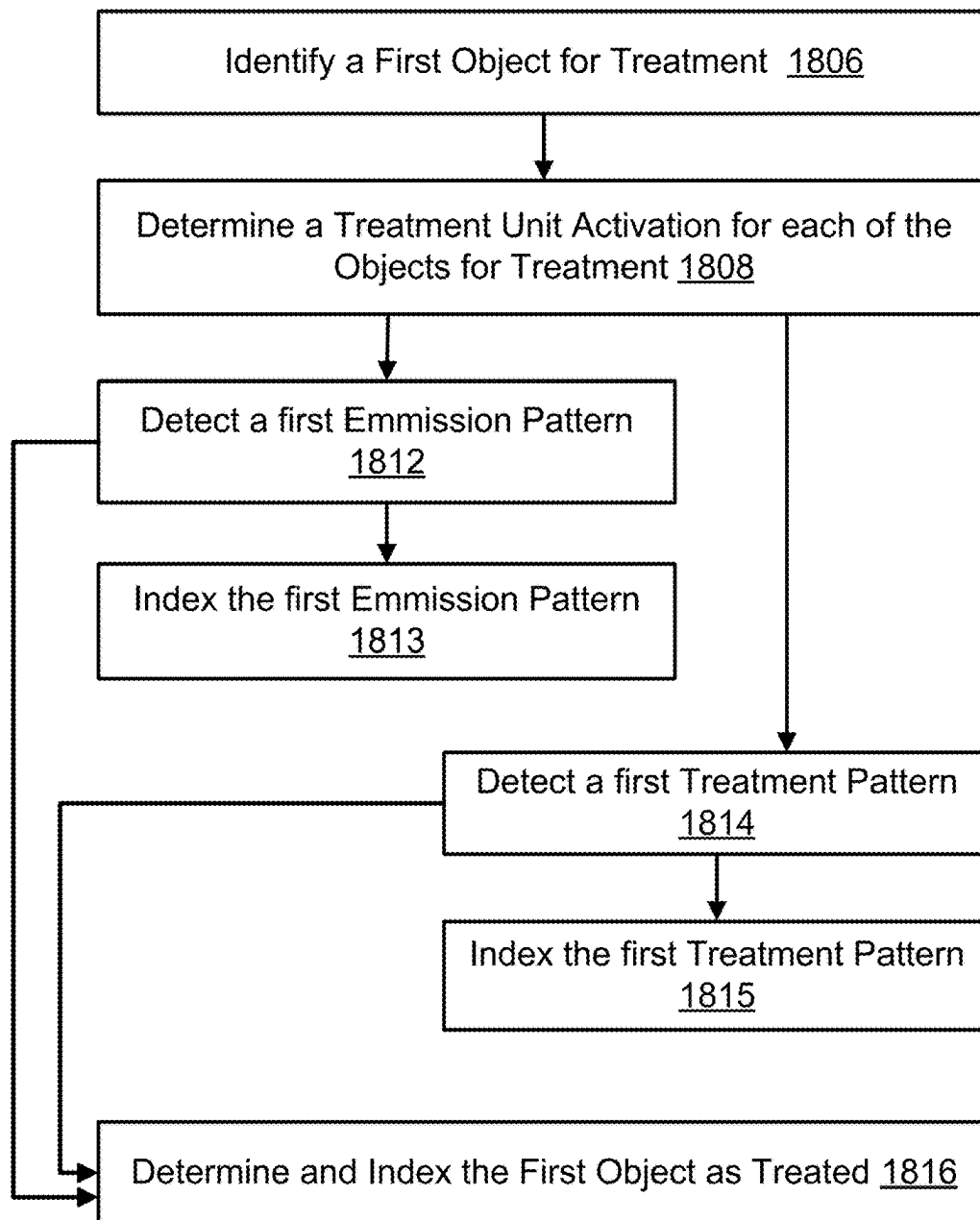


FIG. 18C

1804 **FIG. 18D**

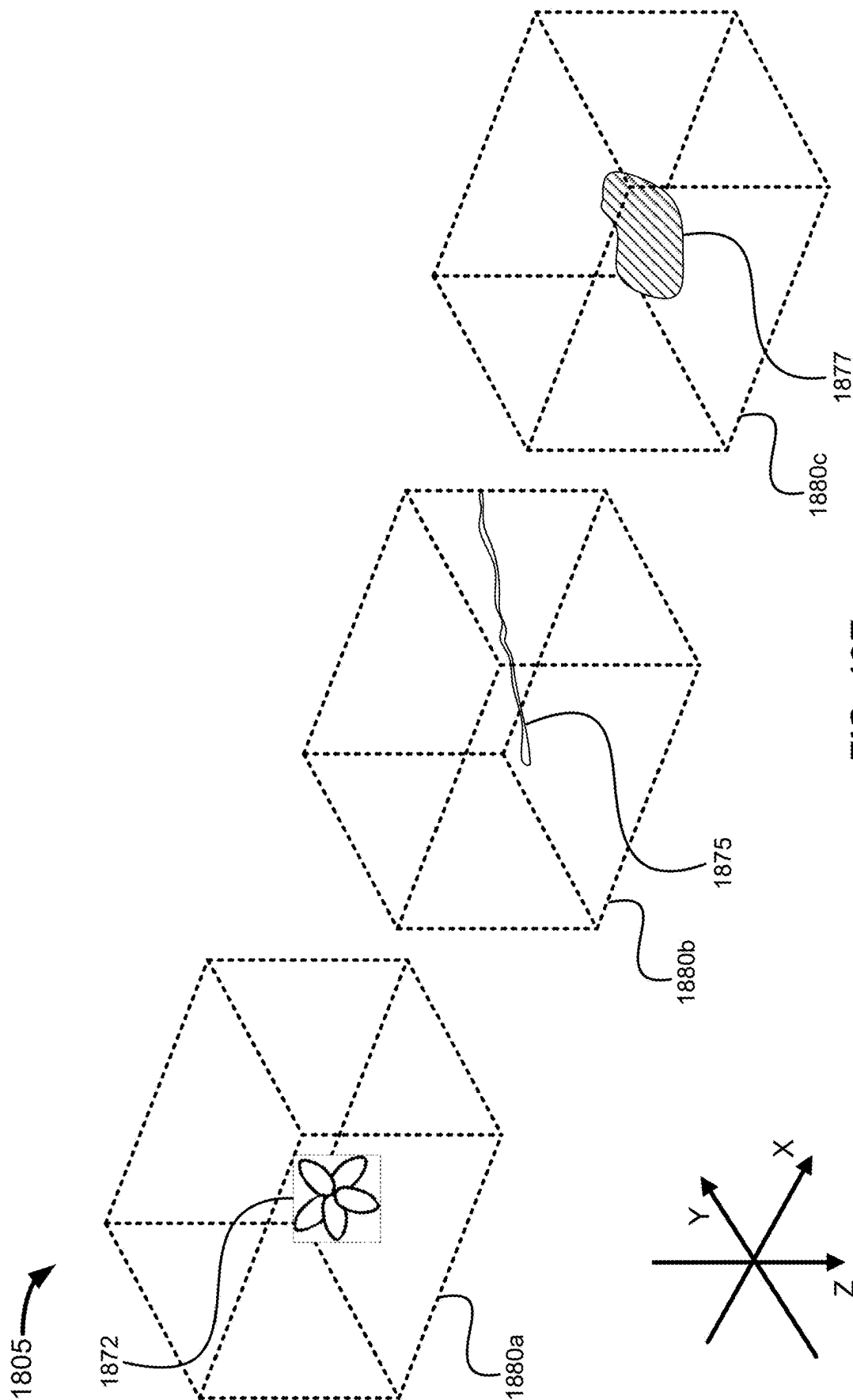


FIG. 18E

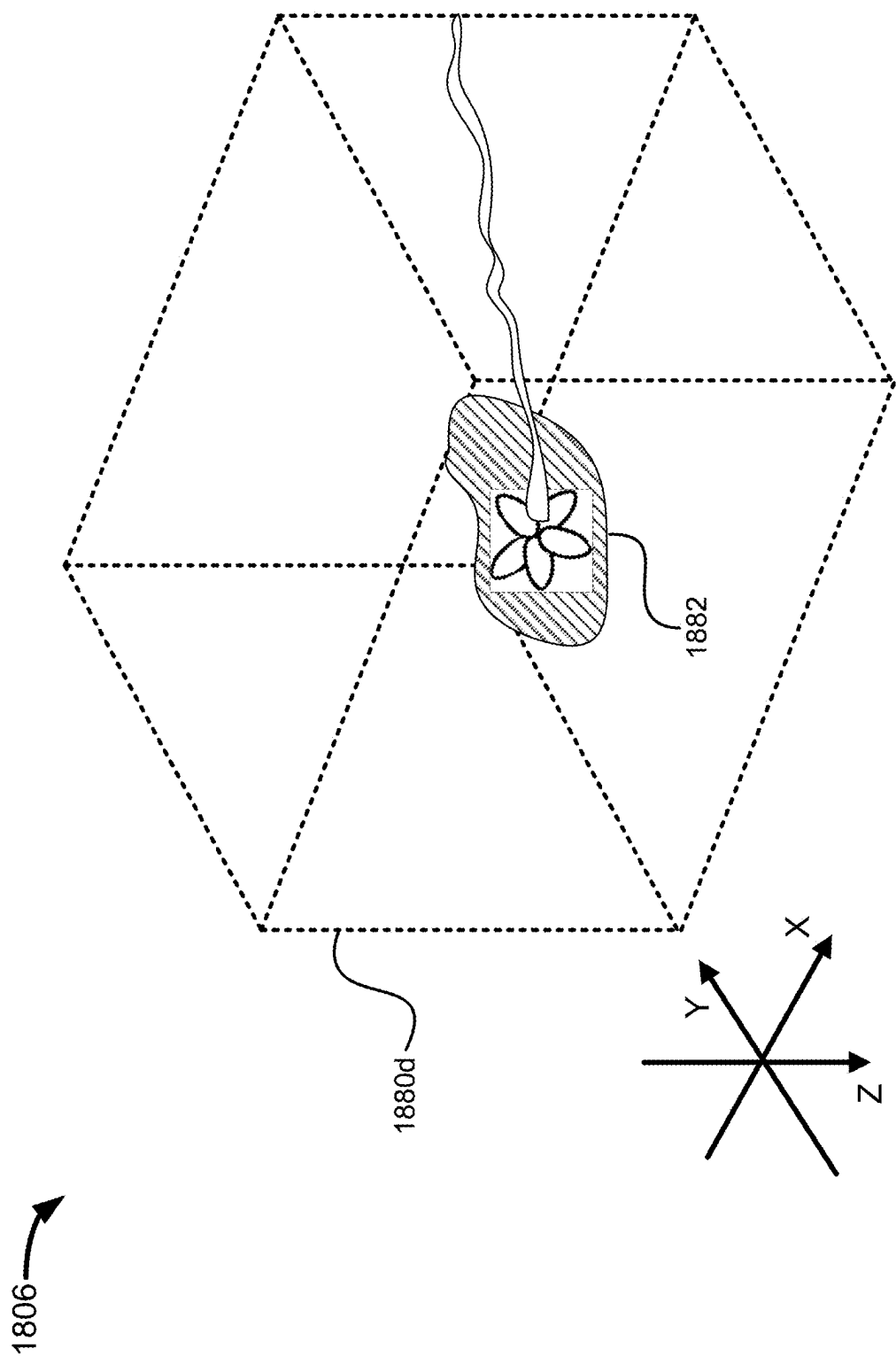


FIG. 18F

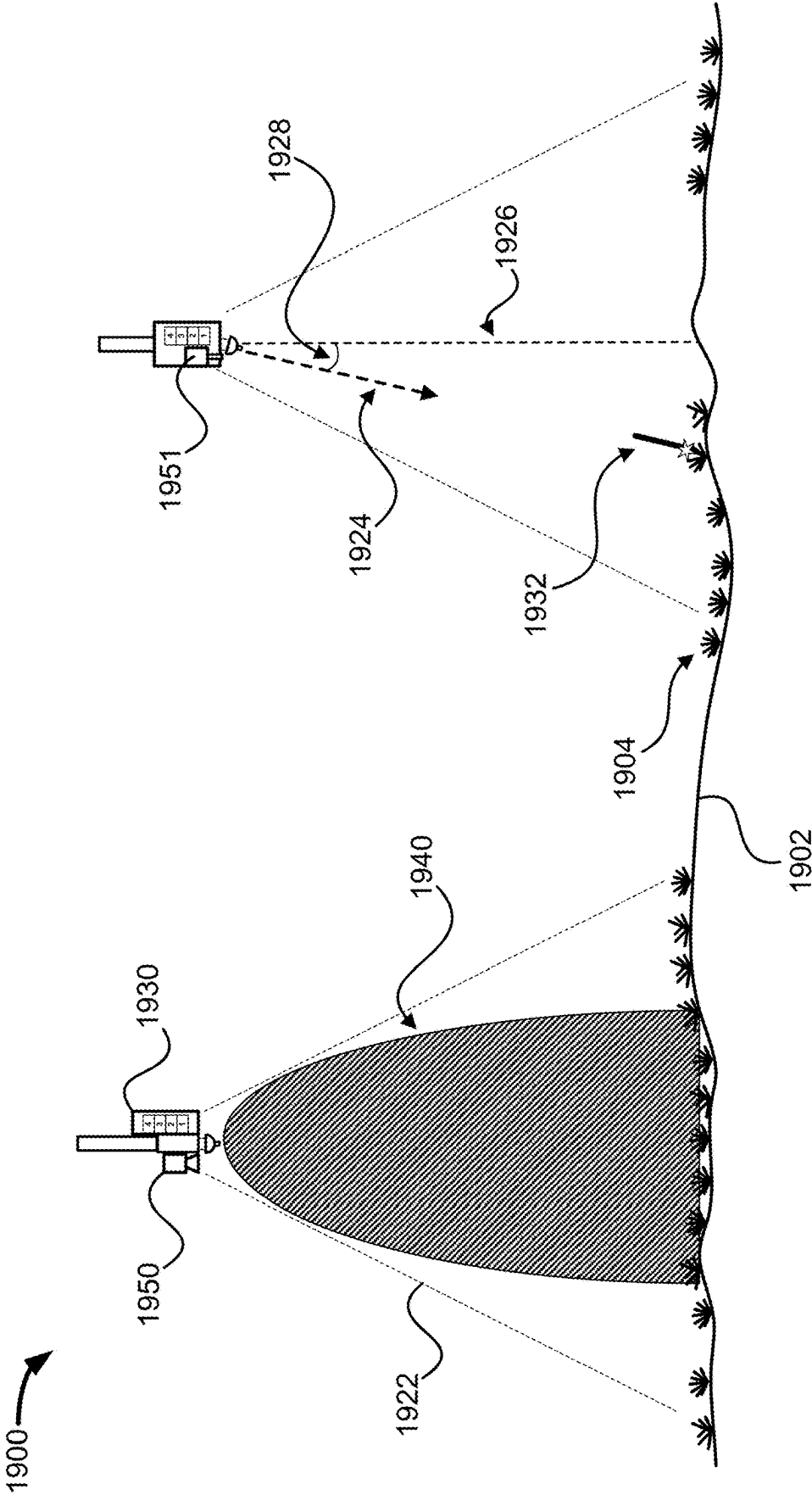


FIG. 19A

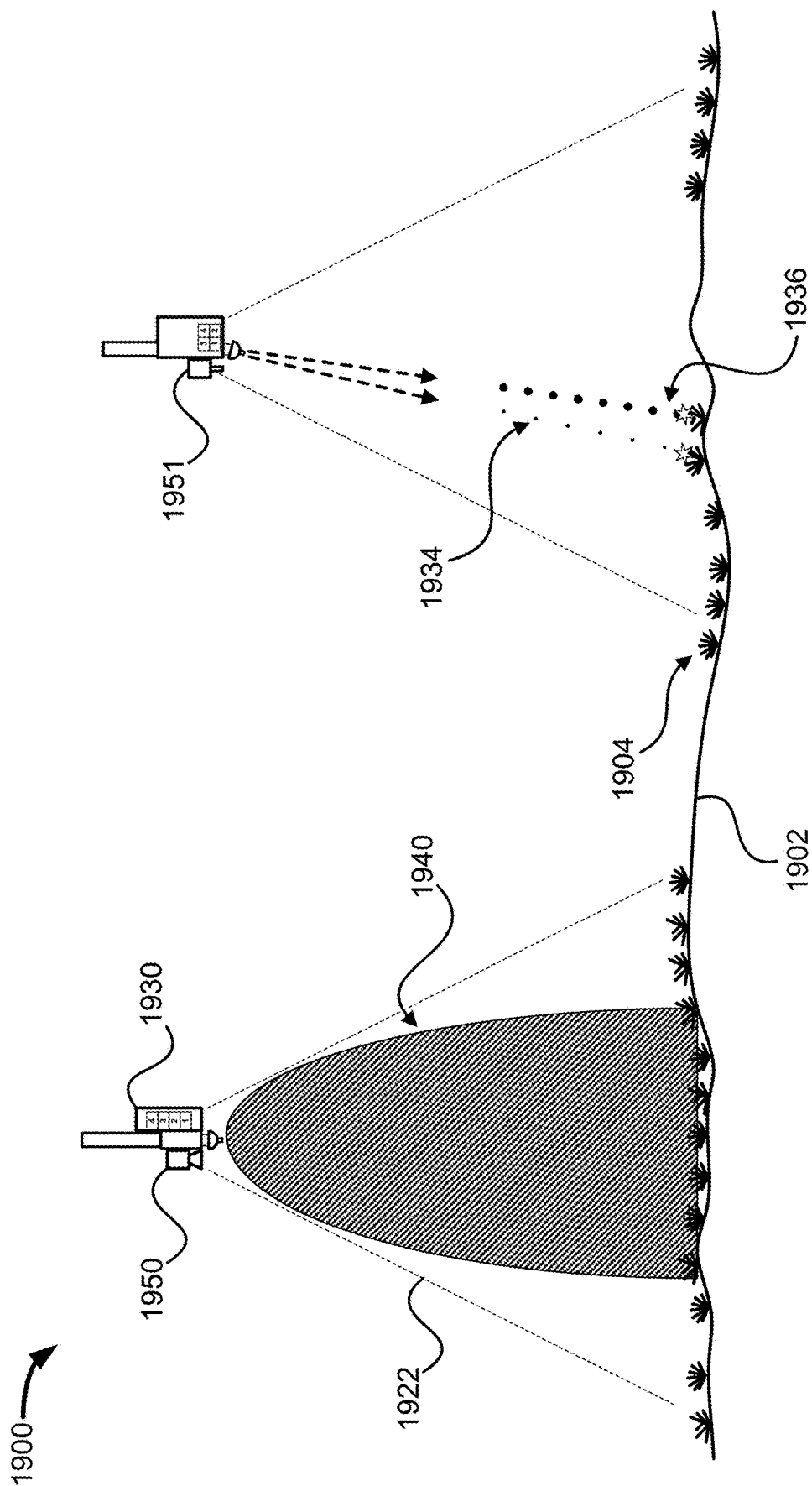


FIG. 19B

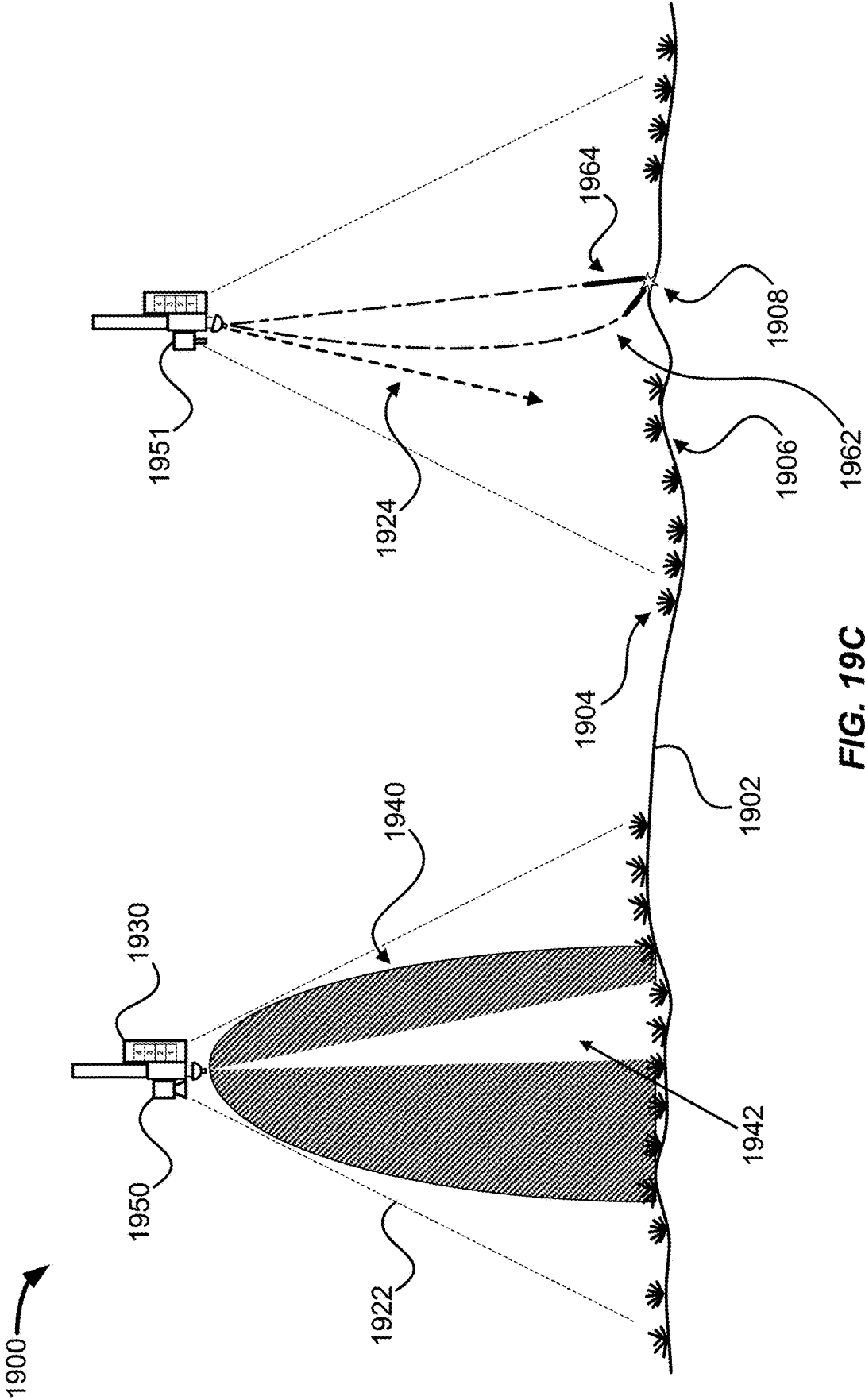
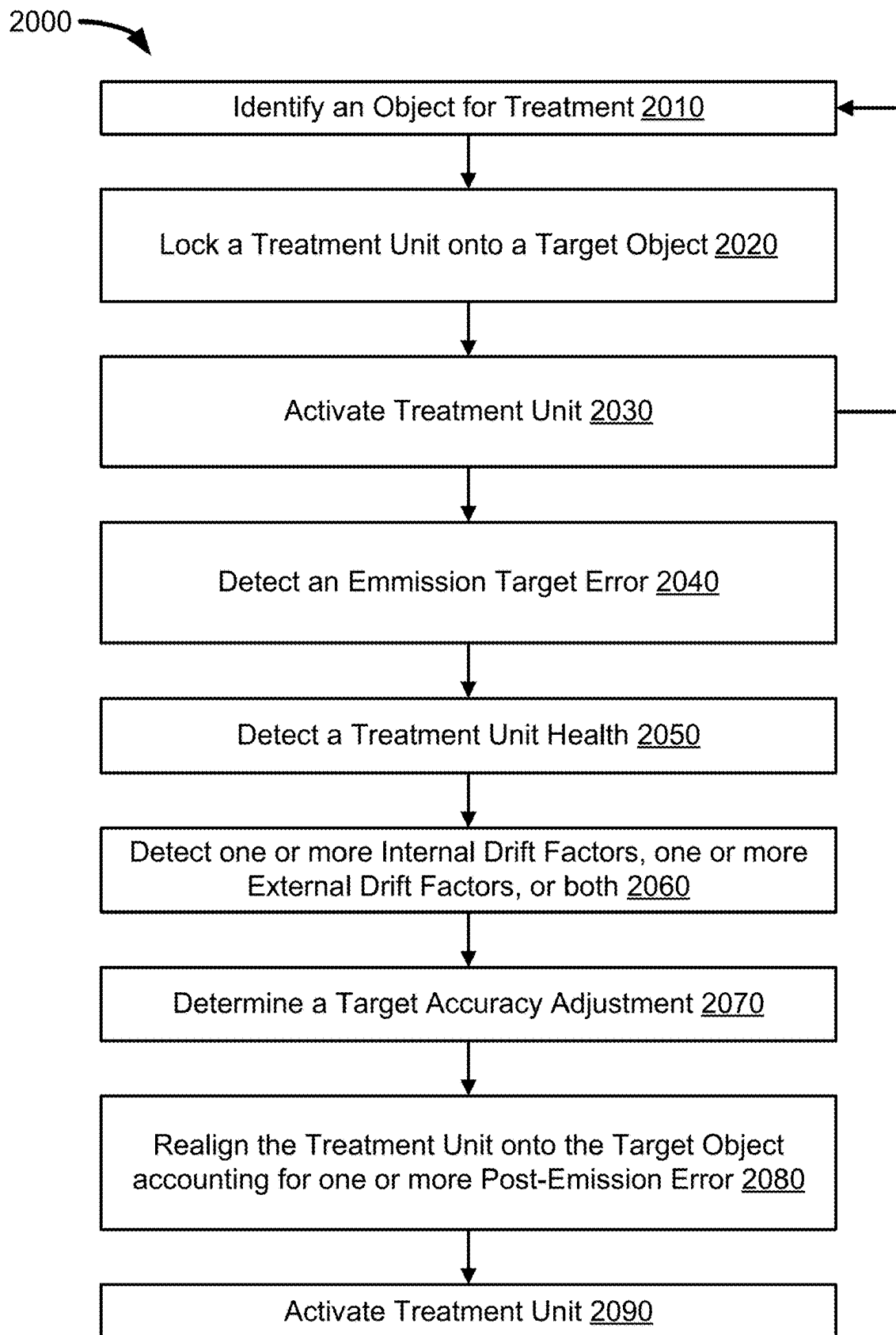


FIG. 19C

**FIG. 20**

2100

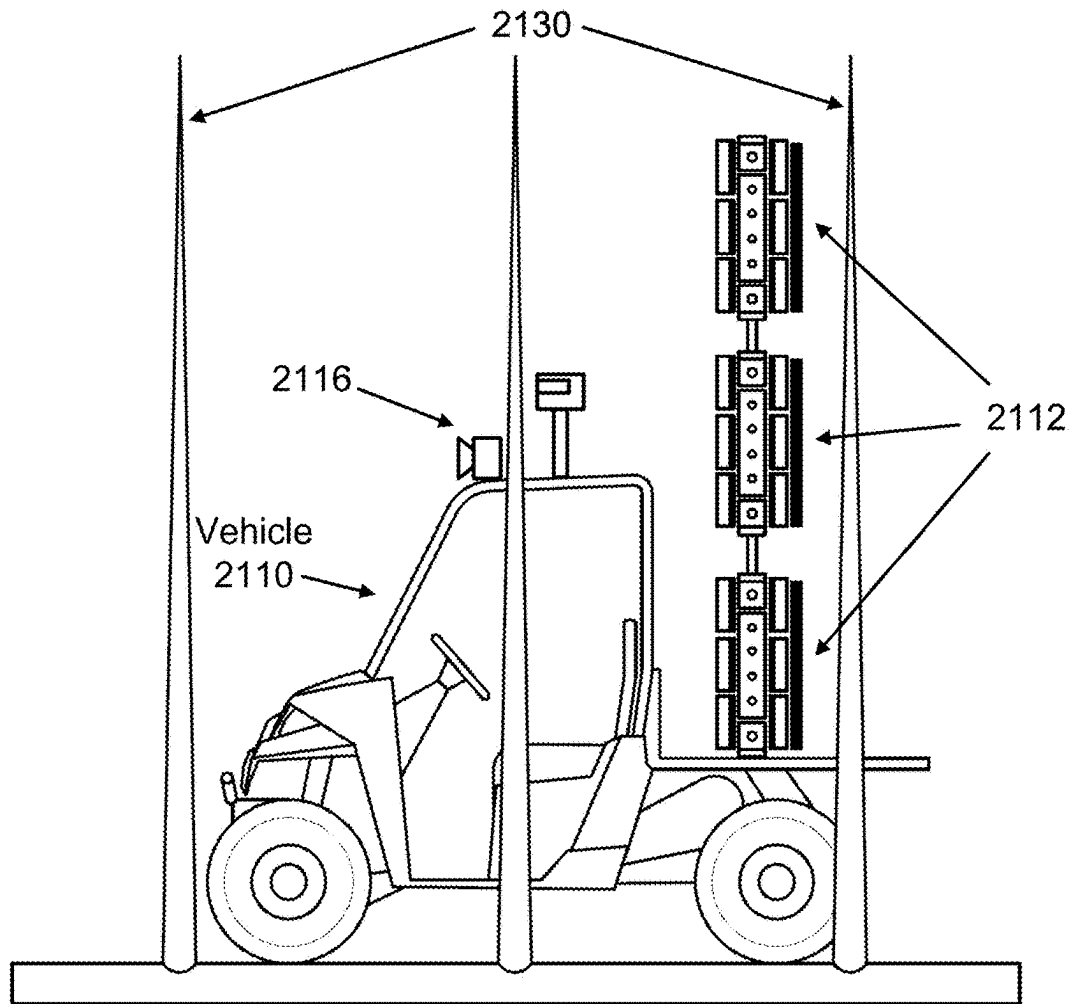


FIG. 21A

2100

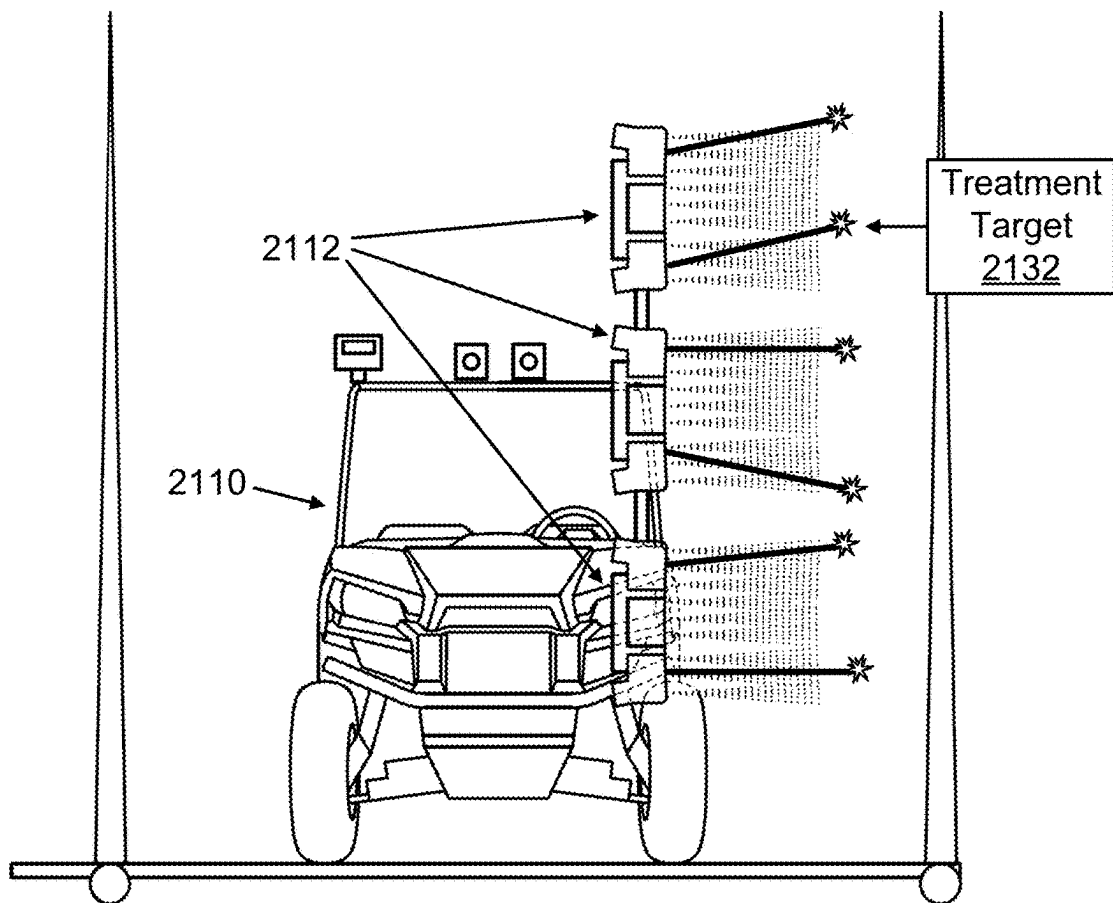


FIG. 21B

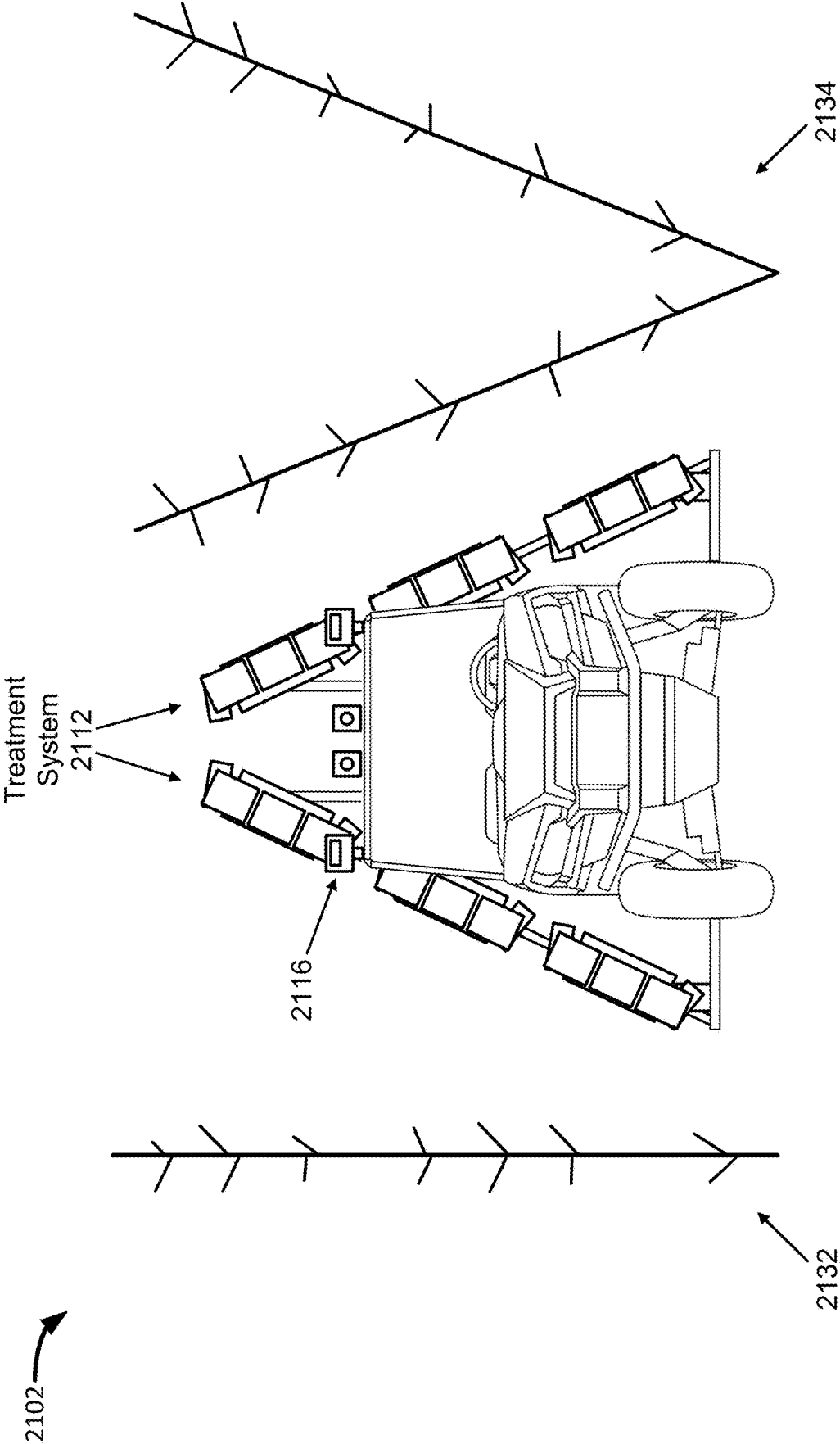


FIG. 21C

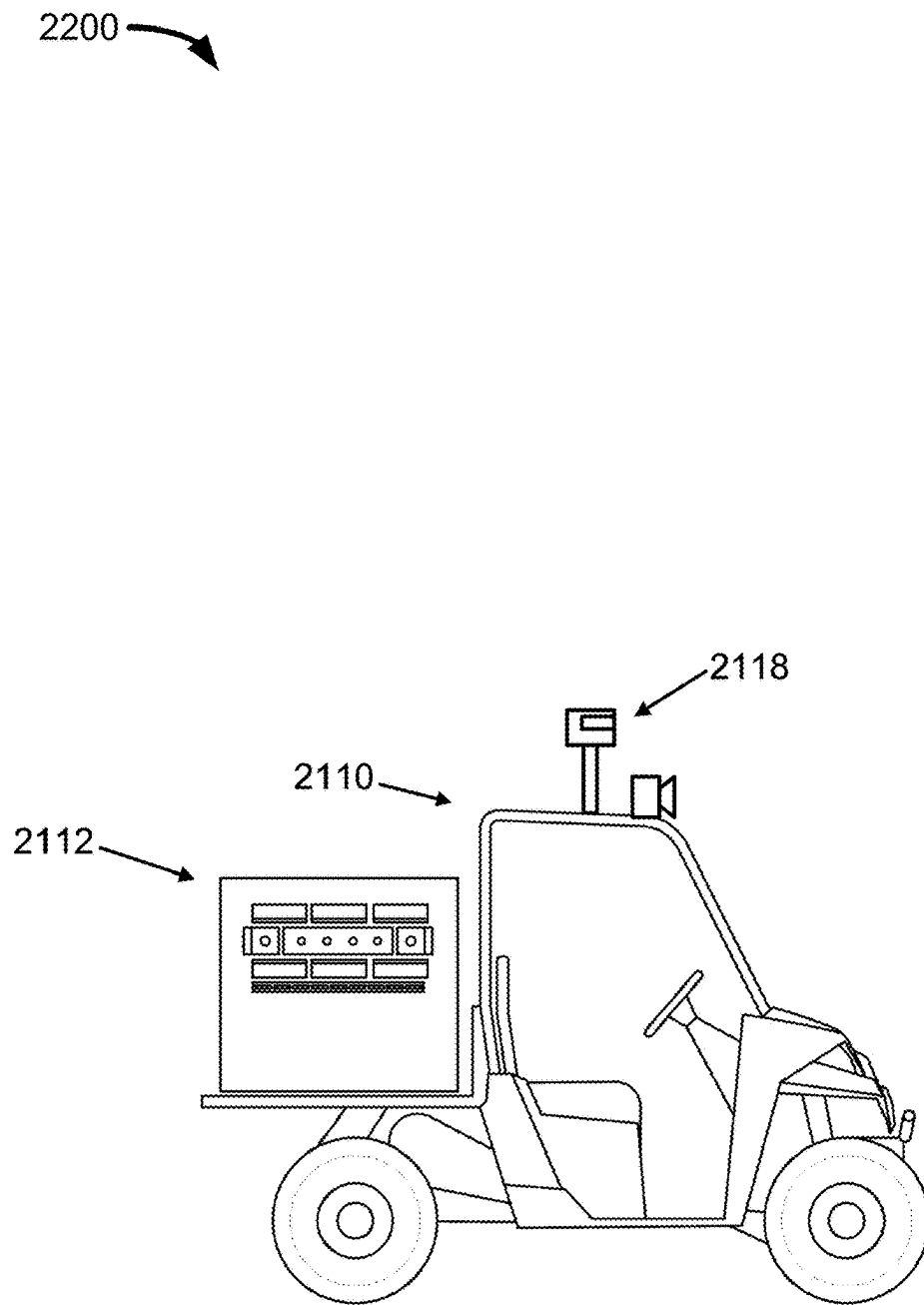


FIG. 22

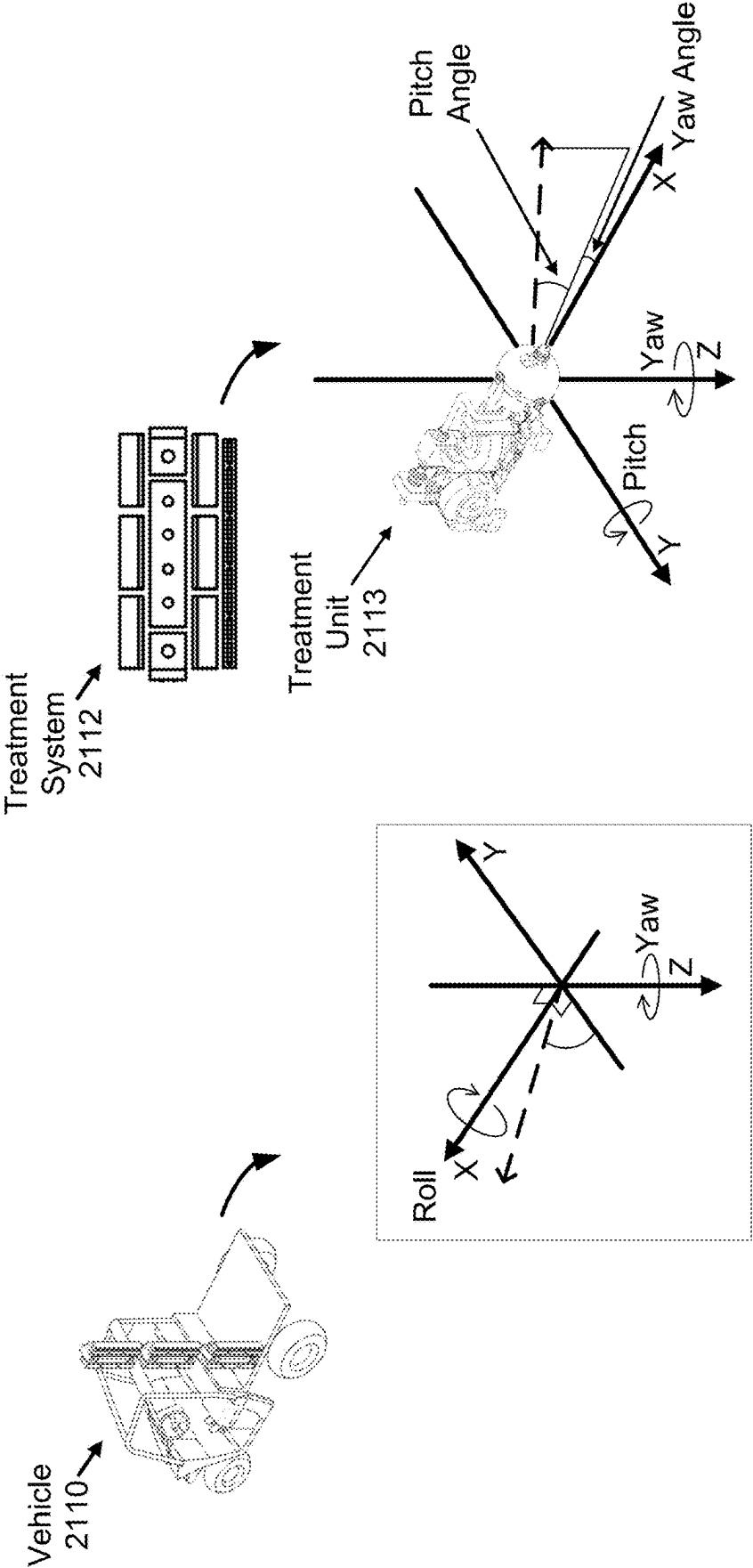


FIG. 23

2400

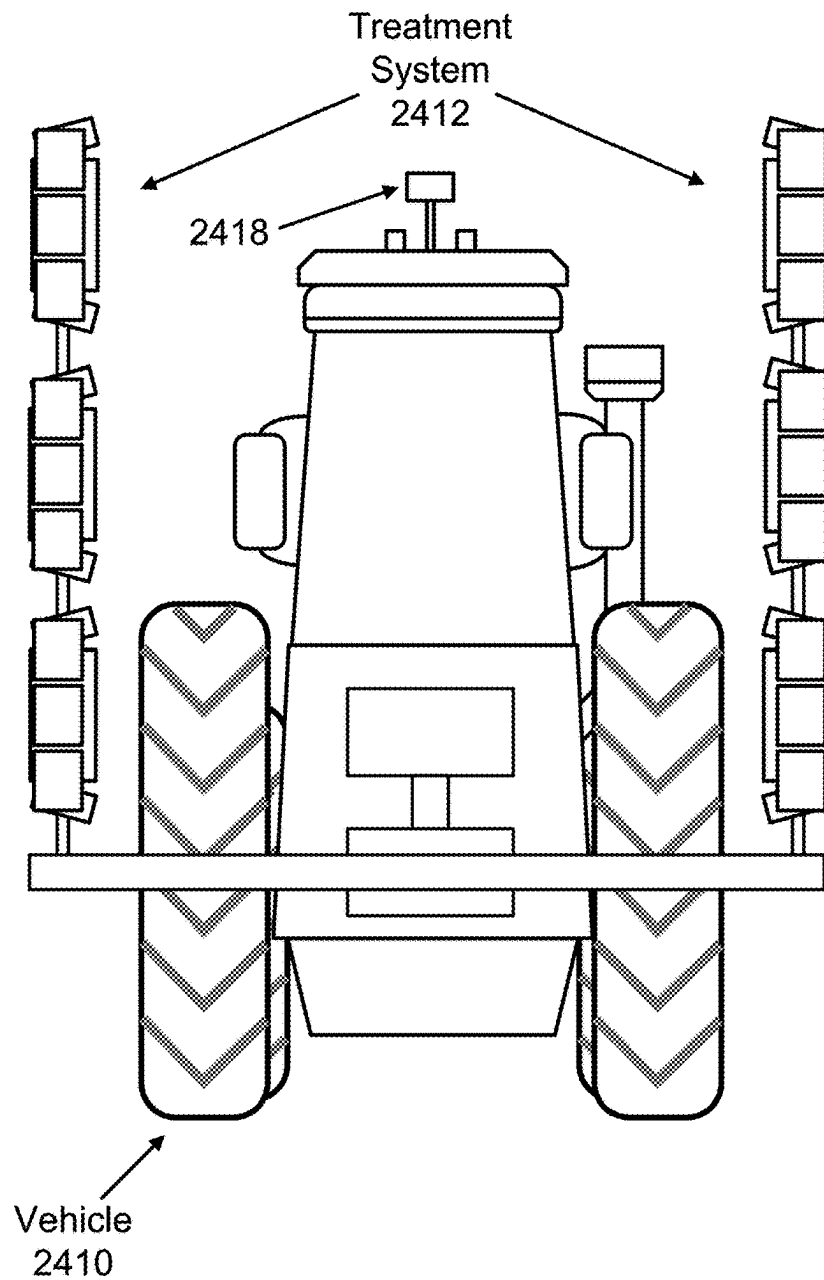


FIG. 24A

2402 

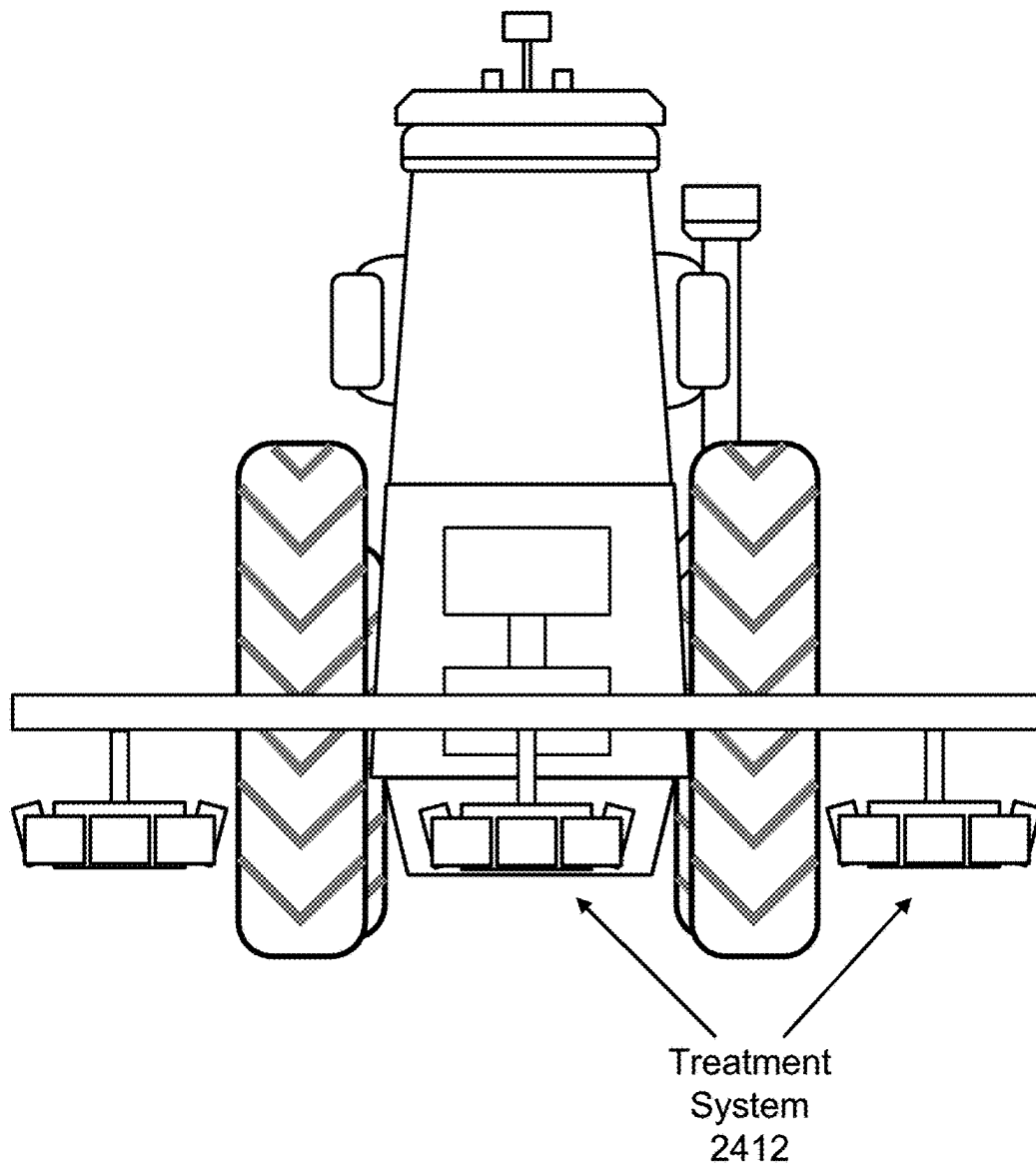
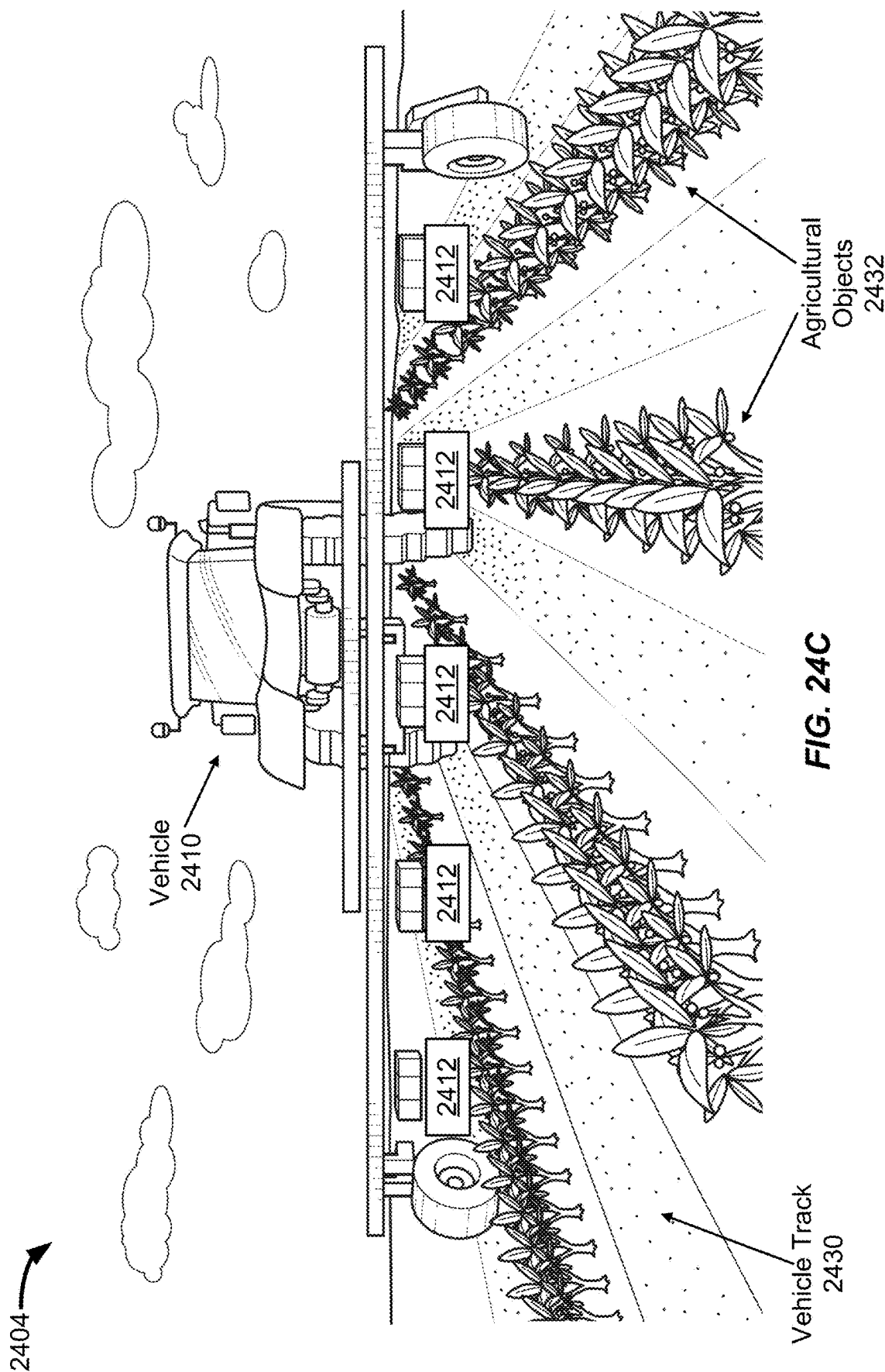


FIG. 24B



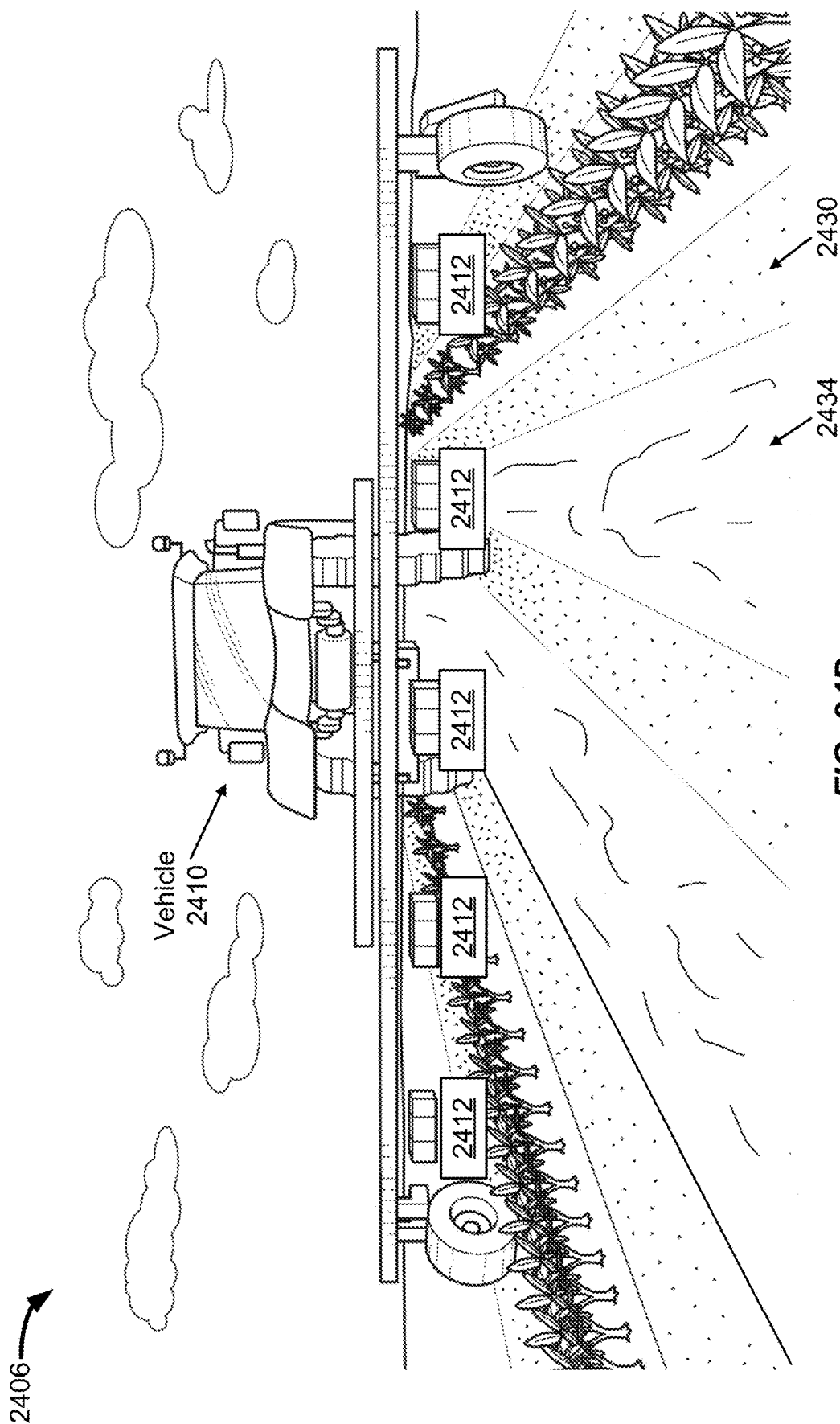


FIG. 24D

2408

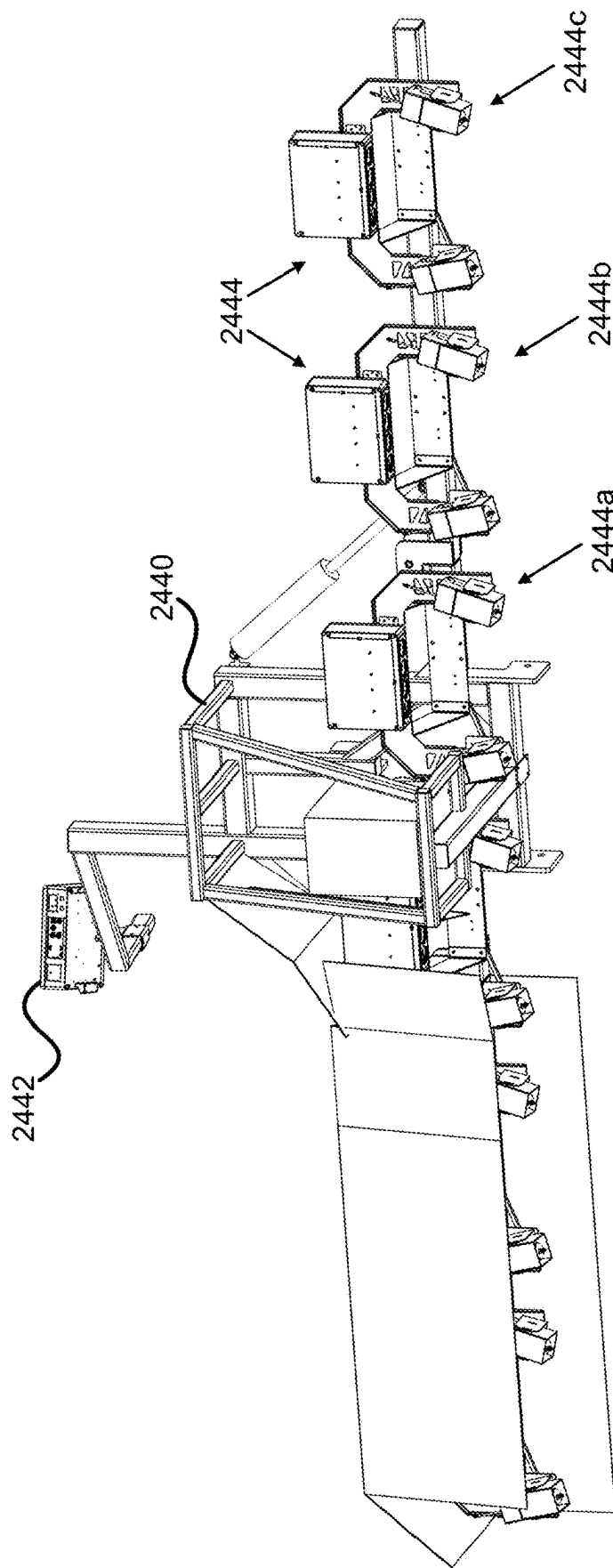
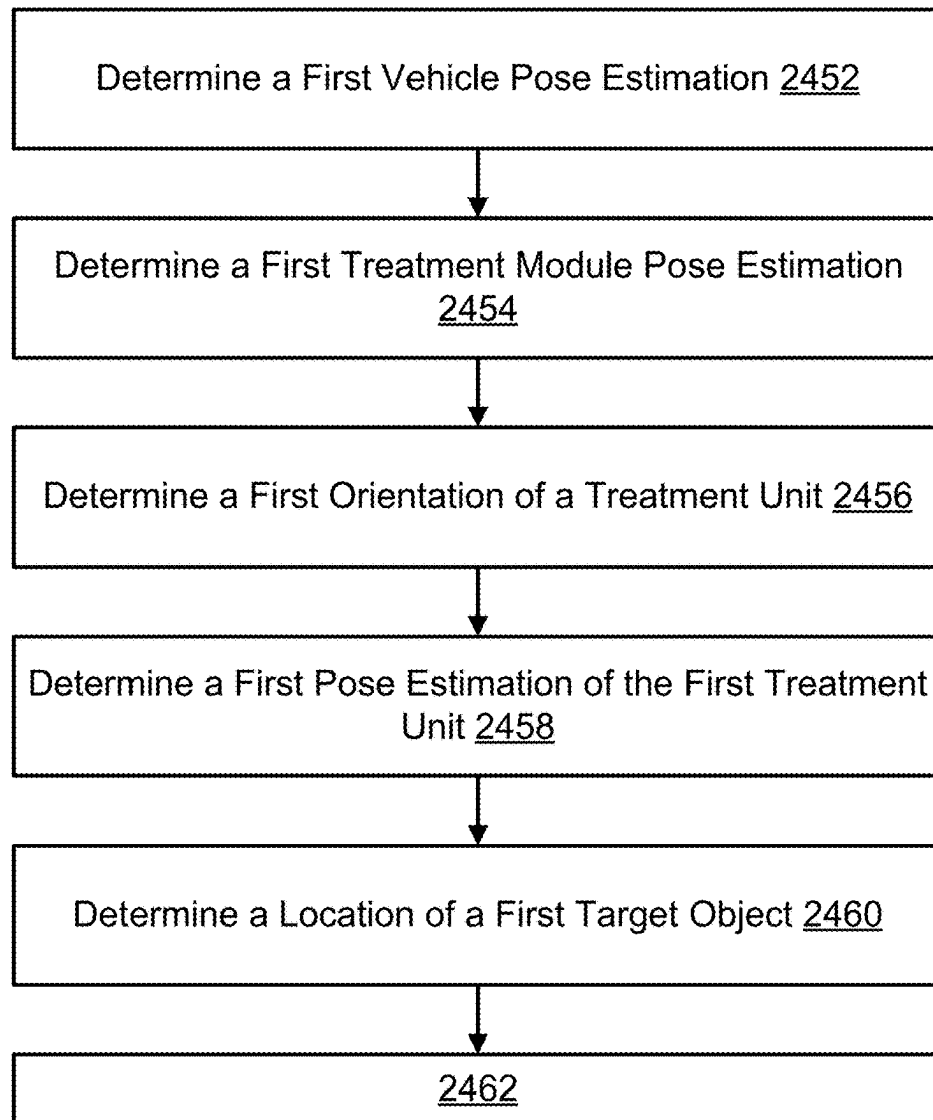


FIG. 24E

2450 **FIG. 24F**

2500

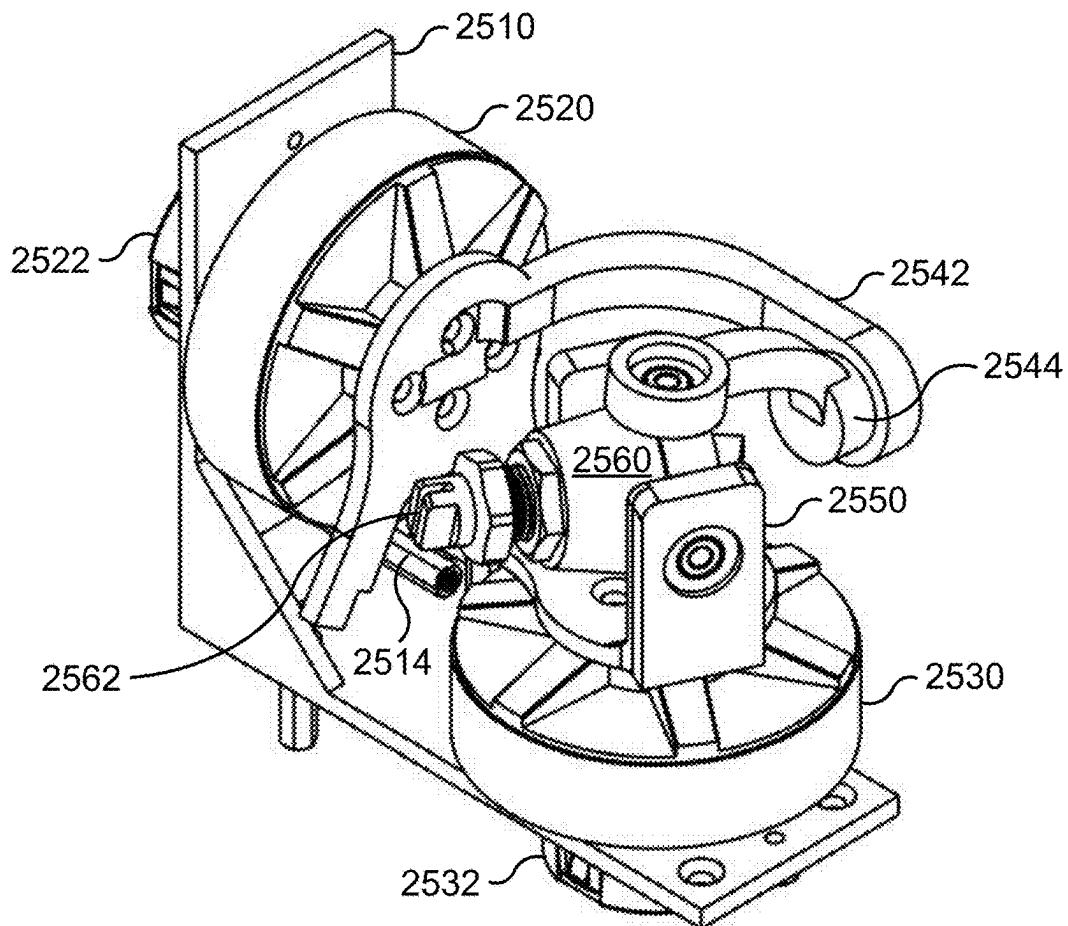
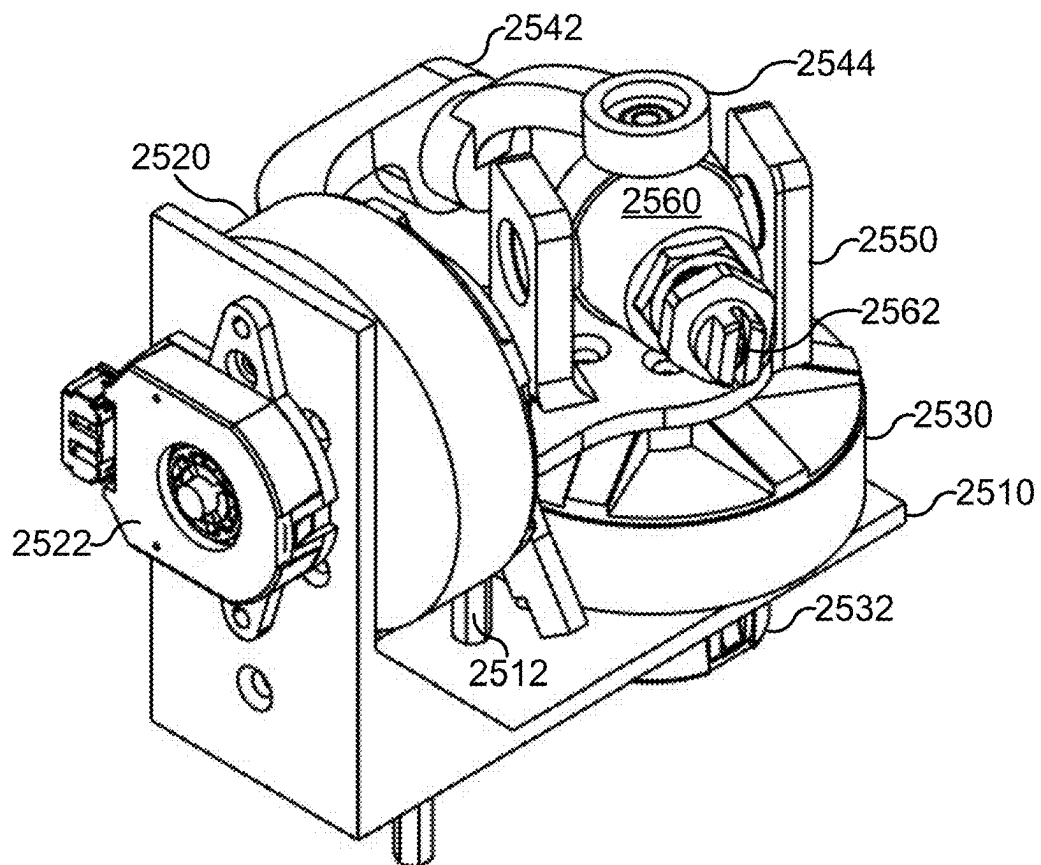


FIG. 25A

2500 **FIG. 25B**

2500 

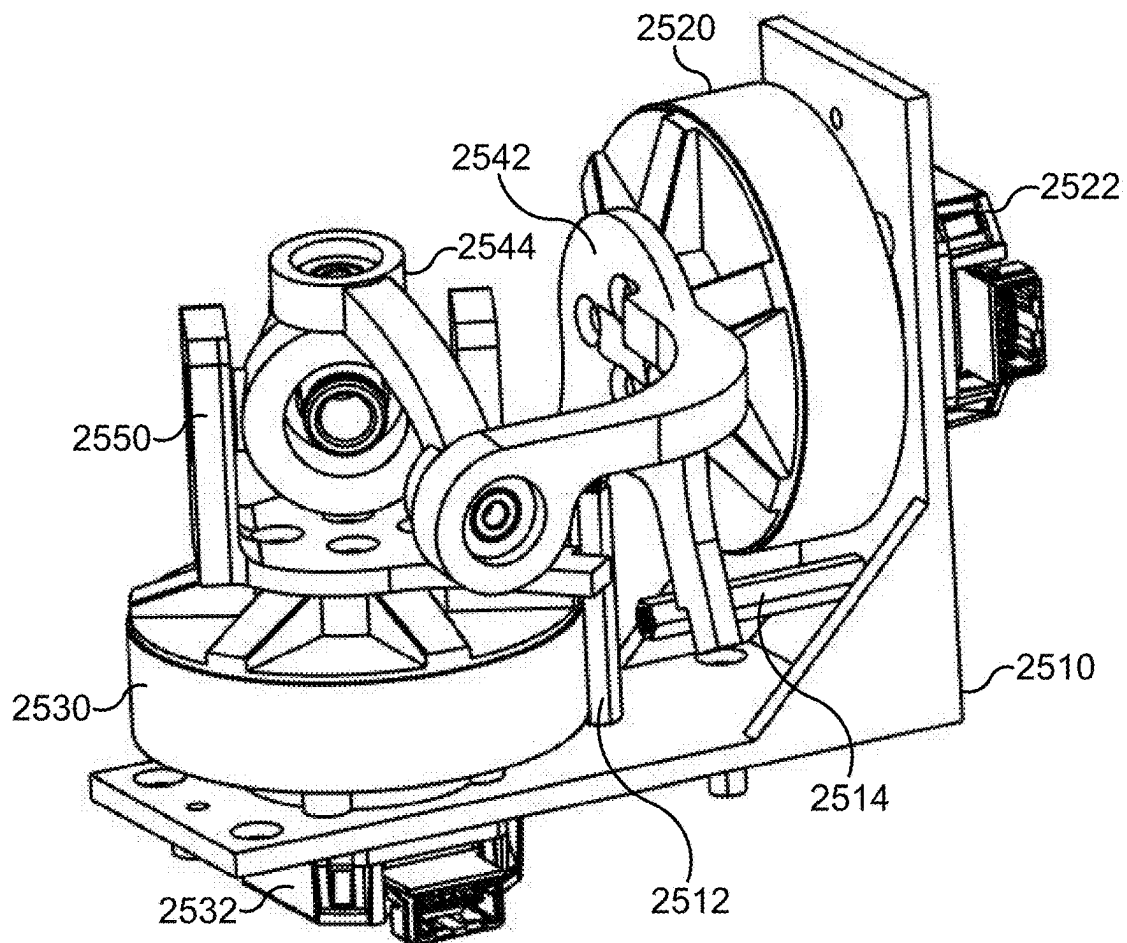


FIG. 25C

2500 →

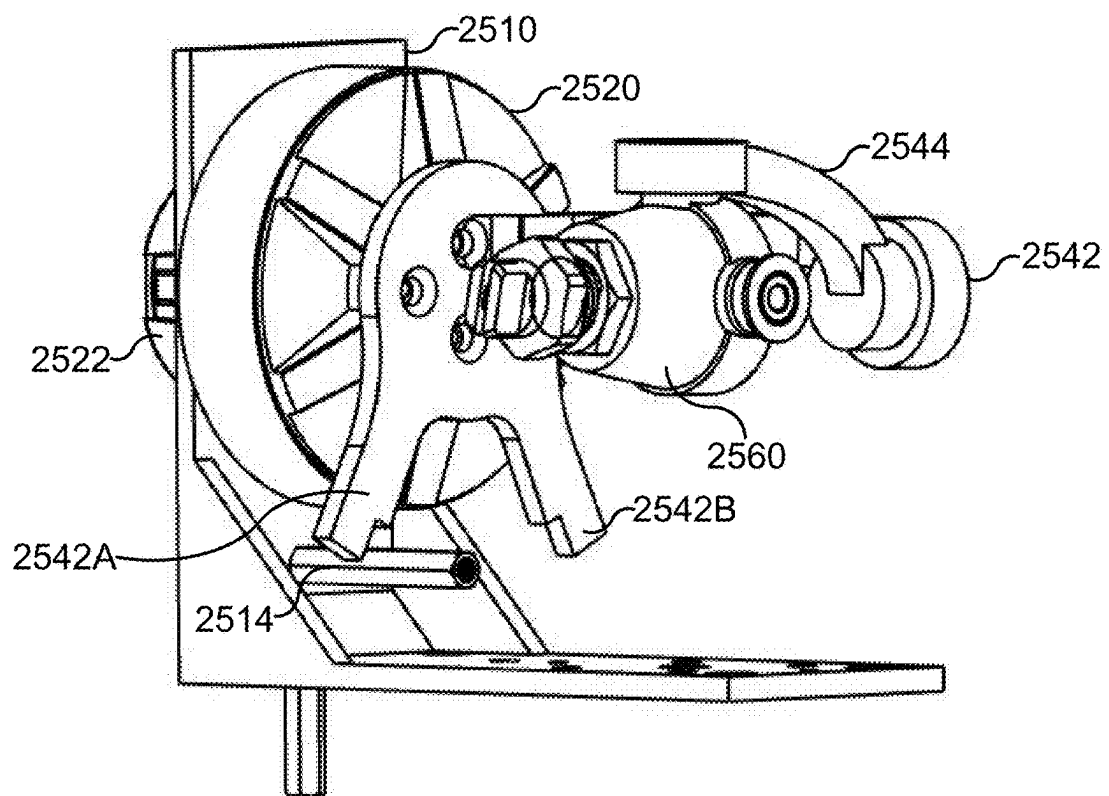


FIG. 25D

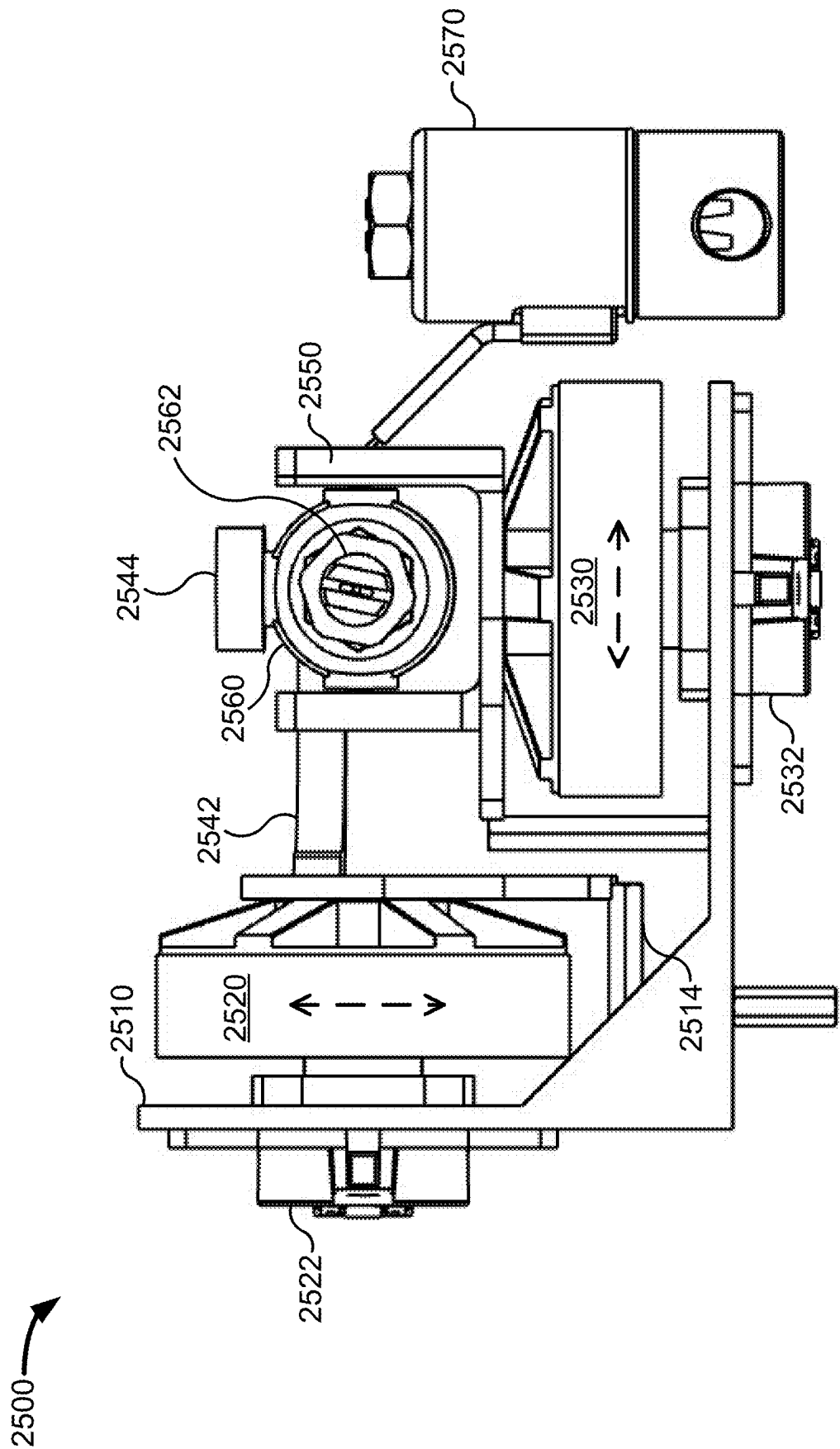
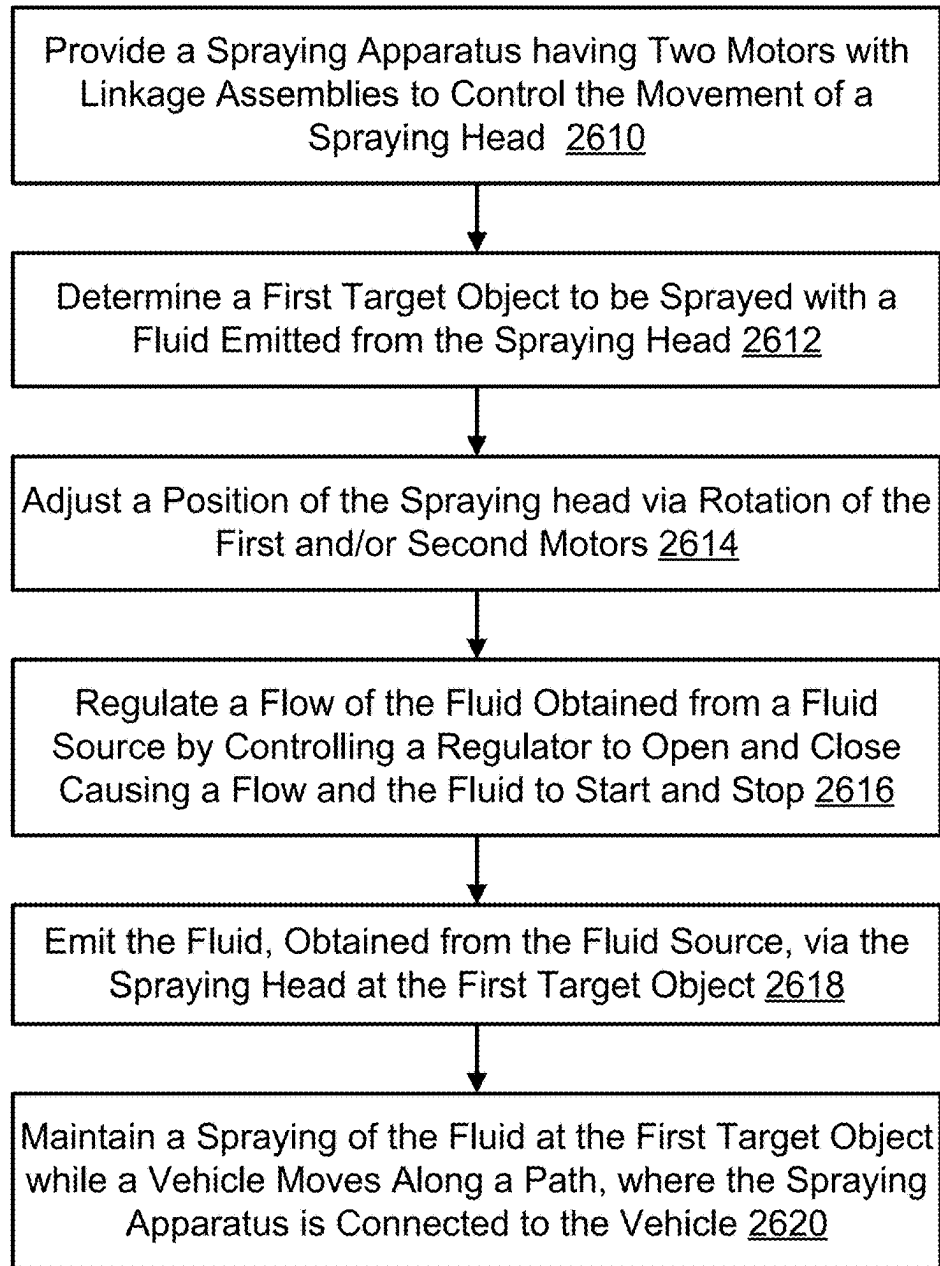
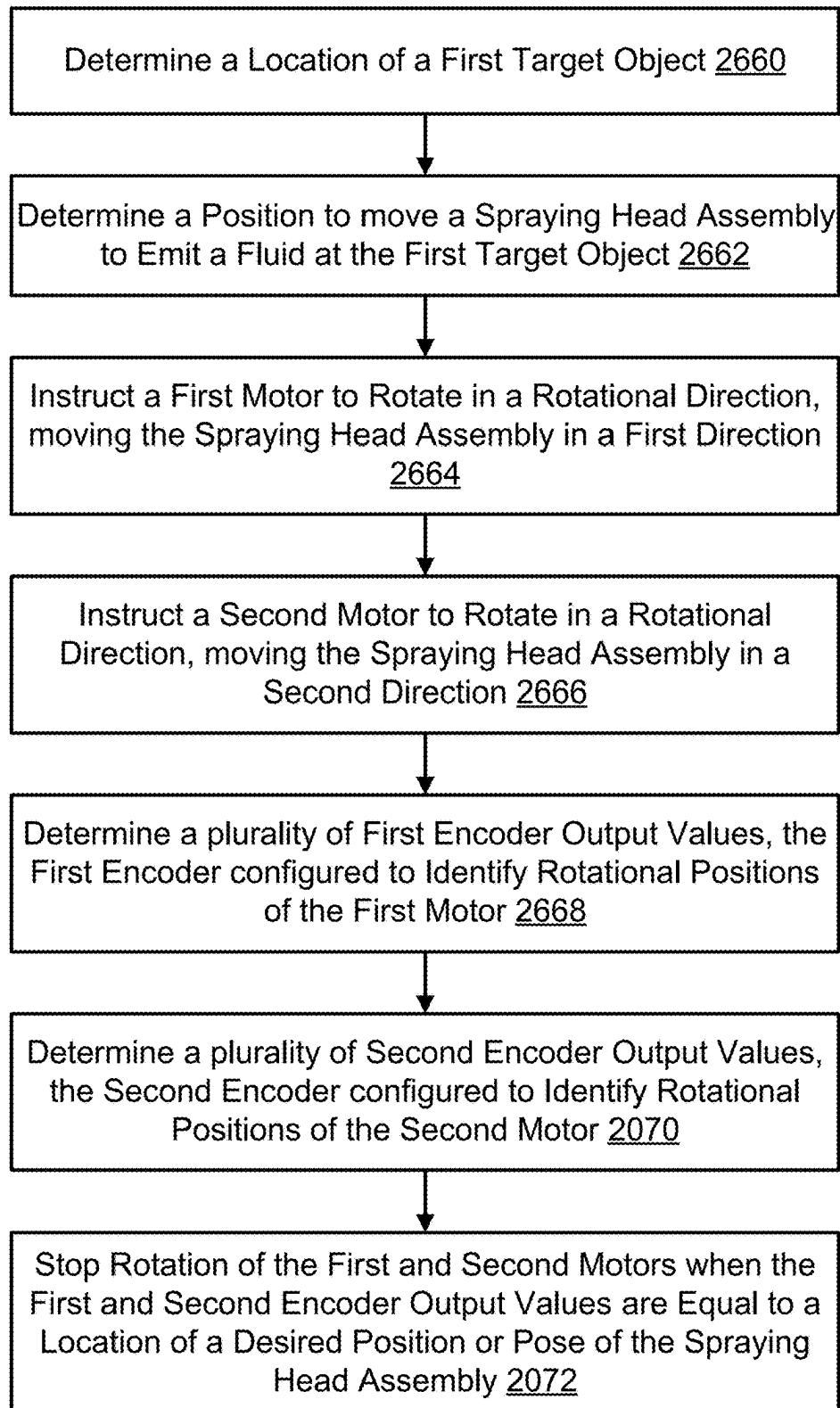
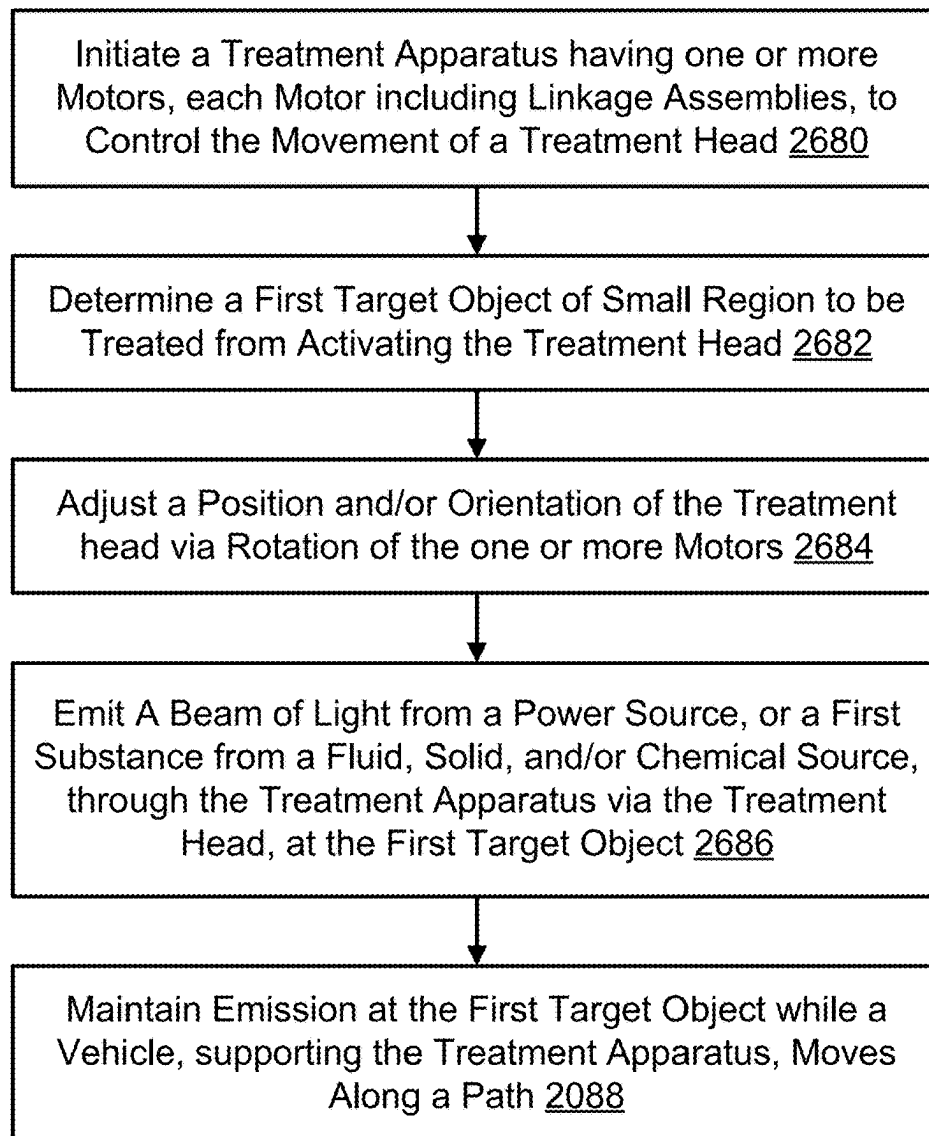


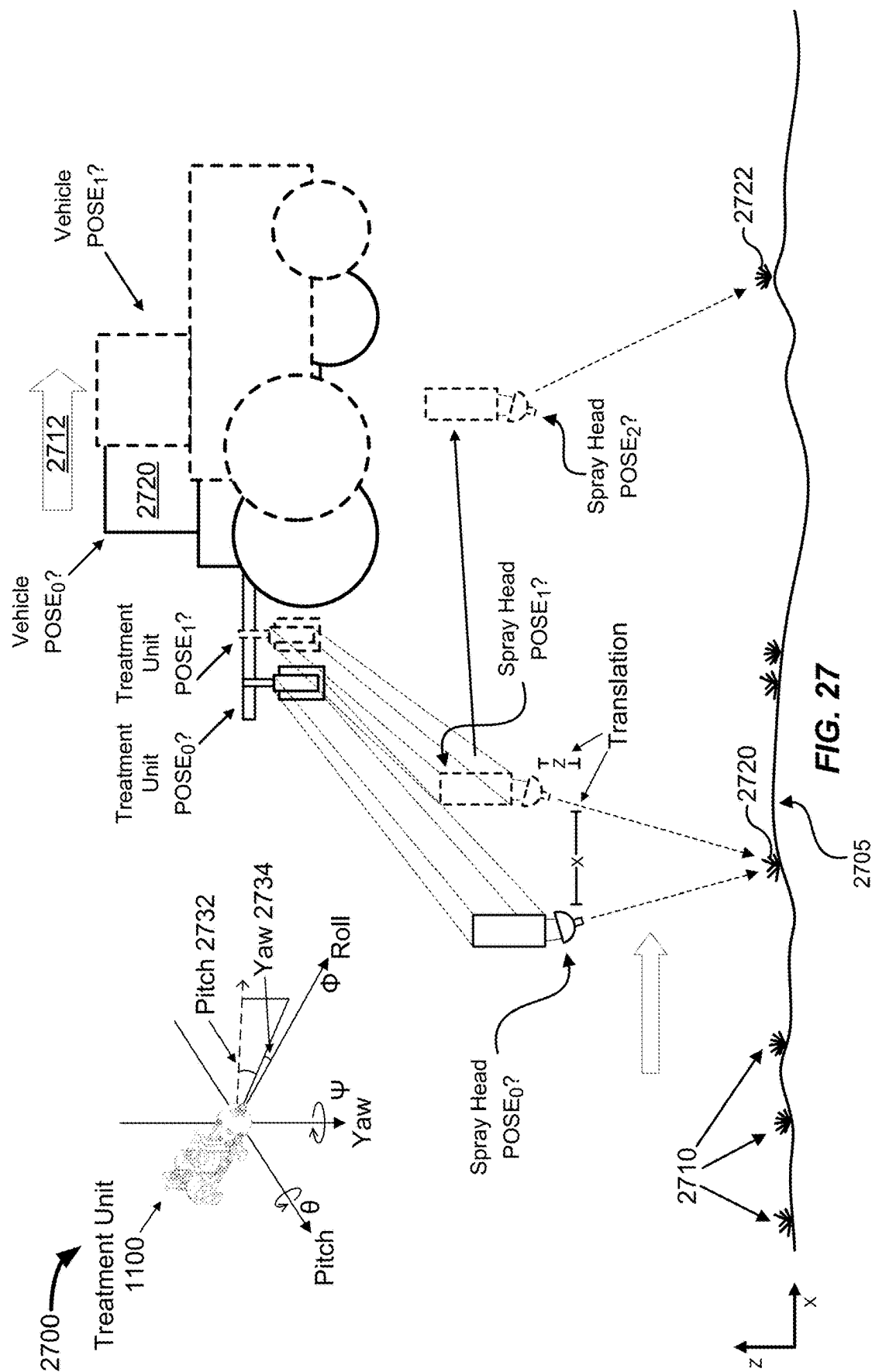
FIG. 25E

2600 **FIG. 26A**

2650

**FIG. 26B**

2602 **FIG. 26C**



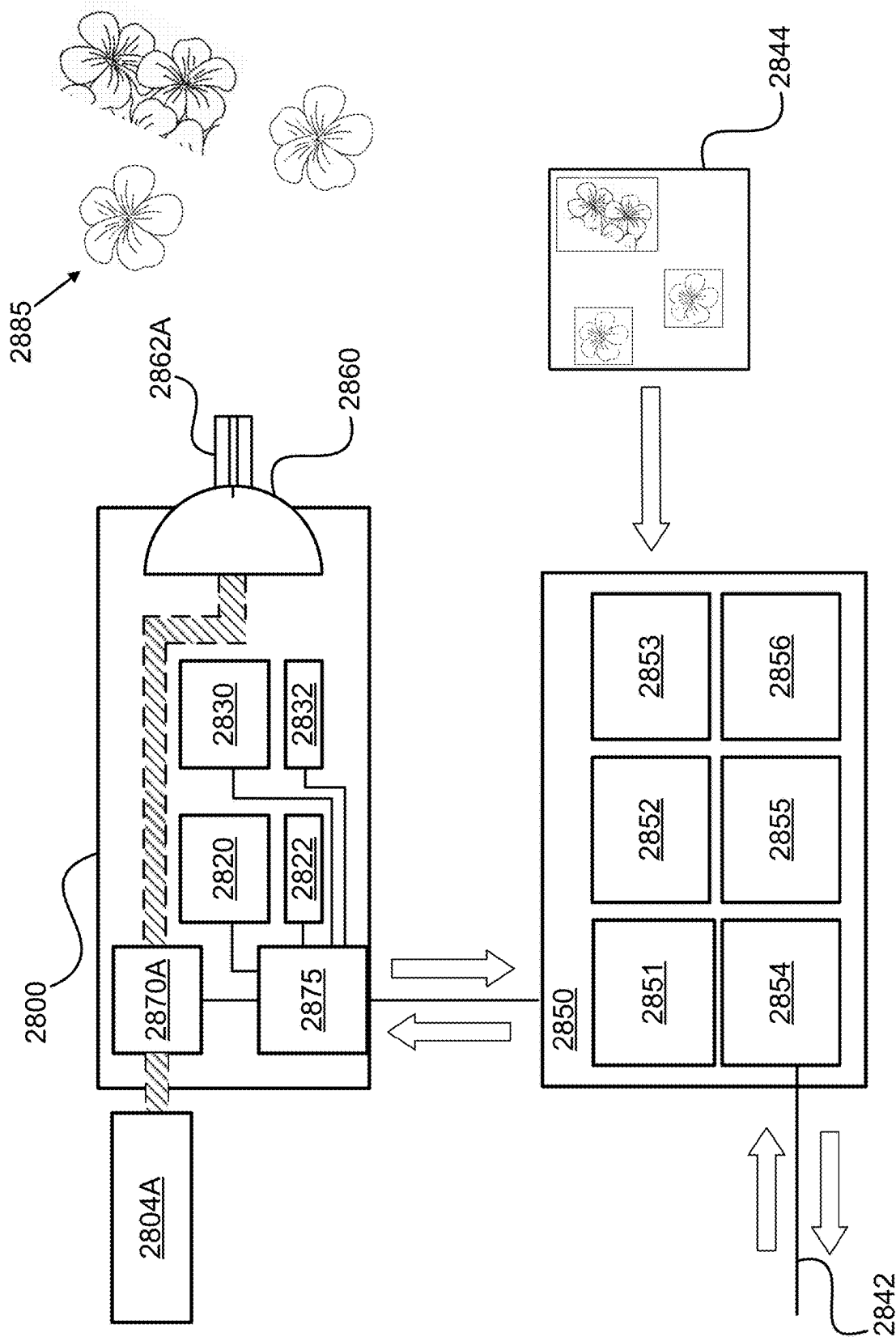


FIG. 28A

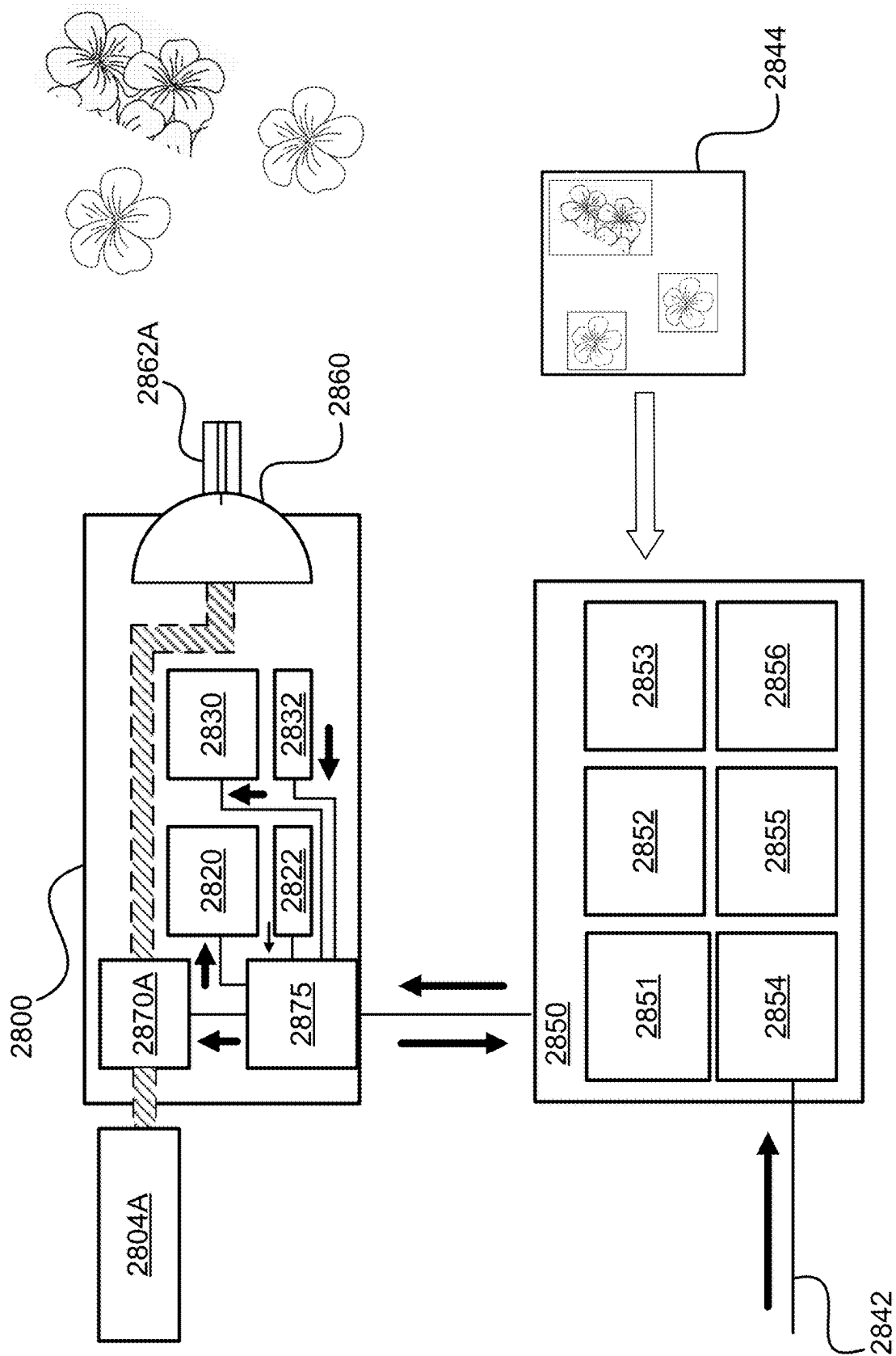


FIG. 28B

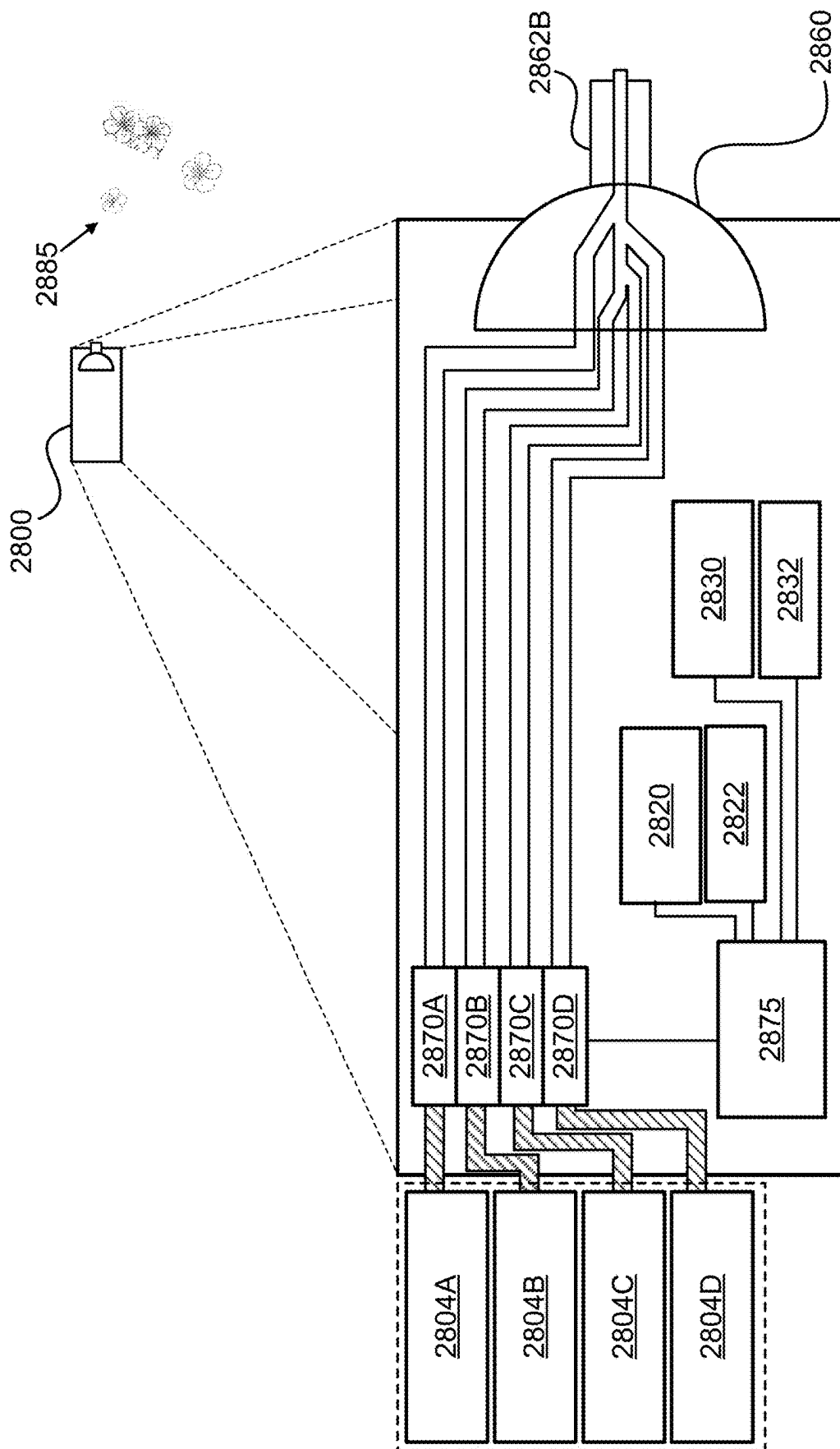


FIG. 28C

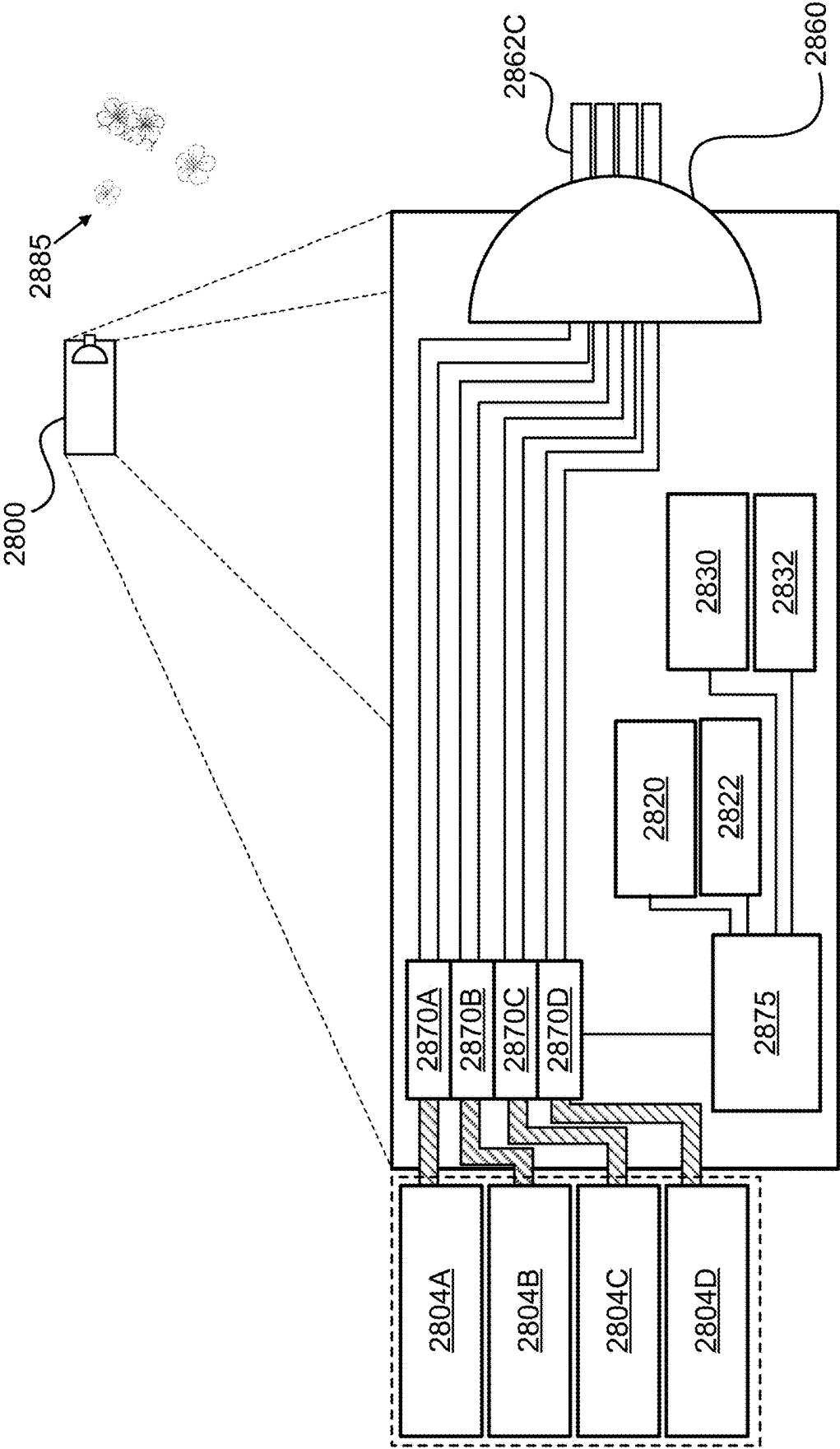


FIG. 28D

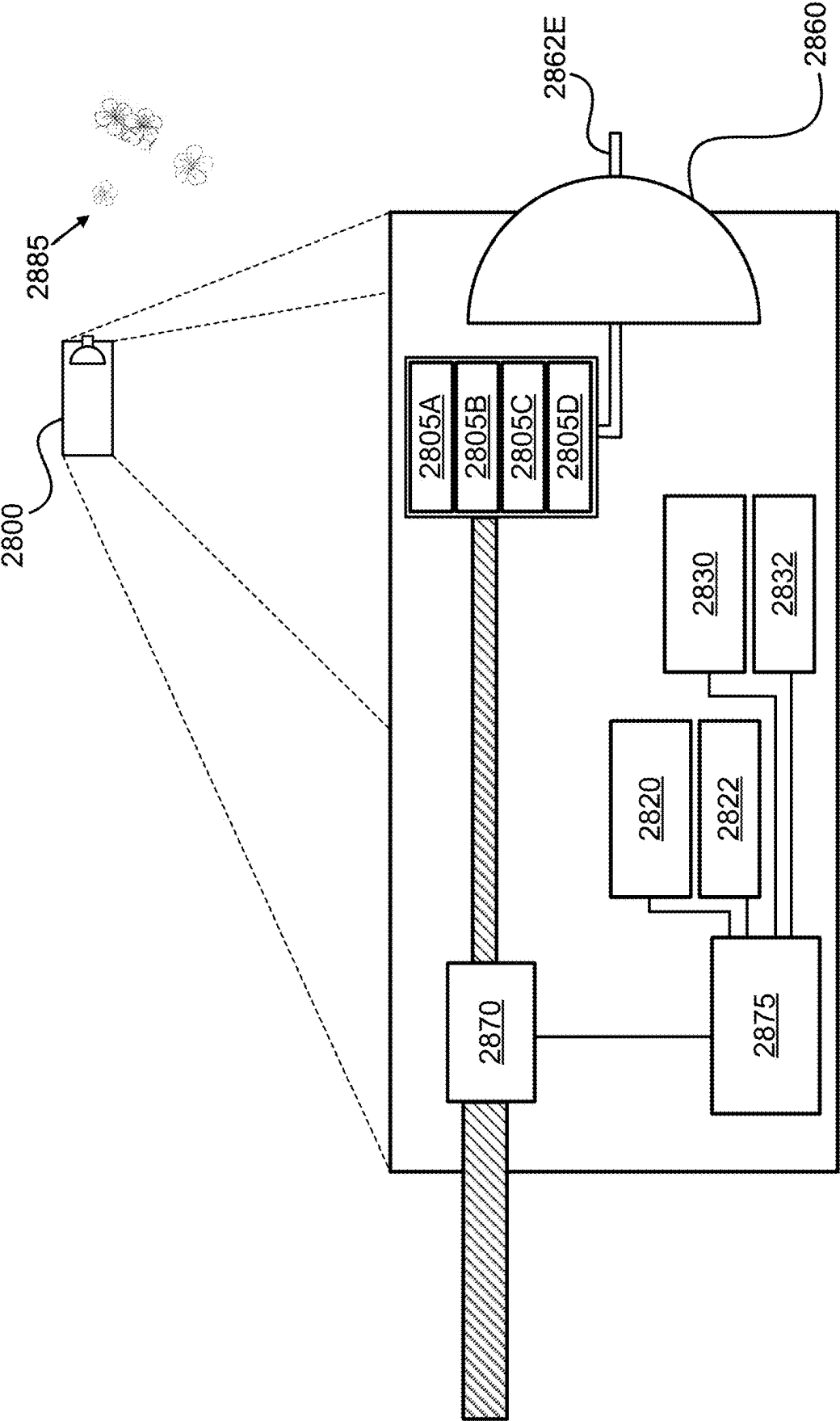


FIG. 28E

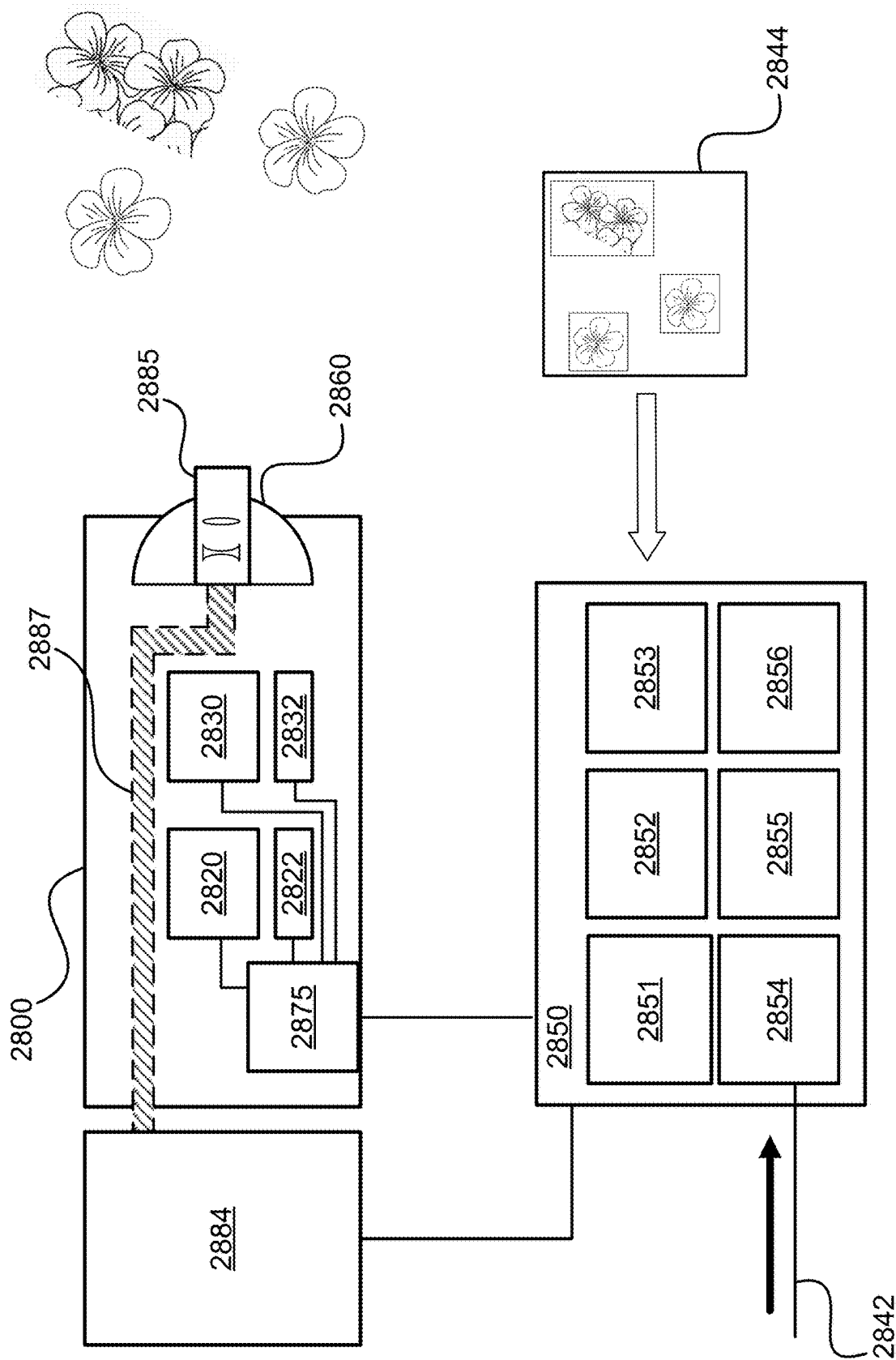


FIG. 28F

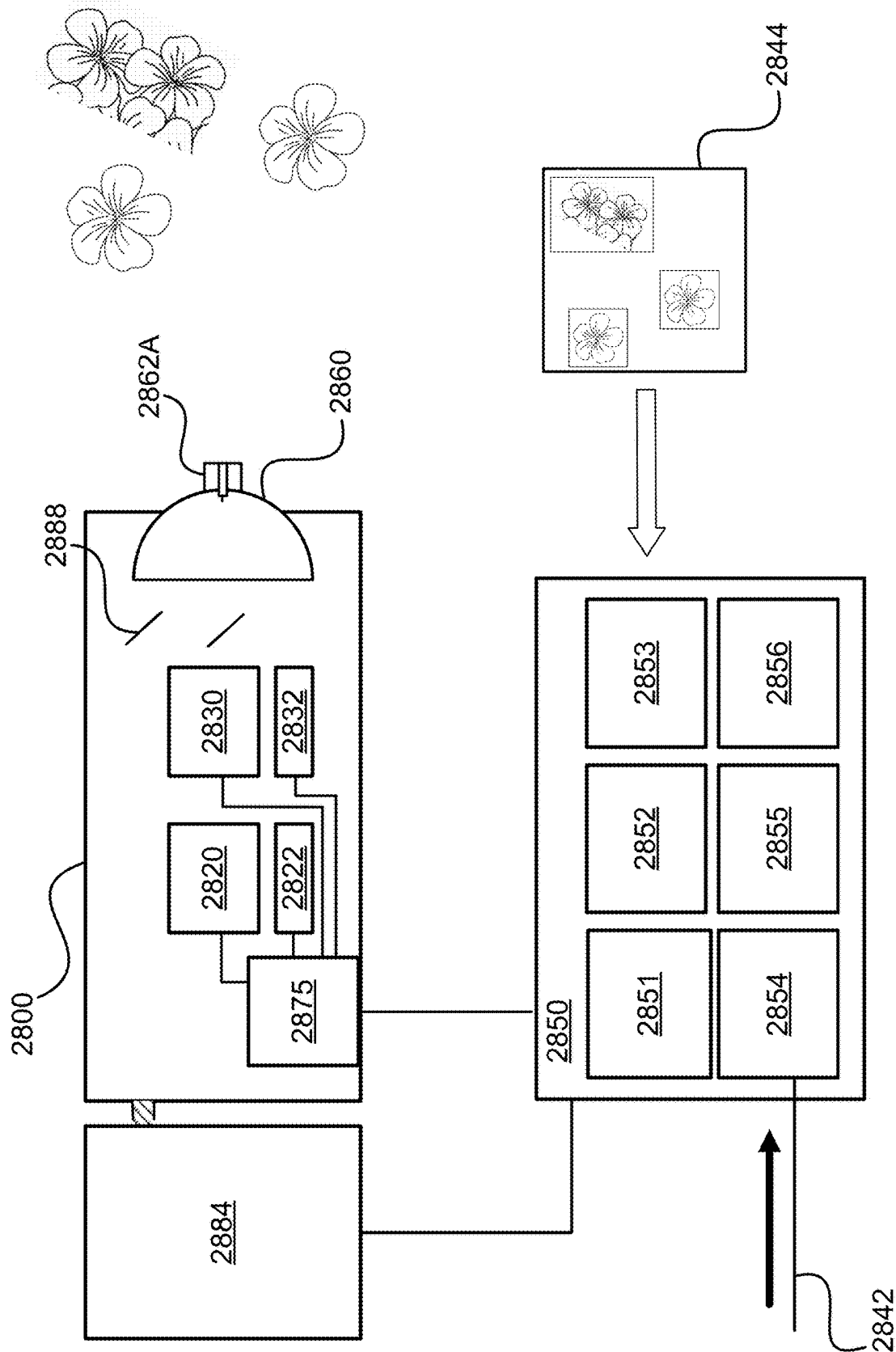


FIG. 28G

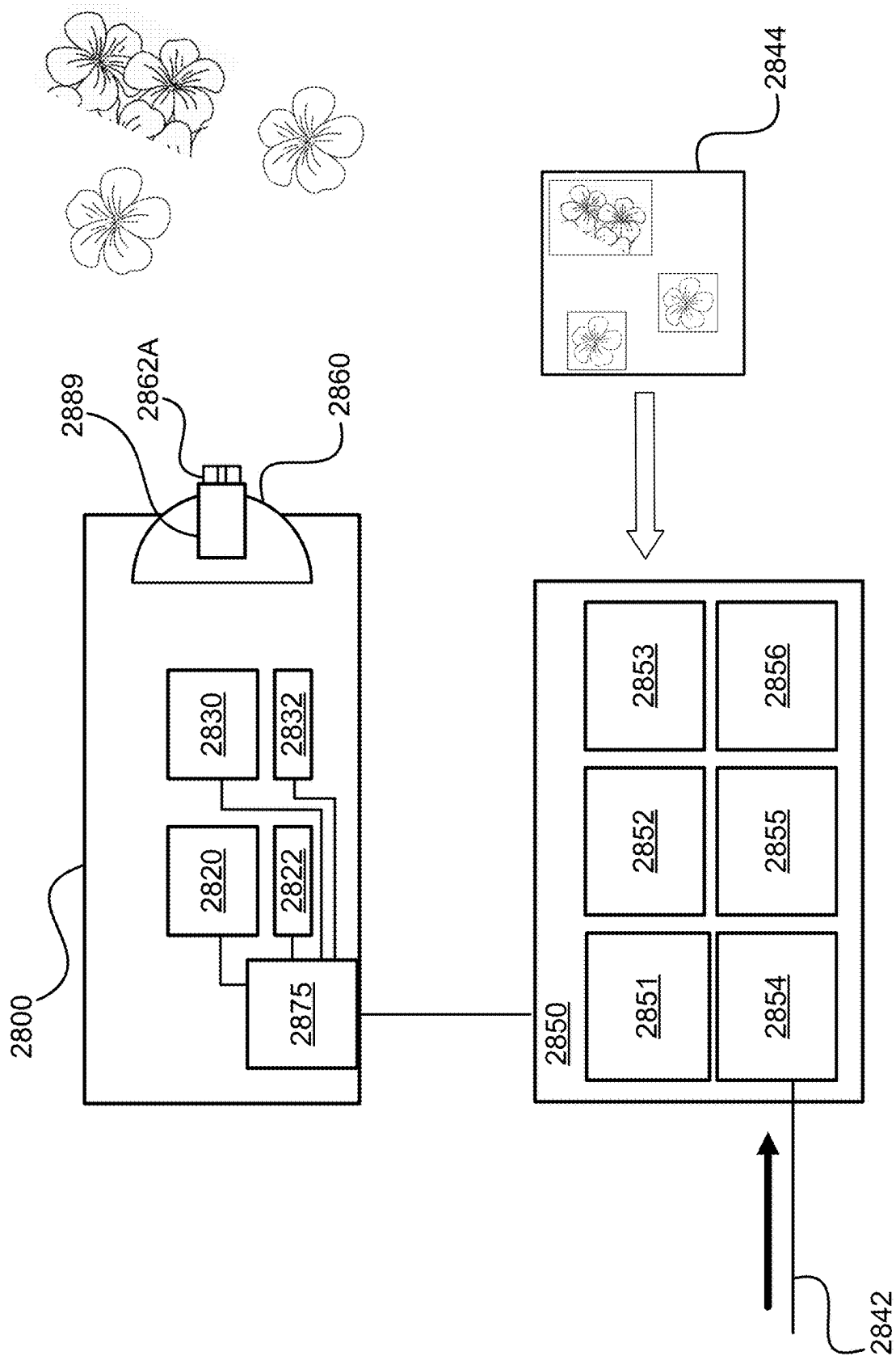


FIG. 28H

2900

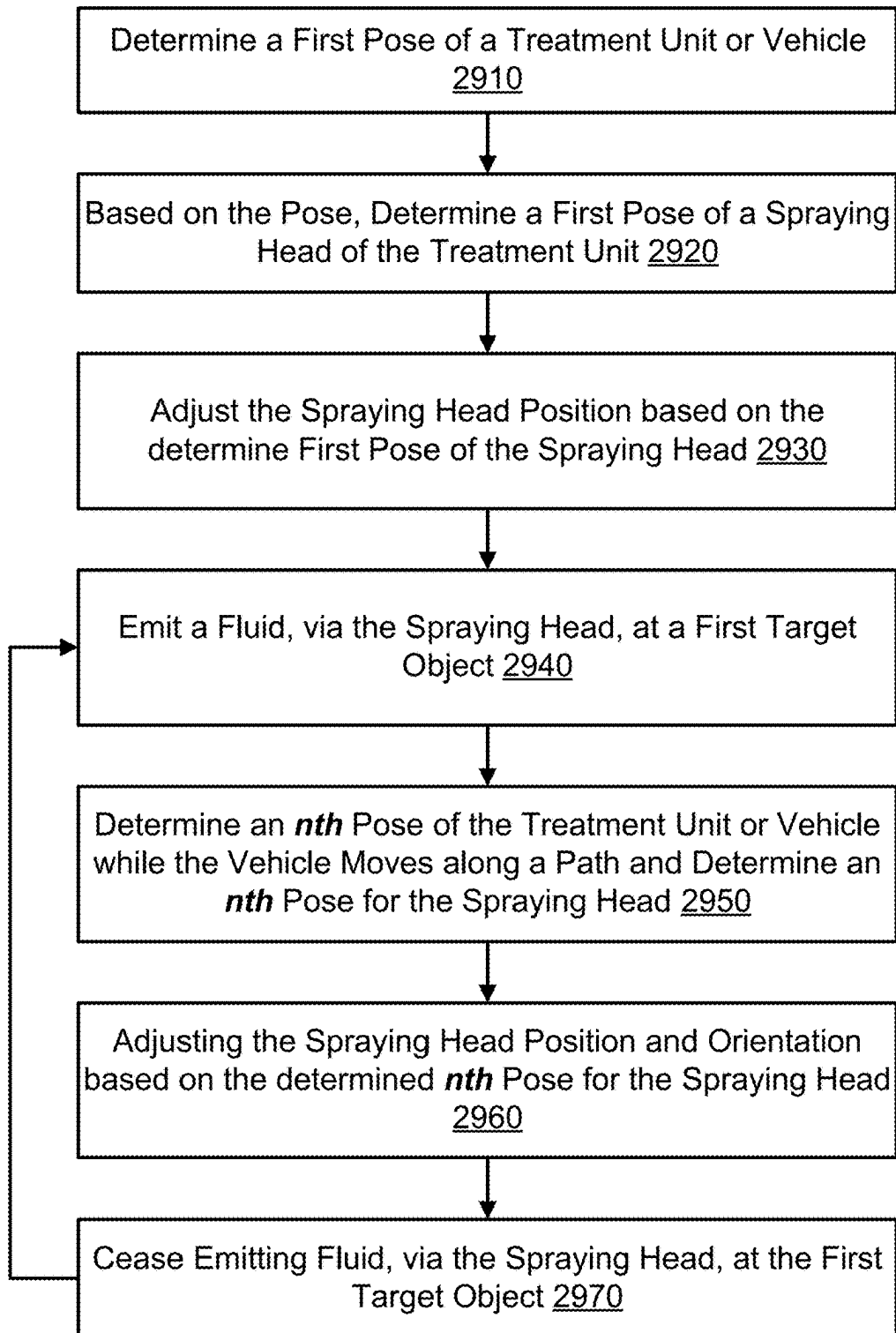
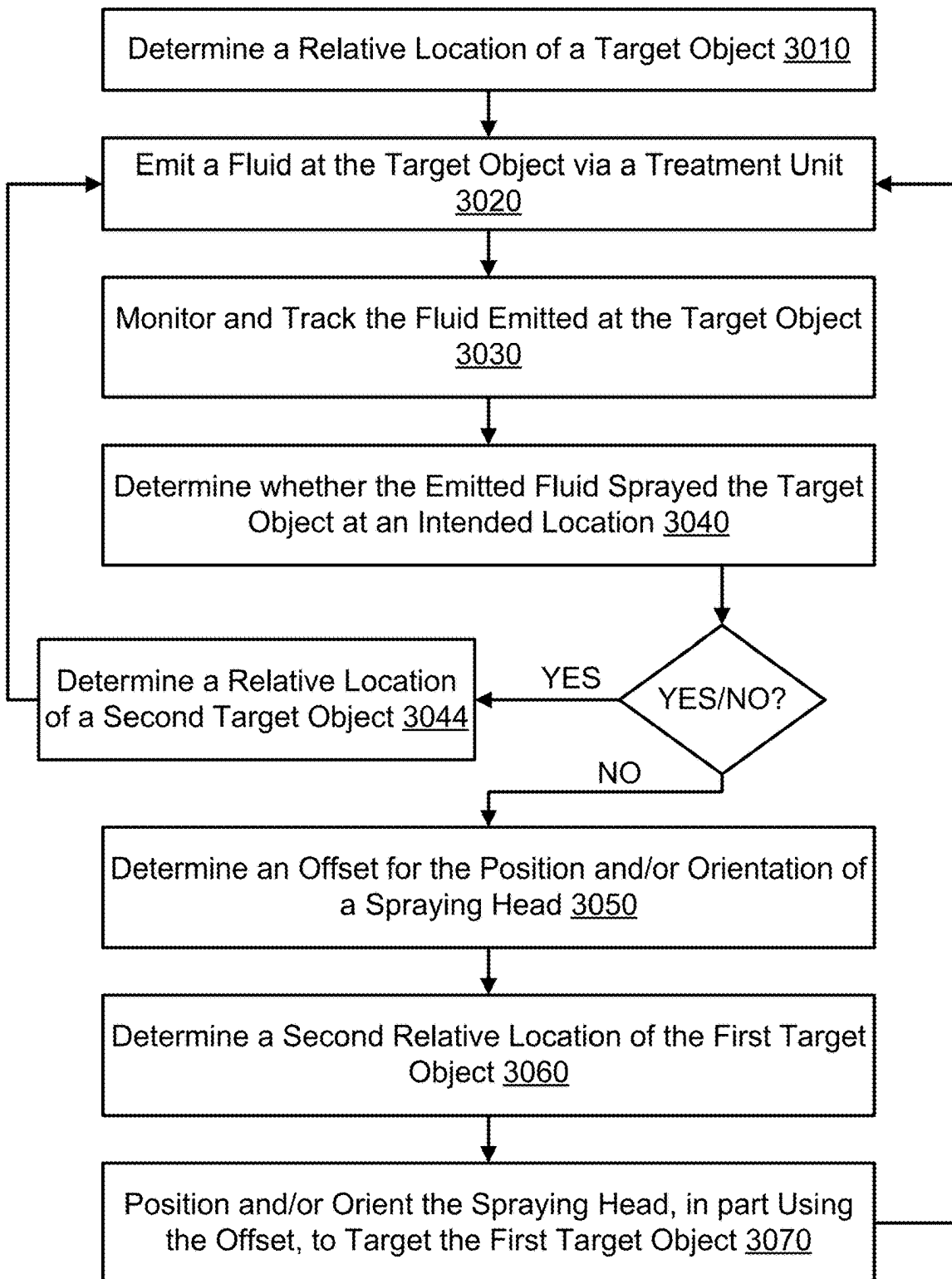
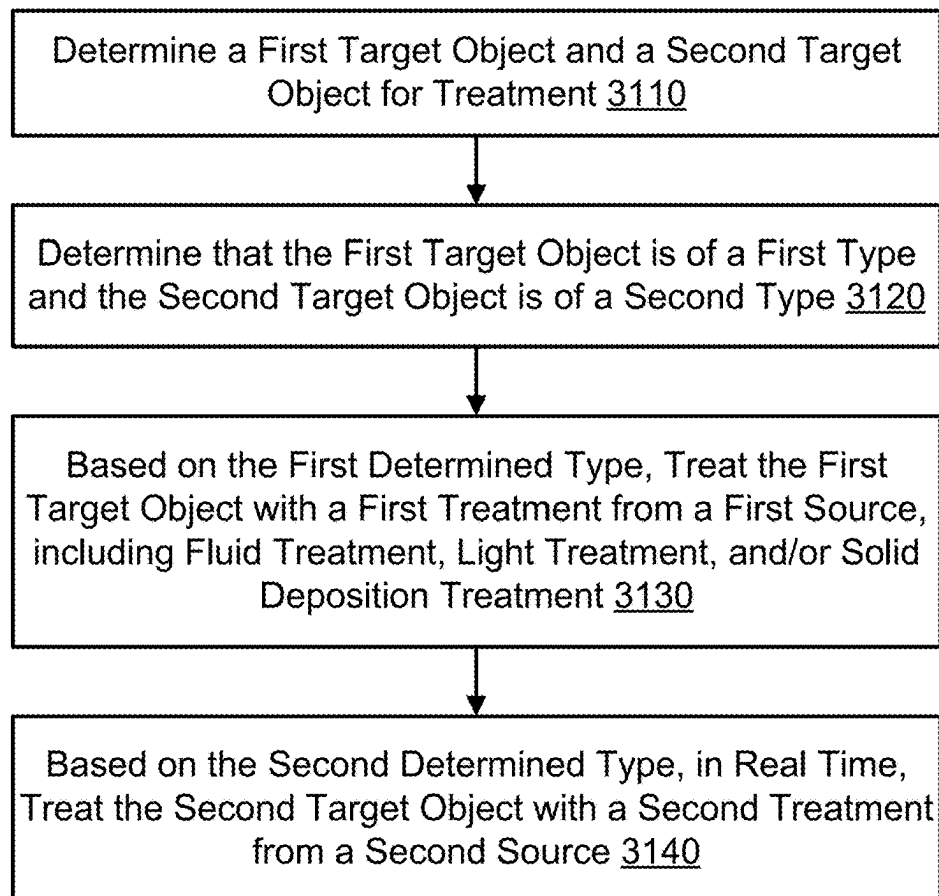
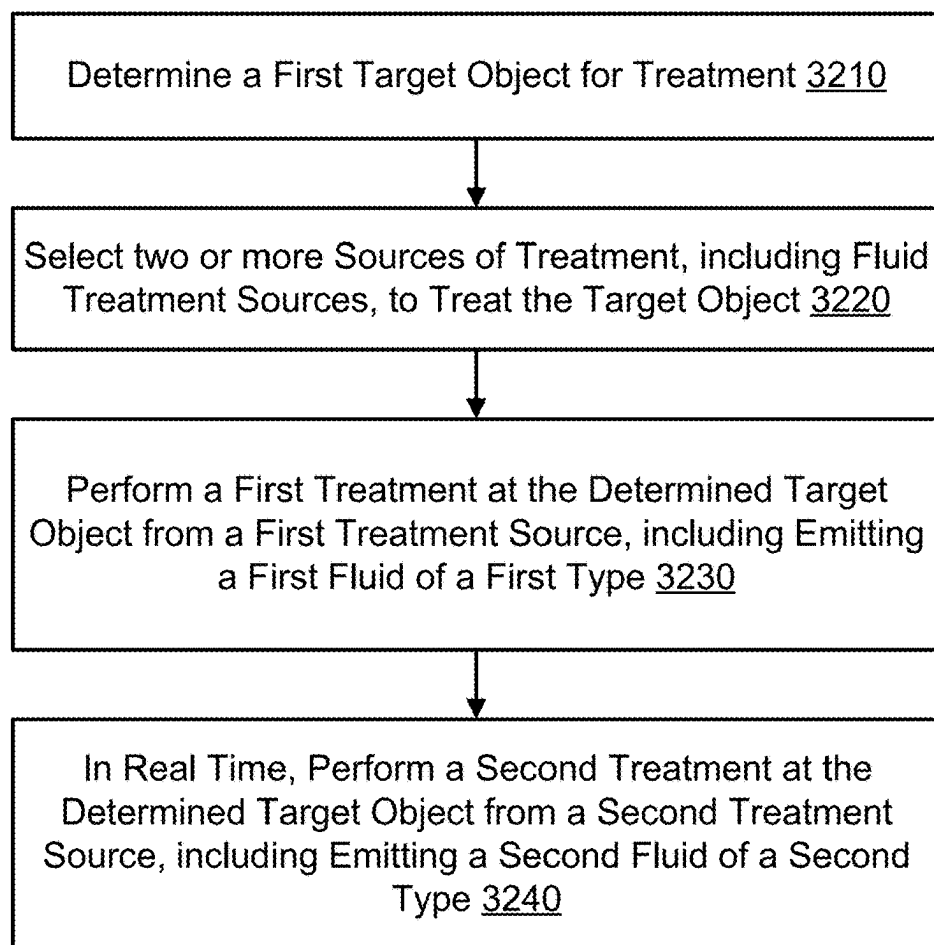


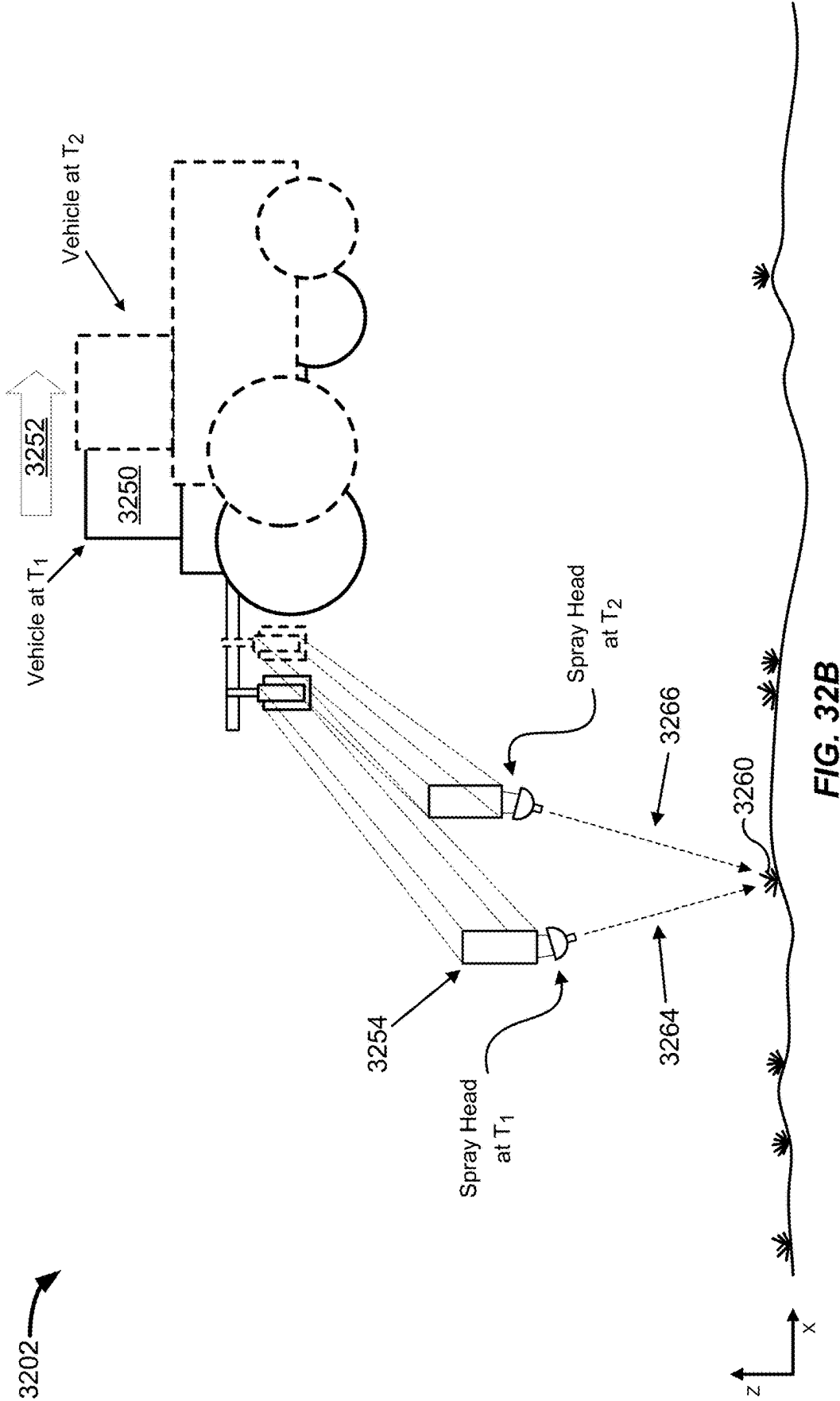
FIG. 29

3000

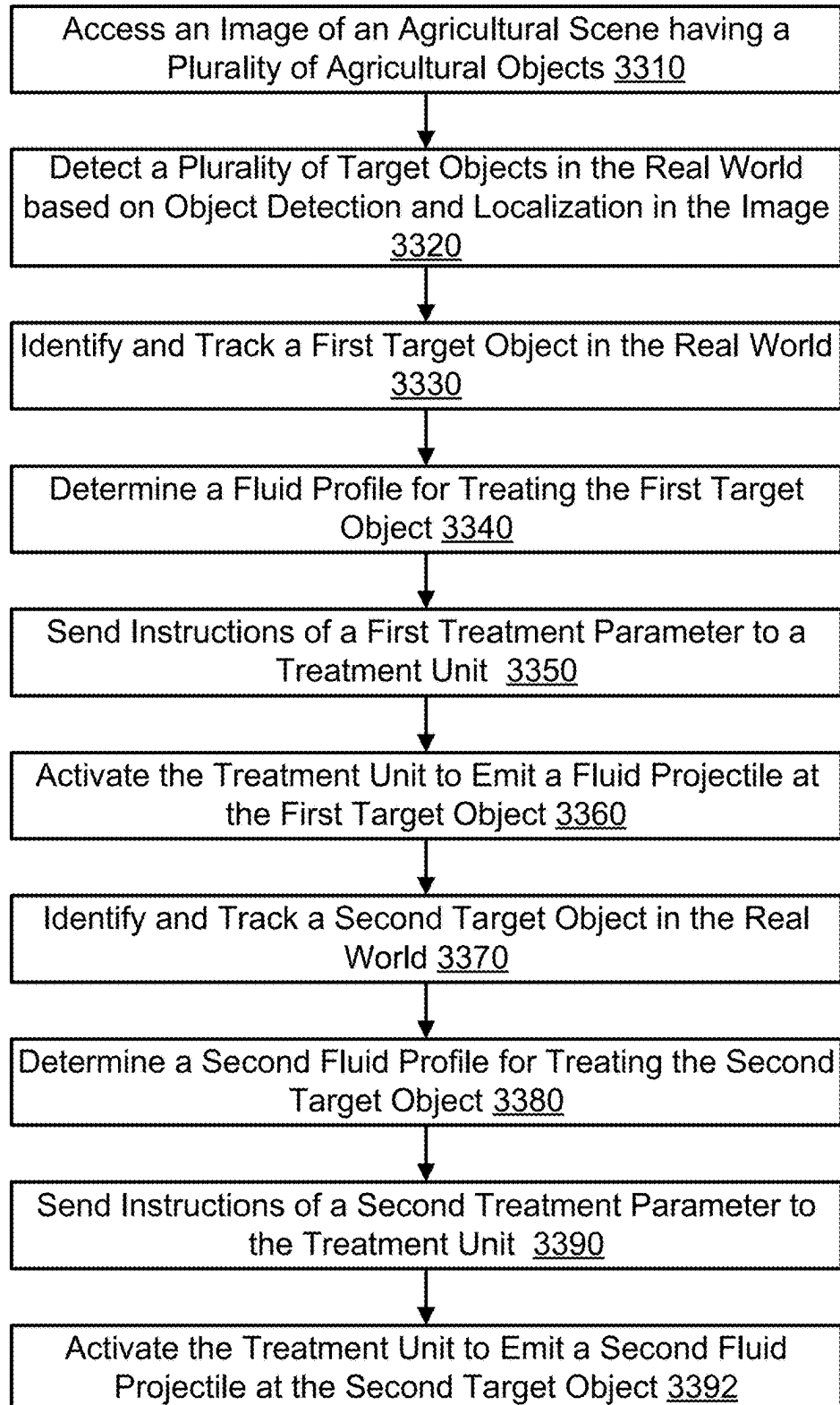
**FIG. 30**

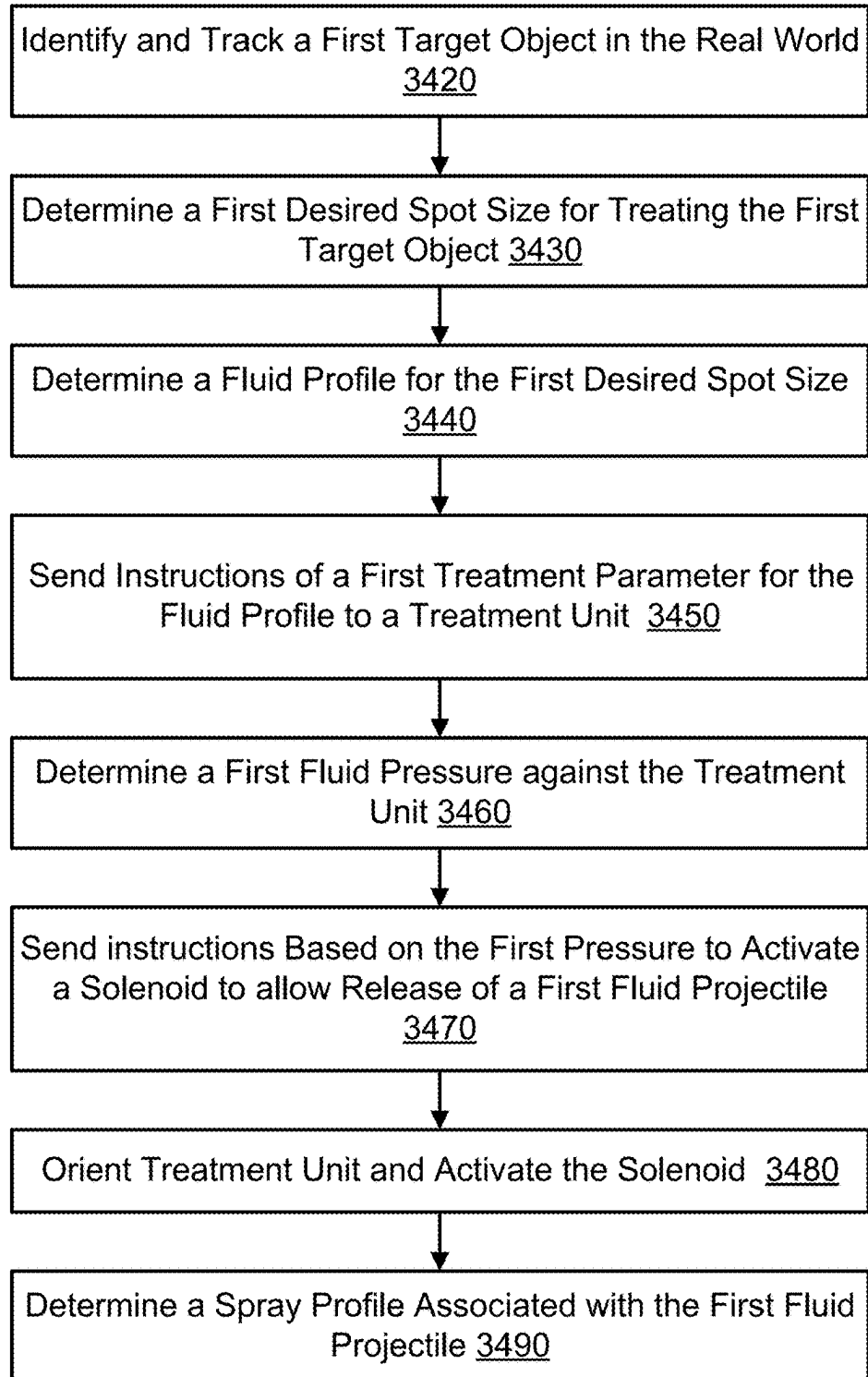
3100 **FIG. 31**

3200 **FIG. 32A**



3300

**FIG. 33**

3400 **FIG. 34**

PERFORMING MULTIPLE ACTIONS ON PLANT OBJECTS ON A MOVING PLATFORM

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/722,285, filed on Apr. 15, 2022, which is a continuation of U.S. application Ser. No. 17/544,152, filed Dec. 7, 2021, which is a continuation-in-part application of U.S. application Ser. No. 17/073,244, filed Oct. 16, 2020, which are hereby incorporated by reference in its entirety.

BACKGROUND

Global human population growth is expanding at a rate projected to reach 10 billion or more persons within the next 40 years, which, in turn, will concomitantly increase demands on producers of food. To support such population growth, food production, for example on farms and orchards, need to generate collectively an amount of food that is equivalent to an amount that the entire human race, from the beginning of time, has consumed up to that point in time. Many obstacles and impediments, however, likely need to be overcome or resolved to feed future generations in a sustainable manner.

To support such an increase in demand, agricultural technology has been implemented to more effectively and efficiently grow crops, raise livestock, and cultivate land. Such technology in the past has helped to more effectively and efficiently use labor, use tools and machinery, and reduce the amount of chemicals used on plants and cultivated land.

However, many techniques used currently for producing and harvesting crops are only incremental steps from a previous technique. The amount of land, chemicals, time, labor, and other costs to the industry still pose a challenge. A new and improved system and method of performing agricultural services is needed.

SUMMARY

In one embodiment, the agricultural treatment system uses a treatment unit for spraying fluid at agricultural objects. The treatment unit is configured with a treatment head assembly that includes a moveable treatment head with one or more spraying tips. A first and second motor assembly are operated by the treatment unit to control the movement of the treatment head. The first motor assembly includes a first motor rotatable in a first rotational axis. A first linkage assembly is connected to the first motor and the treatment head assembly. The first linkage assembly is rotatable by the first motor. The second motor assembly includes a second motor rotatable in a second rotational axis. The first rotational axis is different from the second rotational axis. The second linkage assembly is rotatable by the second motor.

The treatment unit may spray a fluid controlled by a fluid regulator with an operable solenoid to control the release of pressurized fluid. A flexible tube may fluidly be coupled from the fluid regulator to the treatment head assembly. The treatment head assembly may include a spraying head and one or more spraying tips to emit the released fluid. A controller may be configured to control the fluid regulator and the first and second motors thereby adjusting a position of the moveable treatment head and emitting a fluid from the one or more spraying tips. In mode of operation, the rotation

of the first motor causes the treatment head assembly to pivot along the first axis, and the rotation of the second motor causes the treatment head assembly to pivot along a second axis. The agricultural treatment system may determine a first target object to be sprayed with a first fluid emitted from the one or more spraying tips. The agricultural treatment system may adjust the position of the spraying head assembly via rotation of the first and/or second motors, and then emit the first fluid, obtained from a fluid source, at the first target object.

In one embodiment, the agricultural treatment system may include one or more fluid tanks with a first tank configured or capable of holding or storing a first fluid. The system uses a pump in combination with a fluid regulator to obtain and disperse fluid from the fluid tanks. The fluid regulator includes a fluid input port, a fluid output port and a fluid control valve. A first tube fluidly coupled from the fluid pump to the fluid input port. The system includes a moveable treatment head having one or more spraying tips, including a first spraying tip. A second tube is fluidly coupled from the fluid output port to the moveable treatment head. A system includes a controller configured to provide to the fluid regulator to open and close the fluid control valve to release an amount of the first fluid. The controller may be configured to control the fluid regulator causing the fluid control valve to open and close in a manner to emit the first fluid in a predetermined spray amount and a predetermined spray diameter to cover an area of a target object. Additionally, the controller may be configured to control the fluid regulator causing the fluid control valve to open and close in a manner to emit multiple bursts of fluid projectiles of the first fluid. In some embodiments, the controller may be configured to control the fluid regulator to open and close the control valve at different time intervals such that an amount of pressure in the first tube causes the bursts of fluid projectiles of the first fluid to each emit in a predetermined spray volume and a predetermined spray diameter to cover an area of a target object. In some embodiments, the distance by which the control valve opens for each of the multiple bursts may be a different distance so as to maintain a same trajectory, volume and/or droplet size for each burst of the fluid projectiles.

The system includes one or processors configured to perform the operations of determining a first spray profile for treating a first target object; based on the first spray profile, emitting a first fluid projectile of the first fluid from the one or more spraying tips at the first target object; determining a second spray profile for treating a second target object; based on the second spray profile, emitting a second fluid projectile of the first fluid from the one or more spraying tips at the second target object; and wherein a spray size for the first target object is different than a spray size for the second target object.

In some embodiments, the fluid control valve may be maintained in a closed position and the pump creates a fluid line pressure in the first tube in the range of 1-2700 PSI. In some embodiments, the amount of the first fluid projectile and the amount of the second fluid projectile ranges from 1 microliter to 1000 milliliters. In some embodiments, multiple bursts of the fluid projectiles have a same spray distance and a same spray size. In some embodiments, the number of bursts of the fluid projectiles of the first fluid ranges from 2-20 bursts in a period of 1 second.

In one embodiment, the agricultural treatment system determines a vehicle pose of a vehicle as the vehicle moves along a path. The system identifies a first target agricultural object for treatment. Based on the determined vehicle pose,

the system positions a treatment head of a first treatment unit such that a first projectile fluid may be emitted by the first treatment unit at the identified first target agricultural object. The system then causes an emitter to emit a fluid from the treatment head at the first target agricultural object.

Additionally, the system may determine a spraybox pose for a first spraybox. The first spraybox may comprise a treatment module, a component treatment module and/or treatment subsystem. The first spraybox may include one or more cameras integral to the spraybox. The spraybox pose may be based on obtained sensor data from at least the one or more cameras integral to the spraybox. The vehicle pose may be based on obtained sensor data from at least one of a camera, laser, radar sensor, sonar sensor, radio sensor, ultrasonic sensor, rangefinder, photoelectric sensor, global positioning systems (GPS), inertial measurement unit (IMU), gyroscopes, accelerometer, magnetometer, or a combination thereof.

In one embodiment, the vehicle pose may be set to a vehicle local coordinate origin (x_0, y_0, z_0) and a local vehicle orientation (Φ_0, θ_0, Ψ_0), and the first treatment unit pose may be determined as a translated first treatment unit coordinate origin (x_1, y_1, z_1) relative to the vehicle local coordinate origin and as a translated first treatment unit starting orientation (Φ_1, θ_1, Ψ_1) relative to the local vehicle orientation.

The system may identify a second target agricultural object for treatment. The system may determine a spraybox pose for a second spraybox. The second spraybox may include a treatment module, a component treatment module and treatment subsystem. Based on the determined vehicle pose, and the determined spraybox pose, the system may position a treatment head of a second treatment unit such that a second projectile fluid may be emitted by the second treatment unit at the identified second target agricultural object.

Moreover, the system may determine a treatment head pose for the treatment head where the treatment head pose is based on encoders tracking the orientation of the treatment head and spraybox pose. The system may also determine a treatment head pose, based on the vehicle pose, the spraybox pose, and an orientation of the treatment unit to spraybox.

The system may also obtain sensor data from one or more sensors and analyzes the obtained sensor data to determine the vehicle pose. The sensor data may include images of a real-world environment in which the vehicle may travel.

The system may also obtain one or more images depicting a plurality of agricultural objects. The system may analyze the one or more image to identify one of the plurality of agricultural objects to treat. The system may then select the identified one of the plurality of agricultural objects as the first target agricultural object for treatment.

The system may identify one or more agricultural objects positioned about the vehicle. Using the determined vehicle pose, while the vehicle is moving along the path, the system may navigate the vehicle about the identified one or more agricultural objects so as to avoid vehicle impact with the identified one or more agricultural objects.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will become better understood from the detailed description and the drawings, wherein:

FIG. 1 is a diagram illustrating an exemplary environment, according to some examples.

FIG. 2 is a diagram illustrating an exemplary environment, according to some examples.

FIG. 3A is a diagram illustrating an agricultural scene within a geographic boundary, according to some examples.

FIG. 3B is a diagram illustrating image acquisition and digitization of a geographic boundary, according to some examples.

FIG. 3C is a diagram illustrating image acquisition and digitization of objects a geographic boundary measured across time, according to some examples.

FIG. 3D is a diagram illustrating an interface for interacting with a digitized agricultural scene, according to some examples.

FIG. 3E is a diagram illustrating an interface for interacting with an agricultural scene, according to some examples.

FIG. 4 is a diagram illustrating an example agricultural observation and treatment system, according to some examples.

FIG. 5 is a diagram illustrating a component of an example agricultural observation and treatment system, according to some examples.

FIG. 6 is a diagram illustrating an example vehicle supporting an observation and treatment system performing in a geographic boundary, according to some examples.

FIG. 7A is a diagram of a vehicle navigating in an agricultural environment, according to some examples.

FIGS. 7B-7D are block diagrams illustrating an exemplary method that may be performed by a treatment system, according to some examples.

FIG. 8 is a diagram illustrating an additional portion of an example agricultural observation and treatment system, according to some examples.

FIG. 9A is a diagram illustrating an example component of an agricultural observation and treatment system, according to some examples.

FIG. 9B is a diagram illustrating an example component of an agricultural observation and treatment system, according to some examples.

FIG. 10 is block diagram illustrating an exemplary method that may be performed by a treatment system, according to some examples.

FIGS. 11A-H are diagrams illustrating example treatment units, according to some examples.

FIG. 12A is block diagram illustrating an exemplary method that may be performed by a treatment system, according to some examples.

FIG. 12B is block diagram illustrating an exemplary method that may be performed by a treatment system, according to some examples.

FIG. 13A is a diagram illustrating an exemplary labelled image, according to some examples.

FIG. 13B is a diagram illustrating an exemplary labelled image, according to some examples.

FIG. 14A is a block diagram illustrating an exemplary method that may be performed by a treatment system, according to some examples.

FIG. 14B is a block diagram illustrating an exemplary method that may be performed by a treatment system, according to some examples.

FIG. 15 is block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 16 is a diagram illustrating an example image acquisition to object determination performed by an example system, according to some examples.

FIG. 17A is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 17B is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 18A is a diagram illustrating capturing an action performed by an observation and treatment system, according to some examples.

FIG. 18B is a diagram illustrating capturing action and treatment pattern detection, according to some examples.

FIG. 18C is a diagram illustrating capturing action and treatment pattern detection, according to some examples.

FIG. 18D is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 18E is a diagram illustrating capturing action and treatment pattern detection, according to some examples.

FIG. 18F is a diagram illustrating capturing an action and a treatment pattern, according to some examples.

FIG. 19A is a diagram illustrating example treatment patterns performed and observed by an example treatment system, according to some examples.

FIG. 19B is a diagram illustrating example treatment patterns performed and observed by an example treatment system, according to some examples.

FIG. 19C is a diagram illustrating example treatment patterns performed and observed by an example treatment system, according to some examples.

FIG. 20 is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 21A is a diagram illustrating an example vehicle supporting an example observation and treatment system performing in a geographic boundary, according to some examples.

FIG. 21B is a diagram illustrating an example vehicle supporting an example observation and treatment system performing in a geographic boundary, according to some examples.

FIG. 21C is a diagram illustrating an example vehicle supporting an example observation and treatment system, according to some examples.

FIG. 22 is a diagram illustrating an example vehicle supporting an example observation and treatment system, according to some examples.

FIG. 23 is a diagram illustrating axes of movement, rotation, and degrees of freedom of a vehicle and components of an observation and treatment system, according to some examples.

FIG. 24A is a diagram illustrating an example vehicle supporting an example observation and treatment system, according to some examples.

FIG. 24B is a diagram illustrating an example vehicle supporting an example observation and treatment system, according to some examples.

FIG. 24C is a diagram illustrating an example vehicle supporting an example observation and treatment system performing in a geographic boundary, according to some examples.

FIG. 24D is a diagram illustrating an example vehicle supporting an example observation and treatment system performing in a geographic boundary, according to some examples.

FIG. 24E is a diagram illustrating an example observation and treatment system and components of the observation and treatment system, according to some examples.

FIG. 24F is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIGS. 25A-E are diagrams illustrating example treatment units, according to some examples.

FIG. 26A is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 26B is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 26C is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 27 is a diagram illustrating a treatment system observing an environment and performing actions in a geographic boundary, according to some examples.

FIG. 28A is a diagram illustrating an example configuration of a system with a treatment unit having an example configuration of a fluid source and fluid flow mechanisms.

FIG. 28B is a diagram illustrating an example configuration of a system with a treatment unit having an example configuration of a fluid source and fluid flow mechanisms.

FIG. 28C is a diagram illustrating an example configuration of a system with a treatment unit having an example configuration of a fluid source and fluid flow mechanisms.

FIG. 28D is a diagram illustrating an example configuration of a system with a treatment unit having an example configuration of a fluid source and fluid flow mechanisms.

FIG. 28E is a diagram illustrating an example configuration of a system with a treatment unit having an example configuration of a fluid source and fluid flow mechanisms.

FIG. 28F is a diagram illustrating an example configuration of a system with a treatment unit configured with a light source operably connected to a gimbal mechanism.

FIG. 28G is a diagram illustrating another example configuration of a system with a treatment unit configured with a light source operably connected to a gimbal mechanism.

FIG. 28H is a diagram illustrating another example configuration of a system with a treatment unit configured with a light source operably connected to a gimbal mechanism.

FIG. 29 is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 30 is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 31 is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 32A is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 32B is a diagram illustrating another example configuration of a system with a treatment unit moving along a path.

FIG. 33 is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

FIG. 34 is a block diagram illustrating an exemplary method that may be performed by an agricultural observation and treatment system, according to some examples.

DETAILED DESCRIPTION

In this specification, reference is made in detail to specific embodiments of the disclosure. Some of the embodiments or their aspects are illustrated in the drawings.

For clarity in explanation, the disclosure has been described with reference to specific embodiments, however

it should be understood that the disclosure is not limited to the described embodiments. On the contrary, the disclosure covers alternatives, modifications, and equivalents as may be included within its scope as defined by any patent claims. The following embodiments of the disclosure are set forth without any loss of generality to, and without imposing limitations on, the claimed disclosure. In the following description, specific details are set forth in order to provide a thorough understanding of the present disclosure. The present disclosure may be practiced without some or all of these specific details. In addition, well known features may not have been described in detail to avoid unnecessarily obscuring the disclosure.

In addition, it should be understood that steps of the exemplary methods set forth in this exemplary patent can be performed in different orders than the order presented in this specification. Furthermore, some steps of the exemplary methods may be performed in parallel rather than being performed sequentially. Also, the steps of the exemplary methods may be performed in a network environment in which some steps are performed by different computers in the networked environment.

Some embodiments are implemented by a computer system. A computer system may include a processor, a memory, and a non-transitory computer-readable medium. The memory and non-transitory medium may store instructions for performing methods and steps described herein. Various examples and embodiments described below relate generally to robotics, autonomous driving systems, and autonomous agricultural application systems, such as an autonomous agricultural observation and treatment system, utilizing computer software and systems, computer vision and automation to autonomously identify an agricultural object including any and all unique growth stages of agricultural objects identified, including crops or other plants or portions of a plant, characteristics and objects of a scene or geographic boundary, environment characteristics, or a combination thereof.

Additionally, the systems, robots, computer software and systems, applications using computer vision and automation, or a combination thereof, can be configured observe a geographic boundary having one or more plants growing agricultural objects identified as potential crops, detect specific agricultural objects to each individual plant and portions of the plant, determine that one or more specific individual agricultural object in the real world geographic boundary requires a treatment based on its growth stage and treatment history from previous observations and treatment, and to deliver a specific treatment to each of the desired agricultural objects, among other objects. Generally, the computer system provides computer vision functionality using stereoscopic digital cameras and performs object detection and classification and apply a chemical treatment to target objects that are potential crops via an integrated onboard observation and treatment system. The system utilizes one or more image sensors, including stereoscopic cameras to obtain digital imagery, including 3D imagery of an agricultural scene such as a tree in an orchard or a row of plants on a farm while the system moves along a path near the crops. Onboard lights sources, such as LEDs, may be used by the system to provide a consistent level of illumination of the crops while imagery of the crops is being obtained by the image sensors. The system can then identify and recognize different types of objects in the imagery. Based on detected types of objects in the digital imagery, or the same object from one moment in time to another moment in time experiencing a different growth stage which can be

recognized, observed, and identified by the on system, as well as the system associating the growth stage or the different label with a unique individual agricultural object previously identified and located at previous growth stage, the system can apply a treatment, for example spray the real-world object with chemicals pumped from one or more liquid tanks, onto a surface of the agricultural object. The system may optionally use one or more additional image sensors to record the treatment, as a projectile, as it is applied from the system to the agricultural object in proximity to the system.

Referring now to FIG. 1, a diagram of an exemplary network environment in which example systems and devices may operate is shown. In the exemplary environment, clients **141** are connected over a network **145** to a server **150** having local storage **151**. Clients and servers in this environment may be computers. Server **150** may be configured to handle requests from clients. Server **150** may be implemented as a number of networked server devices, though it is illustrated as a single entity. Communications and transmissions between a base station and one or vehicles, or other ground mobility units configured to support a server **150**, and between a base station and one or more control centers as described herein may be executed similarly as the client **141** requests.

The exemplary environment is illustrated with only two clients and one server for simplicity, though in practice there may be more or fewer clients and servers. The computers have been termed clients and servers, though clients can also play the role of servers and servers can also play the role of clients. In some examples, the client **141** may communicate with each other as well as the servers. Also, the server **150** may communicate with other servers.

The network **145** may be, for example, local area network (LAN), wide area network (WAN), networks utilizing 5G wireless standards technology, telephone networks, wireless networks, intranets, the Internet, or combinations of networks. The server **150** may be connected to storage **152** over a connection medium, which may be a bus, crossbar, network, wireless communication interface, or other interconnect. Storage **152** may be implemented as a network of multiple storage devices, though it is illustrated as a single entity. Storage **152** may be a file system, disk, database, or other storage.

In one example, the client **141** may perform one or more methods herein and, as a result, store a file in the storage **152**. This may be accomplished via communication over the network **145** between the client **141** and server **150**. For example, the client may communicate a request to the server **150** to store a file with a specified name in the storage **152**. The server **150** may respond to the request and store the file with the specified name in the storage **152**. The file to be saved may exist on the client **141** or may already exist in the server's local storage **151**.

In another embodiment, the client **141** may be a vehicle, or a system or apparatus supported by a vehicle, that sends vehicle sensor data. This may be accomplished via communication over the network **145** between the client **141** and server **150**. For example, the client may communicate a request to the server **150** to store a file with a specified file name in the storage **151**. The server **150** may respond to the request and store the file with the specified name in the storage **151**. The file to be saved may exist on the client **141** or may exist in other storage accessible via the network such as storage **152**, or even in storage on the client (e.g., in a peer-to-peer system). In one example, the vehicle can be an electric, gasoline, hydrogen, or hybrid powered vehicle

including an all-terrain vehicle, a truck, a tractor, a small rover with bogey rocker system, an aerial vehicle such as a drone or small unmanned aerial system capable of supporting a treatment system including vision components, chemical deposition components, and compute components.

In accordance with the above discussion, embodiments can be used to store a file on local storage such as a disk or solid-state drive, or on a removable medium like a flash drive. Furthermore, embodiments may be used to store a file on an external storage device connected to a computer over a connection medium such as a bus, crossbar, network, wireless communication interface, or other interconnect. In addition, embodiments can be used to store a file on a remote server or on a storage device accessible to the remote server.

Furthermore, cloud computing and edge computing is another example where files are often stored on remote servers or remote storage systems. Cloud computing refers to pooled network resources that can be quickly provisioned so as to allow for easy scalability. Cloud computing can be used to provide software-as-a-service, platform-as-a-service, infrastructure-as-a-service, and similar features. In a cloud computing environment, a user may store a file in the "cloud," which means that the file is stored on a remote network resource though the actual hardware storing the file may be opaque to the user. Edge computing utilizes processing, storage, transfer, and receiving data at a remote server more local to where most, or a desired portion of the data may be processed, stored, and transferred to and from another server, including a central hub or at each geographic boundary where data is captured, processed, stored, transmitted, and received.

FIG. 2 illustrates a diagram 200 of an example system 100 configured to observe a geographic boundary in the real-world, for example a farm or orchard, perform object detection, classification, identification, of any and all objects in the geographic boundary including agricultural objects, determine any individual agricultural object that may require an agricultural treatment based on the agricultural object's growth stage, previous treatments applied, and other characteristics observed, particularly at the point in time of the observation by system 100, and apply a specific treatment to the agricultural object. The system 100 can include and object observation and treatment engine that includes an image capture module 104, a request module 106, a positional data module 108 for capturing, fusing, and transmitting sensor data related to position, localization, pose, velocity, and other position related signals to the rest of the system 100, a vehicle module 110, a deposition module 112 for applying a liquid or light treatment on each individual object detected and determined to require a treatment, a targeting module 114 for targeting and tracking an identified object in the real-world based on sensor data and object detection in an image captured of the real-world while a vehicle is moving, and a user interface (U.I.) module 116. The system 100 may communicate with a user device 140 to display output, via a user interface 144 generated by an application engine 142. In one example, the deposition module 112 can also be a treatment module configured to perform non fluid type deposition treatment including having a mechanical mechanism or end effector, including mechanical arms, blades, injectors, drills, tilling mechanism, etc., that physically interacts with surfaces or roots of plant objects or soil.

The system 100 can also include an image processing module 130, either on board a vehicle supporting the system 100, part of the system 100, embedded in the system 100, or supported by one or more servers or computing devices remote from the vehicle supporting the system 100. The

image processing module 130 can be configured to process any and all images or other sensor data captured by the system 100 including feature extraction, object identification, detection, and classification, image matching, comparing, and corresponding with other images received simultaneously or previously of the same location, labelling unique features in each of the images, as well as point clouds from various other sensors such as that of lidars, or a combination thereof.

While the databases 120, 122 and 124 are displayed separately, the databases and information maintained in a database may be combined together or further separated in a manner that promotes retrieval and storage efficiency and/or data security.

FIG. 3A illustrates a diagram 300a depicting an agricultural scene. The agricultural scene can be any physical environment in the real-world used for agriculture such as, but not limited to, a farm or orchard. The agricultural scene can be contained in a regional geographic boundary or a region without any defined boundaries. The agricultural scene can include agricultural objects including a plurality of one or more different types of plants objects having different plant phenology depending on the season or year on the same agricultural scene. The agricultural objects can be further observed and categorized based on each plant anatomy. For example, diagram 300a can illustrate an orchard having permanent plants, such one or more trees 303. These trees 303 can be permanent trees that can produce crop such as fruit trees or nut trees in seasonal or yearly cycles for multiple years. The plants can also be row crops for harvesting where the plants themselves are for harvest. The agricultural objects observed and potentially treated can be further categorized and identified by the anatomy of the specific type of tree 303. For example, a plant such as a tree 303 can include a trunk, root, branch, stems, leaves, pedals, flowers, plant pistils and stigma, buds, fruitlets, fruits, and many other portions of a plant that make up the plant's anatomy, all of which can be agricultural objects of interest for observation and treatment. For example, the tree 303 in diagram 300a can include one or more agricultural objects 302. These objects can include fruiting flowers or fruitlets that an agricultural treatment system can detect and identify in real-time, and perform an action to treat the flower or fruitlet.

The agricultural scene can also include an agricultural observation and treatment system 311, supported by an example vehicle 310, performing observations and actions in the agricultural scene. In one example, the vehicle 310 can travel inside an orchard along a path 312 such that the agricultural observation and treatment system 311 can sense, identify, perform actions on specific agricultural objects 302 in real time, and index and store the sensed objects 302 and action history, such that the observation and treatment system 311 can use the previously stored information about the specific object 302 that was observed and treated for its next treatment upon detection at a later time or a later phenological stage of the specific object 302. The agricultural observation and treatment system 311 itself can be a component or subsystem of a larger system that can perform computations, store and display information, make decisions, and transmit and receive data from a plurality of agricultural observation and treatment systems performing observations and actions on a plurality of geographic scenes. The larger system can manage a mesh network of individual agricultural observation and treatment systems, each performing online, and onboard a vehicle, in one or more geographic regions, and a mesh network of servers and other

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compute devices in the cloud or edge to perform real time functions, quasi real-time functions or support functions for each online agricultural observation and treatment system, or offline at one or more servers to analyze data such as sensor data, performance activity, perform training one or more machine learning models, updating machine learning models stored on one or more of the agricultural observation and treatment systems located at various geographic regions, as well as a plurality of other tasks and storage capabilities that can generally be performed or maintained offline from the online and real time performing treatment systems. Various examples of agricultural observation and treatment systems, or components of modular agricultural observation and treatment systems are described in further detail below in this disclosure.

In one example, the agricultural scene can be that of an orchard having a plurality of fruiting trees planted in rows as illustrated in diagram 300a. The rows can be further partitioned and categorized by zones 304. In this example, the treatment system 311 can perform a different variety of chemical treatments with varying treatment parameters, such as chemicals used, chemical composition, treatment frequency, and perform A/B type testing (A/B testing) on the agricultural scene by different zones of the same plant type, different chemical trials in the same or different zones or by different individual plant object for harvest, or a combination thereof. The A/B testing for best treatment or best trial discovery can be performed at a microarray level such that varying chemical types can be used in real time and varying chemical compositions and concentrations can be used in real time. These combinations can go up to over a million different combinations of different compositions, concentrations, volume, frequency of chemical treatment on varying plant varieties at different stages of growth. In one example, the agricultural observation and treatment system 311 can apply and log each of these different possibilities of varying treatment parameters and perform A/B testing on each zone, each tree, or each crop level specificity to determine the optimal treatment process for each plant or crop type that has not been previously identified in the industry. For example, as the agricultural observation and treatment system 311 applies different treatment parameters to different objects in the same geographic region throughout the growing cycle, upon harvest, some fruiting objects will have more desirable traits and characteristics as that of others, of the same type of crop. The agricultural observation and treatment system 311 can determine which exact object treated and logged from the beginning of the grow cycle for that particular object of the crop, determine the objects specific treatment history, including treatments used, concentration, volume, frequency, etc. and determine that the particular treatment process based on the treatment history of that particular object, that fruited into the most desired version of the crop, is the optimal process based on the A/B testing.

Additionally, based on the zone 304 of plants that produces the best crops, or the best crop at the individual object or fruit level of each zone 304, the best crops being based on size, health, color, amount, taste, etc. crop, the agricultural observation and treatment system 311 can determine the best method of performing treatment actions, based on a variety of parameters that can be adjusted and customized, and apply the same method of treatment actions on the particular zone 304 that yielded the best crop, for other crops in a new or subsequent crop cycle. In one example, treating each agricultural objects with a different treatment parameter to determine the best method of treating a crop does not have to be partitioned by zones 304. The agricultural observation

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and treatment system 311 can identify, tag, observe, log each unique agricultural object 302 and treat each agricultural object 302 of interest at the individual agricultural object level. For example, instead of treating a first zone 304 with a certain amount or type of chemical of each agricultural object and treating a second zone 304 with a different amount or type of chemical, the treatment system 311 can treat a first agricultural object 302, such as plant bud, and a second agricultural object 302, a different plant bud at the same stage of growth as that of the first plant bud, to observe and discover which bud yields the better fruit.

In one example, the agricultural scene is an orchard having a plurality of rows and trees planted in each row. The vehicle 310 can autonomously travel through each row such that the treatment system 311 can scan one or more trees 303 along a path of the vehicle to detect various agricultural objects including agricultural objects 302 for treatment. Once the treatment system 311's sensing system senses a potential agricultural object, the system 311 can determine whether the agricultural object 302 detected is a new object identified for the first time, a previously identified, tagged, and stored object detected again, a previously identified, tagged, and stored object detected again, that has changed its state or stage of growth in its phenological cycle, a previously identified object that has moved or changed in anatomy, or other objects with varying characteristics detected such as stage of growth, size, color, health, density, etc. Once the object is detected in real time, whether it is of an object previously identified and mapped onto a virtual agricultural scene representing the real agricultural scene, the treatment system 311 can determine, based on a combination of determining the agricultural object's identity, phenotype, stage of growth, and treatment history, if any, whether to perform a unique action onto the agricultural object 302 identified. The action can be that of a chemical fluid projectile emitted from a device as part of the treatment 311 directly onto a portion of a surface of the agricultural object 302. The fluid can be a single liquid projectile similar to that of a shape of a water droplet emitted from a water sprayer, a mist or aerosol, a volumetric spray across a period of time, or many other types of fluid that can be emitted from a device discussed later in this disclosure.

The actions performed by the observation and treatment system 311 can be performed for the purposes similar to that of many actions typically performed in agriculture. These actions can include soil and fertilizer deposition, emitting seeds from the treatment system 311 into soil or dirt, treating individual plant objects including thinning, weeding, pollinating, pruning, extracting, harvesting, among many other actions that can be performed by a treatment system 311 having a device configured to sense an individual object and its stage of growth, access its treatment history, and perform a physical action including emitting a fluid, small object, or shine a light source such as a laser onto the individual object, physically manipulate the object including removing or moving the object for better sense and treatment of another object, destroying the object, pruning or harvesting the object, or a combination thereof.

In one example, the agricultural scene and geographic region can be a farm where the ground or terrain is partitioned into a plurality of rows with row crops for planting, growing, and harvesting and the plants themselves are harvested, unlike that of orchards where agricultural objects are harvested from permanent plants. The observation and treatment can be observed and performed on the crops themselves, or of other plants of interest. For example, weeds can grow in the same agricultural scene as that of a

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crop of interest such that the observation and treatment performed by treatment system 311 can be that of both the crop and the one or more different types of weeds, or just the weeds. In another example, the agricultural scene can be that of a farm, orchard, or any kind of ground terrain that does not yet have any trees or crops, but only of dirt and soil.

FIG. 3B and FIG. 3C illustrate diagrams 300b and 300c each depicting a portion of a virtual and digitized agricultural scene or area similar to that of agricultural scene in diagram 300a. Diagram 300b can depict a virtual scene generated by an agricultural observation and treatment system similar to that of agricultural observation and treatment system 311 or at servers, cloud, or edge computing devices connected to an agricultural observation and treatment system operating and acquiring images and other perception data of the agricultural scene. The virtual scene can be that of a high definition 2D map, 3D map, or both, of an agricultural scene surveyed, observed, treated, logged, or a combination thereof, by a treatment system, such as treatment system 311. The treatment system 311, having perception and navigation related sensors and a plurality of modular treatment modules each having its own sensors, including vision and navigation sensors, compute units, treatment devices or units, illumination devices, can be supported by a vehicle 310 that can drive along a path, and can be configured to scan and observe a geographic scene and build a virtual map of the scene.

In general, the vehicle 310 moves along a path in the real world while the agricultural observation and agricultural treatment system 311 obtains imagery and other sensed readings, including images captured by image capture devices or point clouds captured by LiDAR's, or a plurality of different sensor readings captured by a plurality of different sensors, of the external environment. The observation and treatment system can generate points along the path representing external agricultural objects (e.g., plants, crops, trees, debris, patterns, landmarks, keypoints or salient points, patterns, cluster of features or patterns that are fixed in space, etc.).

For example, as the vehicle 310 passes by a particular agricultural object in the real world, the object determination and object treatment engine can capture images and reconstruct a digital or virtual geographic scene representing the geographic scene as illustrated by diagram 300b. The diagram 300b can include a plurality of mapped data points representing agricultural objects or clusters of objects, including objects that have been treated, objects for observation, objects indexed for marking location of nearby objects in the overall geographic scene or the object itself in the global scene, or a combination thereof, as well as landmarks, patterns, regions of interest, or a combination thereof. The mapped points depicted by objects 320, which can include agricultural objects for treatment, with different identifiers based on the phenology or stage of growth of each individual objects. The points depicted by objects 320 can be generated and/or represented by images taken in the real world of the scene, patches of images, lidar point clouds or portions of points clouds, 3d images modelled by various imaging techniques such as visual constructions of objects in computer vision, including structure from motion or 3d model reconstruction from a single, stereo, multi camera configuration, the cameras being color sensors, black and white sensors, multispectral sensors, or a combination thereof. Multiple images of the same scene or same object can be combined and analyzed as the agricultural observation and treatment system 311 scans and observes the environment multiple times throughout a grow season or

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year. Each object, cluster of objects, or landmarks detected can have a plurality of layers of images or other sensor readings, such as radar and lidar point cloud readings, to form high resolution 2D or 3D reconstructed models of the real-world objects detected. In one example, a stereo vision system in an image capture module can capture images of objects in space and superimpose views of the objects captured in the frames captured in time in stereo into a 3D model of the object. In one example, the generated 3D model of the objects detected, including agricultural objects, with different models at each of its different detected and labelled growth stages, can be positioned in the virtual geographic boundary or geographic scene for a user to scan through a see via the user interface described below. In one example, a geographic scene can be a fruit orchard having multiple trees 303 and agricultural objects 302. The agricultural observation and treatment system 311 can observe the geographic scene, both in real time via compute units disposed on board the vehicle 310 or edge or cloud compute devices or offline. The system can generate a digital or virtual map of the scene having a plurality of objects 320, clusters of objects 320 that can represent a portion of an entire plant 330, for example a tree. Each object 320 can include agricultural objects such as fruits, buds, flowers, fruitlets, or other object types of objects that can be treated by a treatment system, depending on its stage of growth or phenology.

The objects 320 can be digitally indexed objects having a type, identity, stage of growth, and location associated with the objects, which can be represented by individual images, stereo pair images, or portions of images of the real-world equivalent object captured by an image capture device. The object 320 may have an associated geographic data associated to the object, including position data, orientation and pose estimation relative to the geographic boundary view or relative to physical components of the agricultural treatment system 400, including image sensors, or treatment engines, or relative to other agricultural objects. In one example, each of the objects 320 can include images that are full frame images captured by one or more cameras in the agricultural treatment system. The full frames can be 2D or 3D image showing the images captured directly by one or more cameras and/or rendered by the agricultural treatment system 311. The images can include images captured a few meters away from the physical surface and position of agricultural objects in the geographic boundary, which can include images of a plurality of individual agricultural objects, that are potential crops, as well as landmarks including objects or scenery, or other objects of interest including calibration targets and markers or other farming equipment, devices, structures, or machinery typically found on a farm that can be detected for localization of the treatment system and tracking objects in real time and for constructing a map of a scene either in real time or offline. The objects 320 can also include specific patches within captured full frame images. For example, a stereo pair of cameras can each, simultaneously, capture a high-resolution image having hundreds of agricultural objects including target objects for treatment, potential objects for observation for treating at a future time, landmarks, other patterns, etc. The patches can be identified by the agricultural system by detecting, classifying, identifying and extracting features, and labelling specific portions of a full image frame, including labelling agricultural objects and specific stages of growth of agricultural objects. The portion of the full frame image can be extracted as a patch such that each individual patch, which itself is a portion of the full frame image, is a

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visual representation of each individual and unique agricultural object on the geographic boundary, and can be identified and indexed, associated with its position data, any and all treatment history if any of the objects detected are objects detected from a previous trial on the specific marked and identified agricultural object, as well as timestamps associated with the image captured and data acquisition, position captured, treatment applied, or a combination thereof. In one example, each object 320 in the virtual construction of a scene depicted by diagram 300b can be a point of varying size depending on the actual size of the object in the real world, or the size of the patch associated with the object. In another example, each object 320 in the virtually constructed scene can be represented by the real-world object virtual 3d model associated with each of the objects 320. For example, each of the thousands or millions of objects, landmarks, or patterns can be visually represented as 2d or 3d models of each of the specific objects in the real world. Thus, a map, either a 2d or a 3d map can be generated and accessed, visually, illustrating each object, landmark, pattern or region of interest, in the real world such that each object's and/or landmark's visualization, structure, location, treatment and prediction details can be represented and displayed in the map.

A user interface can be accessed to interface with the digitally mapped agricultural scene such that a user can view images, models, model of the specific object, spray history, other data related to agricultural objects including predictions related to yield, size, health, disease, etc., of each object observed and or treated associated with its location in the real world based on its location in the digital map. For example, a cluster of objects 320, including fruits and fruitlets of a fruiting tree, located in a specific area in an orchard, for example in a specific zone 304 in the real world, can be mapped to a specific zone 340 in the map, such that the cluster of objects 320, accessed by a user, in the map also represents the location of the specific real-world objects associated with objects 320 of the virtual map. The user interface is further described in detail below.

In one example, as illustrated in diagram 300c, a user interface 350 may be accessed. The user interface 350 can show the points representing each of the objects 320 of the virtual map. The user interface may provide for the user selection of the any of the points depicted in the virtual scene or map. For example, a unique object 353-a, which in this example can be an individual and unique fruitlet in an orchard, can be detected in the orchard by the agricultural observation and treatment system 311. The object 353-a can be detected in one or more images from a system supported by a moving vehicle. A patch of the image can be identified and extracted depicting just the object 353-a. And additionally, a 3D constructed model of the specific fruitlet, that is object 353-a at the time of capture by the observation and treatment system, can be generated by using one or more computer vision techniques from associated multiple different views of the same object 353-a and associating each view with location data of object 353-a and motion of the image capture device, or by implementing other computer vision techniques.

Upon selection of a specific point in the user interface, for example a point of the map representing object 353-a, the user interface 350 can display multiple visualizations and information associated with the selected object 353-a. For example, the user interface 350 can display an image 352 associated with the point selected, as for this case selecting the point associated with object 353-a. The image 352 can be a pixelated 2-dimensional or 3-dimensional image with

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localization information. The image 352 can also include the specific image patch extracted from the images captured by the treatment system, instead of a constructed image depicting the shape, size, color, or other unique attributes of the object 353-a at the time the observation and treatment system last observed the object 353-a. In one example, the image 352 can be a pixelated 2d or 3d image that represents a model of the specific object 353-a in the real world at a stage of growth, or state, detected by the treatment system. The 2d or 3d model can be generated by using various computer vision techniques by associating multiple views of the object along with depth and/or motion data of the image capture device. Some objects may be occluded such that an image sensor travelling along a path guided by a vehicle may not capture the entire view of the object detected. For example, the object could be hidden further inside of a tree instead of growing closer to the outer surface of a tree. The object could have other objects blocking portions of the object from view such as leaves or other agricultural objects or landmarks of interest. In this example, one or more machine learning algorithms can be applied to process the existing images sensed on the object, whether through a single pass of the image sensor in a single run, capturing a plurality of image frames forming a single video, or in stereo, or by analyzing image frames from multiple passes, each image frame whether captured in a single sequence or multiple sequences from multiple trials, each image frame having at least a portion of a view of the object. The machine learning algorithm can be used to compensate for the occluded portions of the object and construct a high resolution 2D image or 3D model of the object. In cases where the phenological stage of the object changes in time, the agricultural observation and treatment system 311 can generate multiple constructed images or models depicting different stages of growth of the same object with timestamps associating which stage of growth was detected and constructed and the relationship between the different constructed images and models of the same object, for example showing the phenological changes from one constructed image of the object to the next constructed image of the same object. In one example, the computer vision techniques can be performed using machine learning models and algorithms, either embedded on board the agricultural observation and treatment system 311, or offline via edge or cloud computing device. The user interface 350 can also display multiple views of the same patch associated with a specific object selected by the user in the virtual map. For example, the agricultural observation and treatment system 311 can capture more than one view of object 353-a and store all of the different frames that include object 353-a. The system can generate, index and store each of the individual patches of images depicting different views of the object and display it in the user interface 350 as a group of images 354 for the user when the specific object, for example object 353-a, in the virtual map is selected.

Additionally, the user interface 350 can display a time series lapse of the history of images captured on the same object, the images partitioned based on state changes, stage of growth or phenological changes, including bud, leaf, shoot, flower/blossom, fruiting developments, and maturing developments of the same object detected. For example, object 353-a can be fruitlet of an apple fruit, detected as an individual object in the real world with location, position, and/or orientation data, relative to some arbitrary point in the orchard, the exact point in the real world, a relative position to the agricultural observation and treatment system 311 such that the treatment system itself has a location and

orientation data in the real-world. And the object detected at the time can be identified as a fruitlet of a fruiting tree. However, the object 353-a will have been a flower earlier in its life cycle at a prior time, and a bud in its life cycle at an even prior time to flowering, while still being at the same, or close to the same location and position in the orchard, and location and position relative to a defined position of the tree supporting the object 353-a. For example, a portion of the tree, such as a base of the trunk or an arbitrary center point of the tree can be defined with (x_0, y_0, z_0) position data. The object detected can be some $(\Delta x_0, \Delta y_0, \Delta z_0)$ position relative to the base or arbitrarily chosen position (x_0, y_0, z_0) in the geographic scene, zone of the geographic scene, or a particular tree/plant, for example a position (x_1, y_1, z_1) . The coordinate system chosen is just an example wherein a plurality of different coordinate systems and origin points can be used to locate relative positions and orientations of objects relative to other objects.

In this example, the agricultural observation and treatment system 311 may have logged information related to object 353-a before it was labelled as a fruitlet in its most recent timed log. The system may have detected the same object 353-a, at the same or near location, when it was detected with an identifier of a bud associated with the object. And then at a later trial detecting the same object 353-a again, but before it was detected and labelled with the identifier of fruitlet but after it was detected and labelled as a bud, another detection with the identifier, for example, of a flower/blossom. Each of these detections can have location position of, or near location position of (x_1, y_1, z_1) . In this case, the system can associate the different identifications of the same object, based on the objects state changes, or stage of growth or phenological changes, and display, via a series of views across time, the state change in sequence in the user interface 350. Identifying, storing and indexing, and associating portions of images and patches and other sensor readings of objects of the same type with near or the same locations of the same objects identified throughout time from different trials and identifying with different states of the same object in the geographic scene can be performed using various techniques including machine learning feature extraction, detection, and or classification to detect and identify objects in a given image frame as well as generating keyframes based on the objects and landmarks detected. The keyframes can be determined to more efficiently identify and index objects in a frame while reducing redundancy, for example, by identifying common and/or the same landmarks across multiple frames. The machine learning and other various computer vision algorithms can be configured to draw bounding boxes to label portions of images with objects of interest and background, masking functions to separate background and regions of interest or objects of interest, perform semantic segmentation to all pixels or a region of pixels of an given image frame to classify each pixel as part of one or more different target objects, other objects of interest, or background and associate its specific location in space relative to the a component of the treatment system and the vehicle supporting the treatment system.

Additionally, the agricultural observation and treatment system 311 can perform functions to associate portions of image, for example image patches of objects, image frames, key frames, or a combination thereof from different trials where the agricultural observation and treatment system 311 observed, identified, labelled, and stored information about the same object across multiple states and phenological stages. Additionally, the association of the frames or portion of the frames can be packaged into a series of image frames

that can be displayed in sequence as a video displaying the growth, or backwards growth depending on the direction of displaying the images, of the specific object. For example, the series of indexed images or patches of images associated with each other throughout time can be displayed in the user interface 350 in the video or visual time lapse history 356. In one example, the functions can be performed by various computer vision and machine learning techniques including image to image correspondence including template matching and outlier rejection, performed by various techniques including RANSAC, k-means clustering, or other feature-based object detection techniques for analyzing a series of images frames, or a combination thereof. In one example, the above techniques can also be used to generate key frames of a subsequent trial by comparing frames from the subsequent trial with keyframes of one or more prior trials, depending on how many prior trials there are. Additionally, the comparison and the candidate frames or keyframes from a previous trial that may be accessed by the agricultural observation and treatment system 311, or at a server offline, to be used to perform comparisons to identify state and phenological stage change of a same object, such as object 353-a, can be narrowed down for selection based on location data logged at the time of capture, pose data logged at the time of capture, or a combination thereof associated with each of the keyframes, or objects detected in each keyframe. These accessed and selected frames or key frames in the prior trials, having been selected based on its location data associated with the frames or objects detected in the frames, can be used to compare with currently captured frames, or subsequently captured frames from the prior frames, having similar location data associated with the selected frames for key frames to match objects that may have different labels, since different states or phenological stages will have different labels due to the states having different shape, color, size, density, etc. If there is a match, or if there is a threshold reached based on the comparison of the accessed frame or keyframe against one or more frames in a subsequently captured series of frames, the agricultural observation and treatment system 311 can determine that the two, or more, objects of different types and identifiers associated with each of the objects, are the same object and that one, with a first phenological stage, changed into the other having a second label or identifier of a second phenological stage.

In one example, the agricultural observation and treatment system 311 can run a machine learning detector on portions of images, to detect objects of interest in each of the portions of images by performing feature extraction and generating bounding boxes around an object of interest, performing semantic segmentation or performing semantic classification of each pixel of a region of an image to detect objects of interest, or a combination thereof. One or more key frames from a prior trial or in a prior frame of captured frames from a subsequent trial, any trial subsequent to the prior trial, can be propagated into candidate frames captured in the subsequent trial, for example the subsequent trial being the most recent trial, or current trial, with images captured that have not yet been processed and indexed. The frames with the propagated detections or labels can be used to detect whether a machine learning detection was accurate, whether location data associated with the frame is accurate, or detecting other outliers, as a threshold to mitigate false positives and false negatives of features detected but doesn't actually exist or missing but should have been detected, since these frames will have similar location data associated to each of the other frames and detecting and corresponding, above a certain threshold, the same features of more than

one frame in a same trial or across frames having the same location from previous trials may give more confidence that the features detected are real and accurate.

The series of images can be each of the patches displayed in order, similar to that of a video display of images, where a user can view the changes in state of the object from when it was bud, through various growth stages, including small incremental stages from day-to-day in a growing season, until fruiting. In one example, a visual time lapse history 356 of object 353-*a* as it was detected as a bud, to a flower, to a fruitlet, and to a fruit, incrementally, can be displayed in the user interface 350. Additionally, the series of images displayed in the visual time lapse history 356 of an object, for example object 353-*a*, can be reconstructed or generated images from combining multiple images depicting the same object or landmark. These images would be machine learning rendered images generated by associating portions of captured images as well as portions of generated pixels of an image generated by a machine learning model to display a better representation of the selected object to the user. This can include generating higher resolutions from upscaling portions of the captured images when analyzing portions of the captured images to generate an image displayed to the user or generating views of an object detected that were otherwise occluded in the captured images. In this example, upon selection of an object in the user interface, the user interface 350 can display any captured images 352, where the image itself can contain smaller patches within the image 352 containing views of objects, or display one or more rendered images generated from a plurality of captured images associated with the object selected.

In one example, the visual time lapse history 356 can be used to visualize the state changes or visualize its real-world growth from sprouting into crop, or from bud into fruit, depending on the type of crop. This would give the effect, in some instances, of displaying a growth sequence of an agricultural object from a dormant phase, to a fully grown crop of the same or substantially the same location. The location would not be the exact since the object will grow and drop lower due to its weight or can be externally moved by wind. Alternatively, the visual time lapse history 356 can function as a “time machine” visualization of the object. The visual time lapse history 356 can be viewed in reverse time to view what a currently detected, or otherwise the object’s current state in the real world, assuming the agricultural observation and treatment system has captured and detected the object in its current or proximately current state, looked like in the past by visually linking captured sensor readings, including image frames having views of the object, and displaying them in sequence, such as a video.

The agricultural observation and treatment system can associate similarities from an image frame, or a portion of one or more patches within the frame corresponding with an object of interest captured at a first time, with another frame captured at a second time that is close in proximity to the first time, for example a day, such that the state change will be minor and the system can combine location data of the objects detected in frames from frame to frame across time having a same location associated with the object, so the system can have more confidence that, for example, an object from a first frame associated with a first timestamp is a same object from a second frame associated with a second timestamp, because the real world location of both objects from different frames and different timestamps are in proximity with each other above a certain threshold for the system to determine that the two objects are the same. Additionally, the system can determine any relationship

between the images of the same object, such that one object turned from one state detected in the first frame to the other state detected in the second frame. The incremental changes can allow the image correspondence to reach a certain threshold of confidence such that matching an object of a first phenological stage with an object of a second phenological stage as the same object does not have to rely on its detected spatial proximity in the real-world location associated with the object when the object was identified with their respective captured frames and timestamps.

In one example, the user interface 350 can store and display a variety of information, data, logs, predictions, histories, or other information related to each object. The information can be displayed to a user upon selection of information, or upon selection of the object in an interactive virtual map. In one example, the user interface 350 can display a visualization 358 of various data including data related to an object’s treatment history, observation history, or both. This can include information about each of the times the particular object selected, for example object 353-*a*, was detected in the real world and indexed. In one example, the detection of an object across multiple frames or sensor readings in a single trial can be categorized and indexed as a single detection. If a treatment was applied, for example a spraying of a substance, a mechanical interaction with the object with a physical end effector contacting object 353-*a*, or any kind of action other than a treatment that physically affects the object, can be logged in time and location. As the agricultural observation and treatment system 311 performs multiple trials across period of time, the system 311 can associate each observation and/or treatment of a same object with each other, and display the information related to observations and treatments in order. The information can include the type of spray or treatment used, the length of time of the spray or treatment, the time associated with the treatment, timestamp, the phenological stage of the object detected. This can allow the agricultural observation and treatment system 311 to determine the treatment parameters per object. For example, the system can determine, due to its indexing and understanding of each object in a geographic scene, that in an immediate upcoming trial, a first object, if detected, should receive a treatment of a first substance, but a similar second object, proximate to the first object, if detected, does not need to receive a treatment of the first substance, at least during the immediate upcoming trial. Further examples will be provided below in this disclosure. Additionally, the user interface 250 can display a visualization 360 of data related to features, attributes, and characteristics of each object, or the specific object selected. The information can include information of the object relating to its size, color, shape, density, health, or other information related to prediction information relating to yield estimate, future size, shape, and health, and optimal harvest parameters of the specific object. Additionally, since the actions for treating each object, themselves can be sensed, indexed, and stored, a user can access each individual treatment action including its parameters such as type, volume, concentration, dwell time or surface contact diameter for fluid projectile treatments on each individual agricultural object or crop throughout the life cycle of that specific individual crop or object detected. This would allow a user to determine grow, health, and harvest parameters and data per crop or per object.

In one example, as illustrated in diagram 300*d* of FIG. 3*D*, a user can access the user interface 350 and one or more interactive maps through a variety of devices. In one example, the electronic device can be a tablet 380 having a

user interface including user interface **350** and interactive virtual map **382**. The interactive virtual map **382** can be that of the virtual maps discussed above. For example, the interactive virtual map **382** can be that of a virtual map associated with a map of a real-world geographic scene having a plurality of agricultural objects and landmarks. Because the geographic scene changes over a period of time, multiple virtual maps can be generated to index each state of the geographic scene, at a global scene level, such as the broader geographic level including terrain, topography, trees, large objects, etc., and at a local level, such as that of each agricultural object, including target crop objects. Both the local scene comprising a plurality of agricultural objects and the global scene can be combined to generate each virtual map. In one example, each virtual map of the same real world geographic coordinates, or predetermined geofenced location, can be associated with each other such that an interactive changing map can be displayed where one map changing or updating to another map represents the changing state of the geographical scene changing across a grow season. This can include plants sprouting, trees growing in size, or growing fruits. Each trial performed by the agricultural observation and treatment system **311** can include a plurality of sensor readings, including images captured from image capture devices that include 3d structure, location, depth, relative size to objects in the real world, heatmap, etc., such that a virtual map of the area sensed in the trial can be generated.

As more trials are performed, more of the geographic scene can be mapped, and thus used to generate a virtual map, or an index of information associated with objects and landmarks of the geographic scene. For example, a first map can be generated to depict a first geographic scene captured at a first time with a first set of characteristics, the characteristics including global characteristics, such as number of sprouts, number of trees, amount and color of visible dirt, topography, etc., and including local characteristics, such as per crop object of interest and each of its phenological stages, depending on the type of geographic scene such as terrain, row crop farm land, orchard, etc., and a second map can be generated to depict the same first geographic scene captured at a second time having different global and local characteristics.

The system can associate the first and second map maps such that there is a logical link between the first and second map, indexed information related to each of the first and second maps, the indexed and generated maps, or the generated interactive virtual maps, such that the geographic scene having characteristics captured in the first map have turned into the geographic scene, with the same or similar real-world geographic boundary, having characteristics captured in the second map. In one example, the system can generate a single map such that as the system performs more trials and senses and captures more of both the global and local portions of the geographic scene, and thus mapping more details and characteristic changes of the geographic scene from trial to trial, the system can update the same map into one more updated maps having updated global and local attributes and characteristics of the geographic scene across time, instead of or in addition to generating multiple maps.

While the description above discussed virtual maps, the discussion can be applied more generally to indexed information of geographic scenes, including geographic scenes of changing characteristics throughout time, stored in multiple forms and does not necessarily have to be a generated virtual map that can be visualized and displayed in a user interface. The real-world geographic scene can be sensed, and indexed

in a database having information relating to agricultural objects and landmarks of the geographic scene with various sensor readings associated with each agricultural object and landmark, including visual information, location information, etc., such that the information stored in the database can be used to generate a map. The information of each agricultural object and landmark can also be used to generate a visualized virtual map that can be interfaced with a user on an electronic device.

In one example, the tablet **380** can display an interactive virtual map **382** depicting, for example, the most updated map, or the most recently generated map, of a mapped geographic boundary **383**. The mapped geographic boundary **383** can be the most recently captured and sensed state of a real-world geographic region depicted in diagram **300a** of FIG. **3A**, having a plurality of agricultural objects and landmarks sensed, the agricultural objects being in their current state, and indexed, stored, and mapped as mapped geographic boundary **383**. A prior mapping, from a previous trial, on the same real-world geographic region can also be indexed, stored, and mapped and associated with mapped geographic boundary **383**. For example, agricultural object **370-a** can be an individual blossom of an object detected by the agricultural observation and treatment system **311** captured in a recent trial. A user, via tablet **380**, or any other electronic device, can interact with interactive virtual map **382** to select a selectable object **370-a** in the tablet to view information about object **370-a** including any and all views captured of object **370-a** previously, time lapse video and time machine video of object **370-a**'s history as it blossomed from a bud, for example, treatment history, metadata, and crop characteristics of object **370-a** including prediction type information. The interactive virtual map **382** displaying a mapped geographic boundary **383** can have a plurality of selectable objects **320**'s to choose from. For example, object **371-a** can be a different object in the same geographic scene having a different treatment history as that of object **370-a**. The image **352** can be a portion of a larger image captured by one or more image capture devices such as a 4K or 8K image frame, where image **352** is a cropped portion of the 4K or 8K image frame. Additionally, the image **352** can include more than one even smaller patches of the image **352** of a specific object, such as image patch **352-1** of image **352** to display a view of virtual or digitized object **370-a** of some real-world object **302** in the real world, for example.

In one example, the selectable object in the virtual map **382** itself can be an image. Because the virtual map **382** is interactive, the user can zoom in to the specific object in the virtual map **382** to view the specific object inside interactive virtual map **382**. The object zoomed into can be an animated object depicting the specific object sensed and indexed from the real world, or can be an image patch, cropped from an image capture device, having a view of the object in the image patch. The objects and landmarks indexed in the virtual map **382**, are associated with a location in the real world. Each animated agricultural object, or representation of the agricultural object can include data representing at least one image captured by an image sensor of the agricultural object in the real world, a localization data representing the position of the agricultural object relative to the geographic boundary itself, the position of the agricultural object relative to the agricultural observation and treatment system that captured an image of the individual agricultural object, or its position relative to other agricultural objects also with position data associated with the agricultural objects, as well as a timestamp of when the image and location data was acquired.

In one example, one or more agricultural object detected in the real-world will change characteristics, for example phenological stages or changes in size, such that the system 100 can detect a new feature of the agricultural object and assign a label or identifier to the agricultural object that had a different label or identifier previously assigned to the same agricultural object having the same or similar position detected in the geographic boundary. This is due to a portion of a potential crop growing on a plant, for example a lateral, changing characteristics due to the growth stage of the plant. As a simplified example, a fruiting tree can have buds on the tree's laterals which can turn into flowers, and then eventually a fruitlet, and then a fruit, for example. Additionally, each of these features can be associated with each other, particularly for labeled features of agricultural objects that have the same position detected in the real world, or similar image features from a previous trial of when the system 100 captured images of the specific agricultural object, or a combination thereof.

FIG. 3E illustrates a diagram 300e depicting a user, or human 381, interacting in a real-world environment with an electronic device having a user interface and interactive virtual map similar to that of the user interfaces and interactive virtual maps discussed above. A user can have an electronic device with location and image sensing capabilities to detect a location of the device in the real world, the location of the device relative to an identified object, the identified object having location data stored in the device or a location accessible wirelessly by the device, or a combination thereof. As the user physically navigates in the geographic boundary, such as an orchard, the user may come across one more indexed objects in the real world, that may be in plain view or viewable in real time by the electronic device, such as the tablet 380, a phone or smart device 385, or smart glasses 386, or mixed reality smart glasses, or any other wearable or holdable device. In one example, the electronic device can be a drone controlled by the user in real time such as any drone free can be relayed and displayed in real time to the user via a device the user is holding with an interactive interface including a screen.

For example, if the user is near agricultural object 370-a in the real world, the user can access information stored about object 370-a in the electronic device, including most recently views of the object, treatment history, or other information and metadata about the object, particularly those discussed above.

Additionally, an augmented reality or mixed reality environment can be accessed via an electronic device such as a wearable with a display and image sensors, a phone or table with image sensors, or a combination thereof. As the user physically navigates the real-world geographic scene, the electronic device's image sensors can capture and detect objects in its field of view. Each object previously detected, indexed, and stored can be displayed to the user in real time via augmented reality or mixed reality, as the same objects in the real-world are detected by the electronic devices in real time. The user can then interact with the electronic device in a similar way described above. In one example, an entire virtual map can be augmented on the real-world geographic scene so that the user can see information about every object in the user, or electronic devices, field of view. In one example, a virtual reality environment can be generated such that a user, having a virtual reality device can navigate inside the virtual reality environment and interact with each agricultural object and landmark displayed and created in the virtual reality environment. The user can view portions of the entire virtual reality environment changing

across time, either forward or backward. For example, a first virtual reality environment and scene depicting the geographic scene at a given time can change, gradually or instantly, to a second virtual reality environment depicting the geographic scene at a different time. Each of the objects and landmarks can also be selectable so the user can view specific views captured of the objects at various times from different trials. FIG. 4 illustrates a system architecture of an agricultural observation and treatment system, or agricultural treatment system 400, or treatment system. The agricultural treatment system 400 can include a robot having a plurality of computing, control, sensing, navigation, process, power, and network modules, configured to observe a plant, soil, agricultural environment, treat a plant, soil, agricultural environment, or a combination thereof, such as treating a plant for growth, fertilizing, pollinating, protecting and treating its health, thinning, harvesting, or treating a plant for the removal of unwanted plants or organisms, or stopping growth on certain identified plants or portions of a plant, or a combination thereof. In one example, an agricultural observation and treatment system, described in this disclosure, can be referred to as a portion of a system for observing and treating objects that is onboard a moving vehicle. Performances by the portion of the system onboard the moving vehicle, including computations, and physical actions, can be considered online performance or live performance. A portion of the system comprising one or more compute or storage components, that are connected as a distributed system, can be considered the offline portion of the system configured to perform remote computing, serve as a user interface, or storage. In one example, the agricultural observation and treatment system is a distributed system, distributed via cloud computing, fog computing, edge computing, or a combination thereof, or more than one subsystem is performing computations and actions live in addition to the portion of the system onboard a moving vehicle.

The systems, robots, computer software and systems, applications using computer vision and automation, or a combination thereof, can be implemented using data science and data analysis, including machine learning, deep learning including convolutional neural nets ("CNNs"), deep neural nets ("DNNs"), and other disciplines of computer-based artificial intelligence, as well as computer-vision techniques used to compare and correspond features or portions of one or more images, including 2D and 3D images, to facilitate detection, identification, classification, and treatment of individual agricultural objects, perform and implement visualization, mapping, pose of an agricultural object or of the robotic system, and/or navigation applications using simultaneous localization and mapping (SLAM) systems and algorithms, visual odometry systems and algorithms, including stereo visual odometry, or a combination thereof, receive and fuse sensor data with sensing technologies to provide perception, navigation, mapping, visualization, mobility, tracking, targeting, with sensing devices including cameras, depth sensing cameras or other depth sensors, black and white cameras, color cameras including RGB cameras, RGB-D cameras, infrared cameras, multispectral sensors, line scan cameras, area scan cameras, rolling shutter and global shutter cameras, optoelectric sensors, photoelectric sensors, light detection and ranging sensors (LiDAR) including spinning Lidar, flash LiDAR, static Lidar, etc., lasers, radar sensors, sonar sensors, radio sensors, ultrasonic sensors and rangefinders, other range sensors, photoelectric sensors, global positioning systems (GPS), inertial measurement units (IMU) including gyroscopes, accelerometers, and

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magnetometers, or a combination thereof, speedometers, wheel odometry sensors and encoders, wind sensor, stereo vision systems and multi-camera systems, omni-directional vision systems, wired and wireless communications systems and network communications systems including 5G wireless communications, computing systems including on-board computing, mobile computing, edge computing, cloud and cloudlet computing, fog computing, and other centralized and decentralized computing systems and methods, as well as vehicle and autonomous vehicle technologies including associated mechanical, electrical and electronic hardware. The systems, robots, computer software and systems, applications using computer vision and automation, or a combination thereof, described above, can be applied, for example, among objects in a geographic boundary to observe, identify, index with timestamps and history, and/or apply any number of treatments to objects, and, more specifically, of an agricultural delivery system configured to observe, identify, index, and/or apply, for example, an agricultural treatment to an identified agricultural object based on its location in the real-world geographic boundary, growth stage, and any and all treatment history.

In this example, the agricultural treatment system **400** agricultural treatment system **400** can include an on-board computing unit **420**, such compute unit **420** computing unit embedded with a system on chip. The on-board computing unit can include a compute module **424** configured to process images, send and receive instructions from and to various components on-board a vehicle supporting the agricultural treatment system **400** agricultural treatment system **400**. The computing unit can also include an engine control unit **422**, a system user interface, system UI **428**, and a communications module **426**.

The ECU **422** can be configured to control, manage, and regulate various electrical components related to sensing and environment that the agricultural treatment system **400** will maneuver in, electrical components related to orienting the physical components of the agricultural treatment system **400**, moving the agricultural treatment system **400**, and other signals related to managing power and the activation of electrical components in the treatment system. The ECU **422** can also be configured to synchronize the activation and deactivation of certain components of the agricultural treatment system **400** such as activating and deactivating the illumination module **460**, and synchronize the illumination module **460** with one or more cameras of the camera module **450** or one or more other sensors of the sensing module **451** for sensing an agricultural scene for observation and treatment of agricultural objects.

The compute module **424** can include computing devices and components configured to receive and process image data from image sensors or other components. In this example, the compute module **424** can process images, compare images, identify, locate, and classify features in the images including classification of objects such as agricultural objects, landmarks, or scenes, as well as identify location, pose estimation, or both, of an object in the real world based on the calculations and determinations generated by compute module **424** on the images and other sensor data fused with the image data. The communications module **426**, as well as any telemetry modules on the computing unit, can be configured to receive and transmit data, including sensing signals, rendered images, indexed images, classifications of objects within images, data related to navigation and location, videos, agricultural data including crop yield estimation, crop health, cluster count, amount of pollination required, crop status, size, color, density, etc.,

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and processed either on a computer or computing device on-board the vehicle, such as one or more computing devices or components for the compute module **424**, or remotely from a remote device close to the device on-board the vehicle or at a distance farther away from the agricultural scene or environment that the agricultural treatment system **400** maneuvers on.

For example, the communications module **426** can communicate signals, through a network **520** such as a wired network, wireless network, Bluetooth network, wireless network under 5G wireless standards technology, radio, cellular, etc. to edge and cloud computing devices including a mobile device **540**, a device for remote computing of data including remote computing **530**, databases storing image and other sensor data of crops such as crop plot repository **570**, or other databases storing information related to agricultural objects, scenes, environments, images and videos related to agricultural objects and terrain, training data for machine learning algorithms, raw data captured by image capture devices or other sensing devices, processed data such as a repository of indexed images of agricultural objects. In this example, the mobile device **540** can control the agricultural treatment system **400** through the communications module **426** as well as receive sensing signals from the telemetry module **366**. The mobile device **540** can also process images and store the processed images in the databases **560** or crop plot repository **570**, or back onto the on-board computing system of agricultural treatment system **400**. In one example, remote computing **530** component can be one or more computing devices dedicated to process images and sensing signals and storing them, transferring the processed information to the database **560**, or back to the on-board computing device of agricultural treatment system **400** through the network **520**.

In one example, the agricultural treatment system **400** includes a navigation unit **430** with sensors **432**. The navigation unit **430** can be configured to identify a pose and location of the agricultural treatment system **400**, including determining the planned direction and speed of motion of the agricultural treatment system **400** in real time. The navigation unit **430** can receive sensing signals from the sensors **432**. In this example, the sensing signals can include images received from cameras or Lidar's. The images received can be used to generate a grid map in 2D or 3D based on simultaneous visualization and mapping (SLAM) including geometric SLAM and Spatial SLAM techniques, visual odometry, or both, of the terrain, ground scene, agricultural environment such as a farm, etc. The sensing signals from the sensors **432** can also include depth signals from depth sensing cameras including RGB-D cameras or infrared cameras, or calculated with stereo vision mounted sensors such as stereo vision cameras, as well as other signals from radar, radio, sonar signals, photoelectric and photooptic signals, as well as location sensing signals, from having a global positioning system (GPS) unit, encoders for wheel odometry, IMU's, speedometers, etc. A compute module **434**, having computing components such as a system on chip or other computing device, of the navigation unit **430**, or compute module **424** of the compute unit **420**, or both, can fuse the sensing signals received by the sensors **432**, and determine a plan of motion, such as to speed up, slow down, move laterally, turn, change the rocker orientation and suspension, move, stop, or a combination thereof, or other location, pose, and orientation-based calculations and applications to align a treatment unit **470** with the ground, particularly with an object of interest such as a target plant on the ground. In one example, the navigation unit **430**

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can also receive the sensing signals and navigate agricultural treatment system **400** autonomously. For example, an autonomous drive system **440** can include motion components including a drive unit **444** having motors, steering components, and other components for driving a vehicle, as well as motion controls **442** for receiving instructions from the compute module **424** or compute module **424**, or both, to control the drive unit and move the vehicle, autonomously, from one location and orientation to a desired location and orientation.

In one example, the navigation unit **430** can include a communications module **436** to send and receive signals from other components of the agricultural treatment system **400** such as with the compute unit **420** or to send and receive signals from other computing devices and databases off the vehicle including remote computing devices over the network **520**.

In another example, the navigation unit **430** can receive sensing signals from a plurality of sensors including one or more cameras, Lidar, GPS, IMUs, VO cameras, SLAM sensing devices such as cameras and LiDAR, lasers, rangefinders, sonar, etc., and other sensors for detecting and identifying a scene, localizing the agricultural treatment system **400** and treatment unit **470** onto the scene, and calculating and determining a distance between the treatment unit **470** and a real world agricultural object based on the signals received, fused, and processed by the navigation unit **430**, or sent by the navigation unit **430** to be processed by the compute module **424**, and/or another on-board computing device of the treatment system **900**. The images received can be used to generate a map in 2D or 3D based on SLAM, visual odometry including geometry based or learning based visual odometry, or both, of the terrain, ground scene, agricultural environment such as a farm, etc. The sensing signals can also include depth signals, from having depth sensing cameras including RGB-D cameras or infrared cameras, a radar, radio, sonar signals, photoelectric and photooptic signals, as well as location sensing signals from GPS, encoders for wheel odometry, IMUs, speedometers, and other sensors for determining localization, mapping, and position of the agricultural treatment system **400** to objects of interest in the local environment as well as to the regional agricultural environment such as a farm or other cultivated land that has a designated boundary, world environment, or a combination thereof. The navigation unit **430** can fuse the sensing signals received by the sensors, and determine a plan of motion, such as to speed up, slow down, move laterally, turn, move, stop, change roll, pitch, and/or yaw orientation, or a combination thereof, or other location, localization, pose, and orientation-based calculations and applications.

In one example, the navigation unit **430** can include a topography module configured to utilize sensors, computer components, and circuitry configured to detect uneven surfaces on a plane or scene of the terrain which allows the topography module to communicate with the rest of the components of the treatment system to anticipate, adjust, avoid, compensate for, and other means of allowing the agricultural treatment system **400** to be aware of uneven surfaces detected on the terrain as well as identify and map unique uneven surfaces on the terrain to localize the vehicle supporting the navigation unit **430**.

In one example, the agricultural treatment system **400** includes a camera module **450** having one or more cameras, sensing module **451** having other sensing devices, or both, for receiving image data or other sensing data of a ground, terrain, orchard, crops, trees, plants, or a combination

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thereof, for identifying agricultural objects, such as flowers, fruits, fruitlets, buds, branches, plant petals and leaves, plant pistils and stigma, plant roots, or other subcomponent of a plant, and the location, position, and pose of the agricultural objects relative to a treatment unit **470**, camera module **450**, or both, and its position on the ground or terrain. The cameras can be oriented to have a stereo vision such as a pair of color or black and white cameras oriented to point to the ground. Other sensors of sensing module **451** can be pointed to the ground or trees of an orchard for identifying, analyzing, and localizing agricultural objects on the terrain or farm in parallel with the cameras of the camera module **450** and can include depth sensing cameras, LiDAR's, radar, electrooptical sensors, lasers, etc.

In one example, the agricultural treatment system **400** can include a treatment unit **470** with a treatment head **472**. In this example, the treatment unit **470** can be configured to receive instructions to point and shine a laser, through the treatment head **472**, to treat a target position and location on the ground terrain relative to the treatment unit **470**.

The agricultural treatment system **400** can also include motion controls **442**, including one or more computing devices, components, circuitry, and controllers configured to control mechatronics and electronic components of a vehicle supporting the agricultural treatment system **400** configured to move and maneuver the agricultural treatment system **400** through a terrain or orchard having crops and other plants of interest such that, as the agricultural treatment system **400** maneuvers through the terrain, the cameras **350** are scanning through the terrain and capturing images and the treatment unit is treating unwanted plants identified in the images captured from the camera module **450** and other sensors from sensing module **451**. In one example, an unwanted plant can be a weed that is undesirable for growing next or near a desirable plant such as a target crop or crop of interest. In one example, an unwanted plant can be a crop that is intentionally targeted for removal or blocking growth so that each crop growing on a specific plant or tree can be controlled and nutrients pulled from the plant can be distributed to the remaining crops in a controlled manner.

The agricultural treatment system **400** can also include one or more batteries **490** and one or configured to power the electronic components of the agricultural treatment system **400**, including DC-to-DC converters to apply desired power from the battery **490** to each electronic component powered directly by the battery.

In one example, the illumination module **460** can include one or more light arrays of lights, such as LED lights. The one or more light arrays can be positioned near the one or more cameras or sensors of camera module **450** and sensor module **451** to provide artificial illumination for capturing bright images. The light arrays can be positioned to point radially, from a side of the vehicle, pointed parallel to the ground, and illuminate trees or other plants that grow upwards. The light arrays can also be positioned to be pointed down at the ground to illuminate plants on the ground such as row crops, or other plants or soil itself. The light arrays can be controlled by the ECU **422**, as well as by a synchronization module, embedded in the ECU **422** or a separate electronic component or module, such that the lights only flashes to peak power and luminosity for the length of 1 frame of the camera of camera module **450**, with a matched shutter speed. In one example, the lights can be configured by the ECU **422** to flash to peak power for the time length of a multiple of the shutter speed of the camera. In one example, the lights of the light array can be synchronized to the cameras with a time offset such that the

instructions to activate the LED's of the light array and the instructions to turn on the camera and capture images are offset by a set time, predetermined time, or automatically calculated time based on errors and offsets detected by the compute unit 420, so that when the LED's actually activate to peak power or desired luminosity, which will be a moment in time after the moment in time the ECU sends a signal to activate the light array, the camera will also activate at the same time and capture its first image, and then both the lights and cameras will be synchronized and run at the same frequency. In one example, the length of time of the peak power of the activated light is matched and synchronized with the exposure time of each frame captured of the camera, or a multiple of the exposure time. In one example, the cameras can include

For example, the lights of the light array can flash with turning on, reach peak power, and turn off at a rate of 30 to 1000 Hertz (Hz). In one example, the lights can flash at 240 Hz to match one or more cameras that has a rolling shutter speed, global shutter speed, or both, of 240 Hz. In one example, the lights can flash at 240 Hz to match one or more cameras that has a rolling shutter speed, global shutter speed, or both, of 30 or 60 Hz. In one example, the lights can reach a peak power of 2.0M Lumen with a sustained peak power ON for 250 microseconds with a duty cycle of less than 10%. In one example, the color temperature of the light 170 can include the full spectrum of white light including cool, warm, neutral, cloudy, etc. In one example, the color temperature of the light can be around 5000K nm to reflect and artificially imitate the color temperature of the Sun.

In one example, the agricultural treatment system 400 can include a treatment unit 470 with a treatment head 472. In this example, the treatment unit 470 can include a turret and circuitry, electronic components and computing devices, such as one or more microcontrollers, electronic control units, FPGA, ASIC, system on chip, or other computing devices, configured to receive instructions to point and a treatment head 472, to treat a surface of a real-world object in proximity of the treatment unit 470. For example, the treatment unit 470 can emit a fluid projectile of a treatment chemical onto an agricultural object in the real world based on detecting the agricultural object in an image captured and determining its location in the real world relative to the treatment unit 470.

The treatment unit 470 can include a gimbal assembly, such that the treatment head 472 can be embedded in, or supported by the gimbal assembly, effectively allowing the treatment head 472 to rotate itself and orient itself about one or more rotational axes. For example, the gimbal assembly can have a first gimbal axis, and a second gimbal axis, the first gimbal axis allowing the gimbal to rotate about a yaw axis, and the second gimbal axis allowing the gimbal to rotate about a pitch axis. In this example, a control module of the treatment unit can control the gimbal assembly which changes the rotation of the gimbal assembly about its first gimbal axis, second gimbal axis, or both. The compute module 424 can determine a location on the ground scene, terrain, or tree in an orchard, or other agricultural environment, and instruct the control module of the treatment unit 470 to rotate and orient the gimbal assembly of the treatment unit 470. In one example, the compute module 424 can determine a position and orientation for the gimbal assembly to position and orient the treatment head 472 in real time and make adjustments in the position and orientation of the treatment head 472 as the agricultural treatment system 400 is moving relative to any target plants or agricultural objects of interest on the ground either in a fixed position on the

ground, or is also moving. The agricultural treatment system 400 can lock the treatment unit 470, at the treatment head 472, onto the target plant, or other agricultural object of interest through instructions received and controls performed by the control module of the treatment unit 470, to adjust the gimbal assembly to move, or keep and adjust, in real time, the line of sight of the treatment head 472 onto the target plant.

In one example, a chemical selection module, or chemical selection 480, of agricultural treatment system 400 agricultural treatment system 400 can be coupled to the compute module 424 and the treatment unit 470. The chemical selection module can be configured to receive instructions to send a chemical fluid or gas to the treatment unit 470 for treating a target plant or other object. In this example, the chemical selection module can include one or more chemical tanks 482, one or more chemical regulators 484 operable connected to the one or more chemical tanks 484 such that there is one chemical regulator for tank, a pump for each tank, and a chemical mixer 488 which can mix, in real time, chemical mixtures received from each chemical tank selected by the chemical mixer 488. In one example, a vehicle supporting the agricultural treatment system 400 agricultural treatment system 400, including the chemical selection module 480, can support one chemical tank 482, a chemical pump, a chemical regulator 486, a chemical and a chemical accumulator, in series, linking connecting a pathway for a desired chemical or liquid to travel from a stored state in a tank to the treatment unit 470 for deposition on a surface of an object. The chemical regulator 484 can be used to regulate flow and pressure of the fluid as it travels from the pump to the treatment unit. The regulator 484 can be manually set by a user and physically configure the regulator on the vehicle, or controlled by the compute unit 420 at the compute module 424 or ECU 422. The chemical regulator 484 can also automatically adjust flow and pressure of the fluid from the pump to the treatment unit 470 depending on the treatment parameters set, calculated, desired, or a combination thereof. In one example, the pump can be set to move fluid from the storage tank to the next module, component, in the series of components from the chemical tank 482 to the treatment unit 470. The pump can be set at a constant pressure that is always pressurized when the vehicle and agricultural treatment system 400 agricultural treatment system 400 is currently running a trial for plant or soil treatment. The pressure can then be regulated to controlled from the constant pressure at the regulator, and also an accumulator 487, so that a computer does not need to change the pump pressure in real time. Utilizing a regulator and accumulator can cause the pressure needed for the spray or emission of a fluid projectile to be precisely controlled, rather than controlling voltage or power of the pump. In one example, the agricultural treatment system 400 agricultural treatment system 400 will identify a target plant to spray in the real world based on image analysis of the target plant identified in an image captured in real time. The compute unit 420 can calculate a direction, orientation, and pressurization of the treatment unit 470 such that when the treatment unit 470 activates and opens a valve for the pressurized liquid to pass from the chemical selection module 480 to the treatment unit 470, a fluid projectile of a desired direction, orientation, and magnitude, from the pressure, will be emitted from the treatment unit 470 at the treatment head 472. The pump will keep the liquid stream from the chemical tank 482 to the treatment unit 470 at a constant pressure, whether or not there is flow. The chemical regulator 484 in the series of components will adjust and step down the pressure to a

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desired pressure controlled manually before a trial, controlled by the compute unit **420** before the trial, or controlled and changed in real time during a trial by the compute unit **420** either from remote commands from a user or automatically calculated by the compute module **424**. The accumulator **487** will keep the liquid stream in series pressurized to the desired pressure adjusted and controlled by the chemical regulator **484**, even after the treatment unit **470** releases and emits pressurized fluid so that the stream of fluid from the pump to the treatment unit **470** is always kept at a desired pressure without pressure drops from the release of pressurized fluid.

In one example, the chemical can be a solution of different chemical mixtures for treating a plant or soil. The chemicals can be mixed, or premixed, configured, and used as pesticides, herbicides, fungicides, insecticides, fungicides, adjuvants, growth enhancers, agents, artificial pollination, pheromones, etc., or a combination thereof. In one example, water or vapor can be substituted for any of the fluid or chemical selections described above. In one example, the agricultural treatment system **400** agricultural treatment system **400** can apply powder sprays or projectiles as well as foams, gels, coatings, or other physical substances that can be emitted from a chemical spray device.

FIG. 5 illustrates a system **402** for selecting and producing a chemical mixture for spraying. In one example, the system **402** can be a subsystem combined with the agricultural treatment system **400** and mounted or attached to a vehicle. In one example, the system **402** can be implemented in real time such that an emitter **470** of the agricultural treatment system **400** agricultural treatment system **400** can receive instructions to target and spray and a chemical selector **488a** can provide a desired chemical mixture in real time. For example, multiple series of chemical selection components can be configured such that each series of chemical selection components can be run in parallel for a chemical mixer **488a** to mix chemicals, in the form of fluids, liquids, gas, powder, water, vapor, etc., in real time, and send the desired mixed chemical, in both content, proportionality, concentration, and volume to the treatment unit, or an emitter **470**, to be emitted as a projectile, aerosol, mist, or a powder or liquid droplet onto a surface of an object. In one example, a first series of components for chemical selection can include a chemical tank **482a**, a chemical pump **485a**, a regulator **486a**, an accumulator **487a**, and one or more spray tubes and potential circuitry to link each of the chemical tank **482a**, chemical pump **485a**, regulator **486a**, and accumulator **487a** in series to be connected to the chemical mixer **488a**. The chemical tank **482a** can store a desired chemical, which can be a premixed chemical another set of chemicals. For example, the chemical tank **482a** can store chemical-1. In parallel to the series of chemical selection components of **482a**, **485a**, **486a**, and **487a**, is a second series of chemical selection components including a chemical tank **482b**, chemical pump **485b**, regulator **486b**, and accumulator **487b**. The components **482b**, **485b**, **486b**, and **487b** can be connected in series with one or more spray tubes and connected to the chemical mixer **488a**. The chemical tank **482b** can store chemical-2, which can be different chemical mixture or concentration as that of chemical-1. In this configuration the chemical mixer **488a** can select and extract, in real time on the vehicle during an observation and spray trial, either chemical-1, chemical-2, or a combination of both with varying concentrations and volume. The chemical mixer **488a** can then send the mixture of chemical-1 and chemical-2 or any desired mixture of chemicals, or a chemical from only a single channel, to the emitter **470** to emit a

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mixed chemical projectile, droplet, aerosol, etc., at a target object. Further, any number of different chemical mixtures can be stored on-board the vehicle such that the chemical mixer **488a** can extract the chemical mixture and generate a new chemical mixture for treating an object. For example, a third series of chemical selection components, including a chemical tank **482c**, configured to store chemical-3, chemical pump **485c**, regulator **486c**, accumulator **487c**, can be configured in parallel with the other two series of chemical selection components such that the chemical selector can choose from any of the three different chemicals of chemical-1, chemical-2, or chemical-3. Further, the number of chemical tanks stored is limited to only the amount that the vehicle with the agricultural treatment system **400** can support including an nth series of chemical selection components, such as chemical tank **482n**, chemical pump **485n**, regulator **486n**, and accumulator **487n**, linked in series by a spray tube and connected to the chemical mixer **488a**. The chemical mixer **488a** can be configured to select and receive different combinations in volumes of chemical-1, chemical-2, chemical-3, and so forth, to be sent to the emitter **470** and emit a pressurized projectile, aerosol, mist, or a powder or liquid droplet onto a surface of an object. In one example, one of the chemical tanks can store water or vapor such that the selection of the chemical tank with water is used to dilute a solution of mixed chemicals.

In one example, the emitter **470** can emit a projectile, liquid, gas, aerosol, spray, mist, fog, or other type of fluid droplet induced spray to treat a plurality of different plants in real time. An agricultural scene can include a row crop farm or orchard planted with different crops. In this example, each row of plants can include a different type of plant to be cultivated and treated such that the emitter **470** can treat one row with one type of treatment, such as a chemical mixture-1, mixed and sent to the emitter **470** by the chemical mixer **488a**, and another row with another type of treatment to a different crop or plant, such as a chemical mixture-2. This can be done in one trial run by a vehicle supporting the chemicals, and treatment system with emitter **470**. In another example, each row itself, in a row crop farm or orchard, can have a plurality of different type of crops. For example, a first row can include a first plant and a second plant, such that the first plant and second plant are planted in an alternating pattern of a first plant, a second plant, a first plant, a second plant, and so forth for the entire row of a first row. In this example, the chemical selector **488a** and emitter **470** can deposit a first chemical mixture projectile, for precision treatment, to the first plant, and deposit a second chemical mixture projectile, for precision treatment, to the second plant, in real time, and back to the depositing the first chemical projectile to the third plant in the row of crops, the third plant being of the same plant type as the first plant, and so forth. In one example, a plurality of more than two types or species of plants can be planted in tilled soil, and be grown and treated in a row crop with the agricultural treatment system **400** with system **402**.

In one example, the treatment unit of agricultural treatment system **400** can blast water or air, or a water vapor to one or more agricultural objects to wash off any undesired objects detected on the surface or other portion of the agricultural objects. The undesired objects can be unwanted bugs or debris on the agricultural object as well as previously applied chemicals that are no longer desired to leave on the agricultural object. In one example, the treatment unit can then recoat an agricultural object that was previously cleaned with water or air with a new chemical treatment. In one example, one of the chemical tanks can also include

water for the purposes of purging the stream of liquid from tanks to the treatment units of any excess chemical or substance buildup which could affect chemical composition, pressure, spray health, and other controlled factors that could affect desired performance. In one example, one of the tanks can include water for chemigation as water is mixed with substance from a different tank.

FIG. 6 is a diagram 600 illustrating an example vehicle 610 supporting an example observation and treatment system, or treatment system 612, performing in a geographic boundary, according to some examples. In this example, the vehicle 610 can support one or more modular treatment systems 612. The treatment systems 612 can be similar to that of agricultural observation and treatment systems described above. For example, a system can include onboard and offline components performing tasks both in real time while a vehicle supporting the onboard portions of components are performing observations and actions and at edge compute device or remotely both in real time or offline.

For example, the treatment system 612 can be one of a plurality of modular component treatment systems, each component treatment system can include one or more sensors including image capture sensors, illumination devices, one or more treatment units, for example a pair of treatment units each with a treatment head capable of aiming at a target 660 with at least 2 degrees of rotational freedom, a compute unit configured to send and receive instructions of sensors, encoders, and actuators and connected and associated with the component treatment system and the compute unit to time sync all of the components, and other electronics to sync and communicate with other compute units of other component treatment systems. Each of these treatment systems 612 can receive treatment fluids from a common pressurized source of fluid, or each treatment unit is connected to different sources of fluid. The component treatment systems are configured to sense targets 660 in real time while supported by the moving vehicle 612, determine what kind of treatment, or other action, to perform on to a surface of the target 660, target and track the target 660, predict performance metrics of the instructed parameters of the action including projectile location, perform the action, including emitting a fluid projectile or light source, and evaluating the efficacy and accuracy of the action.

Additionally, each of these treatment systems 612, or component treatment systems, can communicate and receive information from a component navigation system or navigation unit which is configured to sense a global scene such that each of the compute units of each of the component treatment systems can sense its local environment from sensors operably connected to the compute unit of the component treatment system, or embedded in the component treatment system, as well as receive information about the global environment in a geographic scene from sensors and analysis performed by sensors and the one or more compute units of the navigation unit connect to each of the local component treatment systems.

The vehicle 610 can be operating in a geographic region such as a farm or orchard. A portion of the geographic boundary, illustrated in FIG. 6, with one or more trees 634 is shown. In this example the vehicle 610 can be operating in an orchard with multiple rows of trees, each having a trunk 636, or other plants for the treatment systems 612 to observe and treat. In one example, the vehicle can be travelling in a straight line along a row of trees and crops on both sides of the vehicle.

One or more treatment systems 612 can be mounted on top, embedded in, suspended underneath, towed, or oriented

in many ways securely onto the vehicle such that the treatment system 612 can be configured and oriented to scan a row of crops or plants or other agricultural scenes in a line while the vehicle 610 is moving.

The vehicle 610 may include functionalities and/or structures of any motorized vehicle, including those powered by electric motors or internal combustion engines. For example, vehicle 610 may include functionalities and/or structures of a truck, such as a pick-up truck (or any other truck), an all-terrain vehicle ("ATV"), a utility task vehicle ("UTV"), or any multipurpose off-highway vehicle, including any agricultural vehicle, including tractors or the like. The treatment systems 612 that may be powered or pulled separately by a vehicle, which may navigate paths autonomously in the geographic boundary.

In one example, a geographic boundary can be configured to have two rows of plants on each side of a single lane for a vehicle to navigate through. On each side of the vehicle will be vertically growing plants such as trees. The treatment system 612 can be mounted on the vehicle in a way that image sensors of the treatment system 612 are pointing directly at the trees on each two left and right side of the vehicle. As the vehicle operates along a lane or path in the orchard, the treatment system 612 can capture a series of images from one side to another of the row of plants as well as treat each agricultural object with a precision treatment.

FIG. 7A illustrates a diagram 700 depicting an agricultural observation and treatment system supported by a vehicle navigating in an agricultural environment. The agricultural environment can be a farm or orchard typically having one or more plants such as trees 303. While the illustration depicts a system in an environment similar to that of an orchard, different, the description below can apply to a system, the same system, performing in multiple different types of environments such as row crop farms where portions of the system include sensors pointing at the ground to detect row crop objects of interest.

The agricultural observation and treatment system can be a modular system similar to that of agricultural observation and treatment system 311 supported by vehicle 310. The system 311 can also include various sensors such as imaged based sensor 313, lidar based sensors 314, or a plurality of non-vision-based sensors as described previously. Similar to that of navigation unit and navigation modules described in this disclosure, the treatment system depicted in diagram 700 can use sensors, such sensor 313 and 314 to perform global registry of the agricultural environment as well as perform localization and pose estimation of the vehicle or portions of the vehicle in a global scene or from a global point of reference. This can include receiving sensor data and generating and building high definition 2-dimensional and 3-dimensional maps, or global maps as opposed to views of individual crops with sensors close up to the individual crops which can be referred to a local scene or local registry of a geographic boundary or scene, of the agricultural environment in real time.

In one example, the agricultural observation and treatment system 311, described in previous discussions, can be configured to perform scene understanding, mapping, and navigation related analysis including localization of the vehicle and/or components of the treatment system and pose estimation of the vehicle and/or components of the treatment system, for example pose estimation of individual image capture devices embedded in each component spray system, each component spray system illustrated in diagrams 900, 902, 2406, and 2408, or each modular subsystem of the agricultural observation and treatment. The sensing can be

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performed with a various suite of image-based sensors, motion-based sensors, navigation-based sensors, encoder sensors, other sensors, or a combination thereof, fused and synchronized together such that at least components of the agricultural observation and treatment system 311 can determine navigational properties of an environment the system is in, including pose and pose estimation of the components in real time as the vehicle, treatment system, and components of the system moves and navigates in such environment.

In one example, the agricultural observation and treatment system 311 can perform mapping of a scene and localizing the treatment system in the scene. This can include mapping a scene with a global frame of reference or point of origin in a given coordinate system, and determining its location relative to a point in the mapped scene, particularly a point of origin in the scene. For example, when a vehicle is navigating in an agricultural environment in FIG. 7A, the system can arbitrarily determine a point of origin in the agricultural environment, for example the portion of the agricultural environment the system has sensed thus far, or preloaded into the system before sensing the environment in real time. For example, a first corner or first edge of a portion or region of a farm or orchard. As the vehicle navigates in the environment, the system can determine where the vehicle and each of the components of the system, due to the components being fixed relative to the vehicle, has moved across time. The treatment system's navigation type sensors, including GPS, IMU, encoders, image capture devices configured to capture a high resolution or low resolution of a global scene and perform techniques and functions in computer vision and machine learning such as visual SLAM and visual odometry (the global scene referring to the farm or orchard, or any kind of agricultural environment itself, and not necessarily those sensors pointing directly at individual plants or crops of plants for high-definition images of individual plants), Lidars to sense a global scene similar to that of the image capture devices, optoelectrical sensors, ultrasonic sensors, radar, sonar, and other image capture devices such as NIR cameras, RGB-D cameras, multispectral cameras, etc., configured to obtain global registry of an environment including mapping the environment at a global level, and can be used to generate and continuously generate a global pose estimation of the vehicle as it moves along a path, and each of its component, relative to a point of origin in the geographic environment, the system can also determine the same global pose of components of the system as well, due to the components being rigidly connected or supported by the vehicle. For example, while a camera sensor that is 15 feet from the ground or 15 feet vertically above a bed of the vehicle may wobble and move more than that of a camera sensor that is 1 foot from the ground or 1 foot vertically above the bed of the vehicle, each poses of the cameras may be different to each other at a local level, relative to a vehicle, and therefore, relative to a point of origin in the geographic scene, the global pose estimation can be estimated to that of the global pose estimation generated for the vehicle by a navigation module and sensors associated with navigation. This is because each of these cameras can be rigidly connected to the sensors configured to perform global registry of the environment, such that physical translation and movement of the vehicle, and particularly movement of the navigation-based sensors (for example, GPS, and navigation-based camera sensors), the local sensors embedded or supported by each component spray module or component treatment module will substantially move the

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same amount. Additionally, each sensor supported by each component treatment module can track local objects, shapes, patterns, or any salient points or patterns local to each of the component treatment modules such that a more accurate pose estimation for each of the component treatment modules, more specifically, pose estimation of sensors of each component treatment modules, can be generated and continuously generated as the component treatment modules are scanning a local scene to observe objects and perform objects on target objects.

In one example, the system, both the navigation system and its components, sensors, and compute units, as well as each component subsystem or component treatment module having its own components, sensors, treatment units, and compute units, can use techniques associate with simultaneous localization and mapping (SLAM) and odometry, particularly visual SLAM, VSLAM, and visual odometry, or VIO, in conjunction with other non-visual based navigation and localization analysis, fused together in real time with sensor fusion and synchronization, to perform pose estimation of the vehicle. Additionally, each modular sub systems of the treatment system including each modular spray subsystem, for example each modular spray subsystem or component treatment module including a structural mechanism, a compute unit, one or more sensors, one or more treatment units, and one or more illumination devices, can perform VSLAM and receive other non-visual based sensor readings, and continuously generate its own localized pose estimation, the pose being relative to specific objects detected by each of the component treatment modules, which can include agricultural objects including target objects or nearby objects or patterns, shapes, points, or a combination thereof that are of a similar size to that of the target objects. The pose estimation of components of each of the component treatment modules will be relative to the location of the objects and patterns detected to be tracked across time and across sensors in stereo for stereo matching points for depth perception. Additionally, the system can perform projection and reprojection, and determine reprojection error, for more accurately determining location of objects and eliminating outliers. Thus, detecting objects and patterns that are known to be fixed in space, for example a ground terrain with unique rocks or dirt patterns, or individual plants, and calculating and identifying the objects' or patterns' 3D location and/or orientation relative to the sensors' 3D location and/or orientation sensing the objects and patterns allows the system to understand navigation, localization, and more specifically local pose estimation of each of those sensors relative to the objects detected. Additionally, since the orientation of the treatment units, and its treatment heads, are in close proximity to the individual component treatment module, and rigidly attached and connected to a structure of the component treatment module (illustrated in at least FIGS. 9A, 9B, and 24E), and also in close proximity and rigidly connected to the sensors associated with that particular component treatment module and compute unit, the location and orientation of the treatment head of the treatment units (the treatment heads having encoders to determine line of sight relative to the body of the treatment unit) can also be continuously generated and determined relative to the target objects or objects near the target objects themselves for better accuracy of treatment.

In this disclosure, while the determined pose estimation can be referred to the pose estimation generated for the vehicle or a component modular spray subsystem, a pose estimation can be determined, using VSLAM, VIO, and/or other sensor analysis, to generate a pose, including a loca-

tion and/or orientation for any component of the vehicle or component of the agricultural observation and treatment system. In one example, a pose estimation can be referred to and generated with coordinates, for example $(x_1, y_1, z_1, \Phi_1, \theta_1, \Psi_1)$ with x, y, z , being the translational location relative to an origin (x_0, y_0, z_0) and starting orientation $(\Phi_0, \theta_0, \Psi_0)$ of the component relative to an origin point and/or orientation, of any component or portion of a component.

In one example, as illustrated in FIG. 7A, a vehicle's navigation module can include sensors such as imaged based sensor 313, lidar based sensors 314, or a plurality of non-vision-based sensors as described previously, fused together to obtain global registry of the scene for mapping the scene as well as in real time navigating in the scene. For example, a pair or more than one pair of image sensors 313, in stereo, can be mounted on the vehicle such that the sensors are pointed out to the real world to observe a global scene, as opposed to down on the ground to observe individual plants or right in front of (a few meters or fewer away) a tree to observe individual plants, crops, or other agricultural objects. Additionally, Lidars, radar, sonar, and other sensors can be mounted in a similar location as that of the cameras to register a global scene.

The agricultural observation and treatment system, at the navigation unit, or a component of the navigation unit, can perform real time VSLAM and VIO to simultaneously map the agricultural environment that it is in, as well as understand the location, localizing, of the agricultural observation and treatment system itself as it is navigating in the environment, whether it is autonomously navigating or navigating with a human driver on the vehicle or remotely. In one example, as illustrated in diagram 700 of FIG. 7A, VSLAM can be performed using keypoint detection and matching across stereo and across time, or surface matching of salient points or regions of surfaces detected. Keypoints, for example keypoint 706 can be generated in real time from image frames captured by image sensors. In one example, stereo image sensors can capture frames at the same time. Keypoints can be generated and identified by a compute unit embedded in the image sensor, or a compute unit of the navigation unit operably connected to each of the stereo image sensors configured to receive images or imagery data from the sensors. Common keypoints are matched such that the system can understand depth of the keypoint from the stereo sensors, and thus the depth of the keypoint from the navigation unit and therefore the vehicle. Thousands of keypoints can be detected, generated, and tracked over time per frame. They keypoints themselves can be cluster of pixel points representing a corner, blob, line, other salient patterns. These points do not necessarily have to be real world identifiable known objects. For example, a keypoint can be generated as a where two objects meet, or where one object and a background meet, for example an edge of a leaf from the background sky. The different in color between a leaf and a sky will create a line or edge between the two colors captured by an image sensor. Other examples can be corners of objects, dots, or lines. For example, three small rocks next to each other can form salient pattern to track, even if the system doesn't understand that it is a 3-rock cluster, meaning the system cannot extract enough features to perform object detection and determine that one or more rocks were identified. But the actual 3 rocks form a rigid and complex pattern that the system can still identify and track.

The system may not be able to detect an identity of an object, but it can detect its contours and edges and track those points. In this example any type of points or pixels, clusters of points or pixels, lines, corners, that the system

may determine as salient points or patters, can be generated as a keypoint. The keypoint may or may not be generated in a different frame captured by a different image sensor, for example a second sensor in stereo with a first sensor. Common keypoints between different frames can be matched using various computer vision techniques such as image correspondence, keypoint matching, template matching where the templates are patches of image including keypoints, dense SLAM techniques, techniques with classifiers, RANSAC and outlier rejection to more accurately detect common keypoints, other statistical modelling techniques, or other corner, line, blob, edge detection techniques including FAST, SIFT, Harris corner detection, Lucas-Kanade tracking, SURF, NERF, ORB, or a combination thereof. Additionally, lines, corners, patterns, or other shapes, whether referred to as keypoints or not, that are generated in each frame can be matched to the similar lines, corners, patterns, and shapes using the same or similar techniques described above.

The keypoints that are matched between two cameras in stereo, or more cameras can be used to determine depth of the object or point or points in the real-world space associated with the keypoints detected. These techniques can be applied similarly with Lidar or used in conjunction with lidar sensed point clouds and fused together for more accurate representations of a scene. The images sense, and keypoints matched between cameras can be used to perform dense reconstructions of environments, as well as perform real time navigation, localization, and pose estimation of the sensors sensing the environment.

In one example, dense visual slam can be performed, whether by a single image sensor, such as a camera, infrared camera, rgb-d camera, multispectral sensors, or other sensor, or by stereo cameras, or two or more of different types of cameras that are synchronized with known fixed distance and orientation relative to each other, to compare image from frame to frame across time. The commonly matched and tracked points can allow the sensor to determine how much movement and amount of translation and orientation change of the sensor based on the shape and depth of the point or object detected and tracked from a previous frame. For example, an agricultural observation and treatment system can sense a point in space, which can be any type of pattern, but for example, can be a pattern of a base of a tree trunk. The system can determine that as a keypoint 706. As the vehicle supporting the agricultural observation and treatment system moves closer to the base of the tree trunk. The pattern of the keypoint 706 from a first frame will change in shape, location, size, etc., in a subsequent frame captured at a later point as the image sensor has moved closer to the point. The system would be able to calculate the amount of movement in space as it tracks the same point in space with keypoint matching or other techniques to perform VSLAM.

In one example, VSLAM can be performed by detecting objects themselves, rather than arbitrary points and patterns detected in an image frame. For example, a whether referred to as keypoints or not, the system can ingest a frame, perform feature extraction and object detection and detect specific known objects with a machine learning model. These objects, rather than salient points such as corners, lines, blobs, or other patterns that can be tracked across frames in stereo and across frames in time, can be objects such as agricultural objects and landmarks, such as leaves, weeds, rocks, terrain, trees, components of irrigation systems, wheels of a vehicle, or any other landmarks where the system can identify the landmark itself, rather than just salient points that may or may not be associated to a

landmark. For example, referring to diagram **1300a** of FIG. **13a**, the agricultural observation and treatment system can detect a plurality of fruitlets, buds, and landmarks in a single frame using a machine learning detector embedded on board the system. From frame to frame, as the treatment system scans the orchard, the image sensors of each of the component treatment modules configured to detect individual objects and landmarks themselves, can detect objects, and match the object detections from frame to frame for the purposes of SLAM and pose estimation of the sensor sensing the object, in addition to determining whether to track an object to perform a treatment action. In another example, as illustrated in diagram **1600** of FIG. **16**, the component treatment module sensing and analyzing the frame or image **1610** can detect every agricultural object in the frame including carrots (or agricultural object not to treat) and weeds (agricultural objects detected for treatment). The system can identify features of each object, and match the features from frame to frame for scene understanding and determining movement and orientation of the sensors capturing the frame (because the objects themselves are not moving in space) relative to those objects, even though the agricultural object and treatment system under certain configurations are tasked to only target, track, and treat the weeds. The objects can be detected using feature extraction and object detection with various machine learning techniques, computer vision techniques, or a combination thereof. Additionally, while the discussion above focuses on image-based sensors, similar techniques can be applied using one or more lidar sensors for point cloud to point cloud matching.

In one example, in addition to performing functions to allow a system to determine pose and therefore navigate in an environment, the system can use the same sensor readings used to determine pose estimation to perform dense reconstruction of a scene and map the agricultural environment. This can be done with VSLAM which takes multiple views across frames in time and in stereo to reconstruct a scene. Other techniques can include dense reconstruction of a scene from SLAM or structure from motion. Other techniques for improvements in scene geometry in a sequence of frames captured by an image sensor can include local bundle adjustment to improve visual SLAM.

In one example, the agricultural objects discussed in this disclosure can be any number of objects and features detected by various sensors including image sensors. The agricultural objects can include varieties of plants, different phenological stages of different varieties of plants, even though the specific object detected in a geographic scene, having different physical features due to its growth is the same physical object in the geographic scene, target plants to treat including treating plants to turn into a crop or treating plants for plant removal or stunting or stopping or controlling the growth rate of a plant. Agricultural objects can include soil or patches of soil for soil treatment. Other objects detected and observed by a treatment system can include landmarks in the scene. Landmarks can be trees and portions of trees including spurs, stems, shoots, laterals, specific portions of the terrain including dirt, soil, water, mud, etc., trellises, wires, and other farming materials used for agriculture. Additionally, landmarks can be any object that can be detected by the observation and treatment system and tracked as a vehicle supporting the treatment system is moving as well as tracked throughout time as the vehicle visits the location across a grow season in multiple trials or runs. For example, agricultural objects described throughout this disclosure can also be considered a landmark for track-

ing the landmark even though the observation and treatment system may not necessarily be targeting the object for treatment or tagging and indexing the object for observation. The landmarks can be tracked, using SLAM or other computer vision and machine learning based or assisted computer vision techniques, to better locate a nearby landmark or object that will be a target for treatment. For example, a plurality of landmarks and potential landmarks that are also target objects can be detected in a given image frame, or a pair of stereo frames, or sensed by a plurality of sensors synched in time. Once the agricultural observation and treatment system has identified

In one example, landmarks that can be tracked are not specifically defined objects in the real world, but patterns or combinations of objects or features such as region in space separating one or more unknown objects from a background. While the system may not be able to perform feature extraction enough to detect an object's identity, it can still detect that an object, or a pattern created by one or more objects, exists in the captured sensor reading, and is that of one that is fixed in space and will not move. These detections can also be referred to as landmarks and used to track the landmarks for real time pose estimation. For example, landmarks detected as illustrated in diagrams **1300a** and **1300b** of FIGS. **13A** and **13B** can be used to generate keyframes for further offline analysis including matching frames taken at different times of the same or similar view to create a time lapse visualization of an object by comparing only keyframes instead of every frame ingested by a sensor, determining which candidate keyframe previously generated and stored by the system is used in real time to perform image correspondence for live/real time detections, targeting, and tracking, as well as used and tracked from frame to frame to perform VSLAM, and generate better pose estimation for the sensor sensing the frames.

An agricultural object of interest can be a target plant for growing into a harvestable crop. In one example, the agricultural object of interest can be a target plant to remove, such as that of a weed, or any plant that is not a crop. In one example, the agricultural object can be portions of a soil of interest to observe and cultivate, such that at least a portion of the cultivating process is treating the soil with one or more fluid chemical treatments or fertilizer.

In one specific example, the agricultural observation and treatment system can perform a variation of SLAM focusing on one or more specific features to extract to more accurately generate pose estimation for agricultural observation and treatment systems performing in agricultural environments. For example, a system can be embedded with a SLAM algorithm, whether assisted by machine learning, other computer vision techniques, or both, to detect tree trunks. In most orchards, tree trunks do not grow in size, change shape, or move in relatively short spans of time. Tree trunks are also spaced apart enough that each tree trunk detected can allow a system to determine a different set of agricultural objects detected in a tree are separated, which can be used to determine which sets of frames in a plurality of frames can be generated or used as a keyframe.

In another specific example, the agricultural observation and treatment system can perform a variation of SLAM focusing on one or more specific features to extract to more accurately generate pose estimation for agricultural observation and treatment systems performing in agricultural environments. For example, a system can be embedded with a SLAM algorithm, whether assisted by machine learning, other computer vision techniques, or both, to detect beds, troughs, furrows, and vehicle tracks of a row crop farm, the

beds being where plants are planted and grow, and tracks being where vehicle wheels can travel. Instead of detecting arbitrary lines in a given frame, a machine learning assisted SLAM algorithm can be configured to detect just the beds and troughs, due to the beds and troughs looking substantially like lines that would take up an entire frame. This can help ease the performance load on performing VSLAM as the vehicle only needs to operate between two lines, detected as a trough, to better minimize drift.

In one example, each of the compute units of each component treatment modules can associate local and global pose. While the compute unit and sensors associated with the navigation unit performs slam and accounts for a vehicle pose (the vehicles components' own SLAM), for navigation purposes, it can also combine to individually map every single plant, because the plant will have a known location due to its relative location to the component treatment modules' sensors, for example stereo image sensors. The stereo image sensors will know where it is relative to a global scene to do the mapping of a local scene and the real time localization of the vehicle itself in the global scene. The combination will allow the system to generate a global map with every single unique and individual agricultural object sensed, and indexed with high precision in the global map itself.

In one example, a drone with one or more image sensors, lidars, GPS, IMU, can be configured to scan and map a scene of a geographic boundary of an agricultural scene and combine the sensor readings of the globally mapped scene from the drone with that of the globally mapped scene from the navigation unit's sensors and compute units of the agricultural observation and treatment system and the locally mapped individual agricultural objects and landmarks imposed onto the globally mapped scene generated by the agricultural observation and treatment system. This would allow any views or scenes not necessarily or accurately captured by the agricultural observation and treatment system on the ground but could be captured by the done to be accounted for to create an even more accurate global map with more views of the map or indexed database of a global map comprising a global scene from the drone with high definition readings of the geographic boundaries and views of the boundaries, a denser global scene of portions of the agricultural environment including trees, rows, beds, troughs, and it's location relative to each other and in the global frame, as well as a local scene of each individual agricultural object and landmark detected by each component treatment module.

In this example, individual crops, plants, agricultural objects, and landmarks can be sensed and registered with location and pose relative to image sensors sensing the individual crops, plants, agricultural objects, and landmarks. This will allow the agricultural observation and treatment system, at each of the component treatment systems having sensors and treatment units to identify more accurately where each object is relative to the local sensors of each component treatment module in real time. Additionally, the location of the individual crops, plants, agricultural objects, and landmarks, relative to the local sensors of each component treatment module can be indexed and stored. Because the local sensors of each component treatment module are rigidly connected to each other with a support structure, and the support structure is also rigidly connected to the vehicle supporting the navigation module and its sensors for obtaining global registry and global pose, The agricultural observation and treatment system can combine the local and global pose to determine where each individual crops,

plants, agricultural objects, and landmarks sensed is located in a global scene. Thus, the agricultural observation and treatment system can be configured to create a global map of a scene with each individual object and/or landmark detected in the global map with sub-centimeter accuracy of where each individual object and/or landmark is located in the global map, or at least digitizing and indexing an agricultural scene without generating a visualizable map.

In one example, a vehicle or global pose estimation can be determined as well as an individual localized pose estimation for each component treatment module can be determined.

For example, each treatment module, having one or more image sensors, one or more illumination sources, and one or more treatment units (being a spray device or a light treatment device) and a compute unit all operable and rigidly attached to each other as a single modular treatment module, for example shown in diagram 900 of FIG. 9, each of the individual treatment modules can compute its own pose estimation. This in effect allows for each treatment unit to have a more accurate pose estimation, for example, a local pose estimation of the treatment unit and the image sensors of the same treatment module supporting the treatment units of the treatment module locally to each other and to the ground row plants or orchard trees that each of the treatment module's sensors are sensing. Additionally, the sensors of the navigation unit or navigation module, for example the navigation unit including a GPS sensor, one or more IMU's, one or more encoders, one or more cameras, for example facing outward into the geographic area as a whole either facing forward or backwards of the direction of the vehicle path, or one or more laser or lidar sensors, can be configured to generate a global pose estimation for the vehicle itself. In this example, each treatment unit of each treatment module can rely on both the pose estimation determined by the treatment module's local sensors and compute units, for the local pose estimation of each treatment module, as well as the pose estimation determined by the navigation unit's sensors and compute unit, for the global pose estimation of the vehicle. The combined and fused pose estimation can be configured to give each treatment module a more accurate localization, orientation, and pose such that as each treatment module detects, targets, and tracks each object of interest detected, the treatment unit can target and treat the agricultural objects of interest more accurately.

FIG. 7B illustrates an example method 702 that may be performed by some example systems, subsystems, or components of systems, described in this disclosure either online, that is onboard a vehicle supporting one or more modular agricultural observation and treatment systems, subsystems, or components of systems, or offline, that is at one or more servers or edge compute devices.

For example, at step 710, an agricultural observation and treatment system can initialize. At step 720, the agricultural observation and treatment system can determine a real-world geospatial location of the treatment system. The determining of location can be performed by location-based sensors such as GPS, image-based sensors, such as cameras including CCD, CMOS, or Lighfield cameras, multispectral cameras, RGB-D cameras, NIR cameras, the same two cameras in stereo or variations of cameras synched in stereo, or more than two synchronized together, Lidar, laser sensors, motion sensors such as IMU, MEMS, NEMS, or motion-based encoders. At step 730, the agricultural observation and treatment system can receive one or more images or point clouds from one or more image capture devices. At step 740, the agricultural observation and treatment system can iden-

tify one or more salient points or salient regions of at least a portion of a first frame. The salient points can be keypoints and the salient regions can be keypoint regions, cluster of pixels in a frame that is associated with a fixed object or points in space that can be compared for similarities, including keypoint matching, across stereo sensors or across time by frames captured by the same sensor from frame to frame. The salient points themselves can be object based, instead of keypoint based, such as objects that can be detected with a neural network using feature extraction and object detection. In one example, the salient points can be points or regions detected by a machine learning detector, keypoints generated using various computer vision algorithms or by a machine learning detector or a machine learning assisted computer vision algorithm, or a combination thereof. At step 750, the agricultural observation and treatment system can identify one or more salient points or salient regions of at least a portion of a subsequent frame. At step 760, the agricultural observation and treatment system can determine a change in position of the treatment system based on comparing the first and subsequent frame. At step 770, the agricultural observation and treatment system can verify or improve the determined change in position with the position determined by the location-based sensors, motions sensors, or both. At step 780, the agricultural observation and treatment system can determine a pose estimation. And at step 790, the agricultural observation and treatment system can send instructions to activate actuators. The points of interest to track for motion estimation and SLAM can be that of real-world objects or patterns detected, or salient cluster of points in an image or point cloud that can be tracked from frame to frame or point cloud to point cloud as a vehicle with image or point cloud sensors move in time. These can be detected by computer vision methods of detecting edges, corners, blobs, lines, etc., or by a machine learning algorithm configured to detect real world objects, such as agricultural objects, for example leaves for sensing systems pointed down at row crops, rocks, dirt, beds, troughs, crops, weeds, etc. For example, if a landmark such as a small rock in the dirt, or a leaf of a crop, in a frame captured by an image sensor, the compute unit can determine that a cluster of pixels of the frame comprise a "rock" detected by a machine learning algorithm, such that an object that is the rock detected can be tracked and matched across stereo vision system and across time, that is from frame to frame, and used to perform motion estimation and pose estimation by tracking relative position of the suite of fixed sensors, and by extension the position of the vehicle, the agricultural observation and treatment system supported by the vehicle and each of its treatment modules, to the rock as the vehicle moves in a direction relative to the rock, or any object detected by the machine learning algorithm. Alternatively, the compute unit can detect that a cluster of pixels in the frame captured is different enough from a detected background of the frame such that the cluster of pixels, while the compute unit may not know, extract enough features to determine and detect, that it is a real-world rock, is still a real-world object that is stationary and can be tracked from frame to frame, and by extension, can be compared to from frame to frame, to perform motion estimation and pose estimation. In one example, the objects detected in real time can be used to determine which areas detected in a geographic scene should be treated. Additionally, the objects detected in real time can also be used to determine motion estimation and pose estimation of the sensor sensing the objects themselves, and by extension the pose of the vehicle

and agricultural observation and treatment system and each subsystem, for example each treatment module, on board the vehicle.

FIG. 7C and FIG. 7D illustrate additional example methods of 702 that may be performed by some example systems, subsystems, or components of systems, described in this disclosure.

Alternatively, or additionally, at step 752, the agricultural observation and treatment system can compare one or more salient points or regions of the first frame with the one or more salient points or regions of one or more subsequent frames. This can be done with a camera by comparing frame to frame, can be done with two or more cameras by comparing left and right, or top and bottom frame for depth, or from frame to frame from a first camera to the next or from the first camera to the second camera, or a combination thereof. This can also be done with different types of cameras, including comparing and matching salient points of an image from a visible color image with that of an infrared image and/or with that of an ultraviolet image and getting structure or the object or salient points, location, and pose from the comparison, not necessarily from motion of from frame to frame in time from the same sensor, but from frame 1 of camera 1, frame 1 of camera 2 and frame 1 of camera 3 at the same time. At step 754, the agricultural observation and treatment system can generate one or more 3D models of objects identified in at least one or the first or subsequent frames, associated with objects in the real world. In one example, the object can be target objects of interest, objects related to sprays such as capturing the spray projectile itself, splash health, and splat detection, referring to whether a spray projectile hit the target by measuring the splat size and location of the projecting creating a "splat" on the object and/or surface of the ground near the object.

Alternatively, or additionally, at step 756, the agricultural observation and treatment system can generate one or more global maps of a real-world geographic area including objects identified in at least one of the first or subsequent frames, or both, associated with objects, landmarks, or both in the real world.

In one example, tracking to find the same feature from a first frame detected to subsequent frames, for example using Lucas-Kanade tracking, under the assumption that the object does not move far away from frame to frame, for tracking an object on a moving vehicle. The feature detected in the first frame for tracking can be real world objects detected and identified by one or more machine learning algorithms on board, or by real time edge compute, the treatment modules of the treatment system performing observation and actions in real time online, objects that can be represented by a cluster of salient points in a frame, such as corners, lines, blobs, edges of an object, detected by computer vision techniques such as FAST, SIFT, SERF, or other techniques described in this disclosure.

In one example, the objects detected in real time can be used to determine which areas detected in a geographic scene should be treated. Additionally, the objects detected in real time can also be used to determine motion estimation and pose estimation of the sensor sensing the objects themselves, and by extension the pose of the vehicle and agricultural observation and treatment system and each subsystem, for example each treatment module, on board the vehicle.

FIG. 8 illustrates an example schematic block diagram of componentry that may be utilized with a system 800 similar to that of agricultural observation and treatment systems described previously in this disclosure. The system 800 may

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include a sub-system **802** that communicates with one or more perches, or treatment modules **804**. The treatment module **804** can be a component of a modular system of one or more treatment devices. In each treatment module **804**, the treatment module **804** can include, one or more image sensors **820** and **822**, and one or more illumination units **824**. In one example, an agricultural observation and treatment system, described in this disclosure, can be referred to as a portion of a system for observing and treating objects that is onboard a moving vehicle. Performances by the portion of the system onboard the moving vehicle, including computations, and physical actions, can be considered online performance or live performance. A portion of the system comprising one or more compute or storage components, that are connected as a distributed system, can be considered the offline portion of the system configured to perform remote computing, serve as a user interface, or storage. In one example, the agricultural observation and treatment system is a distributed system, distributed via cloud computing, fog computing, edge computing, or a combination thereof, or more than one subsystem is performing computations and actions live in addition to the portion of the system onboard a moving vehicle. In one example, treatment modules, a plurality of treatment modules, or a first, second, etc. treatment modules discussed in this disclosure can be treatment module **804** of diagram **800**. The treatment module can be a subsystem having a compute unit, sensors, including image capture sensors, illumination devices, and one or more treatment units comprised of one or more nozzles guided by a gimbal or turret mechanism, local to each treatment module. The totality of multiple modular treatment modules described in this disclosure, including treatment module **804** of diagram **800**, can be part of a greater online or on-board agricultural observation and treatment system having a global compute unit and sensors sensing a greater geographic agricultural environment and communicating, sending, and accessing information and instructions to each modular treatment module subsystems. And the on board agricultural observation and treatment system, supported by a vehicle and/or one or more edge compute devices to perform computing functions, can be a subsystem or a node of a greater agricultural observation and treatment system having a mesh network of onboard observation and treatment systems across a fleet of vehicles operating in multiple geographic areas, for example at multiple different farms and orchards having different crops requiring different observation and treatment services, and backend servers, compute and cloud compute subsystems configured to perform offline functions such as ingesting performance logs and sensor data captured at each farm, perform analysis and quality analysis, perform training on one or more machine learning models that can be uploaded to each on site or on board system, and many other nodes including user interface for a user to engage in the functionalities discussed above.

The treatment module **804** can include a compute unit **806**, which can include a CPU or system on chip, that sends data and instructions to an ECU **818**, or daughterboard ECU, for synchronization of operation of one or more illumination units **824** and operation of image sensors **820** and **822**. The ECU **818** can send/receive data to one or more cameras of image sensors **820**, and/or one or more cameras of image sensors **822**, and one or more illumination units **824** each including a light bar of LEDs, including instructions by the ECU **828** to activate the image sensors **820** and **822** and illumination units **824**.

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The system **800** can also include a navigation unit **802** configured to interface with each treatment module **804**. The navigation unit **802** can include one or more components and modules configured to receive positional, velocity, acceleration, GPS, pose, orientation, and localization and mapping data. In one example, the navigation unit **802** can include a vehicle odometry module **808** with encoders and image sensors to perform wheel odometry or visual odometry and process images and vehicle movement to calculate and determine a position and orientation of the vehicle supporting the system **800**. The navigation unit can also include an IMU module **810** with one or more IMU sensors, including accelerometers, gyroscopes, magnetometers, compasses, and MEM and NEM sensors to determine IMU data. The navigation unit **802** can also include an GPS module **811** to receive GPS location data, for example up to a centimeter accuracy. The navigation unit can also include a SLAM module **812** for performing a simultaneous localization and mapping algorithm and application for mapping an environment including an agricultural geographic boundary such as a farm, orchard, or greenhouse, and determining localization and orientation of a vehicle supporting the system **800**, components of the system **800** relative to the geographic boundary, as well as localization and orientation of agricultural objects and scenes detected by the system **800**. The SLAM module **812** can take sensor data from one or more cameras, including stereo vision cameras, cameras that are omnidirectional, cameras that are moving relative to the vehicle, or other sensors **813** including LiDAR sensors. The LiDAR sensors can be flash LiDAR sensors or static LiDAR sensors, spinning LiDAR sensors, other rangefinders, and other sensors discussed above. As the navigation **802** receives sensing data related to localization and mapping, a compute unit **806**, including a CPU or system on chip, of the navigation unit **802** can fuse the sensing signals and send the data to each of the treatment modules **804** or to a remote compute unit or server through a communications module **840**. The sensing components of the navigation unit **802** can be activated and controlled by an ECU **814**. The ECU **814** can also be configured to interface, including activation and power regulation, with each of the treatment modules **804**.

The treatment module **804** can also include a treatment unit **828** configured to receive instructions from the compute unit and ecu **818** including treatment parameters and treatment trajectory of any fluid projectile that is to be emitted from the treatment unit **828**. The CPU **602** to communications components to send and receive instructions to other components of the system **600** as well as remote devices. A chemical selection unit **826** can include one or more chemical pump(s) configured to receive non-pressurized liquid from one or more chemical tanks **832** and operable to each treatment units of each of the treatment modules **804**, or multiple treatment units **828** of each treatment module **804**. One or more chemical tanks **832** may have different types of chemicals. The chemical pumps can send stored liquid or gas from the one or more chemical tank(s) **832** to one or more regulators **834**, which will further send pressurized liquid to one or more other components in series as the pressurized liquid reaches the one or more treatment units **828** of system **800**. Other components in the series of the chemical selection unit **826** can include an accumulator and chemical mixer **836** (described in previous sections of the disclosure). The treatment unit may emit the liquid at a particular trajectory in order for the fluid projectile to come into contact with an object and at a particular physical location.

In one example, as a vehicle performs a trial on a geographic boundary, each of the treatment modules **804** can perform actions independently of each other. Each treatment module **804** can receive its own image acquisition and processing of images for treatment. The treatment parameters can be determined locally on each treatment module **804**, including object detection and classification of agricultural objects in a scene as well as determining treatment parameters based on the objects and features detected. The processing and be performed by each compute unit **806** of each treatment module **804**. Each of the treatment modules **804** can receive the same data sensed, fused, and processed by navigation, vehicle orientation and position data from the navigation unit **802** since each of the treatment modules **804** will be supported by the same vehicle. In one example, each of the treatment modules **804** can share the same chemical selection component **826**. In one example, multiple chemical selection units **826** can be configured to connect and interface with each treatment module **804** where one treatment module **804** can be configured with one chemical selection unit **826**.

FIG. 9A illustrates an example modular treatment module **900**, or perch. In one example, the modular treatment module **900** may be configured with multiple illumination units **910** mounted to a frame **902**, **903** or supporting structure. The modular treatment module according to various examples may include multiple illumination units **910** of LED lights. Illumination unit **910** may include one or multiple LED lights including an array of LED lights. The LED lights can each be packaged in an enclosure for better mounting of the LED lights to the rest of the modular treatment module. For example, a light enclosure can support 4 individual LED Lights, each LED light can include a plurality of LED diodes to illuminate light. In another example, the LED Lights can be standalone, supported by a structure or heatsink and individually mounted to the rest of the treatment module. In one example, each LED light, having a plurality of LED diodes, can include one or more lenses to focus the illumination intensity, direction, or illumination area. The modular treatment module **900** may include a camera enclosure, or camera bank **904** that includes one or more cameras or other image sensing devices. In one example, the illumination units **910**, treatment units **1100**, supported by treatment unit frame **903**, can all be operably mounted and connected to the camera bank **904** having a camera enclosure. The inner two cameras may be identification cameras to obtain digital imagery of agricultural objects, and the outer two cameras may be cameras used to obtain imagery of agricultural objects being treated including the treatment projectile, treatment profile, splat detection, treatment health and accuracy. In one example, a pair of stereo cameras can be configured to ingest frames at a high frame rate and at a high exposure rate or refresh rate shutter speed. For example, the cameras can ingest images up to 8K definition per frame at 2040 Frames captured per second. In this example, the compute unit embedded and enclosed in the camera bank **904**, configured to send instructions and read inputs from each of the treatment units **1110**, sensors including cameras, illumination units **910**, and so forth, action as the main compute unit of the component modular treatment module **900**, can receive different down-samples of image quality and number of frames per second. For example, one or more FPGA's, ASICs, or one or more microcontrollers can be embedded at the each of the camera modules, such that the camera's exposure and shutter speed receives 8K definition frames at 2400 frames per second. The FPGA, ASICs, or one or more microcontrollers can

automatically sample the images into smaller resolution images into smaller resolution images at fewer frames per second sent to the compute unit. Additionally, it can send more than one of different types of images packets to the compute unit such that the compute unit receives different streams of data captured by the same pair of image sensors. For example, the 8K frame captured at 2400 frames per second can be down sampled to 4K frames at 30 frames per second at the FPGA/ASIC level, and then sent to the compute unit so that the compute unit can partition a task to analyze 4K image frames at 30 frames per second. In one example, it can partition a task to analyze a stereo pair of 4K image frames at 30 frames per second. Additionally, the 8K frame captured at 2400 frames per second can be down sampled to 1080p frames at 240 frames per second at the FPGA/ASIC level, and then sent to the compute unit so that the compute unit can separately partition a task to analyze 1080p image frames at 240 frames per second, both streams of image data coming from the same pair of cameras. This would reduce the need for two sets of stereo cameras enclosed in a single camera bank. The disclosure above is for illustration purposes only, and the FPGA/ASIC and other microcontrollers described can downsample the image stream to any type of quality and speed to send to the compute units for analysis. The camera module itself can account for auto balance, auto white balance, auto exposure, tone, focus, as well as synchronization with the LED lights including the LED light's exposure, temperature, peak to peak exposure time, as well as perform color correction algorithms to the images before the images are sent to the compute unit for analysis. Each LED light may be synchronized to turn on and off with respect to when an identification camera(s) is capturing an image. The number of cameras or other sensing devices, as well as the number of individual LED Lights are for illustration purposes only. In one example, more than two treatment units **1110** can be supported by a single modular treatment module **900** or a part of a single modular treatment module **900**. The modular treatment module **900** can include a varying number of sensing enclosures, illumination modules, and treatment units, all operably connected to each other as one treatment module similar to that of treatment module **804**.

FIG. 9B illustrates an alternate configuration of an example modular treatment module **902**. The module treatment module **902** can include a support structure and components supported by or embedded in the support structure, including a treatment unit and a treatment unit support structure **923**, one or more image sensors **918** including a compute unit and image sensor box or enclosure **916**, and one or more illumination units **920** having one or more LED Lights with one or more lenses.

FIG. 10 illustrates an example method **1000** that may be performed by some example systems, subsystems, or components of systems, described in this disclosure. For example, at step **1010**, the agricultural treatment system can determine a first real-world geo-spatial location of the agricultural treatment system. At step **1020**, the agricultural treatment system can receive one or more captured images depicting real-world agricultural objects of a geographic scene. At step **1030**, the agricultural treatment system can associate the one or more captured images with the determined geo-spatial location of the agricultural treatment system. At step **1040**, the agricultural treatment system can identify, from a group of indexed images, mapped images, previously assigned images, or representations of agricultural objects including at least in part, image data and position data, or a combination thereof, one or more images

having a second real-world geo-spatial location that is proximate with the first real-world geo-spatial location. At step 1050, the agricultural treatment system can compare at least a portion of the identified images with the one or more captured images. At step 1060, the agricultural treatment system can determine a target object based on the comparing at least a portion of the one or more identified images with at least a portion of the one or more captured images. At step 1070, the agricultural treatment system can emit a fluid projectile at a target object in the real-world with a treatment device. The target objects are real-world objects that are intended to be sprayed with the fluid projectile.

The agricultural treatment system may store the group of images in an onboard data storage unit or a remote storage unit. The group of images may include key frame images and sub-key frame images. The key frame images may depict agriculture objects of the geographical scene, and the sub-key frame images may depict a portion of a key frame image, for example a portion of a key frame image can be an image of an agricultural object or cluster of agricultural objects. The key frame images may be images that were previously obtained by image sensors of the system. The captured digital images may be obtained by the same cameras of the system at a time subsequent to when the key frame images were taken. For example, in one trial run, the agricultural treatment system, or similar systems 100, 600, and 800, can perform observations of a geographic boundary including detecting and indexing any and all agricultural objects captured by image sensors, and perform one or more precision treatments on detected agricultural objects on the geographic boundary, such as a farm or orchard. The agricultural treatment system can index each image captured by its on-board vision system including one or more image sensors configured to capture images of agricultural objects or crops, or offline at a remote computing location nearby the physical location of the geographic boundary or at different remote location such that the remote computing units can communicate with the agricultural treatment system. The indexed series of images captured by image sensors can be further indexed, where one or more of the captured images can be assigned as a keyframe, include a unique keyframe marker. Each keyframe can represent image that include one or more unique agricultural object or landmark of interest in the real world. Because of the navigation unit of agricultural treatment system, the keyframes can include location data and a timestamp. For example, the agricultural treatment system, in a trial, can capture a series of captured images as the vehicle travels along a path in the geographic boundary. The series of images captured can be images taken of a row of plants including row crops grown directly from the soil or crops growing off trees. One or more images of the series of images captured can include agricultural objects of interest, either for treatment or for observation where the agricultural object can grow into a stage where it is desirable to select a treatment for the agricultural object. The agricultural treatment system can assign the particular image having the individual agricultural object identified as a keyframe. The keyframe, or any other images captured by the agricultural treatment system can include a location based on image analysis performed by the compute unit of the treatment system. For example, a stereo vision system can use epipolar geometry to triangulate a location of an object identified in an image relative to the location of the image capture device.

Additionally, each portion of the image that includes agricultural objects can be labeled and assigned a unique identifier to be indexed in a database. The data indexed can

be a 2d or 3d constructed image of an agricultural object having a location and position data attached to the image and a timestamp of when the image was taken. In future trials conducted by the agricultural treatment system, the agricultural treatment system may capture images of the same agricultural object at the same or similar location in the geographic boundary. Since the image captured of the agricultural object in the same position was acquired at a future time from the previously captured agricultural object, the agricultural object may have grown to have different features. In one example, the agricultural treatment system can determine that an acquired image of an agricultural object with location and position data, is associated with that of a previously acquired, labeled, assigned, and indexed image or other indexed representation of an agricultural object that is the same agricultural object as the currently detected object. Having associated the two images with location and timestamp data, the agricultural treatment system can determine treatment parameters, including whether to perform a treatment at the given time or trial, determining a mixture, chemical type, volume, concentration, etc., of a treatment, and a precise trajectory for the treatment to be deposited on a surface of the agricultural object. In one example, a user can select in an application the indexed agricultural object, and a user interface of the agricultural treatment object can display information related to the agricultural object including images taken of the agricultural object, including multiple images taken at different locations, and with orientations of the image capture device, for capturing different views of the same agricultural object, as well as multiple images taken at different points in time as the agricultural treatment system conducts multiple trials and captures images of the same or near the same location as previously captured images.

The above example illustrates the agricultural treatment system performing two trials with two sets of images captured at different times, for example a day apart, of the same agricultural object and associating the images of the agricultural object with each other based on image features detected that are common between the images, position, depth, localization, and pose related information from image analysis and computer vision techniques, as well as similar position data captured by the navigation unit of the agricultural treatment system. As more trials are conducted and more images of a same agricultural object are taken, capturing the agricultural object's current growth stage, and associating each captured agricultural object with one or more previously captured images of the same agricultural object, the treatment system can build a unique profile of each unique and individual agricultural object mapped in a geographic boundary, including images associated with each of its growth stages, any and all treatment history to each individual agricultural object. This can allow a user or a treatment system to determine a crop's health, including diseases and stress, for example for fire blight detection, and color change, size, count, growth projection, yield projection and estimation of the crop grown on a farm or orchard and allow a user optimize growing crops on a farm by observing and controlling the growth rate of each individual agricultural object detected on a geographic boundary.

In one example, to identify target objects for spraying, the system may compare at least a portion of the identified images by comparing the sub-key frame image to a portion of one of the captured images. In other words, the agricultural treatment system can compare one or more patches or labeled portions of a previously indexed image of an agricultural object with at least a portion of the currently

captured image. In this example, a patch is an image cropped out of a bigger image having one or more features of interest. The features of interest in the bigger image captured by image sensors can include agricultural objects, landmarks, scenes or other objects of interest to be identified, labelled, and assigned a unique identifier or marker to be indexed. For example, a bounding box of an image, or other shape, can be drawn around a portion of an image, cropped out and separately indexed by the agricultural treatment system and saved as a patch for comparing against captured images taken in the future, for building a digitized map of a geographic boundary, for associating an object captured during one trial with the same object captured at different trials, or a combination thereof. The system determines a confidence level of whether the sub-key frame image matches the portion of the captured image. The system identifies a match where the determined confidence level meets or exceeds a predetermined confidence level threshold value. In one example, various computer vision techniques can be applied to compare and correspond images and determine similar features for matching. This can include template matching for comparing a portion of an image with the region of interest of another image, normalized cross correlation, random sample consensus (RANSAC), scale-invariant feature transform (SIFT), FAST, edge orientation histograms, histogram of oriented gradients, gradient location and orientation histogram (GLOH), ridge and edge detection, corner detection, blob detection, line detection, optical flow, Lucas-Kanade method, semantic segmentation, correspondence matching, and other computer vision and matching techniques. The system may identify that a captured image includes a target object to be treated or a target object that was already sprayed and does not currently need a treatment based on features detected of the agricultural object, based on its treatment history, or a combination thereof. Based on determining the location of the image sensors of the agricultural treatment system, the location of the target object in the obtained image, the system can then configure, orient, and prepare the treatment unit such that a fluid projectile when emitted, would be sprayed in a trajectory to emit fluid onto the real-world targeted agriculture object.

In another example, the system may use landmark features or objects to determine locations of target objects to be sprayed. The landmark objects are real-world objects that aid in determining the location of a target object. The system may identify a landmark object in a captured image and determine a portion of the landmark object in the capture image matches a portion of an image from the group of images. While not intended to be an exhaustive list, examples of landmark object may include a man-made object, a fence, a pole, a structure, a portion of a plant structure, a portion of a tree structure, a leaf formation or a leaf cluster that can be used to mark a specific location of a geographic boundary or distinguish a specific keyframe for having the unique landmark assigned to the portion of the keyframe.

In another example, in one mode of operation, in a first pass along a path along an agricultural environment, the agricultural treatment system obtains a first set of multiple images while the system moves along the path. For example, the agricultural treatment system uses onboard cameras and obtains multiple digital images of agricultural objects (e.g., plants, trees, crops, etc.). While obtaining the multiple images of the agricultural objects, the agricultural treatment system records positional and sensor information and associates this information for each of the obtained images.

Some of this information may include geo-spatial location data (e.g., GPS coordinates), temperature data, time of day, humidity data, etc. The agricultural treatment system or an external system (such as a cloud-based service) may further process the obtained images to identify and classify objects found in the images. The processed images may then be stored on a local data storage device of the agricultural treatment system.

In a second pass along the agricultural environment, the agricultural treatment system using the onboard cameras obtains a second set of multiple digital images using along the path that had been previously taken along the first pass. For example, the agricultural treatment system may obtain the first set of multiple images on day 1, with the images capturing blossoms on a group of apple trees. The digital images depicting the apple trees may be processed for object classification of the types of blooms depicted in the digital images. The agricultural treatment system may retrieve the processed imagery and associated data identifying the objects and classified types. On day 2, the agricultural treatment system may again follow the original path and obtain new imagery of the apple trees. The agricultural treatment system may then use the second set of obtained images in comparison with the received processed images to identify target agricultural objects to be sprayed, and then spray the agricultural objects. The system then can match the landmark objects to aid the system in determining locations of target objects. In other words, the system may use feature matching of objects in the imagery to determine that a prior image is similar to a captured image.

For example, the processed images received by the treatment system, may have associated positional information. As the agricultural treatment system moves along the path in the second pass, the agricultural treatment system may compare a subset or grouping of the processed images based on location information associated with the processed images, and a then current position or location of the treatment system. The agricultural treatment system compares new images to the processed images and determines whether the images or a portion of the images are similar. The agricultural treatment system may then identify a location to spray based on a likely location of a target object in the processed images.

As noted above, the agricultural treatment system may associate images captured by a camera(s) with real-world physical locations of where images of agricultural objects were obtained. For example, while a vehicle with an agricultural treatment system is moving along a path, an electronic control unit of the agricultural treatment system may generate camera data signals and light data signals with synchronized lighting from the lighting devices of the agricultural treatment system and the capturing of digital images. The ECU may synchronize illumination, by one or more lights mounted on the vehicle, of the physical location of an object(s) for generation of the respective captured image(s) that corresponds with that physical location of the object(s). The object determination and object spraying engine sends the camera data signals and light data signals to ECU. The object determination and object spraying engine generates position information that corresponds with a position and an orientation of the vehicle with respect to physical location(s) of the agricultural object(s) and a current route of the moving vehicle. The position information may further be associated with the respective captured image(s) that corresponds with the physical location(s) of the agricultural object(s).

FIGS. 11A-K illustrate example agricultural treatment units, such as treatment unit 1100 of an agricultural treatment system. In this example, the treatment unit 1100 can include a turret assembly with a treatment head 1120, and circuitry, electronic components and computing devices, such as one or more microcontrollers, electronic control units, FPGA, ASIC, system on chip, or other computing devices, configured to receive instructions to point and orient the treatment head 1120, to treat a surface of a real-world object in proximity of the treatment unit 1100. For example, the treatment unit 1100 can emit a fluid projectile of a treatment chemical onto an agricultural object in the real world based on detecting the agricultural object in an image captured and determining its location in the real world relative to the treatment unit 1100.

The treatment unit 1100 can include a gimbal assembly, such that the treatment head 1120 can be embedded in, or supported by the gimbal assembly, effectively allowing the treatment head 1120 to rotate itself and orient itself about one or more rotational axes. For example, the gimbal assembly can have a first gimbal axis, and a second gimbal axis, the first gimbal axis allowing the gimbal to rotate about a yaw axis, and the second gimbal axis allowing the gimbal to rotate about a pitch axis. In one example, the gimbal assembly can have a third gimbal axis to allow roll of the treatment unit, giving the treatment head 120 of the treatment unit 1100 a total of 3 degrees of freedom relative to the treatment unit 1100. In this example, a control module of the treatment unit can control the gimbal assembly which changes the rotation of the gimbal assembly about its first gimbal axis, second gimbal axis, or both. A computing module can determine a location on the ground scene, terrain, or tree in an orchard, or other agricultural environment, and instruct the control module of the treatment unit 1100 to rotate and orient the gimbal assembly of the treatment unit 1100. In one example, the computing module can determine a position and orientation for the gimbal assembly to position and orient the treatment head 1120 in real time and make adjustments in the position and orientation of the treatment head 1120 as the treatment unit 1100 is moving relative to any target plants or agricultural objects of interest on the ground either in a fixed position on the ground, or is also moving. The treatment system can lock the treatment unit 1100, at the treatment head 1120, onto the target plant, or other agricultural object of interest through instructions received and controls performed by the control module of the treatment unit 1100, to adjust the gimbal assembly to move, or keep and adjust, in real time, the line of sight of the treatment head 1120 onto the target plant.

In one example, the treatment unit 1100 can include protective components such as an enclosure 1170 to shield the treatment unit 1100 from dust, water, moisture, sunlight, and other particles that can damage components of the treatment unit 1100, as well as protect the treatment head 1120 from wind or other forces that could disturb the desired orientation of the treatment head 1120. The treatment unit 1100 can also include one or more mounting brackets 1172 to mount the enclosure 70 supporting the treatment unit 1100 to other components of the treatment system or a vehicle supporting the treatment system.

In one example, one or more brushless motors can be configured to change one axes of the treatment head of the treatment unit. A solenoid valve can be configured to open or close the valve where constant pressurized fluid from the pump is pumping fluid into the solenoid valve. The treatment unit can emit a projectile by an amount and time the valve is moved from a closed position to an opened position.

In one example, the treatment unit 1100, having a high-powered laser unit or laser chip embedded in or supported by the treatment unit 1100, can be configured to treat portions of plants that are larger than plant typically only grow a few inches or feet above the ground. These plants can include trees, orchard trees, or other plants with one or more trunks, shrubs, bushes, or other plants grown on trellises or other human made mechanisms such that a horizontally or top mounted treatment unit 1100 is more practical.

In one example, the figures illustrate a regulator 1180 (e.g., a solenoid valve) can be configured to open or close the valve where a constant pressurized fluid from the pump is pumping fluid into the regulator 1180. The treatment unit 1100 can emit a projectile by an amount and time the valve is moved from a closed position to an opened position.

While the figures illustrate a single solenoid valve 1180, the treatment unit 1100 may be configured with multiple solenoid valves that are interconnected with different pumps to obtain fluid from different fluid sources. For example, the agricultural treatment system may have four source tanks to hold different types of fluids. Four pumps may be utilized, each of which can be fluidly interconnected with a respective tank and a separate solenoid valve. The pump(s) may pressurize the fluid, for example in a range of 5-250 PSI. For example, when treating an agricultural object such as carrots, the solenoid may control fluid pressure to about 40 PSI. When treating an agricultural object such as apples, the solenoid may control fluid pressure to about 80-100 PSI. The system may be configured to allow two or more fluid source to be emitted at the same time. For example, the spraying nozzle 1130 of the gimbal assembly 1140 may have multiple spraying ports. Also, the fluid lines may be connected before or after the solenoid valve to mix different fluid sources from the tanks. Pressure sensors may be connected along any part of the fluid channels to determine the pressure of a line.

The system may then increase or decrease the fluid pressure to a desired pressure via opening/closing of the solenoid valve, and by adjusting the pumping of the fluid via a pump. The size and volume of each spray emitted by the spraying nozzle 1130 may be adjusted by the system based on the target object to be sprayed. For example, the system may spray 1.25 mL at 80 PSI for one agricultural object, and at a different volume and PSI for another agricultural object. Moreover, the spray radius may be determined as a distance from the spray nozzle to the targeted object. For example, the spray radius may be about 1/8"-5" diameter with a coverage at about 1 meter with a +/-25-degree sweep at 0.094 m. Additionally, the spray time may be adjustable and intermittent spray burst of fluid may be controlled by the system.

In one embodiment, the system may adjust the operation of the solenoid to release the pressurized fluid to create a desired spray diameter when the spray hits the target agricultural object. For example, the system may adjust the operation of the solenoid so that a target object 3 feet away from the spraying head and a target object 6 feet away from the spraying head would each have a similar spray diameter. Also, the spray diameter may be adjusted such that the diameter of the spray for the closer target object may be a greater or lesser diameter of the spray of the target object farther away from the spraying head. In this example, the system may determine how far away the nozzle of the treatment unit is from the ground, determine what the target object is, the size of the target object and how much (e.g., an area) of the target object to target. The system may target a portion of an agricultural object, such as a root, a leaf, a bud,

etc. The agricultural object may be repeatedly sprayed with fluid from the same and/or separate fluid sources.

In one example, the gimbal assembly 1140 has a detachable spraying nozzle 1130. One or more fluid carrying tubes 1182 may be connected to the gimbal assembly 1140 for carrying fluid from one or more fluid regulators 1180 to be emitted via the spraying nozzle 1130. The spraying nozzle 1130 may have a threaded base portion allowing the spraying nozzle 1130 to be attached to gimbal assembly 1140. In some, configurations the spraying nozzle 1130 has multiple ports that are connected to different fluid lines. This allows the gimbal assembly 1140 to emit fluid pumped from different tanks or fluid sources. For example, the spraying nozzle 1130 may have one, two, three or four ports in the spraying tip. Also, multiple ports of the spraying tip may be connected to the same fluid source. For example, the spraying nozzle 1130 may have two ports. The two ports may be configured such that fluid emitted at target object would spray the target object at two different locations. The spraying nozzle 1130 can be various types of tips, including for example a hypodermic needle with one or more input ports or orifices for different types of fluids, and with one or more output ports or orifices.

In one example, the system may determine a target object to be of a particular type. Based on the particular type of the target object, the system may choose to treat the target object with a fluid source from one or more tanks. For example, the system may determine the target object to be a weed and then cause fluid from a first tank to be emitted for weed treatment. In another instance, the system may determine the target object to be a plant and then cause fluid (e.g., liquid fertilizer) from a second tank to be emitted at the target object.

This treatment unit 1100 is capable of emitting a projectile fluid at a target object in a continuous manner over a period of time, such as many seconds or minutes. The treatment unit may also emit short intermittent bursts of a projectile fluid at a target object. The spraying nozzle 1130 of the gimbal assembly 1140 may be configured to emit the projectile fluid in a stream-like manner where the fluid is kept together in a stream to impact a target object in a focused area. The spraying nozzle 1130 of the gimbal assembly 1140 may be configured to emit the projectile fluid in a spray-like manner where the fluid is fanned out or the fluid separated to impact a target object in a general area. The gimbal assembly 1140 may be configured with one or more spraying tips and/or spraying ports that allow for the emission of either type of fluid spray types. The spraying of fluid from the spraying nozzle 1130 is variable and adjustable based on the target object. In one example, for spraying carrots the distance from the spraying tip to the target object may be about 27 inches. In another example, for spraying apples the distance from the spraying tip to the target about may be about 0.5-1.5 meters. In one embodiment, the fluid may be emitted in a short pulse of less than 10 bursts per second. In another embodiment, the fluid may be emitted in a volume of fluid ranging from 1 milliliter to 1 liter. In another embodiment, the fluid may be emitted in a continuous volume of sprayed fluid over a period of 5 milliseconds to 300 seconds.

The agricultural treatment system may use a controller 1190 to interact with, obtain sensor data and control the motors and the solenoids. In one embodiment, to instruct the gimbal assembly 1140 to a determined pose, the system instructs the motors 1152 to rotate in one direction or the other. Pose determination is described further below. The system may instruct the motors 1152 to rotate at a constant

speed or a variable speed. The system may vary the sample rate based on the rotation speed of the motor, the movement speed of the vehicle, or based on some other value or data obtained by the system. When the system determines that the gimbal assembly 1140 is moved to a desired pose, then the system may instruct the fluid regulator 1180 to open, allowing fluid to flow through the spray tube 1182, and emit a fluid at a target object via the spray nozzle 1130. The system may then instruct the regulator 1180 to close thereby stopping the fluid from being emitted from the spraying nozzle 1130. As described herein, when the either motor 1152 rotates, the coupled linkage assemblies 1154, 1160 cause the gimbal assembly 1140 to rotate or pivot in a particular direction. For example, the system may be configured that when a first motor rotates in a first direction, the gimbal assembly 1140 would pivot or move along a first axis in a first direction, and when the first motor rotates in an opposite second direction, the gimbal assembly 1140 would pivot or move along the first axis in a second direction opposite to the first direction. Moreover, the system may be configured that when a second motor rotates in a first direction, the gimbal assembly 1140 would pivot or move along a second axis in a first direction, and when the second motor rotates in an opposite second direction, the gimbal assembly 1140 would pivot or move along the second axis in a second direction opposite to the first direction.

In one example, one or more brushless motors can be configured to change one axes of the treatment head of the treatment unit. A solenoid valve can be configured to open or close the valve where constant pressurized fluid from the pump is pumping fluid into the solenoid valve. The treatment unit can emit a projectile by an amount and time the valve is moved from a closed position to an opened position.

FIGS. 12A-B illustrate example implementations of method 1200 that may be performed by some example systems, subsystems, or components of systems, described in this disclosure. For example, in one mode of operation, at step 1210, an agricultural treatment system can receive image data from one or more sensors, the image data including one or more agricultural objects. The one or more agricultural objects can be identified as one or more target plants from the image data. At step 1220, the agricultural treatment system can receive agricultural data representing agricultural objects including different crops and target plants. At step 1240, the agricultural treatment system can identify a location of the target plant. At step 1250, the agricultural treatment system can determine treatment parameters of the target plant. At step 1260, the agricultural treatment system can compute a vehicle configuration and treatment unit configuration for treating the target plant. At step 1270, the agricultural treatment system can lock the treatment unit onto the target plant in the real world. At step 1280, the agricultural treatment system can activate the treatment unit and emits a fluid projectile of a treatment chemical onto the target plant.

Additionally, the agricultural treatment system can receive, fuse, compute, compensate, and determine positional, localization, and pose related signals on a geographic boundary. At step 1212, the agricultural treatment system can receive sensor data, from one or more sensors on a vehicle of an agricultural environment. The agricultural environment can be that of a geographic boundary having a plurality of objects typically found on a farm or orchard for cultivating land and growing and harvesting crops. At step 1214, the agricultural treatment system can identify a vehicle position, one or more agricultural objects in proximity of the vehicle, and determine distances of the vehicle

to the agricultural objects. At step 1216, the agricultural treatment system can calibrate the vehicle, including calculating a pose estimation of the vehicle relative to a central or known point in the geographic boundary, pose estimation of components of the agricultural treatment system relative to the vehicle supporting the agricultural treatment system, or agricultural objects detected in space relative to the vehicle. The vehicle can be calibrated by locating one or more calibration targets spread throughout a mapped geographic boundary such that as the agricultural treatment system identifies a physical calibration target and calculates its position relative to the calibration target, the agricultural treatment system can determine, or correct a previous inaccurate determination, a position of the vehicle in the geographic boundary.

FIGS. 13A-B illustrate example images obtained by an agricultural observation and treatment system described in this disclosure. As shown in the diagram 1300a of FIG. 13A, an image received by an agricultural treatment system may include multiple identifiers of different types of objects, for example objects 1302, 1304, 1306, 1308, and/or 1310, of a plurality of objects detected, each having different identifiers portrayed in a captured image. For example, an identifier marked for object 1306 or 1308 may identify a portion of the captured image that portrays a physical landmark of an of an agricultural object or landmarks in an agricultural scene. The object 1306 may further be based on visual characteristics of the object.

The diagrams 1300a and 1300b representing images with one or more detections can either be ingested images by a compute unit of a component treatment module with machine learning, computer vision, or both, based detections performed by feature extraction and object detections in real time while the treatment module is scanning an environment, or representing images with labels performed by human labelers, machine learning detections, or a combination thereof where a machine learning detector scans and detects objects and landmarks in a given frame, and a human labeler verifies the quality of the detections and manually labels missing or incorrectly classified objects.

FIG. 13B illustrates another image 1300b depicting another example real-time captured image with real-time object detection or a received labelled image having the labelling of objects in the received image done offline from the portion of agricultural observation and treatment system supported by a vehicle. Additionally, diagram 1300b illustrates example portions or sub-images of an image obtained by an agricultural treatment system.

In one example, diagram 1300b is a labelled image, either from real-time performed by an agricultural observation and treatment system on the vehicle, or offline at a server, by a human, by a machine learning algorithm, assisted by a machine learning algorithm, or a combination thereof.

Based on visual characteristics of an instance of an apple blossom portrayed by the captured image of an apple tree, the labeled image may include an identifier 1302b for the apple blossom instance. The identifier 1302b may be positioned in the labeled image 1300b at a first pixel position that corresponds to the apple blossom instance's physical location as it is portrayed in the captured image of the apple tree. Based on visual characteristics of an instance of an apple fruitlet portrayed by the captured image of the apple tree, the labeled image may include an identifier 1310b for the apple fruitlet instance. The identifier 1310b may be positioned in the labeled image at a second pixel position that corresponds to the apple fruitlet instance's physical location as it is portrayed in the captured image of the apple tree. Based on

visual characteristics of an instance of a landmark portrayed by the captured image of the apple tree, the labeled image may include an identifier 1308b for the landmark instance, the specific landmark identifier 1308b being that of two branches diverging in the specific pattern, shape, and orientation illustrated in 1300b. The identifier 1308b may be positioned in the labeled image 1300b at a third pixel position that corresponds to the landmark instance's physical location as it is portrayed in the captured image of the apple tree.

In one example, to perform better VSLAM in an agricultural scene, certain objects that are landmarks that are tracked across time can improve the quality of VSLAM and pose estimation, for example, large enough stationary objects typically found in the specific agricultural scene. Landmarks can be used to identify which frames are of interest to store, store as a keyframe (because one does not need so many frames at once all having the same fruits, or detected objects, from frame to frame), and to be used to identify objects in real time and tracked for visual based navigation and mapping including VSLAM. Because there are spatial locations to each of the objects, landmarks, and its unique identifying characteristic. In one example, tree trunk 1336 can be detected, by a machine learning algorithm or programmatically predefined as stationary dark objects that protrude from the ground. Detection and tracking tree trunks in an orchard can allow a system to partition an agricultural environment by the trees themselves, such as to minimize error in detecting one cluster of objects and thinking its origin is at once place, when it should be at another. For example, a system can detect a first tree trunk having a first location in global scene, as well as determine a pose of the system itself relative to the tree trunk detected. The system will also detect a plurality of objects, including its identity as well as whether that unique object was detected before either with the same identifier, or a different identifier, being that the phenological state of the object has changed, but still the same object in space. In this example, the system can associate a cluster of objects detected, being on the same tree, with the tree trunk detected. In this case, if the system incorrectly detects other objects or landmarks at different and nearby trees due to its pattern being similar to a previously identified pattern or object, and its location based sensors are not accurate which the change in location was not detected from a first object, pattern or landmark located near a first tree trunk and a second object, pattern, or landmark located at a second tree trunk, for example if the GPS sensor is off by a few meters or did not update in time, An additional checking point for the system can be detecting a first tree trunk and a second tree trunk. Because the system knows that two different tree trunks must be far enough apart from each other, the system can determine that a previously detected object determined to be a certain location is likely wrong due to the system also determining that the object detected was in proximity to another tree trunk that could not have been located at a different location.

While tree trunks are unique to orchards, any large, stationary objects or patterns that are unique to the specific geographic environment can be programmatically detected to better improve spray performance, navigation performance, and mapping of the scene. For example, detecting beds, troughs, furrows, and tracks of a row crop farm can be used to improve performance of observing and performing actions in the row crop farm. The techniques used can be a combination of computer vision, machine learning, or machine learning assisted techniques in detecting beds, troughs, furrows, and tracks such as long lines in a captured

frame, differences in depth between lines (for example tracks and beds will have substantially the same line pattern because they are next to each other but have different depths), which can be detected with depth sensing techniques and detecting changes in color between beds and tracks, for example.

In one example, the object determination and object spraying engine generates positional data for an instance of the fruit at a particular stage of growth that is portrayed in a captured image based in part on: (i) a pixel position of the portrayal of the instance of a fruit at the particular stage of growth in the labeled image (and/or the captured image), (ii) the position information of the moving vehicle, and/or (iii) previously generated position information associated with a previous captured image(s) of the instance of the fruit and the physical location of the instance of the fruit. Previously generated position information may be associated with captured and labeled images that portray the same instance of the fruit when the vehicle traveled a similar route during a previous time, such as a prior hour of the day, prior day, week and/or month. The agricultural treatment system may generate nozzle signals for the synchronization ECU of the agricultural treatment system on a vehicle based on the positional data for the instance of the fruit at the particular stage of growth. For example, the nozzle signals may indicate a physical orientation of the nozzle to create a trajectory for a liquid. The nozzle signals may represent a change in a current orientation of the nozzle based one or more axial adjustments of the nozzle.

The object determination and object spraying engine sends the projectile from the nozzle towards the physical location of the object according to the trajectory. For example, the object determination and object spraying engine adjusts a current orientation of the nozzle according to the nozzle signals and triggers the nozzle to spray a liquid towards the physical location of the instance of the fruit.

Because not all plants need the same amount, for example by type, volume, frequency, or a combination thereof, of treatment based on the stage of growth of the particular plant, the agricultural treatment system can be configured to scan a row of crops to identify the stage of growth of each individual crop or agricultural object that is a plant or portion of a plant and determine whether the identified crop or agricultural object needs a treatment on the particular trial run, or day, or at the particular moment in time the vehicle with agricultural treatment system is on the field and has detected the individual agricultural object. For example, a row of crops, even of the same kind of plant, can have a plurality of agricultural objects and sub-agricultural objects of the agricultural objects, where the agricultural object may depict different physical attributes such as shapes, size, color, density, etc.

For example, a plant for growing a particular type of fruit, in one agricultural cycle, can produce one or more individual crop units, for example a fruit tree, each taking the shape of a first type of bud, second type of bud, and so forth, a flower, a blossom, a fruitlet, and eventually a fruit, depending on a growth stage of a particular crop. In this example, the agricultural treatment system can label each stage of the same identified object or crop, down to the particular individual bud, on the fruit tree as different agricultural objects or sub agricultural objects, as the object changes in its growth stage including its particular shape, size, color, density, and other factors that indicate a growth into a crop. The different agricultural objects detected and labelled associated with the same object in the real-world space can be associated with each other

For example, a bud detected can be labelled as a unique agricultural object with a unique identifier or label. As time moves forward in a season, the uniquely labelled bud that is mapped on a farm may change shape into a flower for pollination, or from a flower to a fruitlet, and so forth. As this happens, an agricultural treatment system can identify the flower and label the flower as a unique identifier to the agricultural object detected and associate the agricultural object that is the flower with the agricultural object that is the bud previously identified and logically link the two identified agricultural objects as the same object in the real world where one object identified has grown into the other. In another example, the unique real-world flower detected, of a plurality of flowers and other objects in a geographic boundary, can be labelled as a flower but not considered a different agricultural object, and instead be associated with the same agricultural object previously labeled as a bud. In this example, each object detected that can be considered a potential crop can be mapped as the same agricultural object, even though the agricultural object will change shape, size, density, anatomy, etc. The same agricultural object detected in the same space at different times can then have different labels and identifiers as related to the stage of growth. For example, a first agricultural object in space, detected by the agricultural treatment system, can be identified and indexed as a real-world agricultural object #40 with a timestamp associated with the time of day and year that the agricultural treatment system captured one or more images or other sensing signals of agricultural object #40. At the moment in time of identification, the agricultural object #40 can have a first label and assign the first label to agricultural object #40. The first label can be labelled as a bud, or bud #40 since there may be many other buds detected in the geographic boundary such as a farm or orchard. As multiple trials across a span of time are conducted in the geographic boundary on the same agricultural object #40, the agricultural object #40 can turn from a first type of bud, such as a dormant bud, into a second type of bud, or from a bud and bloom into a flower, or many other changes in stages of growth of desired agricultural plants grown for harvest and consumption. In this example, the agricultural object #40 detected as a bud at a given moment in time can be labeled as agricultural object #40 as a first label of bud #40. As time moves forward in a season, the agricultural objects on the farm or orchard, including agricultural object #40 as bud #40 can naturally turn into a flower. At this moment, if and when the agricultural object #40 turns into a flower, the agricultural treatment system can label the agricultural object #40 as a flower #40, associating the bud #40 with flower #40 such that the bud #40 and flower #40 are the same agricultural object #40 in the real world. Not all agricultural objects detected of the same plant may experience the same stages of growth or continue to keep growing. Some agricultural objects may even be removed, for example by thinning. For example, some plants can be thinned such that one or more agricultural objects growing from a single tree or stem can be removed or treated such that the next growth stage will not happen. In this instance, the agricultural treatment system can still detect that a uniquely identified real world agricultural object did not reach, or stopped, at a certain growth stage having unique physical features for a unique object label, or that the agricultural object detected previously is now gone and cannot be detected by the agricultural treatment system due to thinning or other method of removing the agricultural object so that neighboring agricultural objects can continue to grow as desired.

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The description of buds, blooms, flowers, fruitlets, and other agricultural objects and stages of growth of such agricultural objects discussed are only meant to be an example series of objects that can be detected by a treatment system, such as agricultural treatment system detecting fruits and objects associated with the stages of growth of fruits on fruit trees, and not meant to be limiting only to the specific example described above.

For example, as illustrated in FIGS. 13A and 13B, an image depicting an agricultural environment including a fruit tree having one or more spurs, one or more branches and stems, one or more laterals, and one or more potential crops growing on the one or more laterals. At the moment an agricultural observation and treatment system, or an agricultural treatment system, described throughout this disclosure, has observed and labelled each identifiable feature of the image, including detecting agricultural objects and labelling its growth stage, detecting and labelling landmarks including orientations of portions of the tree growing including configurations of leaves, branches, physical manmade materials that can be detected in the image, or other objects and sights of interest in the image that is not a potential crop, the agricultural treatment system can detect that not all identified objects in the image include agricultural objects of the same growth stage. For example, some agricultural objects detected are labelled as buds, some as blossoms, and some as fruitlets. Each of these labels are of agricultural objects of interest to observe and potentially treat, but not necessarily treated the same way depending on the growth stage. The agricultural treatment system can then determine treatment parameters in real time to treat each individually labelled agricultural object with different treatment parameters, or refrain from treating an agricultural object. For example, if a first labelled growth stage does not need to be treated, a second growth stage does need to be treated at least once, a third growth stage does not need to be treated, the agricultural treatment system can scan through a path, capture images such as the one depicted in image 1300a, and treat only the second labelled growth stage. In this specific example, a blossom can be treated with artificial pollen. The agricultural treatment system can detect that there are buds that have not yet blossomed, and fruitlets that have already grown after the blossom, so the agricultural treatment system will refrain from treating the agricultural object 1302 and only treat agricultural objects labelled with the same label as that of agricultural object 1310. In one example, the agricultural treatment system can select different treatment mixtures and emit different treatment projectiles by volume, concentration, mixture type, as well as the type of emission which can be a single spray projectile, a spray projectile with a large surface area travelling towards the surface of the agricultural object, or a mist or fog type spray treatment. In this example, multiple identified agricultural objects at different growth stages can require a treatment with different parameters. Instead of refraining from treating one type of agricultural object at a certain growth stage while treating other agricultural objects having the desired growth stage for a particular trial, the agricultural treatment system can treat multiple types of growth stages of agricultural objects growing on the same tree simultaneously by selecting and receiving a desired chemical mixture for treatment in real time.

The agricultural treatment system can observe, by running a plurality of trials, such that one trial is a sequence of capturing sensor data, depositing treatments, or a combination thereof, along each row of crops on a farm or orchard one time and captures sensor data and has the opportunity to

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deposit a treatment for each crop or agricultural object detected. For example, a trial run, where the agricultural treatment system scans through a farm of one or more row crops in one cycle, can be performed once a day, or twice a day, once during daytime and once during night time in a calendar day. For example, the agricultural treatment system can perform multiple trials or runs on a farm or orchard in a single day, particularly if the growth sequence of a plant is more rapid in one season or series of days over another season, such that the agricultural treatment system can capture more changes in stages of growth by conducting more trials as well as depositing treatments onto surfaces of desired agricultural objects more frequently.

Additionally, each row of crops, whether each row includes the same plant or of different plant types, for example planted in an alternating pattern, can include a plurality of plants that have one or more buds exposed, a plurality of plants that have one or more blossoms exposed, a plurality of plants that have one or more fruitlets exposed for treatment, or a combination of plants having a combination of buds, blossoms, fruitlets, etc., exposed at the same time on a single row. In this example, different agricultural objects at different stages will require different treatments at different volumes and frequencies. The agricultural treatment system can identify the particular stage of growth of each uniquely identified agricultural object mapped in the row of plurality of agricultural objects and give a label or identifier to each agricultural object based on its different and unique growth stage. The agricultural treatment system can then identify the appropriate or desired treatment parameters including treatment chemical mixture, density and concentration, whether a treatment is needed at all for the particular trial if the agricultural treatment system can identify that a particular agricultural object was already previously treated with a treatment deposition such that another treatment at a given trial can be too close in time for the same treatment to be applied again to the same unique agricultural object in the geographic boundary, depending on the stage of growth detected.

The agricultural treatment system can detect a first agricultural object of a plurality of agricultural objects in a row of plants inside a geographic boundary such as a farm or orchard. The agricultural treatment system can determine that the first agricultural object is different from a plurality of other agricultural objects by type or that the first agricultural object detected is among a plurality of the same type of agricultural objects as that of the first and can be indexed by a unique identifier to identify the particular object in the real world so that each unit or object in the real world of the same agricultural object type can be indexed and located in the geographic boundary. For example, a first agricultural object of a plurality of agricultural objects of the same plant type of the same tree or root can be identified on an orchard or row farm. The first agricultural object can be assigned and indexed as agricultural object #400 with a unique identifier that identifies its object type, such as a type of crop, and its location in the geographic boundary and time that the identifier was assigned to the first agricultural object. The agricultural treatment system can also assign a label of the first agricultural object based on the size, shape, color, texture, etc., with a first label, for example fruitlet #400 if the detected first object is a fruitlet of a crop. Because different stages of growth of a same desired plant or crop can require a different type, frequency, volume, or a combination thereof of treatment, the agricultural treatment system can determine treatment parameters, in real time upon detecting the first agricultural object in space and the growth stage of the

first agricultural object either determined in real time or determined based on the growth stage detected on a previous trial. For example, if the first agricultural object detected at a particular time is a flower or cluster of flowers, the agricultural treatment system can label the flower detected in one or more images as a flower and determine treatment parameters for the flower. The agricultural treatment system can apply the same type, mixture, amount, and frequency of a treatment to the each of the same agricultural object type detected at the same growth stage along the same row of plants. The agricultural treatment system can apply a different type, mixture, amount, and frequency of a treatment to each of the same agricultural object type detected at a different growth stage along the same row of plants. In one example, the different growth stage of the plant or portion of a plant can vary by days or hours in one part of a season and vary by weeks or months in another part of a season. For example, a tree of a plurality of trees in a row of the same type of plant yielding the same crop can have portions of the tree, for example shoots, spurs, stems, laterals, or branches with nodes, clusters, buds, or other objects for crops, growing at different stages. A bud for a potential crop can form on one portion of the tree or lateral while other portions of the tree do not have buds. At this stage, the agricultural treatment system can identify the portions of the tree that do have buds and perform any treatment including chemical treatment or light treatment (e.g., laser) that is appropriate for treating a bud of a certain plant. In another example, a tree can have some laterals that have blossoms and some laterals that only have buds. In this example, the blossoms may be treated with a certain treatment and the buds may be treated with a different type of treatment as that of the treatment for blossoms. The agricultural treatment system can identify and distinguish between the various agricultural objects in space having different labels based on their growth stage and apply a treatment appropriate for each unique agricultural object identified and located in the real world.

The agricultural treatment system can also identify and index a treatment history on each unique agricultural object identified in space of a geographic boundary. For example, one or more buds detected on laterals of a tree can be treated with a certain type of chemical or light treatment. At this point in time, certain laterals will have laterals that have yet to form buds. As time moves forward and the agricultural treatment system engages the row of crops for treatment, the laterals that have yet to form buds may now have buds. Additionally, the previously detected buds, that have been treated have not yet turned onto a flower, or even further stage of a bud that may require an additional treatment or different type of treatment. In this example, because the agricultural treatment system has indexed each agricultural object detected by its growth stage, with a label across time, and timestamp for each time the agricultural object was detected and its specific growth stage and image of the growth stage labeled, the agricultural treatment system can determine which agricultural objects in the row requires treatment and which agricultural objects in the row does not require a treatment, either because it was already treated in a previous trial and does not need a treatment every trial, or has not reached a later growth stage detected that will require a different type, frequency, mixture, etc., of treatment.

As with the earlier example, the first real-world agricultural object #400, having one or more images, a location, and object type associated with object #400, based on its labelled stage of growth, for example label #400, can require a first treatment having a specific treatment mixture, type,

volume, concentration, etc., and projectile emission strength. A second agricultural object #401, in proximity to agricultural object #400, for example, being a potential object for harvest of the same tree as that of agricultural object #400, having one or more images, a location, and object type associate with the agricultural object #401, based on its label #401, can require a second treatment having a specific treatment mixture, type, volume, concentration, etc., and projectile emission strength. The difference in treatment parameters such as the mixture, type, volume, concentration, strength of the projectile emitted, or a combination thereof, or abstaining from depositing a treatment at all for the particular trial run conducted by the agricultural treatment system, can be based on the different growth stage detected, even if the agricultural object is of the same type. In one example, different treatment parameters can be applied to a row of crops with the same type of plant but portions of the plant, such as various laterals can have agricultural objects growing on the laterals at different stages and require different treatments. Different treatment parameters can be applied to a row of crops with different plants in the row, for example with alternating crops. In one example, the same treatments with the same treatment parameters can be applied to the same row of crops of each agricultural object having the same or similar stage of growth. In one example, a different concentration or frequencies of treatments deposited can be applied to a row of crops of either the same plant of different plants at different stages of growth. For example, a first bloom of a lateral can require one deposition of chemical-#1 with a certain mixture, concentration, volume, etc. Other portions of the tree or other laterals may not have yet experienced a bloom from the buds so only the first bloom will receive a treatment of chemical-#1. At a later time, and more specifically, at later trial performed by the agricultural treatment system, other laterals will experience a bloom, such as a second bloom. In one example, it would be desirable for the second bloom to receive a single treatment of chemical-#1. Since the first bloom already received a treatment of chemical-#1 and for this particular example growth stage of this particular plant type, this example first bloom only requires one treatment of chemical-#1, the agricultural treatment system can detect that the agricultural object of the second bloom requires a treatment of chemical-#1 of a specified volume, concentration, strength of projectile and apply the treatment of chemical-#1, and detect that the agricultural object of the first bloom does not need a treatment at all for this trial.

For example, a treatment module, with one or more image sensors in real time, can sense and detect both object **1302**, for example a fruitlet, as well as object **1308**, which is a landmark. In one example, a landmark can a specific pattern detected and indexed in the geographic scene, for example of a tree pattern branching into two branches. As the vehicle moves forward in a row of an orchard, the treatment module's image sensors translates and moves relative to the tree, for example from right to left, and scans the tree illustrated in **1300a** in real time. As the treatment module, with its compute unit, detects objects in the tree while the vehicle is moving, the treatment module can track both the object **1302** for targeting, tracking, and treating via the treatment unit, as well as track the landmark object **1308** to generate and obtain a higher accuracy motion estimation. In this example, the detecting, via neural network or computer vision methods such as template matching, correspondence matching, homography estimation, etc. or a combination thereof, and tracking of the target object can be done for treatment but can also be tracked for the motion estimation

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of the treatment module, and by extension the treatment unit and its treatment head, itself. The addition of tracking other objects, including other target objects, landmarks that are real world objects, or real-world objects or salient points in an image that can be tracked, can add accuracy for pose estimation of the treatment module which reduces error or misalignment of treatment when the treatment module's compute unit sends instructions to the treatment unit for treatment.

FIGS. 14A-B illustrate example implementations of method 1400 that may be performed by some example systems, subsystems, or components of systems, described in this disclosure. For example, in one mode of operation,

at step 1410, the agricultural treatment system can obtain a first set of multiple images, on a vehicle, depicting one or more agricultural objects along a path.

At step 1420, the agricultural treatment system can receive localization, velocity, and acceleration data of the vehicle.

Additionally, at step 1480, the agricultural treatment system can generate a pose estimation of the vehicle in a geographic boundary, the agricultural treatment system supported by the vehicle relative to the vehicle, or agricultural objects or other objects detected in the geographic boundary.

At step 1430, the agricultural treatment system can process the set of multiple images to classify objects within the image. The classification can be performed on board the vehicle at the agricultural treatment system. Additionally, the image can be processed with computer vision techniques, image analysis, and machine learning algorithms including deep neural networks for performing feature extraction, object classification, object detection, and object tracking.

At step 1440, the agricultural treatment system can identify a location of the one or more agricultural objects classified. At step 1450, the agricultural treatment system can determine one or more treatment parameters for treating the agricultural object.

Additionally, the agricultural treatment system can detect, target, track, and determine treatment parameters based on previously identified and indexed information of one or more images, including one or more images with the same agricultural object identified and classified in the first set of multiple images.

For example, at step 1442, the agricultural treatment system can receive a second set of mapped images, depicting one or more localized agricultural objects, each with a timestamp associated with the image captured of each mapped image and treatment history associated with each of the localized agricultural objects.

At step 1444, the agricultural treatment system can compare and correspond the classified agricultural objects with the one or more localized agricultural objects. The agricultural treatment system can then activate a treatment unit and emit a fluid projectile onto the target plant that is the agricultural object.

FIG. 15 illustrates an example method 1500 that may be performed by some example systems, subsystems, or components of systems, described in this disclosure either online, that is onboard a vehicle supporting one or more modular agricultural observation and treatment systems, subsystems, or components of systems, or offline, that is at one or more servers or edge compute devices.

For example, in one mode of operation, at step 1510, an agricultural treatment system can receive image data in a real-world agricultural scene from one or more image capture devices. At step 1520, the agricultural treatment system

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can detect one or more agricultural objects in a first image of the image data. At step 1530, the agricultural treatment system can identify a stage of growth associated with a first agricultural object. At step 1540, agricultural treatment system can assign a label of the stage of growth of the first agricultural object. Additionally, at step 1542, the agricultural treatment system can index and store the first image with the labelled agricultural object with a timestamp of the first image taken associated with the label. At step 1550, the agricultural treatment system can determine one or more treatment parameters based on the assigned label of the stage of growth of the first agricultural object. At step 1560, the agricultural treatment system can receive instructions for a treatment unit, of the agricultural treatment system, to receive a fluid mixture from a chemical selector to prepare a treatment. At step 1570, the agricultural treatment system can orient the treatment unit to target the first agricultural object in the real-world agricultural scene and activate the treatment unit to emit a fluid projectile at a surface of the first agricultural object.

In one example, the agricultural treatment system can determine that different chemical concentrations of a chemical mixture are required for different growth stages of the same plant on a row of plants. In one example, the agricultural treatment system can determine that different chemical concentrations of a chemical mixture are required for different growth stages of different plants planted on a same row on a farm or orchard. In another example, the agricultural treatment system can determine that only certain growth stages of agricultural objects detect require a deposition of a particular treatment, and that other agricultural objects detected require a deposition of a different treatment, or no treatment, depending on the stage of growth and treatment history of the particular agricultural object detected in the real world. In one example, a row of plants can have laterals supporting different agricultural objects, or the same agricultural objects with different stages of growth and different treatment histories, such that different treatments are desired for each unique agricultural object in the row. The chemical selection unit can mix different treatment mixtures and concentrations in real time for the agricultural treatment system to accommodate the different requirements of treatments in real time while performing a trial in a particular row of plants. Additionally, the agricultural treatment system can accommodate for applying different treatments to different agricultural objects of different plants in a single row, or other configuration, of crops.

Thus, the agricultural treatment system can, in real time, scan with sensors for agricultural objects and its stage of growth and real-world location in the row, determine whether to apply a particular treatment based on stage of growth detected and the particular agricultural object's treatment history.

In one example, the agricultural observation and treatment system can be configured to detect objects in real time as image or lidar sensors are receiving image capture data. The treatment system can, in real time, detect objects in a given image, determine the real-world location of the object, instruct the treatment unit to perform an action, detect the action (discussed below), and index the action as well as the detection of the object into a database. Additionally, the treatment system, at a server or edge computing device offline, can detect objects in a given image, spray projectiles, spray action, spot of splat detections, and index the object detections and spray action detections. In one example, the agricultural observation and treatment system can perform and use various techniques and compute algorithms for

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perform the object detections including computer vision techniques, machine learning or machine learning assisted techniques, or a combination thereof in multiple sequences and layers such that one algorithm partitions a given image and a second algorithm can analyze the partitioned image for objects or landmarks.

In one example, a machine learning model, embedded in one or more compute units of the agricultural observation and treatment system onboard a vehicle, can perform various machine learning algorithms to detect objects, including object detection including feature detection, extraction and classification, image classification, instance classification and segmentation, semantic segmentation, superpixel segmentation, bounding box object detections, and other techniques to analyze a given image for detecting features within the image. In one example, multiple techniques can be used at different layers or portions of the image to better classify and more efficiently use computer resources on images. Additionally, pixel segmentation can be performed to partition colors in an image without specific knowledge of objects. For, example, for row crop farming, a system can perform color segmentation on a given image to partition detected pixels associated with a desired color from any other pixels into two groups, such as the color segmented pixels and background pixels. For example, a system can be configured to analyze frames by detecting vegetation, which can be a form of green or purple color from background objects, such as terrain, dirt, ground, bed, gravel, rocks, etc. In one example, the color segmentation itself can be performed by a machine learning model configured to detect a specific type of color in each pixel ingested by an image sensor. In another example, the color segmentation can be manually predefined as pixels ranging between a specific range of a color format. For example, vegetation algorithm can be configured to analyze a given frame to partition any pixels having attributes of the color “green” form a Bayer filter. In another example, the algorithm can be configured to detect attributes of “green” under any color model where “green” is defined. For example, a numeric representation of RGB color being (r,g,b) where the value of g>0 in any digital number-bit per channel. The algorithm can itself be a machine learning algorithm to detect “green” or a different color that are of interest.

In one example, machine learning and other various computer vision algorithms can be configured to draw bounding boxes to label portions of images with objects of interest from backgrounds of images, masking functions to separate background and regions of interest or objects of interest in a given image or portion of an image or between two images where one image is a first frame and another image is a subsequent frame captured by the same image sensor at different times, perform semantic segmentation to all pixels or a region of pixels of an given image frame to classify each pixel as part of one or more different target objects, other objects of interest, or background and associate its specific location in space relative to the a component of the treatment system and the vehicle supporting the treatment system.

Multiple techniques can be performed in layers to the same or portions of the same image. For example, a computer vision technique or machine learning technique can be first applied to an image to perform color segmentation. Once a given image is detected and pixels related to a desired or target color is segmented, the separate machine learning algorithm or computer vision algorithm can be applied to the segmented image, for example to an object detection algorithm to draw bounding boxes around the

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segmented image containing weeds and containing crops. In another example, an object detection algorithm can be applied to the entire image to draw bounding boxes around plants of interest, such as crops and weeds. Once the image has bounding box detections draw around each of the detected crop or weed objects in the image, a color segmentation algorithm can be applied to just those bounding boxes to separate pixels bounded by the box that are of a target color, such as green, and those pixels that are considered background. This method can allow a system to more accurately determine which pixels are associated with objects in the real world, such that an image with contours and outlines of a specific object detected in the image, such as a leaf, can be a more accurate depiction of the leaf, and therefore more accurately target the leaf in the real world, than drawing a rectangular box around a leaf where the system determines that any portion inside the bounded rectangular box is associated with the object “leaf”. The example above is just one of many examples, configurations, orders, layers, and algorithms, that can be deployed to analyze a given image for better understanding of objects, that is improved feature detection, performed either online in the field in real time, or offline at a server for other uses, such as creating a time lapse visualization, mapping the object, generating key frames with detections for indexing and storage, diagnosing and improving machine learning models, etc.

In one example, detecting a plurality of agricultural objects and/or landmarks can be used to perform variations of consensus classification. For example, multiple detections of the same agricultural object and/or landmark can be performed to eliminate or reduce false positives or false negatives of object detection. While a machine learning model will be tasked to identify individual objects and landmarks, the closeness of an object to another object in a single frame can also be accounted for an considered by the machine learning detector for detecting an object. For example, if in a first frame, the machine learning detector detects a target object as well as a plurality of nearby target objects, other agricultural objects, or landmarks, but then in subsequent frames, while the vehicle has not moved enough such that the location where the ML detector has detected a target object has not moved out of the next frame, does not detect that same target object, but does detect all of the other nearby target objects, other agricultural objects and landmarks detected in the first frame, the compute unit can determine that the first frame may have had a false positive and flag the frame for review and labelling, at a later time on board the vehicle for a human to label, or offline, without instructing the treatment unit to perform an action at the location in the real world where the system detected a target object to treat based on the first frame.

FIG. 16 illustrates an example diagram 1600 for ingesting an image, performing various computer vision and machine learning algorithms onto various portions or layers of the image to extract and detect features of the image.

As discussed above, multiple techniques can be performed in layers to the same or portions of the same image. For example, an image 1610 can be acquired by an image capture device and loaded onto a local compute unit of a local modular treatment module. For illustration purposes only, the image 1610 captured can be an image of a row crop farm having one or more beds 1612 supporting a plurality of crops, such as carrots, and weeds, and one or more furrows or tracks 1614 for a vehicle’s wheels to run through as a vehicle passes the row. One or more embedded machine learning algorithms and computer vision algorithms in the

compute unit, or accessible by the compute unit in real time via the cloud or edge compute device containing the machine learning algorithm and computer vision algorithm, such as computer vision algorithm 1620 and machine learning algorithm 1630 can be used to partition the image 1610 into analyzed images with features extracted, with the goal of accurately detecting objects in the given image 1610. For example, the first computer vision algorithm 1620 configured to separate beds and furrows can be applied to the analyze and segment classify the image 1610 with portions of the image related to beds such as partitioned image 1613 with portions of the image related to furrows such as partitioned background image 1615. One purpose of deploying this technique is to that the treatment module does not have to run a machine learning detector on the entire image 1610, but only on portions where object of interest may be. The partitioning of beds and furrows, as is the partitioning of green and background, are just many examples of performing a plurality of computer vision and machine learning techniques to an image to reduce computation load while generating accurate detections of features in the real world. Next, the system will have generated a partitioned image 1616 having pixels associated with beds and pixels associated with furrows such as that of partitioned image 1613 and partitioned background image 1615. The machine learning algorithm 1630, which for example can be a machine learning algorithm to detect plant objects of interest, such as crop plants and various species of weeds, can be implemented to further analyze the image 1610 or the partitioned image 1616, and only the portion of the image 1610 that is partitioned image 1613, and not the partitioned image 1615. This would allow the ML detector or machine learning algorithm 1630 to analyze fewer pixels or tiles of pixels, and reduce the load on the system, while the system having a high probability that the machine learning detector is scanning the most important areas of the image 1610. In this example, the detector would run detections on only a portion of the partitioned image 1616, for example a portion of the partitioned image 1613, such as a patch 1632 of the partitioned image 1613. The treatment system can then draw bounding boxes, semantically classify, or perform various machine learning methods deployed by machine learning algorithm 1630, for example detect objects and draw bounding boxes, and generated a machine labelled or machine detected image 1642, which is a labeled image of a portion of the original intake image 1610. The agricultural observation and treatment system can then use those detections to determine which detections are target objects to treat, target the objects in the real world, track the detected objects in subsequent frames, and perform a treatment action to the detected object in the real world. Additionally, using multiple layers of computer vision and machine learning algorithms to optimize the computing load onto a compute unit can be performed to improve VSLAM. For example, vegetation segmentation can be performed to detect green objects. In the VSLAM pipeline for matching keypoints from frame to subsequent frames by the same sensor, the compute module or compute unit associated with the sensors receiving the images, can determine that points associated with green objects are real objects in the world that are stationary and can be tracked via VSLAM by sensors and compute units of each component treatment module for local pose estimation. This would allow the VSLAM algorithm analyze keypoints, keypoints in this case being points related to corners or contours or edges of green objects, with higher confidence that the keypoints generated and analyzed are higher quality than that of arbitrary salient points,

compared to that of known objects, such as objects corresponding to green pixels, since the system will know beforehand that green pixels are of vegetation, which are physical objects in space that are stationary and are of similar size and topography as that of target objects for treatment that will be tracked.

FIG. 17A illustrates an example method 1700 that may be performed by some example systems or subsystems described in this disclosure either online, that is onboard a vehicle supporting one or more modular agricultural observation and treatment systems, subsystems, or components of systems, or offline, that is at one or more servers or edge compute devices.

At step 1710, the agricultural observation and treatment system can initialize the treatment system. At step 1720, the agricultural observation and treatment system can obtain a first image having one or more unique regions of interest. For example, the regions of interest can be regions or portions of images that are specific to a specific geographic boundary such as a row crop farm or an orchard. For example, images where there are tree trunks, images where a substantial portion of the image are either beds or troughs or furrows, images where objects of interest have a certain color and every other portion of the image can be background. At step 1730, the agricultural observation and treatment system can identify the one or more unique regions of interest and one or more regions of background. At step 1740, the agricultural observation and treatment system can partition the first image into the one or more unique regions of interest and the one or more regions of background of the first image. At step 1750, the agricultural observation and treatment system can identify a first region of interest among the regions of interest. At step 1760, the agricultural observation and treatment system can detect one or more objects in the first region of interest. At step 1770, the agricultural observation and treatment system can determine a real-world location of a first object of the one or more objects based on a location of the first object detected in the first image. At step 1780, the agricultural observation and treatment system can determine and prepare one or more actions associated with the first object in the real world. At step 1790, the agricultural observation and treatment system can send instructions to activate actuators. The system can repeat steps 1760 to detect a second object detected in the first region and prepare treatment actions associated with the second object. Once all objects of interest are accounted for in the first region of interest, the system can detect objects in a second region of interest for treatment, or partition the image for a second region of interest.

Additionally, at step 1782, the agricultural observation and treatment system can identify a second region of interest. At step 1784, the agricultural observation and treatment system can detect one or more objects in the second region of interest. At step 1786, the agricultural observation and treatment system can determine a real-world location of a second object based on a location of the second object detected in the second region of interest in the first image. At step 1788, the agricultural observation and treatment system can determine and prepare one or more actions associated with the second object.

FIG. 18A is a diagram 1800 capturing an action performed by an observation and treatment system. In this example, an image capture device can receive a constant stream of images of a local scene having one or more agricultural objects in the scene. Once a target object is detected, targeted, and tracked, the system will instruct a

treatment unit to activate and emit a liquid projectile or a beam of light onto a surface of the target object. This action will take a length of time to release from the treatment unit to exiting the treatment head, travel in space, hit the target if accurately targeted and emission parameters, such as dwell time which is the amount of time the nozzle head is pointed at the target object while the nozzle head is on a moving vehicle, pressure release time which is when a pressure actuator such as a capacitor or solenoid valve opens and closes and allows pressurized fluid to release from the valve and through the nozzle head, nozzle orifice size, and other parameters, and create splash, splat, or a footprint on the ground for row crops where plants, or target plants, are growing out from the ground.

In this example, the image capture system can capture and trace the liquid projectile itself, for example fluid projectile **1830**. Because the projectile is a fluid, it may not flow in an exact straight line. Additionally, the projectile can be comprised of smaller liquid droplets **1850**. The compute unit and image sensors can detect the beam trace directly from detecting the projectile **1830** and its smaller droplets **1850** as the liquid leaves the treatment unit. Additionally, a laser with a laser beam **1840** can be pointed at the intended target object **1820** for the system to detect both the laser beam and trace the projectile beam to determine whether there was a hit, and if there was any error or discrepancy from the desired projectile hit location to the actual trajectory of the projectile.

FIG. **18B** and FIG. **18C** illustrate an example of spray detection, beam detection, or spray projectile detection. In these diagrams **1802** and **1803**, one or more image sensors is scanning a local scene comprising a plurality of plants **1872** including target plants for treatment and crop plants for observation and indexing. As the sensor scans the scene while a vehicle supporting the sensor is moving in a lateral direction, the sensor will capture one or more image frames in sequence from one to another illustrated in image frames **1862**, **1864**, and **1866** where image frame **1864** and **1866** are frames captured by a sensor that captured image frame **1862** subsequently, but not necessarily the immediate next frame captured by the image sensor. During the capturing of images, if component treatment system having sensors and treatment units sends instructions to the treatment unit to perform a spray action, such as emit a fluid projectile, the image sensors would capture the spray action as it comes into the frame and then eventually disappears as the projectile is fully splashed onto the surface of the intended target or ground. In such example, the spray projectile, such as projectile **1875**, can be detected and indexed by the image sensors and the treatment system, as well as the splat area **1877** after the spray has completed. The system can detect the splat size and location.

In one example, the detection of the spray can be performed by various computer vision techniques including spray segmentation, color segmentation, object detection and segmentation, statistical analysis including line fitting, homography estimation, or estimation of a homography matrix, or a combination thereof. For example, the differences between frame **1862** and frame **1864** can be the presence of a spray and the lack of presence of a spray. The rest being the same features in each image. In one example, homography estimation is used to account for change in space across a common plane, such as a bed of a row crop farm. A homography matrix can be used to estimate how much movement in space from a first frame to a subsequent frame. The images will be slightly misaligned from each other due to the camera being on a moving vehicle while the

first frame **1862** is captured and a subsequent frame **1864** is captured. The discrepancy in the frames caused by the motion of the camera can be accounted for with homography estimation, given that the two frames are likely looking at the same plane of equal distance from the camera from the first frame **1862** to the subsequent frame **1864**, at a later time but not necessarily the exact next frame captured by the image capture device. The difference in the two images, other than the discrepancy which can be accounted for by homography estimation, would be the presence of the spray, which can be generated by comparing the two frames and performing spray segmentation, that is the pixels in frame **1864** that has the spray projectile **1875** captured compared to the pixels in frame **1862** that do not have a projectile detected. In this case, one or more statistical and image analysis techniques, including line fitting, and masking function to determine that the pixels detected in frame **1864** but not detected in frame **1862** is a spray projectile. Since spray projectiles are likely line shaped, the pixels related to the spray can be line fitted. Other image differential techniques can be applied to detect the spray beam including outlier rejection and using priors for masking outliers. The priors can be an expected region such as that outline by predicted spray path **1876**. In one example, the difference in pixels profiles detected from a first frame to a subsequent frame, accounting for homography estimation due to changes in translation of the image sensor, can generate a projectile segmentation. Similar techniques can be used to detect the splat or spot detection of the spray outcome onto the surface of the target and ground, for example, seeing the color of the ground and target plant change from unsprayed to sprayed. For example, a liquid projectile hitting a target plant will morph from a projectile having a small cross-sectional diameter to a flat area covering a portion of the dirt or leaf. In this example a liquid projectile may change the color of the dirt surrounding a plant, due to dry dirt turning wet from the liquid projectile hitting the dirt. In this case, the image sensors can detect a color change in the ground and determine that a splat is detected and that a detect target object for treatment has been treated, and logged or indexed by the treatment system. In one example, a stereo pair of cameras can detect sprays in each camera and associated with each other to fit a 3D line such that the system can detect and index a spray in the real world with 3D coordinates.

FIG. **18C** illustrates a diagram **1803** to determine spray accuracy and spray health, spray health being whether external factors outside or correctly detecting target object and lining the treatment head onto the target object and tracking it as the target object moves away from the treatment unit, since the treatment unit is on a moving vehicle, a prior or predicted spray path **1876** can be generated. For example, a sensor, disposed on a moving vehicle, can receive an image frame **1862** having a plurality of crop objects and target objects, including detected target object **1872**. The treatment system will target the target object **1872**, track the object **1872** in subsequent frames, such as that of frame **1862**, and emit a projectile onto target **1872**. In one example, due to external factors not necessarily related to computer vision, such as portions of the treatment unit no longer calibrated to the image sensor, such that targeting at a specific location in the real world from a detection in the image frame may result in a misalignment of the line of sight of the treatment head. For example, the treatment system, given frame **1862** or **1862**, may target the target object **1872** at the correct real-world location, but in doing so and instructing the treatment head to aim its nozzle to target

object **1872** in the real world may in fact be targeting a location **1879** or **1878**, or an incorrect location or misaligned location in the real world, that the treatment systems image sensor would capture. In this case, to quality check the spray targeting and spray action, the treatment system can pre-
 5 determine a predicted spray path **1876** and perform spray segmentation and other computer vision and machine learning techniques described above only in the portion of the image, and therefore compare pixels related to the images contained in the region defined by the predicted spray path **1876**. If the detection is not good enough, such as the line cannot be fitted, the system can determine that the spray did not happen, or happened but not at the intended target. Alternatively, the system can perform spray segmentation on the spray that was detected, whether within the predicted
 10 spray path **1876** or not, and determine whether the end of the spray or the splat detected lines up with the intended target. Thus, seeing where a target object should have been sprayed, and/or should have had a splat detected, and where the actual spray profile was detected, including 3D location, and where the spray splat was detected, can be used to evaluate the specific spray health of that particular spray, and whether intrinsic or extrinsic adjustments needs to be made. The adjustments can be accounting for wind that may have moved the spray, the speed of the vehicle not being
 15 accounted for properly as the system tracks an object from frame to frame, or mechanical defects such that the intended target and the line of sight after sending the correct instructions to orient the treatment head of the treatment unit are misaligned. Upon detecting an inaccurate or incorrect spray projectile, one or more of the discussed defects can be accounted for in real time and a second projectile can be reappplied on to the target object and tracked again for trajectory evaluation and its spray health and accuracy.

FIG. **18C** illustrates an example method **1804** that may be performed by some example systems or subsystems described in this disclosure either online, that is onboard a vehicle supporting one or more modular agricultural observation and treatment systems, subsystems, or components of
 20 systems, or offline, that is at one or more servers or edge compute devices.

For example, at step **1806**, the observation and treatment system or server can identify a first object for treatment. In this example, the observation and treatment system or a server is analyzing the performance of the online observation and treatment system during its latest run, in a location such as an agricultural geographic boundary. The system, online or at a server, can identify each treatment performed or instructed to be performed on the geographic boundary for verification, indexing, and adding the verification to each of the identified target object's treatment history. For example, a treatment system may have identified and initialized a few thousand or a few hundred thousand actions performed in a single run at a field, orchard, or farm, and a server is analyzing the treatment accuracy and efficacy of
 25 each of the actions performed on the field in that particular run. At step **1808**, the observation and treatment system or server can determine a treatment unit activation for each of the objects for treatment. In this optional step, the system or server can determine treatment actions based on the treatment performed and logged previously in real time while the observation and treatment system was on the field performing detection objects and performing treatments. In this example, the server does not have to identify every frame captured and determine which objects detected were treated
 30 for second time, but instead can analyze only those frames captured by image capture devices where each online and

onboard compute unit has already detected. In one example, the determining of treatment activation can include the treatment parameters such as desired spray size, volume, concentration, mixture of spray content, spray time of flight, etc.

At step **1812**, the observation and treatment system or server can detect a first emission pattern. This can be done with techniques described above as well as image correspondence from a previous frame and a subsequent frame to detect a projectile.

At step **1813**, the observation and treatment system or server can index the first emission pattern. This can be stored as a 3D vector, or a 2D or 3D model of the full 3D profile with shape and orientation mapped into a virtual scene.

At step **1814**, the observation and treatment system or server can detect a first treatment pattern. This can be the splat detection from color change in dirt from a first frame to a subsequent frame, performed by similar methods described above.

At step **1815**, the observation and treatment system or server can index the first treatment pattern.

At step **1816**, the observation and treatment system or server can determine and index the first object as treated. For visualization purposes, a target object that has not been accurately treated can have a bounding box with a dotted line indicating a detection of the object itself but no detection of a spray onto that target object. And once a spray or treatment is detected, by the projectile or the splat detection, the dotted line can convert to a solid line, as illustrated in diagram **1803** of FIG. **18C**.

As illustrated in FIGS. **18E** and **18F**, each spray projectile and splat detections can be indexed and visually displayed in a user interface. The 2D or 3D models **1880a**, **1880b**, and **1880c** of each target object **1872**, spray projectile **1875**, and splash **1877** onto a surface of the ground and target object. Additionally, the 3D models can be superimposed on each other to reconstruct the spray action from the targeting of the target object, to the spraying of the target object, to the splash made and splat detected as illustrated in model **1880d** of diagram **1806** of superimposed model **1882**.

FIGS. **19A**, **19B**, and **19C** illustrate diagram **1900** depicting and example treatment pattern performed and observed for treatment accuracy and health. For example, as illustrated in FIG. **19A**, a treatment module **1930** with a treatment unit configured to apply different chemical solutions or water, and different treatment profiles, such as a projectile or a volumetric scan of fluid over a region on a field or tree, having one or more sensors, can be configured to sense objects **1904**, determine treatment parameters, and perform treatment actions illustrated by treatment **1940** and treatment projectile **1932**. In this example, treatment profile **1940** is a blasting spray of fluid covering a sweep angle **1922** and treating a plurality of objects **1904** over a ground **1902**. Alternatively, treatment **1932** is a fluid projectile treatment onto a single object **1904**. The treatment module **1951** can line a treatment unit with a line-of-sight **1924** with a normal angle **1928** from a vertical path **1926**, onto the object **1904**. Additionally, as illustrated in FIG. **19B**, treatment module **1951** can also be configured to emit small fluid droplets **1934** or larger fluid droplets **1936** onto objects **1904**. The above illustrations are merely examples as the treatment units can be configured to emit a plurality of different types of fluid treatments discussed in this disclosure. FIG. **19C** illustrates examples of certain intrinsic or extrinsic factors affecting the desired spray profile or spray accuracy of spray treatments. Of example, with chemical buildup in a spray nozzle or tubing, particularly salt-based substance or surfactant built
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up in the inner walls of a spray nozzle, can affect a volumetric spray profile illustrated in treatment profile **1940** to that of treatment profile **1942**, which may not be the desired treatment profile. Additionally, other extrinsic factors such as wind can affect spray projectiles such that even when the target is accurate and the line of sight of the treatment nozzle is correctly aligned with the target object, the wind can affect the trajectory of the fluid projectile emitted from the treatment nozzle such as projectile trajectory or path **1962** of fluid projectile **1908**. Or even without wind, of substance is built up in the inner walls of tubing or the treatment head nozzle, even when the line-of-sight **1924** is correct, and the projectile shoots straight out of the nozzle undisturbed by wind, the trajectory can be off from the desired trajectory, such as trajectory **1964**.

FIG. **20** illustrates an example method **2000** that may be performed by some example systems or subsystems described in this disclosure either online, that is onboard a vehicle supporting one or more modular agricultural observation and treatment systems, subsystems, or components of systems, or offline, that is at one or more servers or edge compute devices.

For example, at step **2010**, an agricultural observation and treatment system can identify an object for treatment. Multiple objects can be identified from a single image frame and each tracked after the systems targets the objects. At step **2020**, the agricultural observation and treatment system can lock a treatment unit onto a target object. At step **2030**, the agricultural observation and treatment system can activate a treatment unit. At step **2040**, the agricultural observation and treatment system can detect an emission target error. At step **2050**, the agricultural observation and treatment system can detect a treatment unit health. At step **2060**, the agricultural observation and treatment system can detect one or more internal drift factors, one or more external drift factors, or both. At step **2070**, the agricultural observation and treatment system can determine a target accuracy adjustment. At step **2080**, the agricultural observation and treatment system can realign the treatment unit onto the target object accounting for one or more post-emission error. At step **2090**, the agricultural observation and treatment system can active treatment unit to treat the target object found in step **2020**.

FIGS. **21A-24F** illustrate various examples of performing agricultural observation, digitizing a geographic boundary, building a map of each individual agricultural object or crop detected and associating captured images of agricultural objects from one moment in time to another to digitize and map a farm with location and image history of each agricultural object detected, targeting and tracking objects, and treating each individual agricultural object.

The description of buds, blooms, flowers, fruitlets, and other agricultural objects and stages of growth of such agricultural objects discussed are only meant to be an example series of objects that can be detected by a treatment system, detecting fruits and objects associated with the stages of growth of fruits on fruit trees, and not meant to be limiting only to the specific example described above. For example, agricultural objects can include larger objects or portions of a tree that are part of supporting a crop can be detected, classified, and labelled for spraying including spurs, shoots, stems, laterals, other nodes, fruiting clusters, leaves, or other portions of a tree. Different types of plants can be treated by the treatment system including general plants for crops, specialty crops, including fruits, vegetables, nuts, flowers, herbs, foliage, etc. The agricultural treatment systems described in this disclosure can be performed in geographic boundaries typically appropriate for a robotic

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vision and treatment deposition system for observing, treating, harvesting, or a combination thereof, of crops such as farms, orchards, greenhouses, nurseries, or other regionally and topographically bounded locations for agronomy and agriculture, horticulture, floriculture, hydroculture, hydroponics, aquaponics, aeroponics, soil science and soil agronomy, pedology, etc.

FIGS. **21A** and **21B** are side and front views, respectively, of the illustration depicted in FIG. **6**. The vehicle can include spatial and navigation sensors, for localizing the vehicle and objects, as described above. The sensors **2116** can be visual odometry and VSLAM sensors including various types of cameras, rangefinders, and LiDAR. Each modular treatment system supported by the vehicle **2110** can include treatment units **2112** for emitting a treatment projectile or droplet onto a treatment target **2132**, including agricultural objects of interest.

FIG. **21C** illustrates a vehicle supporting or towing one or more treatment systems **2112** configured and optimized for a geographic environment having plants grown on V-shaped trellises **2134** or grown in a "V" shape.

FIG. **22** is a diagram illustrating the vehicle **2110** supporting the treatment system **2112** in an alternative orientation.

FIG. **23** illustrates a vehicle having coordinates associated with rotational movement including that of roll about an X axis, pitch about a Y axis, and yaw about a Z axis, as well as translational coordinates associated with lateral movement including an X, Y, and Z position in a geographic boundary. The vehicle **2110**, illustrated in FIG. **23** can move with at least 6 degrees of freedom. Additionally, the treatment unit **2113** of the treatment system **2112** can also have coordinates associated with rotational movement including that of roll about an X axis, pitch about a Y axis, and yaw about a Z axis, as well as translational coordinates associated with lateral movement including an X, Y, and Z position in a geographic boundary. This can include rotating and moving a gimbal assembly of the treatment unit **1653** to a desired pitch angle **2002** and desired yaw angle **2004** when the treatment unit is configuring and orienting itself to position a nozzle or head of the treatment unit **1653** at a target or aligning a line of sight towards a target for emitting a projectile.

FIG. **24A** illustrates a diagram **2400** including a vehicle **2410**, having one or more sensors **2418** and other electronic devices, supporting and towing one or more treatments systems **2412**. In one example, the vehicle **2410** can be a tractor towing a plurality of modular treatment systems **2412**. FIG. **24B** illustrates the diagram **2402** with an alternate orientation of the treatment systems **2412** being towed by vehicle **2410**.

Further illustrated in FIG. **24C**, a vehicle **2410**, such as a tractor is configured to tow one or more treatment systems **2412** along a vehicle track **2430** having multiple lanes for the vehicle to operate a geographic boundary. Between each vehicle track **2430** are one or more rows of agricultural objects **2432**, such as plant including crop plants and weed plants, for each treatment system **2412** to scan across each row to observe and treat individual plants growing from the ground.

As illustrated in FIG. **24D**, the treatment systems **2412** can be configured to observe a plant, soil, agricultural environment, treat a plant, soil, agricultural environment, or a combination thereof, such as treating a plant for growth, fertilizing, pollinating, protecting and treating its health, thinning, harvesting, or treating a plant for the removal of

unwanted plants or organisms, or stopping growth on certain identified plants or portions of a plant, or a combination thereof.

In one example, the treatment systems can be configured to observe and treat soil for soil sampling and mapping of features and chemical compositions of soil including soil deposition, seed deposition, or fertilizer deposition, nutrient management, for both cultivated and uncultivated soil. The agricultural objects described above for targeting and treating can be of specific patches of soil that can be identified and features and classification labelled by a vision of the treatment system. Each patch or region of the soil detected by the treatment system **2412** and can be indexed and mapped with a timestamp associated with the moment the patch or region was sensed and treatment history detailing each treatment applied to each patch or region of the soil.

FIG. **24E** illustrates a diagram **2408** depicting an example treatment system having a plurality of component treatment modules **2444** supported by a support member **2440** and a navigation unit **2442** (various sensor of the navigation unit **2442** may not necessarily be enclosed in a box illustrated in diagram **2408**).

In one example, each component treatment module **2444**, via its own compute unit and image sensors, and other sensors, can perform VSLAM to continuously map a local environment in the agricultural scene and continuously generate pose estimation, such as a local pose estimation relative to objects or landmarks detected on the ground, such as plant objects, patterns, salient points representing unknown objects near the ground, including target plant objects. Additionally, the navigation unit **2442**, via its own compute unit and image sensors, GPS, IMU, and other sensors, can perform VSLAM and VIO to continuously map a global scene and continuously generate a pose estimation, such as a global pose estimation of a global scene. The compute unit of each treatment module **2444**, can account for both its locally determined pose estimation relative to objects and landmarks on the ground, and the globally determined pose estimation of a global scene relative to a point of origin in an agricultural environment, such as a farm, because each of the treatment modules **2444a**, **2444b**, **2444c**, etc. are rigidly attached to a support structure supported by a vehicle having sensors located throughout the vehicle associated with the navigation unit such that translation of the vehicle, which includes a change in global pose estimation detected, will have substantially the same translation of each of the component treatment module, and therefore also includes a change in a global pose estimate detected and accounted for each component treatment module. In one example, each treatment module or component treatment module **2444** can be referred to as a spraybox, the spraybox including one or more compute units and one or more image sensors disposed within an enclosure of the component treatment module. In one example, the spraybox may refer to the enclosure supporting the one or more compute units and image sensors. In another example, the spraybox may refer to the enclosure as well as illumination modules operably attached to the enclosure and one or more treatment units operable connected to the enclosure. For example, treatment module **2444a** having an enclosure, a pair of treatment units, and one or more illumination devices (now shown) can be referred to as a first spraybox. In another example, just the enclosure housing the compute unit and sensors of treatment module **2444a** can be referred to as the first spraybox.

In one example, tracking multiple poses of each component treatment system, being the local pose generated from

local sensors to the component treatment system, and global pose received from the navigation unit, can account for loss and/or inaccuracies of kinetic motion from the vehicle to each of the component treatment systems, particularly the component treatment systems that are farther away from the vehicle itself, relative to modules supported by the vehicle that are closer to the vehicle. This is especially apparent in farming activities where performance of any agricultural observation and treatment system will likely be performed on rough topography such that movement along a path will cause various magnitudes in bumps, and thus change in height, along the path. For example, as a vehicle navigates in a rough terrain, the component treatment module **2444c** will likely bump up and down more violently than the bumping of component treatment module **2444a**. Thus, it may be impractical for each component treatment system to only determine its local pose estimation from that of the global pose estimation as movement of sensors of the navigation unit **2442**, or that of the navigation unit **2442** box itself, will be different from the movement of, for example, the component treatment module **2444c**. In this case, each of the component treatment modules **2444** can determine its local pose to more accurately detect and track targets in real time for treatment actions.

In one example, when a compute unit of the first treatment module **2444** sends instructions to each of one or more treatment units, for example treatment devices with one or more nozzles on a turret or gimbal mechanism, the agricultural treatment system can determine the specific pose of each of the nozzle heads at the time the treatment module, through local or global poses, global being vehicle and local being at or near each treatment module **2444**, detect and identify an object and its location relative to the treatment module, as well as determine the location of the object in the global scene. At this point, the compute unit of the first treatment module can account for the vehicle's pose of the global scene, that is the global registry of a farm, the pose of the treatment module itself relative to the local first target object, as well as account for the last state of orientation of the treatment unit's nozzle or emitter's line of sight relative to the treatment module. This is because the vehicle, the treatment module, more specifically the treatment module's local sensors, and the treatment unit are all mechanically coupled in a fixed position to each other. Thus, a change in pose estimation generated by sensor signals of the vehicle itself will directly translate to the same change in pose estimation to anything physically supported by the vehicle. —However, calculating for pose at each treatment module as well as accounting for pose of the vehicle, with sensors and computer vision techniques such as performing visual SLAM using machine learning to detect objects to track, particularly for treatment modules that are disposed farther away from the vehicle as compared to other treatment modules closer to the vehicle, and therefore the sensors of the navigation unit **2442**.

In one example, multiple rows where each treatment module **2444** can determine a pose estimation based on determine its own pose local pose with its local sensors embedded or supported by the module **2444**, and that of the vehicle's pose. Thus, each object identified, can be indexed in the real world such that if the vehicle operates on the same geographic area in a subsequent day, or any subsequent time where a break in operation has occurred, an object detected in the subsequent time can be matched and associated with an object previously identified since in at least both cases, the treatment system determined the location of each object identified in the real world, global scene, by approximating

its location in the real world with the treatment system's sensed and determined global map of the geographic boundary and further narrowing down its local position relative to a point in the global map of the geographic boundary to a specific point in the geographic boundary with each of the treatment module's sensed and determined local position of the object relative to the vehicle and/or treatment module.

FIG. 24F illustrates an example method 2450 that may be performed by some example systems or subsystems described in this disclosure either online, that is onboard a vehicle supporting one or more modular agricultural observation and treatment systems, subsystems, or components of systems, or offline, that is at one or more servers or edge compute devices. For example, at step 2452, an agricultural observation and treatment system can determine a first vehicle pose estimation. At step 2454, the agricultural observation and treatment system can determine a first treatment module pose estimation. At step 2456, the agricultural observation and treatment system can determine a first orientation of a treatment unit. At step 2458, the agricultural observation and treatment system can determine a first pose estimation of the first treatment unit. This is done by accounting for the pose estimation of the first treatment module operably and rigidly connected to the treatment unit and knowing the prior orientation, such as the first orientation, of a treatment head of the treatment unit, such as the orientation of the treatment head when it last sprayed a projectile. At step 2460, the agricultural observation and treatment system can determine a location of a first target object.

FIGS. 25A-E illustrate example agricultural treatment units, such as treatment unit 2500 of an agricultural treatment system. The agricultural treatment system in this example, can be similar to that of agricultural treatment system 400 with treatment unit 470 similar to that of treatment unit 2500. Additionally, the treatment system having treatment unit 2500 can be similar to that of agricultural treatment system 800 with treatment module 804 and treatment unit 828, including multiple agricultural treatment module 804's and treatment unit 828's operably connect together in agricultural treatment system 800. In this example, the treatment unit 2500 can include a turret assembly with a treatment head 2560, and circuitry, electronic components and computing devices, such as one or more microcontrollers, electronic control units, FPGA, ASIC, system on chip, or other computing devices, configured to receive instructions to point and orient the treatment head 2560, to treat a surface of a real-world object in proximity of the treatment unit 2500. For example, the treatment unit 2500 can emit a fluid projectile of a treatment chemical onto an agricultural object in the real world based on detecting the agricultural object in an image captured and determining its location in the real world relative to the treatment unit 2500.

The treatment unit 2500 can include a gimbal assembly (including linkage components 2542, 2544 and 2550), such that the treatment head 2560 can be embedded in, or supported by the gimbal assembly, effectively allowing the treatment head 2560 to rotate itself and orient itself about one or more rotational axes. For example, the gimbal assembly can have a first gimbal axis, and a second gimbal axis, the first gimbal axis allowing the gimbal to rotate about a yaw axis, and the second gimbal axis allowing the gimbal to rotate about a pitch axis. In this example, a control module of the treatment unit can control the gimbal assembly which changes the rotation of the gimbal assembly about its first gimbal axis, second gimbal axis, or both. A computing module can determine a location on the ground scene,

terrain, or tree in an orchard, or other agricultural environment, and instruct the control module of the treatment unit 2500 to rotate and orient the gimbal assembly of the treatment unit 2500. In one example, the computing module can determine a position and orientation for the gimbal assembly to position and orient the treatment head 2560 in real time and make adjustments in the position and orientation of the treatment head 2560 as the treatment unit 2500 is moving relative to any target plants or agricultural objects of interest on the ground either in a fixed position on the ground or is also moving. The treatment system can lock the treatment unit 2500, at the treatment head 2560, onto the target plant, or other agricultural object of interest through instructions received and controls performed by the control module of the treatment unit 2500, to adjust the gimbal assembly to move, or keep and adjust, in real time, the line of sight of the treatment head 2560 onto the target plant.

In one example, the treatment unit may have a frame 2510 to which the brushless motors 2520, 2530 are attached. The frame 2510 may be attached or secured to other structures, housings or other frames. An encoder 2520 may be coupled with brushless motor 2520. An encoder 2532 may be coupled with brushless motor 2530. The encoders 2520, 2530 may be used by the treatment system to determine or identify a rotational position of the brushless motors 2520, 2530.

Brushless motor 2520 is coupled to linkage bracket 2542 (may be referred to as a servo horn). The linkage bracket 2542 rotates axially in a rotational axis of brushless motor 2520. Linkage bracket 2542 is rotatably coupled to linkage arm 2544. Linkage arm 2544 is rotatably coupled to spraying head assembly (also referred to herein as a treatment head assembly) 2560. The spraying head assembly 2560 is coupled to the linkage bracket 2544 at one location in a manner allowing the spraying head assembly to pivot in a first axis. The coupling of the linkage arm 2544 to the spraying head assembly allows rotation of the linkage arm 2544 while the linkage arm 2544 causes the spraying head assembly 2560 to move to adjust the spraying head 2562 assembly 2560 position. The spraying head assembly 2560 may be more generally referred to as a treatment head assembly. The treatment head assembly, for example, may be configured as a spraying head assembly to emit a fluid at a target object and/or as a laser head assembly to emit a laser light source at a target object.

In one example, one or more brushless motors 2520, 2530 can be configured to change one axes of the treatment head 2560 of the treatment unit. Brushless motor 2530 is coupled to linkage bracket 2550 (may be referred to as a servo horn). The linkage bracket 2550 rotates axially in a rotational axis of brushless motor 2530. Linkage bracket 2550 is coupled to spraying head assembly 2560. The spraying head assembly 2560 is rotatably coupled to the linkage bracket 2550 at two locations in a manner allowing the spraying head assembly to pivot in a second axis within the linkage bracket 2550. The coupling of the linkage bracket 2550 to the spraying head assembly 2560 allows rotation of the spraying head assembly 2560, thereby moving or adjusting the spraying head assembly position.

The frame 2510, linkage brackets 2542, 2550, linkage arm 2544, spraying head assembly 2560 may formed from various materials, such as CNC'd, milled aluminum, cast or molded metals, cast or molded plastics, three-dimensionally printed parts, and any other suitable construction or manufacturing method.

Each of the linkage bracket 2542, 2550 may have legs that extend from the bracket and interact with bracket stops. For

example, FIG. 25D illustrates linkage bracket 2542 have two legs 2542A, 2542B. A bracket stop 2514 (such as an elongated shaft, post or stud) is attached to the frame 2510. As the brushless motor 2520 causes the linkage bracket 2542 to rotate in one direction or the other, the linkage bracket legs 2542A, 2542B may touch upon the bracket stop 2514. The bracket stop 2514 prevents the brushless motor 2520 from rotating further in a particular direction.

The linkage bracket 2542, 2550 may be configured with different degrees of rotational movement and/or the degree of rotational movement. For example, linkage bracket 2542 may have a total degree of rotation of 120 degrees, while linkage bracket 2550 may have a total degree of rotation of 100 degrees. The degree of rotation of either linkage bracket 2542, 2550 is based on the distance between the inner portions of the two legs of the bracket. The degree of rotation may also be referred to as a sweep angle. For example, the sweep angle of each motor may be about 50 degrees on each axis. A shorter distance between the two legs would provide a smaller range of rotation, as compared to a longer distance between the two legs which would provide a greater range of rotation. The system may rotate a motor from one end of the sweep angle to another end of the sweep angle very quickly, such as within micro or milliseconds.

In one example, FIG. 25E illustrates a solenoid valve 2570 (e.g., a fluid flow regulator) can be configured to open or close the valve where constant pressurized fluid from the pump is pumping fluid into the solenoid valve 2570. The treatment unit can emit a projectile by an amount and time the valve is moved from a closed position to an opened position. Also, the rotational axis of motors 2520, 2530 is indicated by the dashed arrows on the respective motors.

While FIG. 25E illustrates a single solenoid valve 2570, the treatment unit may be configured with multiple solenoid valves that are interconnected with different pumps to obtain fluid from different fluid sources. For example, the agricultural treatment system may have four source tanks to hold different types of fluids. Four pumps may be utilized, each of which can be fluidly interconnected with a respective tank and a separate solenoid valve. The pump(s) may pressurize the fluid, for example in a range of 5-250 PSI. For example, when treating an agricultural object such as carrots, the solenoid may control fluid pressure to about 40 PSI. When treating an agricultural object such as apples, the solenoid may control fluid pressure to about 80-100 PSI. The system may be configured to allow two or more fluid source to be emitted at the same time. For example, the spraying nozzle of the spraying head may have multiple spraying ports. Also, the fluid lines may be connected before or after the solenoid valve to mix different fluid sources from the tanks. Pressure sensors may be connected along any part of the fluid channels to determine the pressure of a line. The system may then increase or decrease the fluid pressure to a desired pressure via opening/closing of the solenoid valve, and by adjusting the pumping of the fluid via a pump. The size and volume of each spray emitted by the spraying head may be adjusted by the system based on the target object to be sprayed. For example, the system may spray 1.25 mL at 80 PSI for one agricultural object, and at a different volume and PSI for another agricultural object. Moreover, the spray radius may be determined as a distance from the spray nozzle to the targeted object. For example, the spray radius may be about 1-2" diameter with a coverage at about 1 meter with a +/-60 degree sweep at 0.094 m. Additionally, the spray time may be adjustable and intermittent spray burst of fluid may be controlled by the system.

In one example, the spraying head assembly 2560 has a detachable spraying tip 2562. One or more fluid carrying tubes may be connected to the spraying head assembly 2560 for carrying fluid from one or more solenoids to be emitted via the spraying tip 2562. The spraying tip 2562 may have a threaded base portion allowing the spraying tip 2562 to be attached to spraying head assembly. In some, configurations the spraying tip 2562 has multiple ports that are connected to different fluid lines. This allows the spraying head assembly to emit fluid pumped from different tanks or fluid sources. For example, the spraying tip 2562 may have one, two, three or four ports in the spraying tip. Also, multiple ports of the spraying tip may be connected to the same fluid source. For example, the spraying tip 2562 may have two ports. The two ports may be configured such that fluid emitted at target object would spray the target object at two different locations. The spraying tip can be various types of tips, including for example a hypodermic needle with one or more input ports or orifices for different types of fluids, and with one or more output ports or orifices.

In one example, the system may determine a target object to be of a particular type. Based on the particular type of the target object, the system may choose to treat the target object with a fluid source from one or more tanks. For example, the system may determine the target object to be a weed and then cause fluid from a first tank to be emitted for weed treatment. In another instance, the system may determine the target object to be a plant and then cause fluid (e.g., liquid fertilizer) from a second tank to be emitted at the target object.

This treatment unit is capable of emitting a projectile fluid at a target object in a continuous manner over a period of time, such as many seconds or minutes. The treatment unit may also emit short intermittent bursts of a projectile fluid at a target object. The spraying tip of the spraying head assembly may be configured to emit the projectile fluid in a stream-like manner where the fluid is kept together in a stream to impact a target object in a focused area. The spraying tip of the spraying head assembly may be configured to emit the projectile fluid in a spray-like manner where the fluid is fanned out or the fluid separated to impact a target object in a general area. The spraying head assembly may be configured with one or more spraying tips and/or spraying ports that allow for the emission of either type of fluid spray types. The spraying of fluid from the spraying tip is variable and adjustable based on the target object. In one example, for spraying carrots the distance from the spraying tip to the target object may be about 27 inches. In another example, for spraying apples the distance from the spraying tip to the target about may be about 0.5-1.5 meters.

The agricultural treatment system may use a controller to interact with, obtain sensor data and control the motors and the solenoids. In one embodiment, to instruct the spraying head to a determined spraying head pose, the system instructs the motors to rotate in one direction or the other. The system may instruct the motors to rotate at a constant speed or a variable speed. The system may monitor the axial position of a motor via the coupled encoder to the motor. For example, as the motor rotates the encoder may provide a feedback signal or value to the system that identifies the distance that the motor has rotate. The sample rate of the encoders may be set to a predetermined sample rate or may be a variable sample rate. For example, the system an encoder may be set to obtain readings every 5 milliseconds. For example, the encoders may operate at various frequencies. In one example, the encoders poll information of the motors at about 8 KHz, while sending data to a microcon-

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troller at about 200 Hz. Alternatively, the system may vary the sample rate based on the rotation speed of the motor, the movement speed of the vehicle, or based on some other value or data obtained by the system. When the system determines that the spraying head is moved to a desired spraying head pose, then the system may instruct the solenoid to open and emit a fluid at a target object. The system may then instruct the solenoid to close thereby stopping the fluid from being emitted from the spraying head.

In one embodiment, the system may instruct the motor to rotate in one direction, and via feedback from the encoder, the system may determine that the motor has rotated to a particular position or distance, and then stop or cease instructing the motor to rotate. As described herein, when the motor rotates, the coupled linkage assemblies cause the spraying head to rotate or pivot in a particular direction. For example, the system may be configured that when a first motor rotates in a first direction, the spraying head would pivot or move along a first axis in a first direction, and when the first motor rotates in an opposite second direction, the spraying head would pivot or move along the first axis in a second direction opposite to the first direction. Moreover, the system may be configured that when a second motor rotates in a first direction, the spraying head would pivot or move along a second axis in a first direction, and when the second motor rotates in an opposite second direction, the spraying head would pivot or move along the second axis in a second direction opposite to the first direction.

The motors may be configured so that the plane of rotation of each motor are substantially perpendicular to each other. For example, the plane of rotation of each motor may be aligned from about 75 to 105 degrees of each other. In one configuration, the plane of rotation of each motor is about 90 degrees to one another.

In one example, the motors may be configured such that the center axis of each motor is aligned with a first and second rotational axis of the spraying head assembly. In a neutral or home position, the spraying head assembly may be positioned such that the spraying head assembly is positioned at an x, y coordinate of 0,0. In this position, a line from the center axis of each motor would extend through the respective rotational axis of the spraying head assembly.

FIG. 26A illustrates example implementations of method 2600 that may be performed by some example systems described above including system 100, agricultural treatment system 400, system 600, and system 800. For example, in one mode of operation, at step 2610, the agricultural treatment system provides a spraying apparatus having two motors with linkage assemblies that are used to control the movement of a spraying head. The spraying apparatus, for example, may be the treatment unit 470, 1653, 1400, 2500 as described herein.

At step 2612, the agricultural treatment system 400 determines a first target object to be sprayed with a fluid emitted from the spraying head. As described herein, the system 400 may determine a location and/or position of the target object in the external environment.

At step 2614, the agricultural treatment system adjusts a position of the spraying head via rotation of the first and/or second motors. As described herein, the system 400 may provide instructions to the motors of the treatment units to position the spraying head such that the spraying head may emit a fluid that would spray from the spraying head onto the target object.

At step 2616, the agricultural treatment system regulates a flow of the fluid obtained from a fluid source by controlling a regulator to open and close causing a flow of the fluid to

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start and stop. For example, the system 400, may regulate the flow and overall pressure of the fluid by opening and closing a valve to release pressurized fluid from the regulator through the spraying head. One or more fluid pumps may be used to pump fluid from a fluid source and pressurize the fluid.

At step 2618, the agricultural treatment system emits the fluid, obtained from the fluid source, via the spraying head at the first target object. The fluid is emitted from the spraying head for a predetermined amount of time. The system 100 may release the fluid in a continuous stream or in intermittent bursts of fluid emitted towards the target object.

At step 2620, the agricultural treatment system maintains the spraying of the fluid at the first target object while a vehicle moves along a path, where the spraying apparatus is connected to the vehicle. As discussed herein, the system 400 may determine a pose of the vehicle while the vehicle is moving along a path. The system 400 translates the pose of the vehicle to commands to instruct movement of the spraying head such that the spraying head is "locked onto" the target object. In other words, as the vehicle moves along a path the system determines adjustments that are to be made so that the spraying head moves in a way that the emitted fluid continues to spray towards the target object. The system 400 calculates the position of where the spraying head should be oriented in relation to the changing pose of the vehicle. The system 400 provides causes the motors to rotate as needed to adjust the spraying head.

FIG. 26B illustrates example implementations of method 2650 that may be performed by some example systems described above including system 100, agricultural treatment system 400, system 600, and system 800. The system may controllers and encoders to control the position of the motors, which in turn control the position of the spraying head assembly.

At step 2660, the agricultural treatment system determines a location of a first target object. For example, the system identifies a target object in a 3-dimensional space to be treated. At step 2662, the system determines a position to move a spraying head assembly to emit a fluid at the first target object. For example, the system may determine a spraying assembly pose to orient a spraying head of the spraying assembly such that the spraying head may emit a projective fluid at the target object. At steps 2664, 2666, the system instructs the motors to rotate in one direction or another. The motors may be instructed to be moved simultaneously or one after another. The system instructs a first motor to rotate in a rotational direction, thereby moving the spraying head assembly in a first direction. For example, via interconnected linkages and/or gimbal structures the rotation of the first motor causes the spraying head assembly to rotate or pivot along a first axis. The system instructs a second motor to rotate in a rotational direction, thereby moving a spraying head assembly in a second direction. For example, via interconnected linkages and/or gimbal structures the rotation of the second motor causes the spraying head to rotate or pivot along a second axis.

While the motors are instructed to rotate, encoders coupled to the motors may read rotational positions of the motors. The encoders may operate simultaneously and read rotational values of the motors. At step 2668, the system determines a plurality of first encoder output values, where the first encoder is configured to identify rotational positions of the first motor. At step 2670, the system determines a

plurality of second encoder output values, the where the second encoder is configured to identify rotational positions of the second motor.

Lastly, the agricultural treatment system stops rotation of the motors when the respective motors have moved the spraying head assembly into a desired position such that the target object may be treated with a fluid and/or laser light source. At step 2672, the system stops rotation of the first and second motors when the first and second encoder output values are equal to a location of a desired or position of the spraying head assembly. In some instances, the first or second motors may stop rotating when the spraying head assembly has reached a respective x/y position. When the spraying head assembly has reached its desired position, the system may instruct the treatment unit to treat the target object. Also, as described previously, while the vehicle is in motion the system may instruct the motors to slightly rotate so that the system maintains the spraying head assembly targeted at the target object. These micro-adjustments allow the spraying head to stay tracked onto a target object and account for changes in the pose of the vehicle and/or treatment.

FIG. 27 is a diagram illustrating pose determination of the agricultural observation and treatment system, according to some examples. The figure illustrates an example of a vehicle 1720 with treatment unit 1653 attached. A vehicle 1720 is shown moving along a path 2712. For example, as illustrated in FIG. 24, a vehicle 1720, such as a tractor may be configured to tow one or more treatment systems along a vehicle track have multiple lanes for the vehicle 1720 and tow support.

If the vehicle 1720 were to remain in a stopped positioned, the system 400 could spray the target object 2720 and then move onto the next target object, and then stop and spray the next target object. However, the system 400 is flexibly configured to allow the continuous movement of vehicle 1720 and make adjustments to the position of the spraying head of the treatment unit 1653. While the vehicle 1720 is moving along the path, the system 400 may determine a pose for the vehicle (e.g., Vehicle POSE₀, Vehicle POSE₁, Vehicle POSE₂ . . . Vehicle POSE_n) and/or for the treatment unit 1653 (e.g., Unit POSE₀, Unit POSE₁, Unit POSE₂ . . . Unit POSE_n). For example, using onboard navigation and IMU sub-systems, the system 400 may determine multiple locations or positions of the vehicle while the vehicle is moving along the path 2712.

As noted above, the treatment unit 1653 emits at fluid at a target object 2720. While the vehicle 1720 is moving, the system 400 determines a translation of Vehicle POSE_n and/or the Unit POSE_n to a Spray POSE_n such that spraying head is oriented or positioned to allow an emitted projectile fluid to spray upon a desired target object. For example, the system 400 may determine that a target object 2720 is to be treated. The system 400 determines a Vehicle POSE₀ and/or a Treatment Unit POSE₀. The system 400 will provide instructions/signals to the motors of the treatment unit 1653 to adjust one or more axis (e.g., pitch 2732, yaw 2734 and/or roll) of the spraying head. As the vehicle 1720 move along the path 2712, the system 400 periodically determines n poses of the vehicle 1720 and/or the treatment unit 1653. The system 400 then translates the periodically determined n poses to an n spraying head pose such that the treatment unit may continuously spray the target object 2720 while the vehicle is moving. The system 100 may evaluate speed, movement, velocity, direction, altitude, location of the vehicle 1720 and/or treatment unit 1653 and determine a pose for the spray head.

As used herein, pose may be understood to be a location and orientation of an object relative to a frame of reference (e.g., x, y, z, phi, theta, psi, where x=an x-axis coordinate in a 3-dimensional coordinate system, y=a y-axis coordinate in a 3-dimensional coordinate system, z=a z-axis coordinate in a 3-dimensional coordinate system, phi=degree or position of roll, theta=degree or position of pitch, psi=degree or position of yaw. For example, the agricultural treatment system may determine pitch, roll and yaw values of the vehicle, treatment unit and or the spraying head assembly. In some embodiments, the agricultural treatment system may not be configured to identify a pitch, roll and/or yaw of the vehicle, treatment unit and or spraying head. In such instances, the value for these variables may be set to zero.

A global frame of reference may be provided for an environment in which the agricultural treatment system operates. For example, a global frame of reference may be set to a particular geospatial location or the fixed reference point on a property (e.g., a corner of a barn, structure, a 5g/wifi/gps tower, etc.). The point of reference may be defined as (x=0, y=0, z=0, phi=0, theta=0, psi=0). The agricultural treatment system may determine multiple poses of the vehicle, in relation to the point of reference, as the vehicle moves about the environment. The pose of the vehicle may be defined as vehicle (x_n, y_n, z_n, phi_n, theta_n, psi_n)_{time_interval}; the system may determine the nth values at a particular time interval being sampled at a particular sample rate (such as 200-5000 times a second). The agricultural treatment system may also determine a pose for a treatment unit, such as treatment unit (x_n, y_n, z_n, phi_n, theta_n, psi_n)_{time_interval}. The agricultural treatment system may also determine a pose for a sprayer head of the treatment unit, such as spraying head (x_n, y_n, z_n, phi_n, theta_n, psi_n)_{time_interval}. The sprayer head may have a pose relative to the vehicle pose, may have a pose relative to the treatment unit pose and/or may have a pose directly in relation to the global frame of reference. The agricultural treatment system may determine a final spraying head (x, y, z, rho, theta, psi) pose to be used to adjust the spraying head to a different position. The final pose can be relative to the body of the treatment unit, the sprayer apparatus components, the vehicle or components thereof, and/or relative to some (0,0,0,0,0,0) location of the farm.

As described herein, the agricultural treatment system may determine the pose of the vehicle and/or treatment unit and translate the pose into commands or instructions to adjust a spraying head assembly to emitted fluid at a desired target object. In other words, the agricultural treatment system may identify a target object to be sprayed, orient a spraying head assembly to the target object and then control fluid spraying operations to emit fluid from one or more fluid sources at the target object. The system can move along a path and make adjustments to the spraying head assembly such that the fluid is continuously sprayed at the target object and/or detect new target object(s) to be sprayed and then position the spraying head assembly to the detected new target object(s).

While the above describes pose determination for a vehicle or treatment unit, the system may determine a pose for any part or object of the system (e.g., a seat, the vehicle, a wheel, treatment unit, spraying head, spray box, turret, nozzle tip, etc.). The pose may be determined with one or more different sensors (e.g., a camera positioned can obtain imagery of different parts or components), and the system can estimate the pose of the parts or components. The system may use computer vision, lidar, radar, sonar, GPS, vslam, wheel encoders, motor encoders, IMU, cameras on a spray

box. In some embodiments, the system may be configured to determine, for example, the vehicle and multiple treatment units. This may be done for example where the vehicle is pulling a trailer with many spray boxes places along a frame or support that has many wheels. Each of the spray boxes may have different poses due to the ruggedness or unevenness of the terrain.

The system may be configured to determine particular poses of the vehicle as a global pose and the treatment units as a local pose. A local pose for each treatment unit may be determined in relationship to the global pose, and/or may be determined individually for the treatment without relationship to the global pose. The system may use the global pose (a.k.a. vehicle pose) as a localization method to determine its relationship to a real-world environment. And the system sensors may obtain information about the real-world environment. The allows the system to build a high map of an agricultural environment (such as a farm). In one embodiment, the system uses a navbox and sensors to determine the global pose.

The system may use the local pose of a particular component for certain operations. As discussed herein, the system may determine a pose for a treatment unit and a spraying head. The system would use the local pose of these components to determine its physical relationship as between the component and a target object. For example, two different treatment units may each have a spraying head. A first treatment unit and spraying head may need to spray a first target object. A second treatment unit and spraying head may need to spray a second target object. In this situation, the system may determine a pose for each spraying unit and each of the spraying heads, and then maneuver or orient the spray nozzle of the spraying heads toward their respective target object. In one embodiment, the system would use the local poses to orient the spraying heads to emit a projectile fluid at the respective target objects.

In one example, the agricultural treatment system determines multiple vehicle and/or treatment unit poses. The system evaluates a first pose, and then periodically determine subsequent poses. The system may calculate the difference or changes of the coordinate values from the first pose and a subsequently obtained posed. In other words, the system may calculate the movement of the vehicle and/or treatment unit. The calculated difference or changes then may be translated to a desired pose for the spraying head. The sample rate of the pose can be configured as set rate or a variable rate. For example, the system may evaluate its pose at predetermined intervals, such as 5 milliseconds. In an alternate configures, the system may use a variable sample rate such that when the vehicle speed increase, the pose determination rate increases. For example, the sample rate for determining a pose may be 5 milliseconds where the vehicle speed is from 1-3 mph, and the sample rate increase to a higher rate, such as every 2 milliseconds, where the vehicle speed is over 3 mph.

In one example, pose for the vehicle may be determined by evaluating data from various sources, such as onboard cameras, GPS, IMU's, wheel encoders, steering wheel encoders, LiDAR, RADAR, SONOAR, and additional sensors that may be used to determine the vehicles position in a real-world environment. The system for example may evaluate the sensor data for example at 200-5000 hz.

In one example, pose for a treatment unit **2800** may be determined in similar manner to the vehicle, as one or more treatment units would be configured in a fixed position in relationship to the vehicle. A change in pose of the vehicle may be considered to be the same change in pose of the

treatment unit. The treatment unit may have one or more processors and microcontrollers to monitor and determine the pose of the treatment. The processors and microcontrollers are configured to keep track of treatment unit pose.

The treatment unit process periodically requests from a vehicle processor system pose information for the vehicle. The treatment unit process may then determine its pose by using the pose of the vehicle and may offset the pose of the vehicle based on a distance value from where a respective treatment unit is positioned relative to the point of where the pose is determined for the vehicle. As discussed herein, the agricultural treatment system may include multiple treatment units. Each of the processor of the respective treatment unit may determine the pose for the treatment unit. Thus, each of the treatment units may have a unique pose relative to the determined vehicle pose.

Each of the treatment unit processors and microcontrollers may determine a spraying head pose. As noted above, each of the treatment unit continually poll or request a vehicle pose from the vehicle's computer system and may determine a treatment unit pose. The treatment unit processors are configured to determine and evaluate the positions of the motors via the encoders coupled to the motors. The treatment unit processor obtains information from the microcontroller **2875** of the treatment unit regarding the encoder output. In other words, the encoders provide data output about motors' position and/or rotational movement. The microcontroller receives the encoder output data and provides the data to the treatment unit processor. Similarly, the microcontroller may receive instructions or data from the treatment unit processor, and the microcontroller in turn may provide or translate the received instructions or data to instructions, voltage and/or commands to that cause the motors to rotate in one direction or the other. The axial rotation of the motor then causes the linkage assemblies to rotate thereby causing the spraying head to change. The treatment unit processor may determine a spraying head pose and provide instructions to the microcontroller **2875** to then make adjustments to each of the motors such the that the spraying head is adjusted to the desired spaying head pose.

In one example, a first 3-dimensional coordinate system may be used for the vehicle and/or treatment unit pose, and a second 3-dimensional coordinate system may be used for the spraying head. Changes in the first 3-dimensional coordinate system may be mapped to the second 3-dimensional system. Distance moved in the second 3-dimensional coordinate system may then be calculated and the distance moved can be translated into instructions/commands to rotate the motors by a certain amount, degree or time to achieve a desired position of the motor.

While the above discussion, focuses on an example of a single treatment unit, the system may determine poses for multiple treatment units and adjust the pose of the respective unit spraying head such that each of the spraying heads may lock on to their respective target objects.

FIG. **28A**-FIG. **28E** are block diagram illustrating various configurations of the system with treatment units **2800** configured for various fluid source and spraying tip options. Referring to FIG. **28A**, the agricultural treatment system has onboard circuitry, processors and sensors that allows the system to obtain imagery of agricultural objects and then identify a target object to be sprayed. Furthermore, the agricultural treatment system has onboard circuitry, process and sensors that allows the system to determine position of the vehicle and/or treatment unit in a three-dimensional space. Moreover, the agricultural treatment system includes

other cameras and computer vision sensor to obtain and process imagery of external real-world objects **2884**. For example, block **2850** illustrates a subsystem having a computer unit **2851**, communication channel **2854**, cameras **2853**, machine learning model and computer vision algorithm **2855**, lights **2856**, and other sensors **2852**. For example, the system may use GPS location data, IMU data to identify inertial movement and distance moved. Over a period of time, the system may determine multiple poses of the vehicle and/or treatment unit and convert/translate these poses that the spraying head would need to be positioned into such that the spraying head would maintain an emit spray at the target object while the vehicle is moving.

The subsystem **2850** interacts with a treatment unit **2800**. While a single treatment unit is shown, the subsystem **2850** may interact with and control multiple treatment units. Generally, the treatment unit **2800** includes a microcontroller that is operably coupled with one or more solenoids **2870**, pumps, multiple motors **2820**, **2830** and multiple encoders **2822**, **2832**. The treatment unit **2800** may draw fluid from one or more source tanks **2804**. The subsystem **2850** may communicate via communications channel **2842** with another computer system. For example, the subsystem **2850** may receive global registry information and data (e.g., global registry information such as GPS location data, IMU data, VSLAM data, etc.).

The microcontroller **2875** may control or interact with the pump, solenoid **2870A**, motors **2820**, **2830** and encoders **2822**, **2832** to position the treatment head assembly **2860** and emit fluid from one or more fluid sources. For example, based on interaction with the subsystem **2850**, the treatment unit **2800** may control the position of a treatment head assembly **2860** to orient the treatment head assembly **2860** such that the treatment head assembly **2860** may emit a fluid at a target object **2885**. FIG. **28A** illustrates a treatment unit with a single fluid source tank **2804A** and a single solenoid **2870A**, and a spraying head **2862A** with a single port.

FIG. **28B** illustrates a treatment unit **2800** with multiple fluid source tanks **2804A**, **2804B**, **2804C** and **2804D**, and multiple solenoids **2870A**, **2870B**, **2870C** and **2870D**. Fluid may be pumped from the respective fluid source tanks, through tubing **2806** and emitted via the spraying head assembly **2860** via spraying tip **2862B** with a single port.

FIG. **28C** illustrates a treatment unit **2800** with multiple fluid source tanks **2804A**, **2804B**, **2804C** and **2804D**, and multiple solenoids **2870A**, **2870B**, **2870C** and **2870D**. Fluid may be pumped from the respective fluid source tanks **2804A**, **2804B**, **2804C** and **2804D** and emitted separately via separate spraying tip ports on the spraying head **2860** assembly. In the example, the spraying head has four channels fluid coupled using separate tubing to the fluid source tanks **2804A**, **2804B**, **2804C** and **2804D**. The microcontroller **2875** may control multiple solenoids **2870A**, **2870B**, **2870C** and **2870D** to control the release of fluid from the fluid source tanks **2804A**, **2804B**, **2804C** and **2804D**.

FIG. **28D** illustrates a treatment unit **2800** with multiple fluid source tanks **2804A**, **2804B**, **2804C** and **2804D**, and multiple solenoids **2870A**, **2870B**, **2870C** and **2870D**. Fluid may be pumped from the respective fluid source tanks **2804A**, **2804B**, **2804C** and **2804D** and emitted via single spraying tip **2862B** via the spraying head **2860** assembly. In the example, the spraying head has four channels fluid coupled using separate tubing to the fluid source tanks **2804A**, **2804B**, **2804C** and **2804D**. The microcontroller **2875** may control multiple solenoids **2870A**, **2870B**, **2870C** and **2870D** to control the release of fluid from the fluid source tanks **2804A**, **2804B**, **2804C** and **2804D**.

FIG. **28E** illustrates an example of the treatment unit **2800** where multiple fluid sources may be combined or mixed with a primary fluid source. The micro controller **2875** may operate a solenoid **2870** to control the flow of a primary fluid source, such as water. The primary fluid source may then be combined with one or more secondary fluid sources **2805A**, **2805B**, **2805C**, **2805D**. The secondary fluid sources may be concentrated chemicals or fertilizers that are mixed with the primary fluid source to dilute the concentrated chemicals. While not shown, each of the secondary fluid sources **2805A**, **2805B**, **2805C**, **2805D** may be controlled via separate solenoids and pumps to cause the secondary fluid sources to disperse fluid from a tank. The combined mixture of the primary fluid source and the one or more secondary fluid sources are then emitted via the spraying head assembly **2860** via spraying tip **2862E** with a single port.

FIG. **28F**-FIG. **28H** are block diagram illustrating various configurations of the system with treatment units **2800** configured for emitting laser light for treatment of agricultural objects. Referring to FIG. **28F**, the agricultural treatment system has onboard circuitry, processors and sensors that allows the system to obtain imagery of agricultural objects and then identify a target object to be sprayed. Furthermore, the agricultural treatment system has onboard circuitry, process and sensors that allows the system to determine position of the vehicle and/or treatment unit in a three-dimensional space. Moreover, the agricultural treatment system includes other cameras and computer vision sensor to obtain and process imagery of external real-world objects **2884**. For example, block **2850** illustrates a subsystem having a computer unit **2851**, communication channel **2854**, cameras **2853**, machine learning model and computer vision algorithm **2855**, lights **2856**, and other sensors **2852**. For example, the system may use GPS location data, IMU data to identify inertial movement and distance moved. Over a period of time, the system may determine multiple poses of the vehicle and/or treatment unit and convert/translate these poses that the spraying head would need to be positioned into such that the spraying head would maintain an emit spray at the target object while the vehicle is moving.

The subsystem **2850** interacts with a treatment unit **2800**. While a single treatment unit is shown, the subsystem **2850** may interact with and control multiple treatment units. Generally, the treatment unit **2800** includes a microcontroller that is operably coupled with multiple motors **2820**, **2830** and multiple encoders **2822**, **2832**. The treatment unit **2800** may cause a laser light source **2884** to generate a laser light. The subsystem **2850** may communicate via communications channel **2842** with another computer system. For example, the subsystem **2850** may receive global registry information and data (e.g., global registry information such as GPS location data, IMU data, VSLAM data, etc.).

An example embodiment of the laser may be a femto or laser diode-array based laser. In one example, the system uses a laser with a power range of 0.5-90 Watts, with an activation speed of 1-5 kHz, with a dwell time of 1 ns-1 S. The frequency may be from Ultraviolet to Infrared. The laser may be class 2, 3 or 4 with a single to array pulse or continuous wave. The spot size generated by the laser may be from 1 micron to 1 meter in diameter. In another example, the system uses a laser with a power range of 0.5-20 Watts, with an activation speed of 10-240 Hz, with a dwell time of 1 microsecond to 20 milliseconds. The frequency may be in the range of 400-1550 nanometers. The laser may be class 4 pulse or continuous wave. The spot size generated by the laser may be adjustable.

While the two example illustrate lasers that may be used by the system, the laser may be configured to operate with different ranges. The laser may have a power range of 0.5-90 Watts, 0.5-20 Watts or 0.5-5 Watts. The system may activate the laser per second in different ranges, for example from 1-5000 Hz, 1-240 Hz or 10-40 Hz. The laser may have different dwell time ranges (i.e., how long the laser stays on a single pulse) such as 1 nanosecond to 1 second, 1 microsecond to 100 milliseconds or 5 to 20 milliseconds. The wavelength of the laser may be UV, visible or Infrared. The laser may have a color range of 400-800 nanometers or 850-1650 nanometers, or from 435-1550 nanometers. The type of the laser may be a continuous wave diode laser, pulse diode laser or a laser diode array. The laser may be class 2, 3, 3b or class 4. The spot size of the laser may be from 1 micron to 1 meter.

FIG. 28F illustrates a treatment unit **2800** with a laser light source **2884**, an optical control mechanism **2802** (e.g., an optical box), and fiber optic cable **2887** a moveable head assembly **2860** and laser emitting tip **2862A**. The agricultural treatment system may generate a laser light via the laser light source **2884**. The optical control mechanism **2802** may include moveable optical components, such concave and convex lenses, to adjust the intensity, focus and or power of the light received from the laser light source via the fiber optic cable **2887**. The optical control mechanism **2802** may be part of the moveable head assembly **2860**. The lenses allow for converging and/or diverging of light by shifting the distance of the lenses to each other thereby adjusting the focus of the laser light. This adjusted laser light source is then emitted via the laser emitting tip **2862A**. The moveable head assembly is adjustable (as described herein) to orient the laser emitting tip towards an agriculture object such that the emitted laser light may treat an agriculture object. For example, the laser light may photoablate plant tissue of an agriculture object. Different laser light sources may be suitable for generating/producing a laser beam to photoablate plant tissue.

FIG. 28G illustrates a treatment unit **2800** with a laser light source **2884**, an optical control mechanism **2802**, and a mirror assembly **2888** a treatment head assembly **2860** and laser emitting tip **2862A**. The laser operates similarly as described with respect to FIG. 28F, however, instead of using fiber optic cabling, the treatment unit uses a mirror assembly to direct the laser light received from the optical control mechanism **2802**, to the laser emitting tip **2862A**.

FIG. 28H illustrates a treatment unit **2800** with a laser light source **2889**, a moveable head assembly **2860** and laser emitting tip **2862A**. In this example, the laser light source is contained or made part of the moveable head assembly **2860**. A small form factor laser light source may be integrated with the moveable head assembly **2860**.

FIG. 29 illustrates example implementations of method **2900** that may be performed by some example systems described above including system **100**, agricultural treatment system **400**, system **600**, and system **800**. For example, in one mode of operation, at step **2910**, the agricultural treatment system determines a first pose of a treatment unit or vehicle. The determination of the pose is described above as to FIGS. 27 and 28. At step **2920**, the agricultural treatment system, translates the first pose of the of the treatment unit or vehicle, and determines a first pose of a spraying head of the treatment unit. At step **2930**, the agricultural treatment system adjusts the spraying head position based on the determined first pose of the spraying head. The spraying head, for example, may be repositioned by instructing one or more motors of the agricultural treat-

ment system to rotate, thereby causing the spraying head to pivot or rotate along one or more axis. At step **2940**, the agricultural treatment system controls a fluid flow regulator (such as a solenoid or another control device) to allow fluid to be emitted from the spraying head at a target object. The vehicle may move along a path, and while doing so the agricultural treatment system may periodically determine n poses. At step **2950**, the agricultural treatment system determines an nth pose of the treatment unit and/or vehicle while the vehicle moves along a path and determines a nth pose for the spraying head. At step **2960**, the agricultural treatment system adjusts the spraying head position based on the determined nth pose of the spraying head. Steps **2940** through **2970** may be repeated. The periodic pose determine process allows the agricultural treatment system to continually adjust the spraying head while the vehicle is moving so as to maintain the emitted spray at the target object.

While the agricultural treatment system moves along a path, the system may continuously evaluate for additional target objects to be sprayed using one or more treatment units. As described above, the spraying head assembly may be positioned such that a fluid may be emitted at an identified target object. After a target object is sprayed with a fluid, the system may instruct the spraying head assembly to reposition to a ready position, such as the neutral position of $x=0$, $y=0$, or at some other ready position. For example, while the vehicle is moving forward a spraying head assembly may be pointed towards the forward path of movement. Doing so would allow (i.e., get ready) the spraying head assembly to be in a ready position when a new target object is detected. The system may instruct the treatment unit spraying head assembly to move into the ready position when the system is initially powered on. Moreover, the system may instruct a particular treatment unit spraying head assembly to move into the ready position after the spraying of a then current target object is completed. Moving the spraying head assembly into a forward ready position allows the agricultural treatment system to readily start spraying subsequent target objects as soon as they are detected without first having to move the spraying head to the target object.

In one example, the treatment unit **2500**, may have a high-powered laser unit or laser chip embedded in or supported by the treatment unit **2500**, can be configured to treat portions of plants that are larger than plant typically only grow a few inches or feet above the ground. These plants can include trees, orchard trees, or other plants with one or more trunks, shrubs, bushes, or other plants grown on trellises or other human made mechanisms such that a horizontally or top mounted treatment unit **2500** is more practical rather than a treatment unit substantially pointing at the ground with rotational freedom.

While the above disclosure contemplates the control of a spraying head assembly for the emission of a projectile fluid, the spraying head assembly may be replaced with a controllable laser head assembly. Also, a laser source may be attached to the spraying head assembly. The system may control the positioning of a laser head assembly to position the laser head assembly to direct an emitted laser beam at a target object. The laser beam may be used, for example, to ablate, burn or otherwise treat the target object with a laser light beam. Additionally, different laser beams of different wave lengths may be configured on the laser head assembly. The laser light may be focused to a desired diameter to treat a target object. In one embodiment, the spraying head includes a spraying nozzle and a laser emitting tip and may

be disposed next to each other such that either a laser or a spray nozzle can activate upon targeting an object of interest for treatment.

The system may treat the target object based on the identified target object. For example, the system may set operative parameters of the laser to treat the target object (such as duration, frequency, wavelength, laser pulse repetition, etc.). Different target objects may be treated with different parameters using emitted laser light from the treatment head.

In one embodiment, the agricultural treatment system may be configured to monitor the health of the spraying head and determine whether the spraying head is accurately emitting a fluid at a target object. In some instances, the spraying tip may build up residue or other particulate. For example, the spraying head may disperse a fluid containing a solution of salts or of other compounds. Over time, salts or other compounds from the solutions may build up on the outer surface of the spraying head tip and cause an emitted fluid to deviate from an intended projected course. In other words, the emitted fluid may miss an intended target object if the emitted fluid deviates in its projected direction.

The system may correct for a deviation of the projected fluid by adjusting the spraying head to account for the deviation. As the fluid is emitted from the spraying head, an onboard camera may obtain imagery of the fluid as the fluid is emitted or projected at an intended target object. The system may determine whether or not the intended target object was actually sprayed by the emitted fluid. The system may calculate an adjustment by determining a distance and position of where the emitted fluid was actually sent, and where the fluid should have landed on the target object. The system then can determine an offset to make a spraying head positional adjustment such that subsequent emitted fluids would land at an intended location of the target object.

In one mode, the system may continuously emit fluid in a spray or in bursts of fluid, and then determine the location of where the fluid is projected. The system may make slight or micro adjustments to the position of the spraying head assembly until the emitted fluid is sprayed at the target object at an intended location. The positional adjustment values then may be used as an offset for subsequent spraying. For example, an emitted spray may be spraying 1.5 inches to the left of an intended location of a target object. The system can then move the spraying head towards the right of the target object and determine when an emitted projectile fluid accurately hits the target object. This allows the system to determine what position or distance the spraying head needs to move to correct for spraying location error.

In one example, the system may use computer vision to track a target object while the vehicle is in motion. The system may evaluate imagery of a target object with onboard cameras. The system may determine the position of features or objects in an image and evaluate the positional changes of pixels of the object moving in the image. The system may translate the pixel movement to adjustments to the spraying head assembly such that the system adjusts the spraying head assembly so that the treatment unit accurately emits a fluid at the target object.

This process of correcting for spraying head projectile deviation may also be used when a new spraying head tip is attached the spraying head assembly. This process allows for initial configuration of a treatment unit to identify and correct for any deviation of an emitted fluid from the spraying tip.

FIG. 30 illustrates example implementations of method 3000 that may be performed by some example systems described above including system 100, agricultural treatment system 400, system 600, and system 800. At step 3010, the agricultural treatment system determines a relative location of a target object. At step 3020, the system may emit a fluid at the target object via a treatment unit. At step 3030, the system monitors and tracks the fluid emitted at the target object. At step 3040, the system determines whether the emitted fluid sprayed at the target object at an intended location. If yes, then at step 3044, the system determines a relative location of a second target object and continues to step 3020. If no, then at step 3050, the system determines an offset for the position and/or orientation of a spraying head. Next at step 3060, the system determines a second relative location of the first target object. Then at step 3070, the system positions and/or orients the spraying head, in part, using the offset, to target the first target object.

For example, the agricultural treatment system may determine a first target object to be sprayed. The treatment unit emits a fluid at the first target object via a spraying head. The system may use an onboard computer vision system to monitor the emitted fluid at the target object. The system determines whether the emitted fluid sprayed the target object at an intended location. The system may evaluate obtained digital images and identify whether or not the emitted fluid actually sprayed the target object. The system may also determine at what distance and location the projectile stream deviated from the target object. The system may determine an offset for the position of the spraying head. For example, the system may calculate a positional adjustment to the spraying head so that the spraying head would spray the fluid at an intended target object. The system then may spray subsequent target objects. The system may determine a second target object to be sprayed. The system may then emit a fluid at the second target object via the spraying head using the offset. For example, the spraying head would be positioned in part using the determined offset.

FIG. 31 illustrates example implementations of method 2600 that may be performed by some example systems described above including system 100, agricultural treatment system 400, system 600, and system 800. The agricultural treatment system may identify and determine that multiple targets are in close proximity to one another and a particular target object can be treated while the spraying head assembly is positioned to emit fluids toward one of the target objects. The treatment unit may be configured such that system may emit a fluid from one source tank at a first target object, and then emit a second fluid from another source tank. At step 3110, the system determines a first target object and a second target object for treatment. At step 3120, the system determines that the first target object is a first type of a target object, and the second target object is of a second type of a target object. For example, the system may recognize the particular type of a target object using various computer vision and object detection techniques. At step 3130, based on the first determined object type, the system may treat the first target object with a first treatment from a first fluid source. The system may cause fluid from a first source tank to be emitted at the first target object. At step 3140, based on the second determined object type, treat the second target object with a second treatment from a second fluid source. For example, two target objects may be identified being close in proximity to one another.

FIG. 32 illustrates example implementations of method 3200 that may be performed by some embodiments of the systems described above including system 100, agricultural

treatment system 400, system 600, and system 800. A treatment unit may pump fluid from different or multiple tank sources and treat a target object by emitting fluid from the different tank sources. For example, it may be desirable in some instances, to treat a particular type of a target object, such as a bud or a flower, with fluid from multiple tank sources, whereas a further developed agricultural object, may only need treatment from one of the tank sources. At step 3210, the system may determine a first target object for treatment. As discussed herein, the system may identify a target object to be treated. At step 3220, based on a first determined target object, the system may select from two or more fluid sources to treat the target object. As discussed herein, the system may include multiple tanks or containers of different fluids that may be used to treat a target object. The treatment unit may be configured to cause fluid pumped from the multiple tanks to be mixed together and emitted from a single spraying tip port, pumped separately and emitted sequentially from one source tank and then another, or may be emitted from a tip that has multiple spraying tip ports (e.g., a 4-port spraying tip). At step 3230, the treatment unit emit a first fluid at the determined target object from a first fluid source. At step 3240, the treatment unit emits a second fluid at the determined target object from the second fluid source. The system may determine that the target object is of a particular type of object, and the select from one or more source tanks to pump fluid and then treat the target object with the fluid.

As discussed above, the system may emit fluids from two different source tanks. This allows the system to perform an action of applying two or more chemicals in real time from the different source tanks. The system may then cause a desired result on an agricultural object to upon contact of the different chemicals being applied to the agricultural object. In other words, the system may apply two or more chemicals to the agricultural object where the two or more chemicals have a chemical reaction that would occur on the agricultural object where the chemicals are applied. In some instances, these chemicals would otherwise create immediate chemical reaction if the chemicals were stored or mixed in the same source tank, and cause the mixed chemical to form undesirable solid, pellet, gas and/or a viscous liquid. Also, some chemicals when mixed have a short effective time range and would be ideally mixed together on the agricultural object. While the foregoing discusses emitted fluids, the source tanks may hold solids, gels, powders, pellets and other non-fluidic chemical that may be emitting via a treatment unit. The system provides for the selective binary deposition of chemicals to be applied onto an agricultural object. In one example, a first chemical is pumped from a first fluid source and emitted from a first spraying tip onto the agricultural object. A second chemical is pumped from a second fluid source and emitted from a second spraying tip onto the agricultural object where the first chemical was sprayed. When the second chemicals contact the first chemical, a chemical reaction is initiated thereby transforming the mixed chemicals into another state, where otherwise premixing two chemicals or chemical compounds into liquid, solid, or gas form would not be desired

For illustrative purposes, the chemicals applied to the agricultural object may include, for example, any of the following which may be held in different source tanks:

1. A reactant and a material. For example, a reactant is a substance that takes part in and undergoes change during the chemical reaction.

2. A reactant and a reactant.

3. A reactant and a reagent. For example, a reagent is a substance that may cause a chemical reaction and is typically consumed during the course of the chemical reaction.

4. A substantially inert material (such as powdered sodium) and one reactant.

5. A catalyst and a solvent. For example, a catalyst is a substance that increases the rate of a chemical reaction without itself undergoing any permanent chemical change. For example, a solvent is a substance that is able to dissolve other substances.

6. Chemicals that cause a pH change (e.g., the change in pH triggers reaction) (for example, a premixed acid that cannot do damage to an agricultural object, or cause harm to a human or mechanical parts of an agricultural system, at its current acidity but will at a certain different acidity. A moving treatment system can then deposit a second fluid projectile that is a base (the base may not treat the agricultural object solely but can change the acid level of the first fluid projectile emitted onto the surface of the same agricultural object (i.e., that have chemical properties to interact with the first chemical's acidity (such as an acidic chemical and a base chemical)).

7. Chemicals that cause a pH trigger (such as an acidic chemical and a basic chemical, where the applied chemical mixture interacts with the agricultural object. For example, with pH adjusting chemicals that may be applied or removed to vary the pH level to activate when applied.

8. Destruction of Barrier. For example, emitting a capsule/microencapsulated chemical coating with some kind of herbicide/pollen/or other premixed chemical for agricultural treatment inside and subsequently emitting a solvent that destroys the coating so that the remaining substance inside the capsule interacts with the target agricultural object for treatment. As another example, a capsule with chemical+ water to dissolve coating.

9. A dry chemical and water may be applied to an agricultural object. This allows a chemical reaction to occur and affects the agricultural object when the chemical is mixed with water. In other words, the dry chemical when mixed with water cause a chemical reaction to occur on the particular agricultural object.

10. A two-part epoxy may be applied to an agricultural object. This application of chemicals is useful for situations where one wants to "glue" a plant receptor shut (so it cannot germinate, pollenate, etc.). For example, a liquid polymer may be applied to the agricultural object, and then a hardener applied to the liquid polymer, causing the liquid polymer to harden.

In one embodiment, a chemical mixture (e.g., a dangerous chemical and other chemical=not dangerous mixture) is stored in a source tank. As the chemical mixture is emitted, the other chemical evaporates while the chemical mixture is in mid-air due to a hot air blowing at it. Then the remaining substance that reaches the plant is just the desired chemical for plant treatment. For example, the chemical mixture may be a premix acid (e.g., an acid and a desired liquid chemical [mixture]=acidic desired liquid mixture) into a liquid acid mixture. The chemical mixture may be emitted from a treatment unit for deposit onto an agricultural object. A hot air blower may be provided as part of the system, to blow hot air close to the chemical mixture being emitted. Also, the spray tip or nozzle may be heated using a heater or heating element. This allows the chemical tubing to remain at an ambient or cooler temperature than the heated spray nozzle or heated air. As the chemical mixture is released from a spray nozzle and travels towards the agricultural object, the acidic portion of the chemical mixture may evaporate,

leaving the desired liquid chemical still travelling towards plant. This provides a level of safety while storing the chemical mixture.

In another embodiment, and inert material may be sprayed onto multiple agricultural objects (e.g., onto multiple plants or crops). The system may then spray onto the pre-treated agricultural objects (having the inert material sprayed onto the objects) a reactive material with precision spray at each of the individual agricultural objects. For example, this allows for an intentional overspray of a first agricultural treatment fluid (for example the inert material), which can allow for a larger error (for example splash error or targeting error, or spray size error) range. For example, in a system where a chemical solution is emitted as fluid projectiles onto desired target object, a slight error in targeting from the agricultural system may cause the fluid projectile to hit another object that should not interact with the fluid, or cause the fluid projectile to splash onto another object that should not interact with the splash. With the two part epoxy, the agricultural system would require two instances of the same slight error in targeting to cause a negative effect on the object that should not interact with the projectile or splash since receiving a hit or a splash from only one of the two projectiles would have no chemical effect on the agricultural object.

In one example, two separate chemicals can be stored in two or more separate tanks or storage units on an agricultural systems. Each of the chemicals themselves, while separate do not trigger any desired effect when in contact with a target agricultural object. In some examples, the separate chemicals themselves are not a hazard to humans or at least are less hazardous to process and store than that of mixing the chemicals together. In one example, when the two separate chemicals mix together, the mixed chemical compound can be considered activated, such that the mixed chemical compound, upon interacting with a target agricultural object can cause a desired effect on the target agricultural object. In one example, the mixed chemical compound has a short shelf life, or can be hazardous to humans, thus, having a treatment system that will allow each of the two separate chemicals to interact with each other only open contacting the surface of the target agricultural object will preserve the potency or effectiveness of the desired result upon treating the target object, as well as reduce hazards.

FIG. 32B illustrates another example configuration of a system shown at different time intervals. A vehicle 3250 is depicted moving in a forward direction at time intervals T_1 and T_2 . At time interval T_1 , the treatment unit 3254 positions a treatment head or spray head at the agricultural object 3260 and emits a first fluid projectile 3264 at the agricultural object 3260. The vehicle 3250 moves in a forward direction 3252. At time interval T_2 , the treatment unit 3254 positions the spray head at the agricultural object 3260, and emits a second fluid projectile 3264 at the agricultural object 3260. As the vehicle moves in the forward direction 3252, the system may repeatedly adjust the spray head at T_n intervals such that the spray head may emit a fluid projectile at the agricultural object at n different positions, which can include treatments of the same type and same fluids or treatments of different types such that different fluid projectiles are emitted at object 3260 in the same pass that the vehicle 3250 is travelling along a path.

FIG. 33 illustrates example implementations of method 3300 that may be performed by some embodiments of the systems described above including system 100, agricultural treatment system 400, system 600, and system 800. At step 3310, the system accesses an image of an agricultural scene

having a plurality of objects. For example, the system obtains, using one or more image capture device, a plurality of images of agricultural objects. At step 3320, the system detects a plurality of target objects in the real world based on an object detection and localization in the image. As discussed herein, the system may maneuver a treatment head and orient the treatment head to emit a fluid project towards the target object. At step, 3340 the system determines a fluid profile for treating the first target object. The system may evaluate the target object and based on type or some other characteristic of the target object, determine a fluid profile for treating the target object. The fluid profile may include treatment parameters as to the fluid to emit from a treatment unit, including, but not limited to, spray volume, fluid source (e.g., which fluid source tank to use), number of bursts to emit, splat size, spot size, fluid temperature and spray duration. At step 3350, the system sends instructions of a first treatment parameter to a treatment unit 3350. At step 3360, based on the fluid profile, and/or the treatment parameters, the system activates the treatment unit to emit a fluid projectile at the first target object. At step 3370, the system identifies and tracks a second target object in the real world. At step 3380, the system determines a second fluid profile for treating the second target object. As step 3390, the system sends instructions of one or more second treatment parameters to the treatment unit. At step 3392, the system activates the treatment unit to emit a second fluid projectile at the second target object. The treatment parameters of the first and second profile may be the same or may be different treatment parameters.

While steps 3370-3392 discuss a second target object, the system may also treat the first target object again, but with a second fluid projectile. The system may determine a second fluid profile for treating the first target object. The system sends instructions of one or more second treatment parameters to the treatment unit. The system activates the treatment unit to emit a second fluid projectile at the first target object. For example, the first fluid projectile may be obtained from a first fluid source tank and the second fluid projectile may be obtained from a second fluid source tank. As discussed previously, the first and second fluid source tanks may hold or contain different chemicals that when applied to the same target object, the chemicals may undergo a chemical reaction. For example, the first fluid projectile may be a liquid polymer and the second fluid projectile is a liquid hardener, and the liquid polymer may harden when the liquid polymer and liquid hardener are mixed on the first target object.

FIG. 34 illustrates example implementations of method 3400 that may be performed by some embodiments of the systems described above including system 100, agricultural treatment system 400, system 600, and system 800. At step 3420, the system identifies and tracks a first target object in the real world. For example, the system obtains, using one or more image capture device, a plurality of images of agricultural objects. As discussed herein, the system may maneuver a treatment head and orient the treatment head to emit a fluid project towards the target object. At step 3430, the system determines a first desired spot size for treating the first target object. At step 3440, the system determines a fluid profile for the first desired spot size. At step 3450, the system sends instructions of a first treatment parameter for the fluid profile to a treatment unit. At step 3460, the system determines a first fluid pressure for the fluid to be emitted by the treatment unit. At step 3470, the system sends instructions based on the first fluid pressure parameter to activate a solenoid to allow release of a first fluid projectile. At step

3480, the system orients treatment unit and activates the solenoid such that the solenoid of the fluid regulator releases the first fluid projectile at the desired fluid pressure. At step 3490, the system determines a spray profile associated with the first fluid projectile. The system emits the first fluid projectile at the first target object. The following discussion further describes these operations.

As discussed above, the system may use a pump to create pressurized fluid from the pump to a solenoid. The system may send specific voltage and pressure instructions to the solenoid such that, accounting for the distance between the turret nozzle to the surface of the target, a $\frac{1}{8}$ inch to 5-inch diameter of the spot of the spray can hit the target. Moreover, the system may variably and incrementally change the liquid projectile for every spray. The system may utilize a predetermined base pressure from the pump to the solenoid, and then open and close the solenoid by providing voltage instructions (for example, 24 or 48 volts). The spot of the spray of the emitted projectile may be controlled by the system to achieve a desired spray amount and spray diameter to cover an area of a target object.

While the solenoid is completely closed, the fluid may be pressurized to a particular psi. For example, a pump may operate to pressurize the fluid in the range of 1-200 psi. A working line psi from the pump to the solenoid may be about 60 psi when the solenoid is completely closed. An emission tubing from the solenoid to the nozzle and/or spray tip would have a psi less than the working line psi when the solenoid is closed. The opening of the solenoid releases the pressurized fluid from the working line and causes the fluid to fill into the emission line and out through the nozzle and/or spray tip. Over a period of time, the emission line may build up pressure and cause the pressurized fluid to emit through the nozzle and/or spray tip. By quickly opening and closing the solenoid, the system may emit intermittent bursts of the fluid from the working line causing the fluid to emit as a projectile via the nozzle and/or spray tip.

As fluid leaves the pressurized working line behind the solenoid, the pressurized working line behind the solenoid (from the pump to the solenoid) loses a small, but negligible amount of pressure. As more and more fluid leaves the working line (for example, 100 bursts or shots), the overall drop in pressure becomes nontrivial. In certain situations, the pump may not be able to compensate for the drop in pressure until a pressure drop becomes significant enough. For example, the pressure may be 60 psi up to the wall of the solenoid. As the treatment unit emits 100 bursts of fluid projectiles, the pressure behind the solenoid may incrementally drop, for example, to 40 psi. At 40 psi, a pressure sensor may inform the pump and solenoid to accommodate for the pressure drop, and the system will increase pressure back to 60 (and in some instances may increase the psi in the working line to more than 60 psi to quickly reach a base 60 psi).

A situation may occur in that dropping the pressure in the working line from 60 psi to the 40 psi, and then suddenly increasing back to 60 psi, incremental bursts (e.g., shots) may be emitted at slightly different starting pressures. So if the system opens the solenoid at the same rate and same voltage every time, the solenoid actually may open at different rates (because the pressure pushing at the wall was different), so emitted fluid projectiles are incrementally different. To account for the variability of the pressure in the working line, the system may account for the incremental drop in pressure due. The system may generate instructions to change voltages sent to the solenoid so to maintain the same droplet size and/or fluid volume. The amount by which

the solenoid opens each time may be slightly different so as to maintain the same trajectory, volume and/or droplet for the size emitted fluid projectile (thereby accounted for the differences in the psi for each burst or shot of the fluid because the pressure pushing at the wall is different). The system may be calibrated and/or configured to open and close the solenoid at different time intervals and different time durations such that the amount of pressure is the same for each shot. Alternatively, the system may be calibrated to open and close at different time intervals and for different durations such that the amount of pressure is similar. The system may use a spray profile that controls the timing and duration of the opening and closing of the solenoid.

For example, the system may receive one or more images of an agricultural scene having one or more agricultural objects, such as plant object. The system may then detect a plurality of target objects based on the received one or more images. The system may identify a first target object in the real world from the detecting of the plurality of target objects in the received one or more images. The system may determine a set of first treatment parameters (e.g., spray profile) for the first target object. Based on the first set of treatment parameters, the system may instruct a treatment unit to emit a fluid projectile at the first target object. The first set of treatment parameters may include one or more of: a spray speed for an emitted fluid, a spray size for an emitted fluid, a spat profile and/or a spray duration. Based on the spray profile, the system controls a solenoid to release a pressurized fluid that has been pumped from a fluid source, the emitted fluid projectile that includes a portion of the released pressurized fluid. The system may adjust the opening and closing of the solenoid to account for hysteresis band of a pressure drop in the pressurized fluid.

The system may continue to treat multiple other target objects, such as agricultural objects that are plant objects. The system may identify a second target object in the real world in the received one or more images. The identification of the other target objects may occur in real time concurrently or after the first target object is detected. The system may determine a set of second treatment parameters (e.g., a spray profile) for a second target object. The set of second treatment parameters may be the same or different from the first treatment parameters for the first target object. For example, the system may determine a second spray size for the second target object that is different from a spray size for the first target object. The system may determine a second spray speed for the second target object that is different from a spray speed for the first target object. The system may determine a second spray volume for the second target object that is different from a spray volume for the first target object. Also, the system may determine a target spat size for the second target object that is different from a target spat size for the first target object. For example, a spat size of a same about of a drop of fluid may have a different impact shape when it impacts a surface. The impact shape may be changed based on the trajectory speed of the drop of fluid when it impacts the surface of the target.

After the second set of parameters are determined, then the system may activate the treatment unit to emit a second fluid projectile at the second target object based on the second treatment parameters. The system may confirm whether the second fluid projectile contacted the second target object (e.g., based on sensor data comprising digital imagery, lidar data, sonar data, radar data or a combination thereof).

While the foregoing describes examples of the pressurized fluid spray with particular pressure and spray profiles,

the system may be configured within various ranges. For example, the emitted spray may be emitted from about 1 millisecond to 1 second between the range of 40 to 80 psi. The system may be capable of generating a pressure of about 1 psi to 2700 psi in the working line. The volume of fluid released and emitted may be from about 1 milliliter to 1000 milliliters. The target splat diameter can be about a dime size at 40 psi to about a quarter size at 60 psi. The projectile/droplet size may be about 1 millimeter to about 100 millimeters diameter in a single drop (not volumetric).

The terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an”, and “the” are intended to comprise the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises”, “comprising”, or a combination thereof, when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

While the disclosure has been particularly shown and described with reference to specific examples thereof, it should be understood that changes in the form and details of the disclosed examples may be made without departing from the scope of the disclosure. Although various advantages, aspects, and objects of the present disclosure have been discussed herein with reference to various examples, it will be understood that the scope of the disclosure should not be limited by reference to such advantages, aspects, and objects. Rather, the scope of the disclosure should be determined with reference to the claims.

Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers, or the like.

It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as “identifying” or “determining” or “executing” or “performing” or “collecting” or “creating” or “sending” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system’s registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage devices.

The present disclosure also relates to an apparatus for performing the operations herein. This apparatus may be specially constructed for the intended purposes, or it may

comprise a general-purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a computer readable storage medium, such as, but not limited to, any type of disk, optical disks, CD-ROMs, and magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, solid state drives, flash drives, SATA drives, NAND and 3D NAND flash drives, or any type of media suitable for storing electronic instructions, each coupled to a computer system bus.

Various general-purpose systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the method. The structure for a variety of these systems will appear as set forth in the description above. In addition, the present disclosure is not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the disclosure as described herein.

The present disclosure may be provided as a computer program product, or software, that may include a machine-readable medium having stored thereon instructions, which may be used to program a computer system (or other electronic devices) to perform a process according to the present disclosure. A machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium such as a read only memory (“ROM”), random access memory (“RAM”), magnetic disk storage media, optical storage media, flash memory devices, etc.

In the foregoing disclosure, implementations of the disclosure have been described with reference to specific example implementations thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of implementations of the disclosure as set forth in the following claims. The disclosure and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

What is claimed is:

1. A treatment apparatus comprising: two or more fluid tanks, including a first fluid tank configured to hold a first fluid, and a second fluid tank configured to hold a second fluid; a first fluid regulator comprising a first fluid input port, a first fluid output port and a first fluid control valve; a first tube fluidly coupled from a first fluid pump to the first fluid input port; a second fluid regulator comprising a second fluid input port, a second fluid output port and a second fluid control valve; a second tube fluidly coupled from a second fluid pump to the second fluid input port; a treatment unit comprising two or more moveable spraying tips, including a first spraying tip and a second spraying tip, configured to be adjustable in at least a first axis and a second axis; a third tube fluidly coupled from the first fluid output port to the first spraying tip; a fourth tube fluidly coupled from the second fluid output port to the second spraying tip; a controller configured to provide instructions to the first fluid regulator to open and close the first fluid control valve to release an amount of the first fluid from the first fluid tank, and the controller being configured to provide instructions to the second fluid regulator to open and close the second fluid control valve to release an amount of the second fluid from the second fluid tank, wherein the controller is configured to control the first fluid regulator to open and close the first

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fluid control valve at different time intervals such that an amount of pressure in the first tube causes bursts of fluid projectiles of a first fluid to each emit in a predetermined spray volume and a predetermined spray diameter to cover an area of a target object; and one or more processors configured to perform the operations of: identifying a first target agricultural object; based on a first spray profile, activating the treatment unit to emit the first fluid at the first target agricultural object, wherein the first fluid regulator is moved to an opened position for an amount of time such that the first fluid is emitted, via the first spraying tip, as a first fluid projectile at the first target agricultural object for a first spray duration; and based on a second spray profile, activating the treatment unit to emit a second fluid at a second target agricultural object, wherein the second fluid regulator is moved to an opened position for an amount of time such that the second fluid is emitted, via the second spraying tip, as a second fluid projectile at the second target agricultural object for a second spray duration.

2. The treatment apparatus of claim 1, wherein the first fluid is obtained from a first fluid source tank and the second fluid is obtained from a second fluid source tank.

3. The treatment apparatus of claim 2, wherein the first fluid is a liquid polymer and the second fluid is a liquid hardener, and the liquid polymer harden when the liquid polymer and liquid hardener are mixed on the first target object, wherein the second target agricultural object is the same as the first agricultural target object.

4. The treatment apparatus of claim 2, wherein the first fluid is a first chemical and the second fluid is a second chemical, and the first and second chemicals undergo a chemical reaction when applied to the first target object, wherein the second target agricultural object is the same as the first target agricultural object.

5. The treatment apparatus of claim 2, wherein the first fluid is reactant that takes part in and undergoes change during a chemical reaction, and the second fluid is a reagent that causes a chemical reaction of the reactant, wherein the second target agricultural object is the same as the first target agricultural object.

6. The treatment apparatus of claim 1, wherein the first spray profile is for a volumetric spray applied to an area of the first target agricultural object, and the second spray profile is for a targeted spray applied to a portion of the area, wherein the second target agricultural object is the same as the first target agricultural object.

7. The treatment apparatus of claim 1, wherein a fluid from the first fluid source is applied to multiple agricultural objects before applying the second fluid at the first target object.

8. The treatment apparatus of claim 1, wherein the emitted first fluid is a volumetric spray applied to an entire surface area of the first target agricultural object.

9. The treatment apparatus of claim 1, wherein the targeted spray is applied to a root portion of the first target agricultural object.

10. The treatment apparatus of claim 1, wherein the first and/or second spray profile is based on a size of a detected agricultural object or a location of the detected agricultural object.

11. The treatment apparatus of claim 1, wherein the one or more processors are further configured to perform the operations of:

determining a desired spot size for treating the first target agricultural object.

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12. The treatment apparatus of claim 11, wherein the one or more processors are further configured to perform the operations of:

in response to a determining the desired spot size, causing the treatment unit to emit a fluid with a first spray diameter cross section to create the desired spot size.

13. The treatment apparatus of claim 1, wherein the one or more processors are further configured to perform the operations of:

based on a spray profile, emitting multiple bursts of fluid at the first target agricultural object.

14. The treatment apparatus of claim 1, wherein the first fluid is emitted for a spray duration based on the first spray profile.

15. The treatment apparatus of claim 1, wherein the one or more processors are further configured to perform the operations of:

determining a first fluid pressure of the treatment unit, wherein the first fluid pressure is in the range of 1-2700 PSI.

16. The treatment apparatus of claim 1, further comprising:

wherein the first spraying tip and the second spraying tip are moveable by a spraying head assembly;

a first motor and a second motor, wherein a rotation of the first motor causes the spraying head assembly to pivot along first axis, and the rotation of the second motor causes the spraying head assembly to pivot along the second axis, and wherein the controller is configured to control the first motor and second motor.

17. The treatment apparatus of claim 1, wherein the first fluid tank holds the first fluid comprising a fertilizer, and a second fluid tank holds a second fluid comprising a herbicide.

18. The treatment apparatus of claim 1, further comprising the operations of:

based on the identified first target agricultural object mixing a concentration of the first fluid and a concentration of the second fluid for application to the first target agricultural object.

19. The treatment apparatus of claim 1, further comprising:

a second pump;
a second fluid tank configured to hold a second fluid;
a chemical mixer configured to receive the first fluid and the second fluid;

wherein the operations further comprise:

instructing the first pump to pump an amount of fluid from the first fluid tank into the chemical mixer; and
instructing the second pump to pump an amount of fluid from the second fluid tank into the chemical mixer for mixing with the first fluid.

20. The treatment apparatus of claim 1, wherein the one or more processors are further configured to perform the operations of:

based on the identified first target agricultural object, determining the first spray profile for treating the first target agricultural object;

sending instructions of a first treatment parameter to the treatment unit;

based on the identified second target agricultural object, determining the second spray profile for treating the second target agricultural object; and

sending instructions of a second treatment parameter to the treatment unit.

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